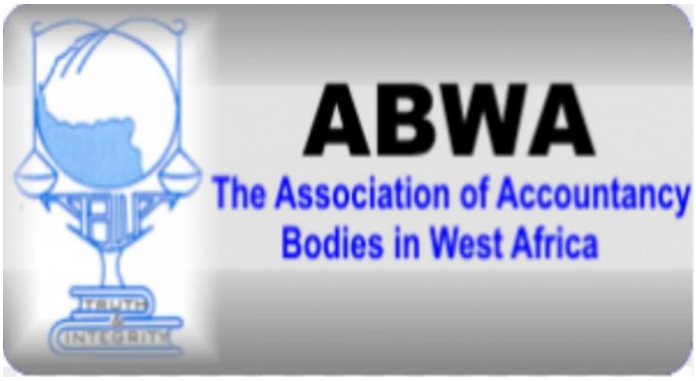



# ATSWA

**ACCOUNTING TECHNICIANS SCHEME WEST AFRICA**

STUDY TEXT

# COST ACCOUNTING

*ASSOCIATION OF ACCOUNTANCY BODIES IN WEST AFRICA (ABWA)*

**ACCOUNTING TECHNICIANS SCHEME WEST AFRICA (ATSWA)**

**STUDY TEXT FOR**

# COST ACCOUNTING

*FIFTH EDITION*

Copyright (c) 2025 by Association of Accountancy Bodies in West Africa (ABWA). All rights reserved. No part of this publication may be reproduced or distributed in any form or by any means, or stored in a database or retrieval system, including, but not limited to, any network or other electronic storage or transmission, or broadcast for distance learning without the prior written consent of the copyright owner.

*Published by*
**ABWA PUBLISHERS**

**DISCLAIMER**

This book is published by ABWA; however, the views are entirely those of the writers.

# PREFACE

## INTRODUCTION

The Council of the Association of Accountancy Bodies in West Africa (ABWA) recognised the difficulty of students when preparing for the Accounting Technicians Scheme West Africa examinations. One of the major difficulties has been the non-availability of study materials purposely written for the scheme. Consequently, students relied on text books written in economic and socio-cultural environments quite different from the West African environment.

## AIM OF THE STUDY TEXT

In view of the above, the quest for good study materials for the subjects of the examinations and the commitment of the ABWA Council to bridge the gap in technical accounting training in West Africa, led to the production of this Study Text.

The Study Text assumes a minimum prior knowledge and every chapter reappraises basic methods and ideas in line with the syllabus.

## READERSHIP

The Study Text is primarily intended to provide comprehensive study materials for students preparing to write the ATSWA examinations.

Other beneficiaries of the Study Text include candidates of other Professional Institutes, students of Universities and Polytechnics pursuing undergraduate and post graduate studies in Accounting, advanced degrees in Accounting as well as Professional Accountants who may use the Study Text as reference material.

## APPROACH

The Study Text has been designed for independent study by students and as such concepts have been developed methodically or as a text to be used in conjunction with tuition at schools and colleges. The Study Text can be effectively used as a course text and for revision. It is recommended that readers have their own copies.

## FOREWORD

The ABWA Council, in order to actualize its desire and ensure the success of students at the examinations of the Accounting Technicians Scheme West Africa (ATSWA), put in place a Harmonisation Committee, to among other things, facilitate the production of Study Texts for students. Hitherto, the major obstacle faced by students was the dearth of Study Texts which they needed to prepare for the examinations.

The Committee took up the challenge and commenced the task in earnest. To start off the process, the existing syllabus in use by some member Institutes were harmonized and reviewed. Renowned professionals in private and public sectors, the academia, as well as eminent scholars who had previously written books on the relevant subjects and distinguished themselves in the profession, were commissioned to produce Study Texts for the twelve subjects of the examination.

A minimum of two Writers and a Reviewer were tasked with the preparation of Study Text for each subject. Their output was subjected to a comprehensive review by experienced imprimaturs. The Study Texts cover the following subjects:

**PART I**

1 Basic Accounting
2 Economics
3 Business Law
4 Communication Skills

**PART II**

1 Financial Accounting
2 Public Sector Accounting
3 Quantitative Analysis
4 Information Technology

**PART III**

1 Principles of Auditing & Assurance
2 Cost Accounting
3 Taxation
4 Management

Although, these Study Texts have been specially designed to assist candidates preparing for the technicians examinations of ABWA, they should be used in conjunction with other materials listed in the bibliography and recommended text.

**PRESIDENT, ABWA**

## STRUCTURE OF THE STUDY TEXT

The layout of the chapters has been standardized so as to present information in a simple form that is easy to assimilate.

The Study Text is organised into chapters. Each chapter deals with a particular area of the subject, starting with a summary and learning objective of sections contained therein.

The introduction also gives specific guidance to the reader based on the contents of the current syllabus and the current trends in examinations. The main body of the chapter is subdivided into sections to make for easy and coherent reading. However, in some chapters, the emphasis is on the principles or applications while others emphasise method and procedures.

At the end of each chapter is found the following:

- Summary;
- Points to note (these are used for purposes of emphasis or clarification);
- Examination type questions; and
- Suggested answers.

## HOW TO USE THE STUDY TEXT

Students are advised to read the Study Text, attempt the questions before checking the suggested answers.

## ACKNOWLEDGMENTS

The ATSWA Harmonisation and Implementation Committee, on the occasion of the publication of the first edition of the ATSWA Study Text acknowledge the contributions of the following groups of people. The ABWA Council, for their inspiration which gave birth to the whole idea of having a West African Technicians Programme. Their support and encouragement as well as financial support cannot be overemphasized. We are eternally grateful.

To The Councils of the Institute of Chartered Accountants of Nigeria (ICAN), and the Institute of Chartered Accountants, Ghana (ICAG), Institute of Chartered Accountants Sierra Leone (ICASL), Gambia Institute of Chartered Accountants (GICA) and the Liberia Institute of Certified Public Accountants (LICPA) for their financial commitment and the release of staff at various points to work on the programme and for hosting the several meetings of the Committee, we say kudos.

We are grateful to the following copyright holders for permission to use their intellectual properties:

- The Institute of Chartered Accountants of Nigeria (ICAN) for the use of the Institute's examination materials;
- International Federation of Accountants (IFAC) for the use of her various publications; International Accounting Standards Board and (IASB) for the use of International Accounting Standards and International Financial Reporting Standards;
- Owners of Trademarks and Trade names referred to or mentioned in this Study Text.

We have made every effort to obtain permission for use of intellectual materials in this Study Text from the appropriate sources.

We wish to acknowledge the immense contributions of the writers and reviewers of this manual.

Our sincere appreciation also goes to various imprimaturs and workshop facilitators. Without their input, we would not have had these Study Texts. We salute them.

***Chairman***

***ATSWA Harmonization & Implementation Committee***

A new syllabus for the ATSWA Examinations has been approved by ABWA Council and the various PAOs. Following the approval of the new syllabus which becomes effective from the September 2024 diet a team was constitutes to undertake a comprehensive review of the Study Texts in line with the syllabus under the supervision of an editorial board.

The Reviewers and Editorial Board members are:

**REVIEWERS**

This Study text was reviewed by:

- Professor T. O. Asaolu — Vice Chancellor of the University of Ilesa
- Mr. S. A. Adegboye — Sunny Adegboye & Co (Chartered Accountants)
- Dr. Olajumoke R. Ogunniyi — Kola Daisi University, Ibadan

**EDITORIAL BOARD**

The Editorial Board Members are:

- Mr. Sylvester Nwanna, FCA — Chairman, ATSWA Examinations Committee
- Dr. Lanre Olasunkanmi, FCA — ICAN, Registrar/Chief Executive
- Dr. Ijeoma Anaso, FCA — ICAN, Deputy Registrar, Technical Services
- Mr. Ejike Ojukwu, FCA — Coordinator, Examinations
- Mr. Kebiru Momoh, FCA — HOD, ATSWA

**COST ACCOUNTING TEXT**

**TABLE OF CONTENTS**


| | | |
|---|---|---|
| | Cover page | i |
| | Publisher | ii |
| | Preface | iii |
| | Foreword | iv |
| | Structure of the study text | v |
| | Acknowledgement | vi |
| | Editorial Board | vii |
| | Table of contents | viii |
| CHAPTER/SECTION | TITLE/ SUBTITLE | |
| 1 | INTRODUCTION TO COST ACCOUNTING | 1 |
| | Chapter Contents | 1 |
| 1.0 | Objectives | 1 |
| 1.1 | Introduction | 1 |
| 1.2 | Definition of Cost | 1 |
| 1.2.1 | Definition of Cost Accounting | 2 |
| 1.2.2 | Cost Objectives | 2 |
| 1.2.3 | The Objectives of Costing Accounting | 3 |
| 1.2.4 | Functions of Cost Accounting/ The Role of Cost Accountant | 3 |
| 1.2.5 | Differences Between Cost Data and Cost Information | 4 |
| 1.2. | Advantages (Benefits) of Cost Accounting | 4 |
| 1.2.7 | Disadvantages (Drawbacks) of Cost Accounting | 5 |
| 1.2.8 | Relationship Between Cost Accounting and Financial Accounting | 5 |
| 1.2.9 | Conditions Necessary for an Effective Cost Accounting System | 7 |
| 1.2.10 | Steps In Installing a Cost Accounting System | 7 |
| 1.2.11 | Problems Inherent in The Installation of a Cost Accounting System | 8 |
| 1.2.12 | Principles of A Good Cost Accounting System | 9 |
| 1.3 | Scope of Cost Accounting | 9 |
| 1.3.1 | Elements of Cost | 9 |
| 1.3.2 | Methods of Costing | 12 |
| 1.3.3 | Techniques Of Costing | 13 |
| 1.3.4 | Information Exclusively Generated from A Cost Accounting System | 13 |
| 1.3.5 | Cost Accounting as Part of The Management Information System | 13 |
| 1.3.6 | Organizations Other Than Manufacturing Companies Where Cost Accounting Can Be Applied | 14 |
| 1.3.7 | Value of Cost Information in Management Decision Making Process | 15 |
| 1.3.8 | Impact of Information Technology on The Generation, Evaluation and Presentation of Cost Information | 15 |

| | | |
|---|---|---|
| 1.3.9 | Roles Required of Accounting Technicians in Collection, Analysis and Presentation of Cost Information | 16 |
| 2 | COST CLASSIFICATION, SEGREGATION AND CODIFICATION | 17 |
| | Chapter Contents | 17 |
| 2.0 | Objectives | 17 |
| 2.1 | Introduction | |
| 2.1.1 | Classification of Cost | 17 |
| 2.1.1.a | Classification According to Elements of Cost | 17 |
| 2.1.1 b | Classification According to Nature | 18 |
| 2.1.1.c | Classification According to Function | 21 |
| 2.1.1.d | Classification According to Behaviour | 22 |
| 2.1.1.e | Classification As Product Costs or Period Costs | 23 |
| 2.1.1.f | Classification According To Controllable and Uncontrollable Costs/Revenue | 23 |
| 2.1.1.g | Classification of Cost As Relevant and Irrelevant Costs | 24 |
| 2.1.1.h | Other Classifications of Cost | 24 |
| 2.1.2 | Cost Segregation | 26 |
| 2.1.3 | Problems Associated With The Conventional Classification of Cost According to Behaviour | 26 |
| 2.1.4 | Typical Cost Patterns | 27 |
| 2.1.5 | Analysis of The Effect of Changing Activity Levels on Costs | 30 |
| 2.2 | Cost Centre and Cost Unit | 30 |
| 2.2.1 | Cost Centre | 30 |
| 2.2.2 | Cost Unit | 31 |
| 2.2.3 | Profit Centre | 31 |
| 2.3 | Codification | 31 |
| 2.3.1 | Cost Coding System | 31 |
| 2.3.2 | Characteristics of A Good Designed Coding System | 31 |
| 2.3.3 | Methods of Coding | 31 |
| 2.4 | CHAPTER SUMMARY | 33 |
| | MULTIPLE CHOICE AND SHORT ANSWER QUESTIONS | 33 |
| | EXAMINATION TYPE QUESTIONS | 34 |
| 3 | ELEMENTS OF COSTS: MATERIALS | 37 |
| | Chapter Contents | 37 |
| 3.0 | Objectives | 37 |
| 3.1 | Introduction | 37 |
| 3.2 | Basic Classifications of Materials | 37 |
| 3.2.1 | What are Materials | 37 |
| 3.2.2 | Significance of Inventories | 38 |
| 3.3 | The Store Functions and Activities | 39 |
| 3.3.1 | Purchasing | 39 |
| 3.3.2 | Operating The Store | 40 |
| 3.3.3 | Inventory Control | 40 |
| 3.3.4 | Summarized Overview of Stores Function | 40 |
| 3.4 | Factors That Facilitate Effective Materials Cost Control | 40 |
| 3.5 | Procurement of Materials and Documentation | 41 |

| | | |
|---|---|---|
| 3.5.1 | Responsibilities of Purchasing or Procurement Unit | 41 |
| 3.5.2 | Methods of Purchasing | 41 |
| 3.5.3 | Settlement Methods | 42 |
| 3.5.4 | Summary of The Stages of The Purchasing Function | 42 |
| 3.6 | Receipt And Storage of Materials | 44 |
| 3.6.1 | Store Operations | 44 |
| 3.6.2 | The Factors That Influences The Design of Procedures For Operating The Store | 45 |
| 3.6.3 | Receiving Goods | 45 |
| 3.6.4 | Holding Inventory | 45 |
| 3.6.5 | Principles of Storage | 46 |
| 3.6.6 | Issuing Goods | 46 |
| 3.6.7 | Store Location And Layout | 46 |
| 3.6.8 | Factors That Facilitate Effective Materials Cost Control | 47 |
| 3.7 | Pricing of Issues And Valuation of Inventory | 48 |
| 3.7.1 | Introduction | 48 |
| 3.7.2 | Methods of Pricing Issues And Valuing Inventory | 49 |
| 3.7.3 | FIFO | 49 |
| 3.7.4 | LIFO | 49 |
| 3.7.5 | The Simple Average Method | 50 |
| 3.7.6 | Weighted Average Price Method | 50 |
| 3.7.7 | Standard Price | 51 |
| 3.7.8 | Replacement Price | 51 |
| 3.7.9 | Specific Identification | 52 |
| 3.8 | Inventory Controls Levels | 52 |
| 3.8.1 | Re-Order Level Of Inventory | 52 |
| 3.8.2 | Minimum Inventory Level | 53 |
| 3.8.3 | Maximum Inventory Level | 53 |
| 3.8.4 | Re-Order Quantity | 54 |
| 3.8.5 | Average Inventory Level | 56 |
| 3.9 | Inventory Control Records And Procedures | 57 |
| 3.9.1 | The Need For Inventory Control Records | 57 |
| 3.9.2 | Inventory Control Records | 57 |
| 3.10 | The Perpetual Inventory System | 71 |
| 3.11 | Inventory Taking | 71 |
| 3.11.1 | Periodic Inventory Taking | 71 |
| 3.11.2 | Continuous Inventory Taking | 72 |
| 3.11.3 | Effective Inventory Taking | 72 |
| 3.11.4 | Causes of Stock Discrepancies | 73 |
| 3.12 | CHAPTER SUMMARY | 73 |
| | MULTIPLE CHOICE AND SHORT ANSWER QUESTIONS | 74 |
| | EXAMINATION TYPE QUESTIONS | 75 |
| 4 | ACCOUNTING FOR LABOUR | 80 |
| | Chapter Contents | 80 |
| 4.0 | Objectives | 80 |
| 4.1 | Introduction | 80 |
| 4.2 | Labour Administration and Documentation | 80 |
| 4.2.1 | Labour : A Factor of Production. | 80 |
| 4.2.2 | Labour Recruitment Process | 80 |
| 4.2.3 | Labour Induction and Placement | 81 |
| 4.2.4 | Labour Timing and Assessment | 81 |

| | | |
|---|---|---|
| 4.2.5 | Labour Behaviour and Control | 82 |
| 4.2.6 | Job Evaluation | 82 |
| 4.2.7 | Merit Rating | 82 |
| 4.2.8 | Work Study | 83 |
| 4.3 | Labour Remuneration Methods | 83 |
| 4.3.1 | Time Rate Methods of Remuneration | 84 |
| 4.3.2 | Piece Rate (Payment by Result) Method | 84 |
| 4.3.3 | Differential Piece Rate | 86 |
| 4.3.4 | Premium Bonus Schemes | 86 |
| 4.3.4.1 | Individual Incentive Schemes | 87 |
| 4.3.4.2 | Advantages and Disadvantages of Individual Incentive Schemes | 90 |
| 4.4 | Overtime Remuneration Schemes | 90 |
| 4.5 | Idle (Waiting)Time | 91 |
| 4.6 | Advantages and Disadvantages of Group Incentive Schemes | 96 |
| 4.7 | Direct and Indirect Cost of Labour | 96 |
| 4.8 | Labour Cost Minimization Techniques | 100 |
| 4.9 | Labour Turnover | 100 |
| 4.10 | Payroll Preparation and Accounting | 105 |
| 4.10.1 | Types of Payroll for Labour | 105 |
| 4.10.2 | Labour Payroll Preparation | 105 |
| 4.10.3 | The Differences Between Labour Cost Accounting and Payroll Accounting | 109 |
| | CHAPTER SUMMARY | 110 |
| 4.11 | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 110 |
| | EXAMINATION TYPE QUESTIONS | 111 |
| 5 | ELEMENTS OF COST : OVERHEADS | 116 |
| | Chapter Contents | 116 |
| 5.0 | Objectives | 116 |
| 5.1 | Introduction | 116 |
| 5.2 | Definition And Classification of Overheads | 117 |
| 5.3 | Types And Classification of Overheads | 117 |
| 5.3.1 | Overhead Allocation and Apportionment Methods | 119 |
| 5.3.2 | The Concept of Overhead Absorption | 122 |
| 5.3.3 | Methods of Overhead Analysis | 122 |
| 5.3.4 | Elimination Method | 125 |
| 5.3.5 | Continuous Allotment | 127 |
| 5.3.6 | Simultaneous Equations | 128 |
| 5.3.7 | Overheads Absorption Methods | 130 |
| 5.4 | Calculation of Overhead Absorption Rate | 134 |
| 5.5 | Why predetermined rates are used | 136 |
| 5.6 | Blanket Overhead Absorption Rates Versus Cost Centre Overhead Absorption Rates | 136 |
| 5.7 | Determination and Treatment of Overheads: Over Absorbed or Overheads Under Absorbed | 137 |
| 5.8 | Activity – Based Costing | 140 |
| 5.8.1 | Stages in ABC | 141 |
| 5.9 | CHAPTER SUMMARY | 143 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 144 |

| | | |
|---|---|---|
| | EXAMINATION TYPE QUESTIONS | 146 |
| 6 | INTERLOCKINGANDINTEGRATEDACCOUNTS | 147 |
| | Chapter contents | 147 |
| 6.0 | Objectives | 147 |
| 6.1 | Introduction | 147 |
| 6.2 | Interlocking Accounting System and Cost ledger | 148 |
| 76.3 | Cost Ledger | 148 |
| 6.4 | Relevant Cost Accounting entries under interlocking system up to WIP Control Account. | 148 |
| 6.5 | Reconciliation of Costing and Financial Accounting Profits | 156 |
| 66.5.1 | Appropriations of Profit not dealt with in Cost Accounts | 156 |
| 6.5.2 | Financial Matters that are outside the scope of Cost Accounts | 157 |
| 6.5.3 | Items whose Costing treatment differ from those adopted in the Financial Accounts | 157 |
| 6.6 | Integrated Accounting System | 159 |
| 6.7 | Accounts Classification and Codes | 165 |
| 6.8 | Computer Assisted Costing Techniques | 166 |
| 6.9 | CHAPTER SUMMARY | 167 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 168 |
| | EXMINATION TYPE QUESTIONS | 169 |
| 7 | COSTING METHODS: | 179 |
| | SPECIFIC ORDER COSTING METHODS | 179 |
| | Chapter Contents | 179 |
| 7.0 | Objectives | 179 |
| 7.1 | Introduction | 179 |
| 7.2 | Specific Order Costing Methods | 179 |
| 7.2.1 | The Concept of Order Costing | 179 |
| 7.2.3 | Job Order Costing Procedure | 180 |
| 7.2.4 | Overhead Cost Absorption Methods in Job Costing | 181 |
| 7.2.5 | Batch Costing | 184 |
| 7.3 | Contract Costing and Accounting | 186 |
| 7.4 | CHAPTER SUMMARY | 194 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 194 |
| | EXAMINATION TYPE QUESTIONS | 196 |
| 8 | COUNTINUOUS OPERATION COSTING: | 201 |
| | PROCESSCOSTINGAND JOINTPRODUCTS COSTING | |
| | Chapter Contents | 201 |
| 8.0 | Objectives | 201 |
| 8.1 | Introduction to Continuous Operation Costing | 201 |
| 8.1.1 | Principles of Process Costing | 201 |
| 8.1.2 | Application Areas of Process Costing | 202 |

| | | |
|---|---|---|
| 8.1.3 | Distinction between Process and Order Costing | 202 |
| 8.1.4 | Production Stages in Process Costing | 203 |
| 8.1.5 | Treatment of Process Losses and Wastages | 204 |
| 8.1.6 | Determination of Equivalent Units | 207 |
| 8.1.7 | Process and Cost of Production Report | 210 |
| 8.2 | Joint Products and By-Products | 216 |
| 8.2.1 | The Concept of Joint Products | 216 |
| 8.2.2 | Accounting Treatment of By-Products | 217 |
| 8.2.3 | Methods of Apportionment of Joint Cost | 217 |
| 8.2.4 | Total Cost Determination Using NRV Method | 220 |
| 8.3 | Non Specific Order Costing Methods | 221 |
| 8.3.1 | Service Costing | 221 |
| 8.4 | CHAPTER SUMMARY | 223 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 223 |
| | EXAMINATION TYPE QUESTIONS | 225 |
| 9 | COSTING TECHNIQUES – MARGINAL COSTING | 233 |
| | Chapter Contents | 233 |
| 9.0 | Objectives | 233 |
| 9.1 | Marginal Costing Versus Absorption Costing | 233 |
| 9.1.1 | Introduction | 233 |
| 9.1.2 | The Concepts of Marginal Costing and Absorption Costing | 234 |
| 9.1.3 | Major Differences between Marginal Costing and Absorption Costing | 235 |
| 9.1.4 | Basic Equation, Advantages and Limitations of Marginal Costing | 236 |
| 9.2 | Preparation of Income Statements | 238 |
| 9.2.1 | Income Determination: Under Absorption and Marginal Costing | 238 |
| 9.2.2 | Role of Contribution | 241 |
| 9.3 | Relevant Cost and Relevant Revenue | 244 |
| 9.4 | Practical Applications of Marginal Costing | 246 |
| 9.4.1 | Make or Buy Decisions | 246 |
| 9.4.2 | Decision Criteria | 246 |
| 9.4.3 | Acceptance /Rejection of Special Order | 251 |
| 9.4.4 | Marginal Costing and Pricing Decision | 252 |
| 9.4.5 | Marginal Costing and Profit Planning | 254 |
| 9.5 | CHAPTER SUMMARY | 256 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 257 |
| | EXAMINATION TYPE QUESTIONS | 259 |
| 10 | COST-VOLUME-PROFIT ANALYSIS | 264 |
| | Chapter Contents | 264 |
| 10.0 | Objectives | 264 |
| 10.1 | Cost-Volume-Profit Analysis | 264 |
| 10.1.1 | The Concept of Cost-Volume-Profit | 264 |
| 10.1.2 | Importance of Cost-Volume-Profit Relationship | 265 |
| 10.2 | The Concept of Break-even Analysis | 265 |
| 10.2.1 | Assumptions of Break-even Analysis | 266 |
| 10.2.2 | Computation of Break-Even-Point, BEP | 266 |
| 10.2.3 | Profit-Volume Ratio, P/V Ratio | 269 |

| | | |
|---|---|---|
| 10.2.4 | The Concept of Margin of Safety | 272 |
| 10.2.5 | C.V.P Graphs | 275 |
| 10.2.5 (a) | The Break-Even Point Graph | 276 |
| | Graphical demonstration | 276 |
| 10.2.5.(b) | The Profit-Volume Chart, P/V Chart | 279 |
| 10.2.6 | Limitations of Break-Even Analysis | 281 |
| 10.3 | CHAPTERSUMMARY | 280 |
| | MULTIPLE-CHOICE AND SHORT ANSWERQUESTIONS | 281 |
| | EXAMINATION TYPE QUESTIONS | 282 |
| 11 | SIMPLE INVESTMENT APPRAISAL TECHNIQUES | 288 |
| | Chapter Contents | 288 |
| 11.0 | Objectives | 288 |
| 11.1 | Introduction | 288 |
| 11.1.1 | The Concept of Capital Budgeting | 288 |
| 11.1.2 | The Objectives of Capital Budgeting | 288 |
| 11.1.3 | Data for Capital Expenditure Decisions | 289 |
| 11.1.4 | Opportunity and Interest Costs | 289 |
| 11.2 | Capital Budgeting Methods | 290 |
| 11.2.1 | Payback Period Method (PBP) | 290 |
| 11.2.2 | Advantages of Payback Period Method | 294 |
| 11.2.3 | Disadvantages of Payback Period Method | 295 |
| 11.3 | Accounting Rate of Return Method (ARR) | 296 |
| 11.3.1 | Advantages of Accounting Rate of Return Method | 296 |
| 11.3.2 | Disadvantages of Accounting Rate of Return Method | 296 |
| 11.4 | Discounted Cash Flow Method (DCF) | 296 |
| 11.5 | Net Present Value Method (NPV) | 297 |
| 11.6 | CHAPTER SUMMARY | 300 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 300 |
| | EXAMINATION TYPE QUESTIONS | 301 |
| 12 | PERFORMANCE MEASUREMENT: BUDGETINGANDBUDGETARYCONTROL | 305 |
| | Chapter Contents | 305 |
| 12.0 | Objectives | 305 |
| 12.1 | Introduction | 306 |
| 12.2 | Definition of Budget | 306 |
| 12.2.1 | What a Budget is | 306 |
| 12.2.2 | Misconceptions about Budgets | 306 |
| 12.3 | Types of Budgets | 307 |
| 12.3.1 | Long Term Budgeting | 307 |
| 12.3.2 | Short Term Budgeting | 307 |
| 12.3.3 | Annual Budgets | 307 |
| 12.3.4 | Functional Budgets | 308 |

| | | |
|---|---|---|
| 12.3.5 | The Cash Budget | 309 |
| 12.3.6 | Master Budget | 309 |
| 12.3.7 | Principal Budget Factor | 309 |
| 12.3.8 | Forecasting in Budgeting | 309 |
| 12.4 | The Objectives of Budgeting | 309 |
| 12.4.1 | Planning | 310 |
| 12.4.2 | Co-ordination | 310 |
| 12.4.3 | Communication | 310 |
| 12.4.4 | Motivation | 310 |
| 12.4.5 | Control | 310 |
| 12.4.6 | Performance Evaluation | 311 |
| 12.5 | The Budgeting Process | 311 |
| 12.5.1 | Introducing Budgeting for the First Time | 311 |
| 12.5.2 | Necessary Conditions for Successful Budgeting | 311 |
| 12.5.3 | The Budget Committee | 311 |
| 12.5.4 | The Budget Manual | 312 |
| 12.6 | The Process of Preparing Initial Budgets | 313 |
| 12.6.1 | The Initial Budget Proposal | 313 |
| 12.6.2 | Budget Negotiation | 314 |
| 12.6.3 | Review and Approval of the Budget | 314 |
| 12.6.4 | Revision of the Budget | 314 |
| 12.6.5 | Behavioural Factors in Budgeting | 314 |
| 12.7 | Preparation of Annual Budgets | 315 |
| 12.8 | Cash Budget | 323 |
| 12.8.1 | Benefits of Cash Budget | 323 |
| 12.8.2 | Preparation of the Cash Budget | 323 |
| 12.9 | Master Budget | 327 |
| 12.10 | The Meaning of Budgetary Control | 332 |
| 12.10.1 | Objectives of Budgetary Control | 333 |
| 12.10.2 | The Budgetary Control Process | 333 |
| 12.11 | Fixed and Flexible Budgets | 334 |
| 12.11.1 | Fixed Budget | 334 |
| 12.11.2 | Flexible Budget | 334 |
| 12.12 | Budgetary Control Improvement Techniques | 338 |
| 12.12.1 | Zero Based Budgeting ( ZBB) | 338 |
| 12.12.2 | Approaches to Zero Based Budgeting | 339 |
| 12.13 | Mutually Exclusive Decision Packages | 339 |
| 12.13.1 | Incremental Decision Packages | 339 |
| 12.13.2 | The Ranking Process in ZBB | 339 |
| 12.13.3 | Advantages of Implementing Zero Based Budgeting | 340 |
| 12.13.4 | Disadvantages of Implementing Zero Based Budgets | 340 |
| 12.14 | Incremental Budgeting | 341 |
| 12.15 | Activity Based Costing Systems in Budgeting (ABC) | 342 |
| 12.15.1 | The process of Activity Based Costing System | 342 |

| | | |
|---|---|---|
| 12.15.2 | Periodic or Continuous Budgets | 343 |
| 12.16 | CHAPTER SUMMARY | 343 |
| | MULTIPLE-CHOICE AND SHORT ANSWER QUESTIONS | 344 |
| | EXAMINATION TYPE QUESTIONS | 348 |
| 13 | STANDARD COSTING TECHNIQUES | 363 |
| | Chapter Contents | 363 |
| 13.0 | Objectives | 363 |
| 13.1 | Introduction | 363 |
| 13.2 | Standard Costing | 363 |
| 13.2.1 | Standards and Standard Costing | 363 |
| 13.2.2 | Types of Standards | 364 |
| 13.2.3 | Standard Cost Defined | 365 |
| 13.2.4 | Uses of Standard Costing | 366 |
| 13.2.5 | Merits of Standard Costing | 366 |
| 13.2.6 | Limitations of Standard Costing | 366 |
| 13.3 | The Process of Standard Costing | 367 |
| 13.4 | Setting Standard Costs | 367 |
| 13.4.1 | Setting Standards for Material | 368 |
| 13.4.2 | Setting Standards for Labour | 368 |
| 13.4.3 | Setting Standards for Overheads | 369 |
| 13.4.4 | Setting Standards for Selling Price | 369 |
| 13.4.5 | Setting Standards for Sales Margin | 369 |
| 13.5 | Computation Of Standard Cost Card | 369 |
| 13.6 | Variance Analysis | 370 |
| 13.6.1 | The Concept of Variance Analysis | 370 |
| 13.6.2 | Analysis of Variances | 370 |
| 13.6.3 | Basic Cost Variances | 371 |
| 13.6.4 | Material Cost Variances (MCV) | 371 |
| 13.6.5 | Labour Cost Variances | 373 |
| 13.6.6 | Total Overhead Cost Variances (TOCV) | 376 |
| 13.6.6.1 | Variable Overhead Cost Variances | 377 |
| 13.6.6.2 | Fixed Overhead Variances | 379 |
| 13.7 | Other Variance Analysis | 385 |
| 13.7.1 | Sales Margin Variances | 385 |
| 13.7.2 | Reconciliation Of Budgeted Profit and Actual Profit | 390 |
| 13.8 | Computation Of Activity, Capacity and Efficiency Variances | 397 |
| 13.8.1 | Activity Ratio | 397 |
| 13.8.2 | Capacity Ratio | 398 |
| 13.8.3 | Efficiency | 398 |
| 13.9 | Causes of Variances | 400 |
| 13.10 | CHAPTER SUMMARY | 400 |
| | MULTIPLE CHOICE AND SHORT ANSWER QUESTIONS | 401 |
| | EXAMINATION TYPE QUESTIONS | 403 |
| 14 | RECENT DEVELOPMENTSINCOST ACCOUNTING | 411 |
| | Chapter Contents | 411 |
| 14.0 | Objectives | 411 |
| 14.1 | Just-in-time (JIT) | 412 |
| 14.2 | Kanban inventory control | 413 |
| 14.3 | Back flush Accounting (Back flush Costing) | 415 |
| 14.4 | Life Cycle Costing | 416 |

| | | |
|---|---|---|
| 14.5 | Target Costing | 416 |
| 14.6 | Advanced Manufacturing Technique (AMT) | 417 |
| | MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS | 441 |
| 15 | COST REDUCTION | 444 |
| | Chapter Contents | 444 |
| 15.0 | Objectives: | 444 |
| 15.1 | Introduction | 444 |
| 15.2 | Cost Reduction | 445 |
| 15.2.1 | Comparison Between Cost Control and Cost Reduction | 445 |
| 15.2.2 | Where to Apply Cost Reduction | 445 |
| 15.3 | Value Analysis | 446 |
| 15.4 | Value Engineering | 447 |
| 15.5 | Organisation and Methods (O & M) | 448 |
| 15.6 | Work Study | 449 |
| 15.7 | Cost Audit | 450 |
| 15.8 | Differences between Cost Audit and Financial Audit | 450 |
| 15.9 | Cost Audit Programme | 451 |
| | MULTIPLE-CHOICE AND SHORT ANSWER QUESTIONS | 454 |

# COST ACCOUNTING STUDY TEXT

## CHAPTER ONE: INTRODUCTION TO COST ACCOUNTING

## CHAPTER CONTENTS

a. Definition of Cost Accounting and related terms
b. Nature, purpose and scope of Cost Accounting
c. The role of the cost accountant
d. Difference between cost data and cost information
e. Value of Cost information (financial and non-financial) in management decision-making process
f. Impact of information technology on the generation, evaluation and presentation of cost information.
g. Benefits and drawbacks of cost accounting
h. Differences between Cost Accounting and Financial Accounting
i. Designing a Cost Accounting system

### 1.0 OBJECTIVES

After studying this chapter, readers should be able to:

a. define cost accounting;
b. state the objectives of cost accounting,
c. explain the role of the cost accountant;
d. differentiate between cost data and cost information;
e. explain the scope of cost accounting;
f. explain the benefits and drawbacks of cost accounting;
g. explain the different ways of cost classification; and
h. explain the various patterns of cost behaviour.

### 1.1 INTRODUCTION

Business organisations exist to provide goods and services at a profit. In order to earn profit, organisations must know the cost incurred in producing the goods and services so that a desired profit margin will be added to the cost to determine the selling price. This chapter explains what constitutes cost, the forms of classifying cost and different patterns of cost behaviour. It also explains the role of the cost accountant as well as the benefits and drawbacks of cost accounting.

### 1.2 DEFINITION OF COST

Cost is the amount of expenditure incurred on or attributable to a specified thing or activity. Mathematically, cost is the product of the quantity of a given resource used and the price per unit of the quantity of resource.

Costs are usually ascertained in respect of cost units or cost objectives.

### 1.2.1 **Definition of Cost Accounting**

*Cost Accounting is the application of accounting and costing principles, methods and techniques in the ascertainment of cost and the analysis of savings and or excesses as compared with previous experience or standard. (**CIMA Terminology).***

The points to note about this definition are that:
Cost accounting involves:

- The application of costing principles, methods and techniques
- The ascertainment of cost
- Analysis of savings or overages

Again cost accounting could be defined as:

"That part of management accounting which establishes budgets and standard cost and actual cost of operations, processes, departments or products and the analysis of variances, profitability or social use of funds". (**CIMA Terminology)**

The issues to remember about this definition are that, cost accounting is

- Part of management accounting
- Concerned with establishing budgets, standard costs, actual costs and variances
- Concerned with standard costs and revenues
- Concerned with actual costs and revenues
- About establishing costs and revenues relating to
- Operations
- Processes
- Departments
- Products

Whether such costs or revenues are standard or actual

- The analysis of variances
- The analysis of profitability of ventures and projects
- The analysis of the social use of funds.

Cost accounting can also be defined as the ascertainment of cost. That is, finding out how much is incurred to produce a given product or unit of service.

### 1.2.2 **Cost Objective**

A cost objective is any activity for which a separate measurement of cost is desired. A cost unit is a cost objective. However, there are some cost objectives which are not cost units. Examples of cost objectives include a product, a service, a department or segment of the business, a function, a process or activity or indeed anything for which one wants to measure the cost of resources used.

Sometimes it is impossible to trace some costs direct to cost units. Under such situations, costs are first traced to cost centres for further spreading to cost units.

### 1.2.3 **The Objectives of Cost Accounting**

The objectives of cost accounting include the following:

a. To ascertain cost and facilitate pricing
b. To provide information to assist planning
c. To provide information to facilitate decision making
d. To provide information to aid control
e. To provide motivation and to promote cost consciousness at all levels of the management hierarchy
f. To provide a formal means of gathering detailed information on operations.

These objectives are achieved through the processes of cost ascertainment and cost control. These processes form the subject matter of this study.

### 1.2.4 **Functions of Cost Accounting / The Role of the Cost Accountant**

The functions of cost accounting include the following:

a. The application of costing principles, methods and techniques in ascertaining the cost of units (products or services or process). This helps in pricing decisions.

b. The provision of information to management for the purpose of planning. Planning is the process of establishing goals and suitable courses of action to achieve the goals.

   The cost system generates information such as:

   - resource availability e.g. materials, labour, other facilities
   - cash flow pattern within a given time horizon
   - expected returns from alternative projects etc.

   Information such as these will enable management to make a commitment about the way scarce resources may be utilized and set good priorities.

c. The provision of information to management for decision making. Decision making entails choosing from among a given set of alternatives to achieve defined objectives. Cost accounting system generates a pool of information that facilitates informed decisions. Examples of such decisions include various costs, their behaviour, composition and nature, that is, whether the costs are fixed or variable, sunk cost or opportunity cost, incremental or avoidable or unavoidable cost, etc. With information such as these, prudent choices may be made regarding whether to make or buy a component, sell or further process an item, continue or discontinue a product line or division, accept or reject a special order, etc.

d. Provision of information to management to control the activities of the business organisation. Control is the process of ensuring that plans are achieved. Control may be broken down into three essential elements.

i. Set targets and goals in a budget statement (planning)
ii. Measure actual achievements during the control period and compare same with the set goals and targets.
iii. Any resultant variance should be analysed, investigated and corrective action taken to remedy the situation.

### 1.2.5. Difference between Cost Data and Cost Information

Cost data means factual information concerning the cost of materials, labour, and overheads and other cost elements which are expected to be incurred or have actually been incurred in a business venture.

Sources of Cost Data

i. Potential suppliers for the purpose of submitting proposals and bids
ii. Existing suppliers with which the business has developed a solid basis for financial control
iii. Cost models

Cost information is the detailed analysis of costs, the calculation of production costs, quantification of loss and the estimation of work efficiency to provide a solid basis for financial control.

Cost information is used in making decisions and planning future operations with the knowledge of the cost of production and services. It also assists in establishing company performance standards based partially on past cost history.

Cost data therefore are like inputs from which cost information are generated.

### 1.2.6 Advantages (Benefits) of Cost Accounting

a. It helps to identify profitable and unprofitable products and services. This kind of information helps management to make prudent decisions regarding the adoption of a new product or discontinuing a product line, etc.

b. It helps to identify efficient and inefficient methods, and helps departments to take corrective measures to improve efficiency
   - areas of wastage are indicated
   - idle time is highlighted
   - loopholes and system weaknesses are identified.

c. It helps to generate comparative information on jobs, products, projects, etc. over a period of time allowing for temporary comparative analysis for a given cost centre and cross sectional comparison from among a number of cost centres. This makes for effective control.

d. It facilitates production control. Costing principles are used to maintain costs consisting of material cost, labour cost and expenses at desirable levels. This enhances production planning and scheduling.

e. It facilitates inventory control costing and establishes procedures for the procurement, storage and issue of inventory items within the organisation.

f. It facilitates accurate estimation of cost of various cost units. Cost estimation in turn is essential for the pricing of the cost units involved.

### 1.2.7 Disadvantages (Drawbacks) of Cost Accounting

a. The analysis of cost data involves some amount of expenses and cost.

b. Some costing systems are complex and cumbersome to handle and are thus not well understood by the managers who need to operate the system.

c. The production of basic data requires the completion of many forms, sometimes inaccurate data is provided. Under such conditions, the foundation of subsequent analysis is weakened.

### 1.2.8 Relationship between Cost Accounting and Financial Accounting

The definition of cost accounting has already been noted. Financial accounting is the recording, classification and analysis of financial transactions and the ascertainment of the financial position of a business enterprise.

Alternatively, financial accounting can be defined as that part of accounting that records actual transactions of an entity in monetary terms in accordance with established concepts, principles, accounting standards and legal requirements, and presents as accurate a view as possible, of the effects of those transactions over a period of time and at the end of that time.

It involves the recording of financial transactions of an enterprise as and when they take place and their final inclusion in a periodical financial statement for the use of persons outside the business to assess the financial performance and position of the business. It deals with an enterprise's relationship with creditors, debtors, banks, revenue authorities, etc.

*Weaknesses of financial accounting*

Financial accounting has been criticised for the following deficiencies:

a. Information is not given about individual jobs, products or processes.

b. It is historical, and financial information is not available until the end of the accounting period. That is, cost data relates to the past.

c. It does not require costs to be classified into direct and indirect, controllable and non-controllable, normal and abnormal, variable and fixed, etc.

d. Detailed records are not kept of inventory movement. This inhibits the efficient control of inventory.

e. Generally, there is no budget plan.

Fig 1.1 **Differences between Cost and Financial Accounting**

| Factor | Cost accounting | Financial accounting |
|---|---|---|
| Nature of costs / revenue | Predetermined estimates, standards and budgets | Historical and past costs / revenue. |
| Users of Information | Management of a business enterprise. | Management as well as external stakeholders such as shareholders, creditors, debtors, tax authorities, investors, etc. |
| Objectives | Provide information to aid planning, decision making and control | Satisfy the stewardship function of management i.e. account for the use of Resources entrusted to management |
| Conformity to concepts and standards. | No need to comply with standards and concepts | Absolute need to comply meticulously with accounting concepts and standards. |
| Scope and form of presentation | Determined by management on the basis of relevance and cost-benefit considerations | Determined by concepts, standards and legal provisions. |
| Precision of Information | Approximations are permitted due to uncertainties in making estimates, but must be sufficiently precise and accurate | Information must be more precise. There is usually an audit by independent external auditors to confirm the reliability of information |

### 1.2.9 Conditions Necessary for an Effective Cost Accounting System

The purpose of cost accounting can only be achieved when an effective cost accounting system is in place. The system should be tailor-made to suit the environment of the business. It should be simple, economical and practical.

The following are some conditions for a successful cost accounting system:

a. There must be co-operation and co-ordination among all staff members of the organisation including management.

b. There must be a proper system of stores and inventory control

c. There must be adequate payroll procedures

d. There should be an improved system of data capture, in this regard, it is recommended that standardised pre-printed forms be used for recording:

   a. receipts of materials
   b. issue of materials
   c. labour attendance time and hours worked
   d. wages calculations etc.

There must be a sound plan for the collection of all indirect costs under suitable headings and for its absorption to products or service departments on a predetermined basis.

A costing department or section should be established and the responsibilities and duties of the cost accountant should be clearly defined.

The cost accounts and financial accounts should be maintained in such a way that their results can be reconciled easily.

### ii. Steps in Installing a Cost Accounting System

The installation of a cost accounting system is a substantial task and thus requires very detailed discussions with management and staff and a careful analysis of correct procedures and records.

The following stages are generally recommended:

- Establish the objectives and scope of the proposed system through discussions and agreement with management

- a. Analyse the current system:

  - Are there cost centres?
  - Is there a coding system?

- What accounting records are maintained by the existing system? etc.

b. Study the current system:

- Staff composition by qualification, experience and competence

- Production system and methods:
  - Is it a jobbing or engineering business?
  - Is it a contract business?
  - Is it a service provider? etc.

- Store keeping systems

- Wage systems etc.

- Start to think about the type and nature of the cost accounting system required:
  - Is it a job costing system or batch costing system?
  - Is it a contract costing system?
  - Is it a process costing system?
  - Is it a service costing system?
  - What approach or technique of costing is required? Is it full absorption costing, marginal costing or standard costing?
  - What methods will be used to price store issues?
  - What methods of remuneration are desirable?
  - What will be the basis of overhead absorption? etc.

- Design the system in collaboration with management and the staff who will operate it.
- Prepare an operating manual for all personnel showing procedures, objectives and codes.
- Arrange for training and instruction for all staff who will be involved in implementing the system.
- Implement the system with full publicity.
- Monitor the system closely and periodically report to management.
- Build a feedback mechanism into the system

**iii. Problems Inherent in the Installation of a Cost Accounting System**

1. The following factors define different problems for cost accounting system:

   a. Size of organization
   b. Type of product/service
   c. The production process
   d. The methods of manufacture
   e. Availability of staff

2. Other specific problems include:

   a. definition of responsibility
   b. definition or designation of cost centres
   c. compilation of a comprehensive cost system
   d. availability of staff
   e. labour hours worked
   f. machine utilization time

3. Scrap treatment

4. Rectification cost, etc.

5. The difficulty of accurate classification of cost

**iv. Principles of a Good Cost Accounting System**

For a cost accounting system to be effective, certain principles must be observed. These include:

1. The system itself must be appropriate to the organisation and its processes or business.

2. Reports, statements and analyses produced by the system should contain relevant information for the intended purpose. Information is relevant when it has value, which helps to resolve a problem.

3. Reports, statements and analyses should be produced between appropriate time intervals and promptly, to make them useful and relevant. Reports submitted late lose relevance.

4. Reports should be addressed to the appropriate responsible persons who have the capacity to act on the subject matter of the report.

5. Information generated from the system must be sufficiently accurate to make them reliable else decisions will be made on totally invalid foundations or premises.

6. Reports should be so presented as to make them simple and understandable. Jargons and complex terminologies should be avoided as much as possible.

7. The benefits from generating information should outweigh the cost before such information is generated.

## 1.3 Scope of Cost Accounting

### 1.3.1 Elements of Cost

There are three elements of cost, namely: materials, labour and expenses. These result in the following classes of cost:

a. materials cost
b. labour cost
c. expenses

*a. Material Costs*

These are the costs of materials or commodities other than fixed assets, introduced into a product or consumed in the operations of an organisation. In other words, they are the cost of materials input into the production of goods and services. For example, the cost of:

i. raw materials
ii. component parts
iii. work in progress
iv. primary packing materials
v. lubricating oil
vi. consumable tools
vii. stationeries
viii. cleaning materials

*b. Labour Cost*

This is the cost of employee remuneration. In other words, payments made to and on behalf of employees for offering labour services in the production function.

*c. Expenses*

These are costs other than material costs and labour costs. For example, the cost of:

a. Hiring special equipment and maintaining such equipment
b. Royalty payments
c. Copyrights and patent payments
d. Utilities such as electricity and water charges
e. Rent etc.

Each of the elements of cost can either be direct or indirect

- Direct Cost

These are costs that can be directly identified and charged to a cost unit without apportionment. Direct costs comprise:

- Direct material cost
- Direct labour cost
- Direct expenses

a. *Direct material cost*

These are the costs of material that can be physically identified with a specific cost unit. They are the cost of materials entering into and becoming constituent elements of a product or saleable service. Direct materials costs are allocated directly to cost units.

Note:
Some materials which could qualify as direct materials may be treated as indirect materials for purposes of materiality and convenience.

The following groups of material are examples of direct materials:

i. Materials specifically purchased for a particular job order or process.
ii. Materials requisitioned from store for a particular production order
iii. Work in progress and component parts
iv. Primary packing materials e.g. cartons, wrapping material, etc.

b. *Direct labour or wages*

This is the cost of remuneration for employees' efforts and skills applied directly to a product or saleable service. Such wages are allocated directly to cost units. Examples of direct labour cost are:

i. Wages of production operatives who are involved in transforming the raw materials into finished goods.
ii. Wages of waiters who serve meals at a hotel.
iii. Wages of sales assistants involved in selling goods in a retailing shop, etc.

c. *Direct Expenses*

These are costs other than material cost and labour cost which can be identified with and charged or allocated to a cost unit. In other words, they are costs other than material cost and labour cost which are incurred for a specific product or saleable service.

Examples of direct expenses are:

i. the cost of hiring a special equipment for a particular production order.

ii. the maintenance cost of special equipment hired for particular production orders
iii. royalty payments.
iv. travelling expenses to site of contracts, etc.

The sum of all direct costs is prime cost.

d. Indirect Cost

All costs that cannot be identified with and allocated to a cost unit but that has to be apportioned to a number of cost centres and further absorbed by cost units are described as indirect cost. Another term for all indirect costs is overhead costs. Indirect costs comprise of:

- Indirect material cost
- Indirect labour cost
- Indirect expenses

The sum of all indirect costs is overhead.

i. *Indirect Material Cost*

These are the costs of material items that cannot be identified with any one product because they are used for the benefit of all products rather than for any one specific product. In other words, these are material costs which cannot be allocated to cost units but are apportioned and absorbed by cost units.

For example;

- The cost of materials required for operating and maintaining plant & equipment such as:
    - lubricating oil
    - consuming tools
- cost of stationeries
- cost of cleaning materials
    - soap and detergents
    - rugs and dusters
    - brooms and brushes, etc.

ii. *Indirect Labour Cost*

These are wages of employees who do not work on the product itself but who assist in the manufacturing process. Such costs cannot be allocated to cost units; rather they are apportioned to cost centres and further assigned to cost units through absorption.

For example:

- salaries of factory supervisors
- wages of the store department employees
- wages of cleaners, etc.

iii. *Indirect Expenses*

These are expenses incurred in general, not for the production of a specific cost unit.

For example
selling and distribution expenses
- advertising
- sales promotion, etc.

administrative expenses
- stationeries
- audit & consultancy fees
- secretarial charges

production expenses
- rent
- insurance
- electricity

### 1.3.2 **Methods of Costing**

The cost of products or services is determined using several methods. The use of a given method is dictated by such factors such as: the nature of cost units, the production process, the mode of cost accumulation, the duration of work, etc.

The following are the well-established methods of costing:

a. Job / Batch costing
b. Contract costing
c. Process costing
d. Service costing

All these methods will be discussed in details in subsequent sections.

### 1.3.3 Techniques of Costing

Irrespective of the type of costing method being applied there are various approaches that could be adopted. These are:

- Absorption Costing
- Marginal Costing
- Standard Costing using
- Absorption Costing
- Marginal Costing

These approaches will be discussed in detail later on.

### 1.3.4 Information Exclusively Generated from a Cost Accounting System

a. Statements of material, direct labour and machine utilisation showing trends and values.

b. Cost of products and services broken down to material cost, labour cost and other expenses showing comparison with estimates.
c. Cost control statements for each section and department showing actual results compared with budgets.
d. Periodic inventory valuation reports.
e. Scrap and rectification cost reports.
f. Budgets for planning and control purposes including cash budgets for monitoring liquidity.

### 1.3.5 Cost Accounting as Part of the Management Information System

Management requires information for decision making, planning and control. To obtain the information, there is the need for a management information system.

Cost accounting and for that matter management accounting, form an integral part of the management information system. As part of the management information system, cost accounting provides information required by management for such purposes as:

a. formulation of policy
b. planning and controlling the activities of the enterprise
c. decision making
d. disclosure to stakeholders external to the enterprise
e. disclosure to employees
f. safe guarding the assets of the enterprise

The cost accounting function is particularly useful in the managerial function in the following ways:

i. *Policy formulation*

The cost accounting system provides information to help in deciding:

- what products to sell
- in what markets to operate
- at what selling prices

ii. *Planning*

Information is also provided by the cost accountant to help in the evaluation of proposals for capital expenditure.

The cost accountant provides data on past performance that may guide future performance. The cost accounting function is at the centre of the establishment of budgetary procedures and budget time tables and the coordination of short term plans from all sections of the business to ensure their harmony. The role of the cost accounting function in the preparation of the master budget of the business cannot be over emphasized.

iii. *Control*

The cost accounting function aids the control process by producing performance reports that compare the actual results with intended results for each responsibility centre. Management attention is drawn to trouble spots.

iv. *Decision Making*

The analysis of costs and revenues contained in most cost accounting reports is very useful in guiding decision making.

iv. *Others*

Indeed, the cost accounting function provides a rich database for making disclosures to all stakeholders of an enterprise and then ultimately assists the enterprise to achieve its objectives and safeguard its assets.

### 1.3.6 Organisations Other Than Manufacturing Companies Where Cost Accounting Can Be Applied

a. Local Authorities: District, Municipal or Metropolitan Assemblies. e.g. Accra Metropolitan Authority, Kumasi Metropolitan Authority, Kano Municipal Local Government or Lagos Island Local Education Authority.

b. Health Sector Organisations:
   - Hospitals
   - Clinics, etc.

c. Educational institutions
   - Colleges
   - Polytechnics
   - Universities

d. Financial institutions
   - Banks
   - Insurance companies, etc.

e. Transport businesses

f. Consultancy firms
- Accounting and audit firms
- Economic and management consultancies
- Legal chambers, etc.

g. Utility companies
- Water company
- Electricity company

h. Farming ventures

i. Trading outlets
- Retailing
- Wholesaling
- Departmental stores
- Super markets, etc.

In respect of the organisations outlined, the following are some information that could be generated to assist their control:

a. Cost per unit of a product / service.
b. Cost of running the organisation analysed into: sections, departments, etc.
c. Wages cost per unit and material cost per unit.
d. Cost per patient per consulting hour, per bed, etc.
e. Budgetary control statements.
f. Cost per full time equivalent student
g. Branch operating statement

### 1.3.7 **Value of Cost Information in Management Decision-Making Process**

Business activities, being dynamic, require decisions by management almost on a daily basis. There is need to regularly review decisions which are currently being implemented to ensure that they are still relevant in the face of changing circumstances. There is also the need to take fresh decisions for the immediate and distant future in line with industry trends if a business must remain afloat.

Inflation for instance, renders cost structures obsolete as a decimal point shift in the indices requires a reworking of the elements. Economic items like a change in foreign exchange rates may adversely affect the costing of products in industries which depend on imports or exports. If the cost of products changes, so also must the selling prices, and such changes take place sometimes as rapidly as on a daily basis or even more frequently. Without cost information, management may not be able to respond to the demands for on- the-spot decisions which today's dynamic business situations require.

### 1.3.8 **Impact of Information Technology on the Generation, Evaluation and Presentation of Cost Information:**

Because of the need to keep abreast of current trends in business, management cannot afford to obtain information late because an untimely piece of information is more damaging than no information at all.

Financial information most times comes in arrears; sometimes the damage it reveals for a particular period may have continued midway into the succeeding period because the period being reported on must necessarily expire before the complete figures could be compiled. Whilst this is acceptable for historical purposes, it may spell doom for a business facing stiff competition or operating under very tight conditions. A day's delay may be critical to the survival of such a business.

Information Technology comes handy to support businesses in the area of online, real-time presentation of reports to management. With a good information network, top management can access figures almost as soon as they are being generated at the lowest level of the shop floor.

In some cases, controls are already built into the system either to raise danger signals or to require authorisation of some actions at some points in the business process. For example, where credit limits and credit periods are established and built into the system, there would be need for a very high level of authorisation before supplies could be made to debtors who have exceeded the limits set for them.

Also, with information technology, time wasted on generating papers and obtaining physical authorisations are reduced to the lowest minimum as the system does most of the groundwork and passes the results to the various levels of management on-line.

### 1.3.9 Roles Required of Accounting Technicians in Collection, Analysis and Presentation of Cost Information

The accounting technician, as the name implies, is the person who drives the nitty-gritty of information presentation. Any laxity in the performance of his/her tasks has a profound adverse effect on the quality and timeliness of the information generated for use by management in decision making.

Specifically, the accounting technician is expected to perform the following roles:

a. Ensure that established formats for data compilation are being correctly followed at the various levels.
b. Monitor the regularity of entry of data to ensure timeliness. Hourly, daily, weekly and monthly reports are to be given by the responsible officers when they are required.
c. Confirm the accuracy of figures supplied through reasonableness checks or other independent sources.
d. Raise flags where extraneous or extraordinary figures are presented.
e. Prepare summaries of raw data collected showing the results from various perspectives to give a total picture. For example, sales by branch, sales by period, sales by salesman, sales by season, etc. would reveal the total sales figures from different perspectives.
f. Draw summaries in a simple, straightforward and concise manner which would not be too voluminous for a busy executive, without also losing its relevance.
g. Pre-empt questions by providing explanatory notes on the trends and movement observed in the reports.

## CHAPTER TWO

### COST CLASSIFICATION, SEGREGATION AND CODIFICATION

### CHAPTER CONTENTS

a. Cost classification according to nature, functions, elements, responsibilities and behavioural patterns
b. Cost segregation and cost estimation.
c. c. Analysis of the effect of changing activity levels on unit costs
d. Cost coding systems

## 2.0 OBJECTIVES

After studying this chapter, readers should be able to:

a. Define the classification of cost.
b. Know the method of cost classification.
c. Explain the various patterns of cost behaviour.
d. Explain the meaning of cost segregation.
e. Explain the meaning of cost estimation.
f. Analyse the effect of changing activity levels on unit costs; and
g. Explain cost codification.

## 2.1 INTRODUCTION

Cost accounting is concerned with controlling operations, products, or services. Cost control is possible, subject to the introduction of cost classification into business activities. CIMA terminology defines cost classification as 'Arrangement of cost into logical groups, concerning their nature (fixed, variable, value adding), function (production, selling) or use in the business of the entity'

### 2.1.1 Classification of Cost

Classification is the practice of grouping costs according to their common characteristics. Therefore, cost may be classified variously for different purposes.

**a.** Classification according to elements of cost

Cost can be classified according to the three basic elements of cost as: Material; Labour; and Expenses.

**i. Material Costs.**

These are costs of materials and commodities other than fixed assets, introduced into a product or consumed in the operations of an organisation. Alternatively, material costs may be described as the costs of material input into production of goods and services.

Typical examples of material costs are:

- Raw materials
- Component parts
- Work in progress
- Primary packing materials
- Lubricating oil

- Consumable tools
- Cleaning materials
- Stationery, etc.

**ii. Labour Costs**

These are the costs of employee remuneration or payments made to and on behalf of employees for offering services in the production or service function.

**iii. Expenses**

These are costs other than material costs and labour costs. Typical examples of expenses are:

- Cost of hiring special equipment and maintaining it;
- Royalty payments;
- Copyrights and patent payments;
- Cost of utilities such as electricity and water; and
- Rent, etc.

b. **Classification according to Nature**

This is a important method of cost classification. Costs can be classified as Direct costs or Indirect costs based on their nature or identification with units/jobs/processes/cost centres.

**Direct Costs**

These are costs that can be directly identified and charged to a cost unit without apportionment. Direct costs are made up of:

- Direct material cost;
- Direct labour cost; and
- Direct expenses.

**i. Direct material costs**

These are costs of material that can be physically identified with a specific cost unit. They are costs of materials entering in to and becoming constituent elements of a product or saleable service. Direct material costs can be conveniently allocated to cost units.

Typical examples of direct material costs would include:

- Materials specifically purchased for a particular job order or process;
- Materials requisitioned from store for a particular production order;
- Work-in-progress and component parts; and
- Primary packing materials e. g. cartons, wrapping materials, etc.

It is important to note that some materials which could qualify as direct materials may be treated as indirect materials for purposes of materiality and convenience.

ii. **Direct Labour (Wages) Costs**

These are costs for employees' efforts and skills applied directly to a product or saleable service.

Direct labour/wages costs are allocated directly to cost units. Typical examples of direct labour costs would include:

- Wages of production operatives who are involved in transforming raw materials into finished goods;
- Wages of waiters who serve meals at restaurants, and
- Wages of sales assistants in supermarkets, etc.

iii. **Direct Expenses**

These are costs other than material cost and labour cost, which can be identified with and charged or allocated to a cost unit. They can also be described as costs (other than material cost and labour cost) incurred for a specific product or saleable service. Typical examples of direct expenses are:

- Cost of hiring a piece of special equipment for a particular production order;
- Maintenance cost of special equipment hired for particular production orders;
- royalty payments; and
- travelling expenses to contracts sites, etc.

The aggregate of direct costs, direct labour and direct expenses is termed PRIME COST.

Direct material
+
Direct labour
+
Direct expenses
} = PRIME COST

**2. Indirect Costs**

All costs that cannot be identified with and allocated to a cost unit but would have to be apportioned to a number of cost centres and further absorbed by cost units are described as indirect costs or overheads. Indirect costs comprise of:

- Indirect material cost
- Indirect labour cost
- Indirect expenses

**i. Indirect material costs**

These are cost of material items that cannot be identified with any particular product because they are used for the benefit of all products instead of a specific product. Indirect material costs cannot be allocated to cost units , hence they are apportioned and absorbed by cost units. Typical examples of indirect material costs would include the costs of those materials required for operating and maintaining plant and

Equipment such as:

- Lubricating oil;
- Consumable tools;
- Cost of stationery;
- Cost of cleaning materials;
- Soap and detergents;
- Rugs and dusters; and
- Brooms and brushes, etc.

ii. **Indirect labour costs**

These are wages of employees who do not work on a specific product itself, though they assisted in the manufacturing process. Such costs cannot be allocated to cost units, but they are apportioned to cost centres and further related to cost units through absorption.

- Typical examples are:
- Salaries of factory supervisors;
- Wages of the store department employees; and
- Wages of cleaners, etc.

**iii. Indirect Expenses**

These are expenses incurred in general and not for the production of a specific cost unit.

Typical examples are:

- Selling and distribution expenses
- Advertising
- Sales promotion, etc.
- Administrative expenses
- Stationary
- Audit & consultancy fees
- Secretarial charges
- Production expenses
- Rent
- Insurance
- Electricity

When classification of costs according to element and nature is combined, a cost build-up /cost card would appear as shown below:

**COST CARD**

| | |
|---|---|
| Direct material cost | xx |
| Direct labour cost | xx |
| Direct Expenses | xx |
| Prime cost | xxx |
| Indirect material cost | xx |
| Indirect labour cost | xx |
| Indirect expenses | xx |
| Overhead cost | xxx |
| | xxx |

c. **Classification according to function**

All indirect costs or overhead cost can be classified according to function of the following:
Production (Factory/Works/ Manufacturing) Overheads
- Selling overheads
- Marketing overheads
- Distribution overheads
- Administrative overheads
- Research and development overheads.

**Production Overheads**
These are the indirect costs of manufacturing a cost unit
They are made up of indirect materials consumed in the factory, indirect factory wages and other indirect expenses incurred in connection with production.

**Selling Overheads**
These are marketing costs incurred in securing orders, e.g. sales promotion cost.

**Marketing Overheads**
These are the costs incurred in publicizing and presenting products to customers in suitably attractive forms and at acceptable prices. They include costs of all relevant research work, those incurred in the course of securing orders and delivery of goods to customers and costs of after sales services or processes. Thus, marketing costs comprise of:
- Selling costs;
- Distribution cost; and
- Other related costs.

**Distribution Overheads**
These are costs incurred in making the finished goods ready for dispatch and delivery to customers e.g., cost of carriage outwards.

**Administrative Overheads**
These are costs of formulating policies, directing and controlling operations that are not directly related to production, selling, distribution or research and development.

**Research and Development Overheads**
These are costs of seeking new ideas, alternate materials, improved production methods, improved products, and development/and design of innovative ideas.

When classification of costs according to elements of costs, nature and function are combined, a revised cost build up/ cost card would appear as shown below:

| | | |
|---|---|---|
| Direct material cost | xxx | |
| Direct labour cost | xxx | |
| Direct expenses | xxx | |
| Prime cost | xxx | |
| Indirect production material cost | xxx | |
| Indirect production labour cost | xxx | |
| Indirect production expenses | xxx | |
| Production overheads, | xxx | |
| Production cost, | xxx | |
| Selling overheads, | xxx | |
| Distribution overheads | xxx | xxx |
| Administrative overheads | xxx | |
| Total cost, | | xxx |
| Profit Margin, | | xxx |
| Selling Price | | xxx |

***d. Classification according to behaviour***

Costs may behave differently with respect to activity. Some costs change in sympathy with activity, while others remain unchanged. Costs may therefore be classified according to variability or behaviour as:

- Fixed cost
- Variable cost
- Semi-fixed, semi-variable or mixed cost
- Stepped fixed cost

i. Fixed Costs
These are costs that do not vary with changes in activity levels. They usually change with the passage of time. For example, rent and rates, the managing director's salary, etc. They are sometimes referred to as period costs.

ii. Variable Costs
These are costs which vary indirect proportion with changes in activity levels. For example, the cost of raw materials, direct wages and direct expenses such as royalties.

iii. Semi-fixed/semi-variable or mixed costs
These are costs which contain both fixed and variable elements, that is, for a given range of levels, the cost may remain constant and beyond there level ranges, costs may then vary in direct proportion with changes in activity level. For example, the cost of utilities such as electricity, water and telephone.

iv. Stepped fixed costs
These are costs which are fixed for a given range of activity level and would change discretely for ranges of activity levels beyond the given range.

e. ***Classification as product costs or period costs***

**Product Costs**

These are costs that are identified with goods produced or purchased for resale. They may also be referred to as production or manufacturing costs. They are costs used for the valuation of inventory and work in progress. Examples of product costs are:

- Cost of raw materials
- Production wages
- Production overheads such as electricity, depreciation of plant, rent of factory premises, etc.

**Expired product costs**

These are a portion of the product cost which relates to products that have realised revenues and do not have future revenue-generating potential.

**Unexpired Product Costs**

These are the costs of resources acquired, which are expected to contribute to future revenue. Typical examples of such are the cost of materials not yet sold. When the inventories are eventually sold and revenue realised, the cost is said to have expired and would be recorded in the profit and loss account.

**Period Costs**

These are costs incurred and charged against profit for a period, and not included in the cost for inventory valuation purposes. These are usually non-manufacturing costs. Typical examples are selling and distribution overheads and administrative overheads. Period costs are charged in full to the profit and loss account for the relevant period. They are not included for inventory valuation purposes. Product costs can be analysed as expired costs or unexpired costs.

***f. Classification according to controllable and uncontrollable costs/revenue***

These are costs/revenues that are reasonably amenable to regulation by a given responsibility centre manager. From a control liability perspective, costs may be classified as controllable and uncontrollable.

Controllable costs /revenue is such that it is reasonably amenable to regulation by a given responsibility centrecentre manager. In other words, they are costs/revenues whose incidence is influenced by the actions or inactions of a given responsibility centre manager. Typical examples would include costs of raw materials (when purchased in bulk), variable costs that are prone to control.

Uncontrollable costs are those costs which cannot be influenced by actions of a specified responsibility centre manager. Examples are factory rent, managers salaries, etc.

All costs/revenues are controllable at some level of management. However, lower down the hierarchical structure of management, some costs/revenues are not controllable. In preparing control reports, it is very necessary to have cost/revenue is classified as controllable and non-controllable.

**g. Classification of costs as relevant/irrelevant costs.**

**Relevant Costs/Revenue**

These are future costs/revenues that can be altered by a given decision. Examples of Relevant Costs include:

- Future costs/predetermined;
- Opportunity costs;
- Avoidable costs; and
- Incremental costs, etc.

**Irrelevant Costs/Revenues**

These are costs/revenues that will not be affected by a given decision. Irrespective of what decision is taken, the cost will change. Examples of Irrelevant Costs include:

- Past costs;
- Sunk costs;
- Unavoidable cost;
- Fixed costs (except for incremental fixed costs)

**h Other Classifications of Cost would include:**

**Avoidable Costs**

These are costs that may be saved by the adoption of a given alternative option.

**Non-Avoidable Costs**

These are costs that cannot be saved or eliminated by the adoption of a given alternative line of action.

**Normal Costs**

These are costs planned for and expected at given levels of activity under specified conditions. For example, normal scrap, loss of materials, cost of expected idle time, etc.

**Abnormal Costs**

These are costs not planned for and therefore they are not expected to be incurred at a given level of activity given the conditions in which the level of activity is normally achieved, e.g., cost of excessive crap and abnormal idle time pay, etc.

**Sunk Costs**

These are the costs of resources already acquired. They are costs created by decisions made in the past and which cannot be altered by decisions to be taken in the future. The written-down value of a plant previously acquired is a typical example of sunk costs.

**Opportunity Costs are the values of benefits forgone or sacrificed in favour of the alternative courses of action.**

**Incremental Costs/Revenue**

They are additional costs or revenues that arise from the production or sale of a group of additional units. They are sometimes called differential costs.

**Marginal Costs/Revenues:** These are additional cost or revenues that arises from the production of one additional unit of output or service.

**Future Costs**
These are estimated costs that are reasonably expected to be incurred in the future.

**Replacement Costs**
These are estimated costs at which an identical item can be acquired or produced.

**Conversion Costs**
These are the costs of transforming raw materials into finished goods or costs of converting raw materials from one stage of the production cycle to the next. Conversion costs are the total cost of production less the cost of bought-in materials.

**Attributable Costs**
These are costs that can be directly associated with a particular project, program or cost centre.

**Discretionary Costs**
These are non-essential costs implying that a business unit can survive without them.

**Development Costs**
These are costs of research (exploration) into improving the production of goods and services.

**Pre-Production Costs**
These are the part of development costs relating to making trial production- run as a preliminary step to formal production

**Committed Costs**
These are costs that are expected to be incurred and for which resources have been ear-marked or allocated as a result of a contractual agreement or an earlier decision taken to have the cost incurred.

**Value Added**
This is the increase in market value of a product as a result of changing the form, location, etc. of that product. It is the total market value of the product less cost of bough- in - materials and services.

**Pre-Determined Costs**
These are costs estimated and computed in advance of production process on the basis of specifications of all the factors affecting cost.

**Standard Costs**

These are costs estimated and expected to be incurred per unit of activity under efficient production conditions.

**Budgeted costs**

These are costs estimated and planned for a given activity level, function or segment of the organization within a specified time horizon.

**Actual costs**

These are the costs actually incurred in the production process. Actual costs are not estimates but are historical or past costs.

2.1.2 **Cost Segregation**

In this section, you will learn about the different behavioural patterns of cost. After the completion on of this section, you should be able to:

a. Explain the problems associated with the conventional methods of cost classification
b. Explain the different cost patterns
c. Explain the factors that influence cost behaviour.

2.1.3 **Problems associated with the conventional classification of cost according to behaviour**

a. The overall simplistic assumption of cost linearity for variable cost per unit is not practical. In addition to the possibility that variable costs may be linear, it is also possible that they may be curvilinear or non-linear.
b. The behaviour of cost should be established by analysis of the cost in question and not by some overall simplistic assumption.
c. Fixed cost can change as activity levels change. Some cost could be classified as fixed or variable in different organisations e.g., depreciation cost.

It is better to sub-divide fixed cost into four categories:

i. Time-related fixed cost, e. g. rent and rate
ii. Volume-related fixed costs-
These are the same with a step cost, e.g. cost of the plant.
iii. Policy related fixed cost e.g., advertising cost, research and development cost.
These types of fixed cost are known as programmed fixed cost.
iv. Joint fixed cost.

The classification of cost as variable and fixed and may not hold for certain items of cost. Some cost items contain variable and fixed elements.
Some methods could be applied to separate fixed and variable costs.

-Statistical methods
-Nonstatistical methods (judgment)

Costs do not necessarily behave in regular patterns. For example, a variable cost may be linear between 90% - 115% of normal activity level and thereafter have a curvilinear function.

Variable costs are assumed to vary with activity levels, but there are multiple causes of this variation.

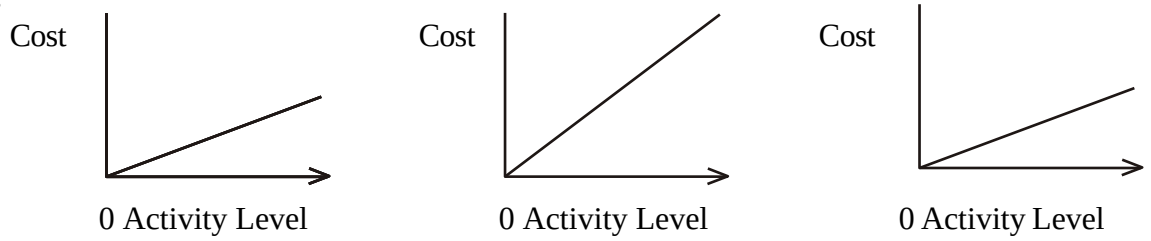
Different variable costs react to different activity measures, e.g. Direct wages may vary with the number of direct workers engaged, distribution costs may vary with the deliveries made, etc. The simplistic assumption that all variable costs vary with the same measure of activity level is misleading.

In summary, the conventional classification of fixed and variable costs, being defined by one factor (production volume), is simplistic and at best, a crude

Approximation of reality.

### 2.1.4 **Typical Cost Patterns**

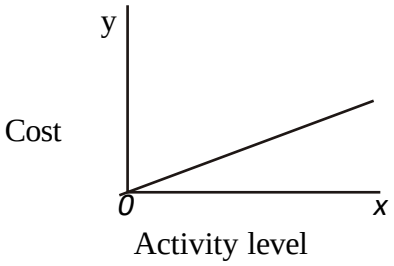
a. ***Variable cost patterns linear***




Examples of costs that behave in a linear pattern are:

Direct materials cost
Royalties per unit of production
Power usage (without the fixed element)

A perfectly linear variable cost overall activity levels is extremely unlikely. More realistically, a cost may be linear only over the relevant range of activity levels.




y = Total Cost
*x* = Activity Level
b = Variable cost per unit

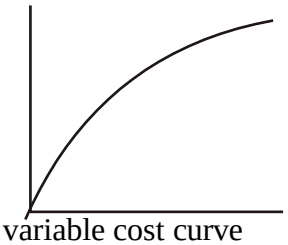
The cost function for a linear variable cost is:
y = b x (where b is the slope of the curve, i.e. variable cost per unit.)

b. **Variable cost patterns- Curvilinear**

These are cost patterns wherein the costs do not vary indirect proportion to activity level. e.g.

Convex Curve

Cost



0 variable cost curve

Activity level

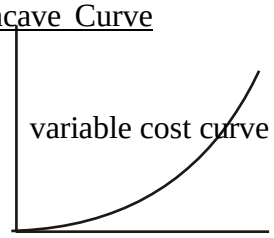
**Implications**

An extra unit of activity causes less than less-than-proportionate increase in cost. i.e. Economics of scale operate.

Concave Curve

Cost

L




0 Activity level

**Implications**

The above functions are termed "para Mathematically, the function of a par is given as:

$Y = bx + Cx^2 + \ldots + Px^n$

Where b, c, and p are constants, x is the activity level, and y is the variable cost.

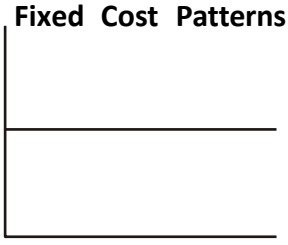
An extra unit of activity causes a more than proportionate increase in cost. i.e. diminishing returns operate

**Other Cost Patterns**

**i) Fixed Cost Patterns**

Cost

a



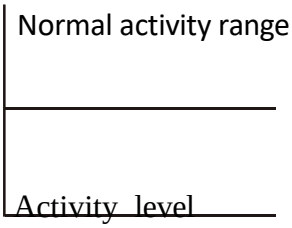
Activity level

**ii)**

Cost

Normal activity range

a



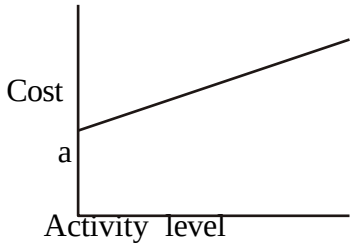
Activity level

e.g., rates, rent, timebased depreciation, salaries, etc.

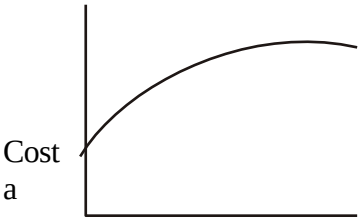
Mathematically, fixed cost function is given as $y = a$

b. **Semi variable cost patterns**

**i) linear semi variable cost**




**ii) curvilinear semi variable cost**




Activity level

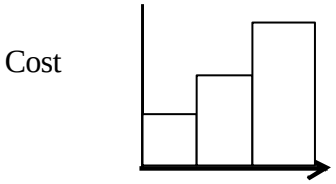
Examples include:

- Cost of electricity
- Telephone charges
- Computer bureau costs
- Some direct wages schemes with piecework and a guaranteed
- Minimum payment system

Mathematically, a semi variable function is given as: Lineary = a + bx

curvi-lineary = $a + bx + cx^2 + ... + Px^n$

(e) Stepped Costs




Examples

-Supervisory salaries, bought in materials when there are discounts.

f. **Factors Influencing Cost Behaviour**

i. Activity level
   - Sales volume
   - Production volume
   - Hours of work
   - Number of invoices processed, etc.

ii. Time- the passage of time

iii. Non- volume factors
   a) Technologies
   b) Methods of production
   c) Levels of efficiency
   d) Price levels

### 2.1.5 Analysis of the Effect of Changing Activity Levels on Costs

The concept of "levels of activity" may be addressed by some other alternative terms, depending on the nature of the business concerned. Typical terms include through put, output, volume and capacity.

Unit costs do not remain the same over different activity levels, a great deal of care is required in monitoring the behaviour of costs at each level of activity, because it forms the basis upon which costs are classified into fixed and variable.

It is instructive to note that fixed cost only remains fixed up to a certain level of activity and thereafter changes. For example, factory rent is fixed up to the level of the maximum capacity of that particular factory; any output beyond that may necessitate the acquisition of an additional factory cost, hence more costs.

Even for variable costs, some costs may change at some activity levels. For example, unit material costs may drop at high volumes due to quantity discounts. The accounting technician should therefore recognize this fact and make appropriate changes if his/her cost information is to remain reliable.

## 2.2 Cost Centre and Cost Unit

### 2.2.1 Cost Centre:

This is defined in the terminology as 'production or service location, function, activity or item of equipment for which costs are accumulated'.

Therefore, a cost centre can be described as a location, a person, an item of equipment or a group of these to which costs are incurred and further related to cost units. For example, the factory, canteen, maintenance section, the foreman, the managing director, the plant, the pool of computers, etc. Cost centres are broadly classified into:

1. Production cost centres
2. Service cost centres

The manager in any cost centre is accountable for the outcome achieved in his domain. This means that each centre is evaluated in terms of performance achieved.

### 2.2.2 Cost Unit:

The CIMA terminology defines cost unit as 'a unit of product or service about which costs are ascertained.

So, a cost unit is a quantitative unit of a product or service to which costs are ascertained. For example, tons of cocoa, bags of maize, bags of rice, barrels of beer or cartons of minerals, kilowatt hours, passenger per mile, ton per mile, etc.

Some of the selected popular cost units are provided below:

| Nature of Industry | Cost Unit |
|---|---|
| Brewing | Barrel |
| Coal Mining | Tunnel/ton |
| Electricity | Kilowatt hour (kw/hr.) |
| Oil | Barrel/Tonne, Liter |
| Education | Courses, Enrolled Student |
| Paper | Ream |
| Hospitals | day |
| Telephone | Calls made, number of extensions |

### 2.2.3 Profit Centre

This is a centre which not only incurs cost but also generates income.

## 2.3 CODIFICATION

### 2.3.1 Cost Coding System

Cost Classification of controlling cost is not complete until codes are applied to classified cost items for simplicity purpose. The terminology defines code as 'a brief, accurate reference designed to assist classification of items by facilitating entry, collation and analyses.

### 2.3.2 Characteristics of a well-designed coding system

A good coding system should have the following characteristics

a. Simple for the organisation to operate
b. Flexible in nature
c. Easy to remember
d. Each item should have a fixed coding length (e.g.,00000. The coding length in this example is six (6) digits
e. Concise in nature

### 2.3.3 Methods of Codification

The three popular methods of codifying classified items are:

a. **Mnemonic Method:** This method applies symbols to classified items such as:
   FAU applied to Factory Unit,
   ADM applied to Administration

DMA applied to Direct Material
DLA applied to Direct Labour

b. **Decimal Method:** This method applies a whole number to a master group and a decimal to the primary and secondary groups in each of the master groups, i.e

1. Direct Materials
2. Direct Labour
3. Direct Expenses
4. Factory Overheads

    4i. Indirect material in the factory
    4ii. Consumable stores etc

5. Indirect Labour
    5.1 Factory Labour
    5.2 Factory Cleaners, etc

c. **Numerical Method:** This method applies fixed digits to classified items, i.e if an item such as wood has varied dimensions of width and thickness. If the same wood type is either iroko wood or mahogany wood, then the code designed for different types of woods, such as iroko wood or mahogany wood, with each type having dimensions involving length, width and thickness should possess a fixed length. Supposing eight digits of 00000000 are designed. Then it can be shown as under:

d.

| **Type of wood** | **Length** | **Width** | **Thickness** |
|---|---|---|---|
| 0 | 000 | 00 | 00 |

Equals 0 0 0 0000 0 (Eight digits)

The code for Iroko wood with dimensions of 12 feet length; 2 inches width; and 1 foot thickness after the design could be:

Iroko wood 12 feet by2
inches by1 inch Code 1 1
4 4 0 4 0 4

Note that the names and dimensions of the wood will be kept secret. Only code 1 1 4 4 0 4 0 4 will be disclosed.

Supposing mahogany of the same dimensions of 12 feet length; 2 inches, width and 1 inch thickness. Then, the code may be: Code 2 14 4 0 4 0 4

Note that codes are not only applicable to material's type but could equally be designed for material cost, labour cost, and over head.

## 2.4 CHAPTER SUMMARY

Chapter 2 has been discussed in three parts. Part 1 focused classification of cost, part 2 explained cost segregation while the part 3 dealt with cost codification.
In conclusion, this chapter has given you affirm foundation of the various concepts, and principles in cost accounting to aid the comprehension of other chapters ahead.

**MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS**

1. Cost accounting involves
   A. Drawing up the balance sheet
   B. Writing off costs
   C. Ascertaining of cost
   D. Preparation of statement of value added
   E. Annual audit of financial statements

2. Cost accounting is an integral part of
   A. Financial accounting
   B. Forensic accounting
   C. Treasury accounting
   D. Historical accounting
   E. Management accounting

3. Which of the following is **NOT** an objective of cost accounting?
   A. Provision of information to aid control
   B. ascertainments of cost in order to facilitate pricing
   C. provision of information for decision making
   D. investigation on of fraud
   E. Provision of assistance in planning

4. Material costs do **NOT** include cost of
   A. Fixed assets
   B. Raw materials
   C. Work in progress
   D. Packing materials
   E. Cleaning materials

5. Which one of the following is a direct expense?
   A. Cost of hiring a special equipment for a particular production order
   B. Advertising expenses
   C. Electricity expenses
   E. Insurance premiums

6. Classifying costs by elements includes materials, labour and ……..
7. The addition of all direct costs is known as ……..
8. Costs which are fixed for a given range of activity level but which change discretely for ranges of activity levels beyond the given ranges are called …….

9. Costs which may be saved by the adoption of a given alternative option are known as .....

10. Costs which vary indirect proportion with changes inactivity levels are called .....

**SOLUTIONS TO MULTIPLE CHOICE AND SHORTANSWER QUESTIONS**

1. A
2. E
3. D
4. A
5. B
6. Expenses
7. Prime costs
8. Stepped fixed costs
9. Avoidable costs
10. Variable costs

**EXAMINATION TYPE QUESTIONS**

1. You have been appointed to the newly created position of a Cost Accountant in a small manufacturing company. Having installed a costing system successfully, the Managing Director would like to have a detailed explanation on the kind of information the newly installed system would provide. You are required to write a report to the Managing Director setting out SIX principal items of information obtainable from costing system. **(16 marks)**

2. "Installation of a costing system is a major move in a business" You are required to discuss the inherent problems associated with installing ofcosting system.

**(8 Marks)**

3. Outline the possible problems which may be encountered as a result of the introduction of a system of cost control into an organization. **(10 Marks)**

4. Enumerate the differences, if any, between financial and cost accounting. **(8 Marks)**

5. The Managing Director of a manufacturing concern once said "since the main object of our firm is to know the annual profit, there is no need retaining the cost accounting staff; let us pay them off and retain only the financial accounting staff in order to reduce the administrative costs".

**Required**

Write a report to the Managing Director to convince him of the need to retain the cost accounting staff (10 marks )

6. Describe **THREE** different methods of cost classification and explain the usefulness of each method.

(**11 marks)**

7. Describe **THREE** different methods of cost classification and explain the usefulness of each method. (**11 marks)**

CACA Level I Costing, June 1989

8. The following are some of the cost classifications used in any costing system

(i) Period costs
(ii) Product costs
(iii) Variable costs
(iv) Opportunity costs

**Required:**
Citing two examples in each case, explain each of these classified costs. **(17 marks)**

CACA Level I Costing, June 1987

9. "Cost may be classified in a variety of ways according to their nature and the information needs of management." With relevant illustrative examples, you are required to provide a detailed explanation of the above statement.

(22 marks)

ICSA Management Accounting, December 1984

10. An existing cost system lacks the essential aspects of cost control. List the main matters to be introduced to make good the system's deficiencies.

11. Company HIMCO produces different products through automation. The company incurs large amount of overheads in the factory on power alone that is semi-variable in nature.

The details of Power Costs for six (6) months ending September 2024 are as provided below:

| Months | Machine Hours | Power Cost |
|---|---|---|
| 2024 | Hours | ₦ |
| April | 50,000 | 400,000 |
| May | 60,000 | 460,000 |
| June | 40,000 | 340,000 |
| July | 45,000 | 370,000 |
| August | 70,000 | 520,000 |
| September | 75,000 | 550,000 |

**Required:**

ai. Differentiate between fixed cost and variable cost
ii. Summarise the meaning of semi-variable cost

b. Segregate the semi-variable costs into total fixed costs and unit variable costs
c. Estimate the power cost for October if the firm plans to use 88 machine hours

**CHAPTER THREE CONTENTS**

**ELEMENTS OF COST: MATERIALS**

a. Definition and basic classifications
b. Procedure and documentation
c. Purchasing Control
d. Stores Planning Operation: Re-ordering procedures and Economic Order Quantity (EOQ)
e. Inventory valuation using FIFO, LIFO, Weighted Average, Standard Cost and Replacement Cost
f. Storage Control: Bin cards and store ledgers, ABC analysis Techniques, Physical and Continuous inventory taking
g. Just-In-Time (JIT) production and purchasing

3.0 O**bjectives**

After studying this chapter, you should be able to:

a. explain the stores function and the factors that facilitate effective inventory control
b. explain the procedures and documentation involved in the procurement process;
c. explain some principles of good storage system;
d. explain the methods of pricing issues and inventories;
e. compute and explain the inventory control levels; and
f. record stores information in inventory control records.

## 3.1 INTRODUCTION

Material cost is a significant portion of total costs for many businesses, especially for manufacturing businesses. To minimize cost, material cost control is vital. This chapter therefore outlines mechanisms for controlling material cost and provides a structure for determining material cost.

## 3.2 BASIC CLASSIFICATION OF MATERIALS

### 3.2.1 What are materials?

Other names given to materials include:
Inventory, stores, merchandise, etc. Inventory has been defined as: "all the tangible material assets of an organization other than its fixed assets." Inventory therefore comprises of the following:

a. raw materials;
b. work-in-progress;
c. finished goods;
d. merchandise ready for sale;
e. any parts or materials to be incorporated into a finished product (component parts);
f. consumables such as - stationery, oil, grease, fuel, gas;
g. jigs, fixtures and special tooling;
h. by-products, scraps;
i. works supplies; and
j. packaging materials.

3.2.2 Significance of Inventories

Inventories form a significant amount of the assets of organizations. For example, on average, two thirds of a manufacturing business' working capital will usually be in inventories. Retailers and wholesalers will often have close to nine-tenth or 90% of their working capital in inventories.

The value of the assets of a business and the quality of service offered to customers by the business is heavily and severely affected by the effective management of inventories. The importance of quality of service to customers needs no emphasis.

The manner in which stores are managed can either define success or failure for the business.

There is no organization which can exist without the use of some items of stores or material. This includes non-profit making organizations as well. Thus every organization will need to use some items of stores. This need is even crucial for profit making businesses whose operations may include the following:

i. manufacturing of some goods for sale;

- cement manufacturing;
- breweries; and
- food processing industries which include;

  - cocoa processing;
  - cassava processing;
  - canned food processing; and
  - canned meat processing, etc.

- restaurants;
- garment industry;
- construction industry;
- agricultural activities, and
- printing industry

ii. the purchase and sale of goods.

This involves commercial activities like:

- supermarkets
- retail shops
- distribution outlets
- bookshops, etc.

iii. the provision of services;

- transportation
- education and training
- Medicare
- personal service
    - hair dressing / barbering
    - dress making and tailoring
- repairs and maintenance services, etc.

For most businesses, the cost of stores and materials account for over 60% of total cost. The efficient management of the stores function is therefore very critical to minimizing cost and maximizing profits for the organisation.

Even for non-profit oriented organisations, available resources must be judiciously applied, hence a requirement of efficient management of the stores function.

## 3.3 THE STORE FUNCTIONS AND ACTIVITIES

### 3.3.1 Purchasing:

a. Identifying and defining need for items of stores and supplies;
b. Identifying and evaluating available suppliers of these items;
c. Negotiating with selected suppliers; and
d. Making contracts and placing orders for the needed items to be supplied

### 3.3.2 Operating the Store:

a. Receiving and accepting (rejecting) the items ordered
b. Holding or keeping the items until they are used
c. Issuing the items out for use

3.3.3 Inventory Control

a. Recording inventories;
b. Checking inventories;
c. Planning replenishment of inventories;
d. Valuation of inventories

3.3.4 Summarised Overview of Stores Functions

a. the purchasing function:
   i. Requisition for inventories;
   ii. Supplier analysis and selection;

   - negotiation with suppliers; and
   - placing orders.

b. operating the store:

   i. receipt of orders; and
   ii. materials handling, comprising;

   - holding inventories
   - issuing inventories for use

c. inventory control

   i. Inventory records;
   ii. Inventory audits; and
   iii. Re-ordering of inventories.

It is worth noting that the stores function is a complete process. Many people take part in the process at various levels of the management hierarchy and in several functional departments.

## 3.4 Factors That Facilitate Effective Materials Cost Control

To control material cost to desirable levels, an enabling environment must be created. The following factors are essential elements of an effective material control system:

a. A budget of material usage and purchases;
b. The handling of purchases by competent and qualified personnel. In this regard, a central purchasing system may be preferred;
c. Availability of sufficient and effective storage facilities;
d. A system to clearly classify materials either as direct or indirect with unique codes and a consistent assignment of cost to cost units or cost centres as may be appropriate;
e. The documentation, accounting systems and controls at each stage should be well

designed and effective. In this regard, the use of standard forms is recommended;

f. An efficient and well organised inventory taking and a reconciliation of inventory taking results with inventory records;
g. An effective coordination among all departments involved in materials handling e.g. buying, receiving, storage and usage; and
h. Internal audits.

## 3.5 Procurement of Materials and Documentation

### 3.5.1 Responsibilities of Purchasing or Procurement Unit include:

a. To purchase all materials and stores not internally produced which involves the following tasks:

   i. Supplier selection;
   ii. preparation of purchase orders / contracts;
   iii. monitor suppliers' performance; and
   iv. renegotiate or terminate purchase orders/contacts when conditions change.

b. Provide information to and participate in management planning sessions on subjects related to purchases of materials;

c. Review purchase specifications and assist operating departments in the selection of required materials;

d. Protect the organisation from unnecessary commitments resulting from inappropriate contracts with suppliers; and

e. Dispose of all obsolete materials, equipment, or scraps that are no longer required for the organization's operations.

Items of stores may either be bought for inventory or are bought just-in-time to meet manufacturing schedules.

### 3.5.2 Methods of Purchasing include:

a. verbal contracts of purchase e.g. purchase at auction sales;
b. standard purchase order forms;
c. simplified order forms (similar to a purchase requisition);

- cash purchases, etc.;
- spot contracts for immediate delivery;
- future contracts for delivery at a stipulated future date;
- blanket orders/period contracts; and
- call-off orders.

### 3.5.3 **Settlement Methods**

Settlement methods include:

a. cash with order;
b. part-payment when the contract is made;
c. stage payment during the course of the contract; and
d. payment upon complete delivery, etc.

The choice of settlement method depends on the nature of store item, the type of market, etc.

### 3.5.4 **Summary of the Stages of the Purchasing Function**

| Stage | Forms used: |
|---|---|
| Originating | a) requisition;<br>b) buy list;<br>c) schedule requisition; and<br>d) travelling requisition |
| Supplier selection | a) enquiry;<br>b) Request For Information (RFI)<br>c) Request For Quotation (RFQ) and<br>d) Tender. |
| Ordering | a) purchase order;<br>b) period contract /blanket order;<br>c) call off order;<br>d) standing order; and<br>e) scheduled order. |
| Completing | a) advice note;<br>b) packing note;<br>c) Goods Received Note (GRN)<br>d) Goods Inspection Report;<br>e) invoice; and<br>f) statement. |

**Originating purchases**

Depending upon the nature of the system and the item needed, the person or officer requiring the item writes out a purchase requisition for it. The purchase requisition will have to be authorised by a designated officer before it is acted on. The purchase requisition authorises the expenditure.

In respect of items which are purchased into inventory, as the items get used up, they may need to be replenished through the issue of purchase requisitions.

The process in selecting suppliers is as follows:

i. make a short list of possible suppliers by gathering information about suppliers of the particular item from sources such as;

- trade magazines;
- calls by company representatives;
- catalogues;

ii. membership of professional associations;
iii. a directory of suppliers;
iv. print and electronic media adverts;

- from a long list of suppliers, scale the list down by requesting for information through;

v. Request for Information;
vi. Prequalification questionnaire; and
vii. Tender forms

- Prepare a short list of suppliers
- Make a request for quotations from the short list
- Evaluate suppliers' quotations or tenders.

ii. Factors to consider in the evaluation price analysis and cost analysis which include:

- specification and design standard;
- delivery terms / reliability;
- durability and maintenance cost i.e. quality of supplies;
- improvement in productivity;
- capability, i.e. ability to meet volume requirements;
- set-up costs;
- tooling charges;
- transport cost;
- payment arrangements / terms; and
- discount offers, etc.

iii. Select and negotiate terms with the supplier; This may be preceded by an inspection visit to the supplier's facility n highly regulated industries. e.g. food, drinks and medical.

iv Negotiations must be aimed at a mutually satisfactory agreement. All aspects of the purchase contract which are not standard should be negotiated. The negotiations should also aim at a win-win relationship between the organisation and the suppliers.

v. Ordering: The purchase order is often used for ordering. The use of a standard purchase order is not only to meet a legal requirement but also to satisfy organizational requirements. To ensure that all orders are properly authorized and are made against authorized requisitions.

The order is an offer to buy on the terms stipulated by the organization. When the supplier agrees to supply then there is a contract.

Various methods of ordering exist among which are:

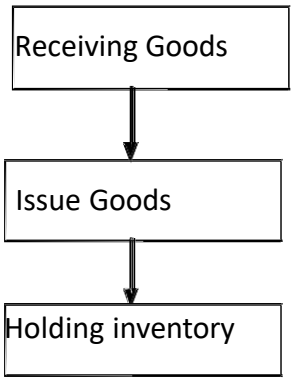
- period contracts;
- blanket orders;
- schedule orders; and
- standing orders.

vi. Completing the order: The purchasing is concluded when the items ordered are received into store. There is a need to control delivery. This implies making sure that, suppliers will meet delivery schedules.

## 3.6 RECEIPTAND STORAGE OF MATERIALS

### 3.6.1 Store operations

Store opeartions include:



3.6.2 **The Factors that Influence the Design of Procedures for Operating the Store include:**

a. the volume of materials handled;
b. the frequency of transactions;
c. the range and variety of inventory items;
d. the kind of demand on the store;

e. the type of store;
f. the inventory control system;
g. the materials handling system;
h. the bulkiness and weight of goods; and
i. the value of the goods.

3.6.3 **Receiving Goods**

When goods are received, they are checked for quantity and description against the Delivery Note, Advise Note and Purchase Order. If they are satisfactory a Goods Received Note is made. If the goods are damaged, defective or they are not of the prescribed description, they should if possible, be returned by the vehicle which brought them. Failing that, they are stored at a separate location clearly labelled until they are returned. Both the carrier and supplier should be notified immediately. Rejected goods should be covered by the Inspection Report and a Debit Note.

3.6.4 **Holding Inventory**

After acceptance, the goods should then be moved into their proper location in the stores or to the department which requires them.

There are different methods used for locating goods in the store. The three methods are:

a. Fixed location - every regular inventory item has a fixed address or inventory location. This is expensive in terms of store space. Accessing inventory is easy;

b. Random location - any item can be stored anywhere. Here location records are essential. This makes better use of space; and

c. Zoned location - a hybrid of fixed and random locations.

Allocate groups of goods to particular zones. Location within a zone may either be fixed or random.

d. **Principles of Storage**

**Some principles of storage include:**

e. frequently used items should be nearest to the issue point;

f. complementary inventories should be stored together;

g. keep things which are similar to each other near each other;

h. store things in issue batches where possible; and

i. isolate dangerous items like:

i. Inflammables

ii. explosives such as oil and gas;

j. change location when demand changes; and in loading racks;

i. put light bulky goods near the top;

ii. locate small items from waist to shoulder height;

iii. put heavy things in big compartments low down; and

iv. very heavy things are best on the floor not on the rack.

3.6.5 **Issuing Goods**

The organization should design a procedure for issuing stores. A specimen routine will be:

a. The foreman of the unit that requires the materials writes a requisition, inserts job number, date and signs; The requisition should be duly authorised.

b. The section's labourer takes requisitions to stores issue counter;

c. Store man collects the requisition and issues out specified items, labourer takes them to section;

d. The completed requisition is passed to store clerks who adjust inventory records to show new balance after issue and copy price from inventory record to requisition; and

e. The priced requisition is then passed to the cost clerks.

3.6.6 **Store Location and Layout**

Stores should be located as close as possible to the point of use of materials in order to reduce material handling charges.

This may be difficult to implement where the same material is used at more than one location within a factory or where special storage conditions are required. Where possible, specialised stores should be located close to the point of use.

### 3.6.7 **Factors that Facilitate Effective Materials Cost Control**

To ensure efficient storage, the following factors need consideration:

a. Provide speedy receipt and issue of materials. Some factors relevant in this regard include: -

- locating the store close to the point of entry of materials;
- locating store close to user departments;
- locating fast moving inventory items close to issue point;
- having separate receipt and issue points to reduce possible congestion; and
- separate issue points for different groups of materials.

b. Ensure easy location of materials;

c. Facilitate easy identification through proper classification and coding;

d. Keep adequate records of quantities held e.g. Bin cards, Tally cards, and Store ledger cards;

e. Adequate security measures to safeguard materials. Materials should be adequately stored in such a way that they are appropriately protected against: -

- fire,
- rust,
- theft,
- Deterioration; and
- effects of weather.

f. Ensure efficient use of storage space. In this regard, the most appropriate
g. method of storage should be used for the materials concerned e.g. shelves, bins, etc.

The appropriate type of store should be used. There are two types of store namely: centralised and decentralised stores.

i. Centralised Stores
Materials are kept in one central warehouse and issued from that point only.

**Advantages**

a. reduces storage space;
b. less number of staff required;
c. reduced administrative work and costs;

d. easier inventory control;
e. lower risks of duplication of items; and
f. economy of staff and concentration of experts in one department.

**Disadvantages**

a. increased transportation costs;
b. increased handling charges, breakages, etc.;
c. increased risk of loss by fire; and
d. likely delays in issue to distant usage points.
e. Decentralised stores
f. Materials are held and issued by stores in each user department.

**Advantages**

a. lower transport costs;
b. lower handling charges and breakages;
c. lower risk of loss due to fire; and
d. less delays at delivery to usage points.

**Disadvantages**

a. duplication of cost;
b. more number of staff required;
c. cumbersome inventory taking;
d. more administrative work and cost; and
e. inability to use experts in all sub-stores.

The layout of stores should ensure the following:

a. Ease of access for the movement of materials in and out of stores;
b. The issue of perishable materials on a FIFO basis;
c. The segregation of toxic or dangerous materials in a separate location; and
d. Security of materials by restriction of access to authorized personnel only.

## 3.7 PRICING OF ISSUES AND VALUATION OF INVENTORY

### 3.7.1 Introduction

When materials are needed from stores, the prospective user prepares a materials requisition to request for the materials.

When the materials are issued out they should be priced. The problem usually is the price to use. This is because materials inventory may have been purchased at varying prices from time to time. There are several methods that could be used to price issues.

The objectives of material pricing are:

a) To charge to production on a consistent and realistic basis, the cost of materials used; and
b) To provide a satisfactory basis of valuing inventory at the end of the period.

### 3.7.2 Methods of Pricing Issues and Valuing Inventory

There are several methods. Examples are:

a. First In First Out (FIFO);
b. Last In First Out (LIFO);
c. Simple Average method;
d. Weighted Average method;
e. Standard Price;
f. Replacement Price;
g. Specific Identification; and
h. Retail Method.

### 3.7.3 FIFO

This method uses the price of the first batch of materials received for all issues until all units in the first batch have been issued, after which the price of the next batch received becomes the issue price.

Inventories are thus valued using the prices of the last batches of materials received.

**Merits**

a. It reflects the normal movement of materials in store issuing materials in order of receipt;
b. Inventory is valued using recent prices;
c. The method is simple and easily understood; and
d. It uses actual prices.

**Demerits**

a. The comparison of the cost of one job with another is difficult because different issue prices are often used. The cost of a job depends on when materials are requisitioned from stores;
b. The issue price may not reflect current economic values;
c. Inventory items with low turnover will tend to be priced at old prices when they are eventually issued. These prices could well be unrealistic; and
d. In times of inflation, product cost may be low whilst replacement cost is high. Thus profit may be overstated.

### 3.7.4 LIFO

This method uses the price of the last batch of materials received for all issues until all units from this batch have been issued and then the price of the previous

batch is used. Note that if a new delivery is received before the first batch is fully issued, the new issue price at once becomes the last in price and is used to price issues until either the batch is exhausted or a new delivery is received.

Inventories are valued using the prices of materials delivered first.

**Advantages**

a. The issue price is near to current economic value;
b. It is fairly simple to operate when materials are slow moving;
c. It uses actual prices; and
d. During a period of rising prices, inventory values are conservative.

**Disadvantages**

a. It is not realistic as it is contrary to normal issue procedures;
b. The frequency of calculations causes much clerical work;
c. Inventory values may significantly be understated;
d. It can become complicated with a fast moving items since not all of one receipt may be exhausted before a differently priced replacement order arrives.

### 3.7.5 The Simple Average Method

Under this method, the issue price is the co-efficient of the different prices of all the materials in store and the number of material items in that total.

The assumption is that materials issued out of inventory at any time are drawn from a mixed group in the store without any deliberate attempt to identify materials or inventories as being from the earliest or the latest batch.

**Advantages**

a. It is simple to operate; and
b. It evens out effects of any price fluctuations.

**Disadvantages**

i. It does not use actual prices;
ii. Where prices fluctuate, inventory values may be misleading; and
iii. It gives very false issue and valuation figures.

### 3.7.6 Weighted Average Price Method

This method averages prices after weighting by their quantities. With each receipt of materials, the weighted average price is re-calculated and subsequent issues are priced at the calculated weighted average price until a further receipt of goods necessitates the average price to be re-calculated.

**Advantages**

a. It smoothens out fluctuations in issue prices; and
b. The comparison of different jobs is easier because it assumes that values of identical items are all equal.

**Disadvantages**

a. Calculations have to be made to approximately four decimal places to achieve a fair degree of accuracy; and
b. The issue price may not reflect current economic values.

3.7.7

3.7.8 **Standard Price**

Standard price is a predetermined estimate of the price or cost of a material item under specified conditions.

**Advantages**

a. It is easier to operate as there is no variation in the issue prices over long periods;
b. The efficiency of purchasing is checked as actual prices are compared with standard prices; and
c. It eliminates price fluctuations from costs enabling satisfactory manufacturing cost comparisons to be made.

**Disadvantages**

i. It requires careful initial determination;
ii. Issues may not be valued at current economic values; and
iii. It disregards price trends.

3.7.9 **Replacement Price**

This method uses current replacement Costs/Price to value all issues

**Advantages**

a. Issues are at current prices;
b. It is easier to operate; and
c. Inventory values are at current prices.

**Disadvantages**

i. The issue price is not at cost, so where there is a long interval between each receipt of material, estimates have to be made; and
ii. There is often difficulty and much work involved in ascertaining current market prices.

### 3.7.10 Specific Identification

Items issued are priced at the actual prices at which they were actually bought by specifically identifying the materials to their actual prices. Thus closing inventory is also valued at the specific prices they were bought.

## 3.8 INVENTORY CONTROL LEVELS

**Introduction**

The level of inventory held will depend upon a number of variables which will each have cost implications. Management must make decisions about the control of inventory levels with a view to minimising the cost to the business whilst achieving maximum efficiency in the availability of materials to fulfil planned usage requirements.

The following inventory levels need to be considered:

a. Minimum level;
b. Maximum level;
c. Re-order level;
d. Re-order quantity or Economic Order Quantity; and
e. Average inventory level.

**Illustration3.1**

The following data relates to Good Works Company Ltd with respect to material AY4

i. 12,000 units of the material will be used every day for a 360-day year.
ii. It will cost 50,000 cedis to place each order
iii. The cost of one unit of AY4 is 12,000 cedis and it will cost 10% of this amount to hold each unit of AY4 in store
iv. Daily usage of material AY4 will not exceed 12,500 units and will not be less than 11,500 units
v. The most reliable supplier takes a maximum period of 4 days to deliver but the shortest delivery period could be 2 days.

**NOTE:** We shall use this activity for illustrative purposes

### 3.8.1 Re-Order Level of Inventory

This is the level at which an order will be placed for additional supplies of material so that delivery will be made before the business runs out of inventory. The factors that influence the re-order level are:

a. rate of consumption
b. delivery period - reliability of supplier.

Mathematically, it is computed as:
Re-order inventory level = Maximum consumption x Maximum delivery period

Using the data in the illustrative example:

Re-order level (ROL)

= maximum usage X maximum delivery period
= 12,500 x 4 days
= 50,000 units

### 3.8.2 **Minimum Inventory Level**

This is the lowest level at which inventory may be allowed to fall. It is not prudent to allow inventories to fall below the minimum level:

The following factors determine the minimum inventory level:

a. The rate of consumption;
b. The delivery period normally allowed; and
c. The re-order level of materials.

Mathematically, the minimum inventory level may be calculated as:

Minimum inventory level = Re-order - (average consumption x average delivery period)

Using the illustrative example;

Minimum inventory level:

= Re Order Level - (average usage x average lead time)
= 50,000 units (12,000 x 3)
= 50,000 - 36,000
= 14,000 units

### 3.8.3 **Maximum Inventory Level**

This is the largest possible quantity of inventory that may be in store at any given time. It is not prudent to maintain a quantity of inventory above this level.

The following factors determine the maximum inventory level:

a. The storage space available;
b. The nature of inventory such as its perishability and seasonality. In other words, the risk of deterioration;
c. The cost of storing above normal level;
d. The rate of consumption;
e. The re-order quantity;
f. The delivery period; and
g. The re-order level.

Mathematically, it is computed as Maximum inventory level =

$$\text{Re-order level} + \text{Re-order quantity} - \left( \text{minimum consumption} \times \text{minimum delivery period} \right)$$

Using the data from illustrative example;
Maximum inventory level
= ROL + ROQ - (minimum usage x minimum lead time)

= 50,000 + 18,974 - (11,500 x 2)

= 68,974 - 23,000 = 45,974 units

**Note: Check the computation of the Re-Order Quantity from 2.8.4 below.**

3.8.4 **Re-Order Quantity**

This is the optimum quantity that should be ordered each time an order is being placed. It is often referred to as the economic order quantity. It is set in such a way as to optimise material costs.

The cost of material inventory is made up as follows:

a. Costs of holding or carrying inventories:

   i. forgone interest on capital invested in inventory;
   ii. storage charges e.g. rent, lighting and cooling, refrigeration, air conditioning, etc.; and
   iii. stores administration cost.

- staff salaries
- equipment maintenance and handling charges

iv. Handling costs e.g. cost of packing and unpacking inventories;
v. Inventory verification costs or perpetual inventory costs;
vi. Insurance, security, etc.;
vii. Deterioration and obsolescence; and
viii. Pilferage, damages, etc.

b. Cost of obtaining inventories or ordering costs:
i. Administrative costs associated with purchasing, accounting and receiving goods;
ii. Transport costs e.g. carriage inwards;
iii. Set-up and tooling costs of production runs for internally manufactured goods; and
iv. The purchase price of the inventories.

c. Stockout costs:
i. Lost contribution due to lost sales arising from stockout;
ii. Loss of future sales because customers will go elsewhere;
iii. Loss of customer goodwill;
iv. Cost of production stoppages e.g. idle time pay, not using plant optimally;
v. Labour frustrations over stoppages; and
vi. Extra costs of urgent replenishment purchases.

Economic Order Quantity (EOQ) = $\sqrt{\frac{2DCo}{C_C}}$

Where:
D =Annual demand
Co = Cost of order
Cc = Carrying cost per unit
From the illustrative example above; Economic Order Quantity (EOQ)

$$\sqrt{\frac{2DCo}{C_C}}$$

Where:
D = annual demand = 12,000 x360 = 4,320,000 Co = Cost of order =50,000
Cc = Carrying cost per unit =1,200

EOQ = $\sqrt{\frac{2(4,320,000)\ (50,000)}{1,200}}$

= 18,974 units

3.8.5 **Average Inventory Level**

This is the midway between the minimum level and the maximum level. Mathematically, it is computed as:

$$\text{Average inventory level} = \frac{\text{minimum level + maximum level}}{2}$$

Using the data in the illustration **2.1**;

Average inventory level

$$= \frac{\text{maximum level + minimum level}}{2}$$

$$= \frac{45{,}974 + 14{,}000}{2}$$

$$= 29{,}987 \text{ units}$$

**Illustration 3.2**

Using the information provided below, calculate the maximum, minimum, re-order and average inventory levels for Boateng Ltd.

Boateng Ltd. manufactures a special product for the domestic market. Records available at the stores department indicated the following:

| | | |
|---|---|---|
| Maximum usage | - | 1,200 units per week |
| Minimum usage | - | 500 units per week |
| Re-order quantity | - | 1,500 units per week |
| Delivery period | - | between 2 to 4 weeks |

8 marks

ICAG, May 2000

**Solution to Q2**

ROL = Max usage x max lead time

= 1,200 x 4

= 4,800 units

Minimum Inventory level = Reorder level (average usage of average lead time)

$$= \left(4{,}800 - \frac{1200 + 500}{2}\right) \times \frac{2 + 4}{2}$$

= 4800 - (850 x 3)

= 4800 - 2550

= 2250 units

Maximum Inventory level = reorder level + reorder qty - min usage x min lead time

= ⟩4,800 + 1,500 - ⟨500 x 2 ⟩
= 6,300 – 1,000
= 5,300 units

Average Inventory = $\frac{\text{max inventory + min inventory}}{2}$

= 5,300 + 2,250 ÷ 2
= 3,775 units

## 3.9 Inventory Control Records and Procedures

### 3.9.1 The Need for Inventory Control Records

A comprehensive inventory control record system is necessary in order that:

a. Materials can be recorded on receipt in relation to both quantity and price;
b. Issue of material for production can be recorded so that the residual balance of inventory both in quantity and monetary terms can be calculated;
c. Checks can be implemented on a regular or random basis to minimize losses due to pilferage or damages in inventories;
d. Records can be examined to highlight slow moving items which may deteriorate or become obsolete;
e. New items can be ordered when the re-order level is reached;
f. Material costs are charged to the appropriate cost centres when materials are issued from store;
g. Returns to store and to suppliers can be properly recorded; and
h. The valuation of inventories for balance sheet preparation and profit measurements purposes can be accurately implemented.

### 3.9.2 Inventory Control Records

The following are examples of inventory control records.

a. The Bin Card;
b. The Stock Tally Card; and
c. The Stock Ledger Card.

i. Bin Card

This is a document which assists the store keeper in the recording of receipts and issues of materials and the control of inventory. The

specific format of the bin card will depend on the record system which is in use.

Below is a specimen of a bin card. Fig 2.1 Bin Card

XYZ LIMITED

| | | | |
|---|---|---|---|
| Description | : | Bin No. | : |
| Code | : | Max Level: | |
| Re-order Quantity: | | Min Level: | |
| Stores Ledger Folio: | | Re-order Level: | |

| Date | Order | | Allocated | | Inventory Balance | |
|---|---|---|---|---|---|---|
| | Qty | Received | Qty | Issued | | |
| | | | | | | |

**Illustration 3.3**

The following transactions occurred in respect of Material JD42 during the four months ending 31July.

| **Month** | **April** | **May** | **June** | **July** |
|---|---|---|---|---|
| Allocated units | 1,000 | 1,500 | 1,300 | 3,200 |
| Receipts in units | 600 | 900 | 1,800 | 1,500 |
| Issued units | 500 | 800 | 1,400 | 1,200 |

The following information is also available:

a. Balances as at 1st April were as follows:
Balance on Order 1,000 units
Inventory on hand 500 units
Balance on Allocation 700 units
Free balance 800 units

b. Orders are placed in lots of 1,000 units when the free balance falls to 600 units or below. You are required to present the above data in a Bin Card.

Solution

FIG. 3.1 BIN CARD

| Date | Order | | | Allocated | | | Free balance | | | Inventory | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Qty | Rcvd | Bal | Qty | Issued | Bal | Order | | Bal | Rcvd | issued | Bal |
| April 1 | - | - | 1,000 | - | - | 700 | - | - | 800 | - | - | 500 |
| April | | | | | | | | | | | | |
| May | | | | | | | | | | | | |
| June | - | 600 | 1400 | 1000 | 500 | 1200 | 1400 | 1200 | 1000 | 600 | 500 | 600 |
| July | | | | | | | | | | | | |
| | 2000 | 900 | 2500 | 1500 | 800 | 1,900 | 2500 | 1900 | 1600 | 900 | 800 | 700 |
| | 1000 | 1800 | 1,700 | 1,300 | 1,400 | 1,800 | 1,700 | 1,800 | 1,500 | 1,800 | 1,400 | 1,100 |

**Note**

1. inventory balance = balance + received issued
2. allocated balance = balance b/d + quantity allocated - quantity issued
3. order quantity is determined so as to ensure that free balance does not fall to or below 600 units. It is in multiples of 1,000 units
4. order balance = balance b/d + quantity ordered + quantity received
5. free balance = balance b/d + order balance - allocated balance
6. free balance b/d = stock balance + order balance - allocated balance
7. it is assumed that all materials issued were first allocated before actually being issued

2. Inventory Tally Card: This is another type of inventory record form. It is often used for inventory items involving small values. Below is a sample of a Tally Card.

**Fig 3.2**

STORE TALLY CARD

XYZ LIMITED

Description: Max. Inventory level:
Code: Min Inventory level
Re-order level: Economic order qty:

| Date | Receipt | Issued | Balance | Signature |
|---|---|---|---|---|
| | | | | |

3. Store Ledger Card

The stores ledger card shows similar information as that contained in a bin card or a tally card and additionally records the unit price and total value of all inventories received and issued.

Below is a specimen of a store ledger card

**Fig 3.3**

Store Ledger Card XYZ LIMITED

Description: Max. Inventory level :
Code : Min Inventory level :
Re-order level:
Re-order quantity :

| Date | Receipts | | | Issued | | | Balance | | |
|---|---|---|---|---|---|---|---|---|---|
| | Qty | Unit Price | Value | Qty | Unit Price | Value | Qty | Unit Price | Value |
| | | | | | | | | | |

**Illustration 3.4**

Mr. Kofi Abebrese decided on 1 July, to invest his insurance compensation of 4,000,000 cedis in a retail business to buy and sell second hand shovels. The following transactions took place from that month to December.

| **Purchases** | | | **Sales** | | |
|---|---|---|---|---|---|
| **Date** | **Quantity** | **Cost** | **Date** | **Quantity** | **Value** |
| July 5 | 200 | 720,000 | Aug 2 | 500 | 2,500,000 |
| Aug 1 | 400 | 1,520,000 | | | |
| Sept 3 | 600 | 2,400,000 | | | |
| Oct 4 | 400 | 1,400,000 | Oct 12 | 600 | 2,700,000 |
| Dec 7 | 500 | 1,400,000 | Dec 12 | 400 | 1,500,000 |

**Required**

(a) Calculate the cost of shovels issued during the period and the cost of shovels on hand on 31/12 using the following methods of pricing

i. FIFO **(3 Marks)**
ii. LIFO **(3 Marks)**
iii Weighted average (calculation to 2 decimal places) **(4 Marks)**

(b) Calculate and discuss the effect, each of the pricing methods will have on the reported profit of the business **(3 Marks)**

(c) Examine critically the performance of the business during the period **(2 Marks)**

**(Total 15 Marks)**

**Solutions**

**Table 3.2 STORES LEDGER CARD- FIFO**

| Date | Receipts | | | Issued | | | Balance | | |
|---|---|---|---|---|---|---|---|---|---|
| | Quantity | Unit Price ¢ | Value ¢ | Quantity | Unit Price ¢ | Value ¢ | Quantity | Unit Price ¢ | Value ¢ |
| July5 | 200 | 3,600 | 720,000 | | | | 200 | 3,600 | 720,000 |
| August 1 | 400 | 3,800 | 1,520,000 | | | | 600 | | 2,240,000 |
| August 2 | | | | 200 | 3,600 | 720,000 | | | |
| | | | | 300 | 3,800 | 1,140,000 | 100 | | 380,000 |
| Sept 3 | | | | 500 | | | 700 | | 2,780,000 |
| | 600 | 4,000 | 2,400,000 | | | | | | |
| Oct 4 | | | | | | | 1,100 | | 4,180,000 |
| | 400 | 3,500 | 1,400,000 | | | | | | |
| Oct 12 | | | | 100 | 3,800 | 380,000 | | | |
| | | | | 500 | 4,000 | 2,000,000 | | | |
| | | | | 600 | | 2,380,000 | 500 | | 1,800,000 |
| | | | | | | | 1,000 | | 3,200,000 |
| Dec 7 | 500 | 2,800 | 1,400,000 | | | | | | |
| Dec 12 | | | | 100 | 4,000 | 400,000 | | | |
| | | | | 300 | 3,500 | 1,050,000 | | | |
| | | | | 400 | | 1,450,000 | 600 | | 1,750,000 |

(i) Stores ledger card (FIFO)

Closing inventory

October 4 100 units @ 3,500 = 350,000

| | | | |
|---|---|---|---|
| December 7 | 500 units @ 2,800 | = | 1,400,000 |
| 600 | | = | 1,750,000 |

Cost of sales:

| | Units | Cost |
|---|---|---|
| August 2 | 500 | 1,860,000 |
| October 12 | 600 | 2,380,000 |
| December 12 | 400 | 1,450,000 |
| | 1,500 | 5,690,000 |

Therefore, the cost of the 1,500 shovels sold using the FIFO method is ¢5,690,000 and the value of the remaining 600 shovels is ¢1,750,000.

**Table 3.3** : **STORES LEDGER CARD (LIFO)**

| Date | Receipts | | | Issued | | | Balance | | |
|---|---|---|---|---|---|---|---|---|---|
| | Quantity | Unit Price ¢ | Value ¢ | Quantity | Unit Price ¢ | Value ¢ | Quantity | Unit Price ¢ | Value ¢ |
| July5 | 200 | 3,600 | 720,000 | | | | 200 | 3600 | 720,000 |
| August 1 | 400 | 3,800 | 1,520,000 | | | | 600 | | 2,240,000 |
| August 2 | | | | | | | | | |
| | | | | 400 | 3,800 | 1,520,000 | | | |
| | | | | 100 | 3,600 | 360,000 | | | |
| | | | | 500 | | 1,880,000 | 100 | | 360,000 |
| | | | | | | | 700 | | 2,760,000 |
| Sept 3 | 600 | 4,000 | 2,400,000 | | | | | | |
| | | | | | | | 1,100 | | 4,160,000 |
| Oct 4 | 400 | 3,500 | 1,400,000 | | | | | | |
| Oct 12 | | | | 400 | | 1,400,000 | | | |
| | | | | 200 | 3,500 | 800,000 | | | |
| | | | | 600 | 4,000 | 2,200,000 | | | |
| | | | | | | | 500 | | 1,960,000 |
| | | | | 400 | | 1,120,000 | 1,000 | | 3,360,000 |
| Dec 7 | 500 | 2,800 | 1,400,000 | | | | 600 | | 2,240,000 |
| Dec 12 | | | | | 2,800 | | | | |

Closing inventory

| | Units ¢ | unit cost | value ¢ |
|---|---|---|---|
| | | | 360,000 |
| | 100 | 3,600 | 1,600,000 |
| July 5 September 3 | 400 | 4,000 | 280,000 |
| December 7 | 100 | 2,800 | 2,240,000 |
| | 600 | | |

Cost of sales

| Date of sales | unit sold ¢ | Cost of units sold ¢ |
|---|---|---|
| August 2 | 500 | 1,880,000 |
| October 12 | 600 | 2,200,000 |
| December 12 | 400 | 1,120,000 |
| 1,500 | | 5,200,000 |

**Table 3.4 STORES LEDGER CARD- WEIGHTED AVERAGE**

| Date | Receipts | | | Issued | | | Balance | | |
|---|---|---|---|---|---|---|---|---|---|
| | Quantity | Unit Price | Value ¢ | Quantity | Unit Price | Value ¢ | Quantity | Unit Price | Value ¢ |
| July 5 | 200 | 3,600 | 720,000 | | | | 200 | 3,600 | 720,000 |
| August 1 | 400 | 3,800 | 1,520,000 | | | | 600 | 3,733 | 2,240,000 |
| August 2 | | | | 500 | 3,733 | 1,866,500 | 100 | 3,735 | 373,500 |
| Sept 3 | 600 | 4,000 | 2,400,000 | | | | 700 | 3,962 | 2,773,500 |
| Oct 4 | 400 | 3,500 | 1,400,000 | | | | 1,100 | 3,794 | 4,173,500 |
| Oct 12 | | | | | | | 500 | 3,794 | 1,897,100 |

Therefore, the cost of 1,500 shovels sold using the LIFO method is ¢5,200,000 and the value of the remaining inventory of 600 units is ¢ 2,240,000.

(iii)Stores ledger card (Weighted Average)
Closing inventory
600 units @ 3,297 = 1,978,300

| Cost of sales: Date of sales | units sold | cost of sales ¢ |
|---|---|---|
| 2/8 | 500 | 1,866,500 |
| 12/10 | 600 | 2,276,400 |
| 12/12 | 400 | 1,318,800 |
| | 1,500 | 5,461,700 |

b)

Sales revenue

| Date of sales | sales revenue ¢ |
|---|---|
| 2/8 | 2,500,000 |
| 12/10 | 2,700,000 |
| 12/12 | 1,520,000 |
| | 6,720,000 |

Profit statement

| | FIFO method ¢ | LIFO method ¢ | Weighted average method ¢ |
|---|---|---|---|
| Sales | 6,720,000 | 6,720,000 | 6,720,000 |
| Less cost of sales | 5,690,000 | 5,200,000 | 5,461,700 |
| Gross profit | 1,030,000 | 1,520,000 | 1,258,300 |

From the gross profit figures above, it can be seen that the profit from the LIFO is the smallest. The weighted average method profit falls between the FIFO and LIFO gross profits. This position is because the cost of shovels have tended to decline. In a situation where the cost of materials experiences an upward trend, FIFO will always produce the greatest profit and LIFO the smallest profit.

b) The company has been able to make some margin on its costs. The sales price of the company is however on a downward trend from ¢5,000 per unit to ¢3,800 per unit for the last sales in December. Its cost has also experienced decline.

The company is operating in an industry faced with falling prices. This could imply a depression for the industry.

**Illustration 3.5**

Edwuma Pa Ye Manufacturing Company Ltd is reorganizing its costing system. A materials consultant has recommended that the company adopts one pricing method to cost the issues of materials to production or specific jobs. He recommended that the choice should be either the FIFO or LIFO method. Based on the consultant's recommendation, Edwuma Pa Ye has requested that you assist its newly employed cost accountant to prepare cost statements for its Job 2499. The following data for material 'A' used for the job in May 2025 have been provided.

| Date | Particulars | Kilos | Cost ¢ |
|---|---|---|---|
| 2nd May | Balance | 40,000 | 12,000,000 |
| 5th May | Receipts | 10,000 | 400 per kilo |
| 12th May | Receipts | 24,000 | 500 per kilo |
| 16th May | Issues | 48,000 | |
| 19th May | Receipts | 34,000 | 450 per kilo |
| 26th May | Issues | 40,000 | |

**Required:**

(a) i Using FIFO and LIFO methods of pricing, cost the materials issued to job 2499 as well as the materials on hand on 26th May, 2025. **(7 Marks)**

ii Using the two methods, work out the price to be charged for Job 2499 based on the following additional information:
Direct labour for the job is ¢10,000,000
Overhead is absorbed at 110% of materials.
It is company policy to make a profit margin of 10% on all jobs. **(6 Marks)**

(b) Edwuma Pa Ye Manufacturing Company Ltd further sought the materials consultant's advice on inventory management for production and the commitment of funds for such a purpose.

The consultant recommended two policy strategies:

i. that the company should ensure that sufficient amount of materials is available to reduce any subsequent shortage risk and subsequent interruptions in production
ii. that working capital should as much as possible not be locked up in excess materials.

The company has decided to implement the above recommendations and had provided the following data on materials B based on its order schedule and production capacity for the year.

i) 400 Kg of materials B will be used every week for 50 weeks
ii) It will cost ¢30,000 to make an order
iii) Each unit of the material will cost ¢1200 and it will cost the company an additional 33 1/3% of that cost every year to carry a kilo of the material
iv) Consumption of material will not exceed 600 kilos in a week, it will not fall below 400 kilos. Usually, it takes between 1 to 3 weeks for suppliers to deliver orders that are made.

**Required:**

The company has requested you to compute:

i) the economic quantity that ought to be ordered
ii) the level at which the company should put in a new order
iii) the maximum level that material B should carry
iv) the minimum level that material B should carry (5 marks)

(Total marks:18) ICAG, NOV. 1997 (adapted)

Solutions

**Table 3.5 STORES LEDGER CARD (FIFO)**

| Date | Receipts | | | Issued | | | Balance | | |
|---|---|---|---|---|---|---|---|---|---|
| | Quantity | Unit Price | Value ¢ | Quantity | Unit Price | Value ¢ | Quantity | Unit Price | Value ¢ |
| May 2 | | | | | | | 40,000 | 300 | 12,000,000 |
| May 5 | 10,000 | 400 | 4,000,000 | | | | 50,000 | | 16,000,000 |
| May 12 | 24,000 | 500 | 12,000,000 | | | | 74,000 | | 28,000,000 |
| May 16 | | | | | | | | | |
| | | | | 40,000 | 300 | 12,000.000 | | | |
| | | | | 8,000 | 400 | 3,200,000 | | | |
| | | | | | | | | | 12,800,000 |
| | | | | 48,000 | | 15,200,000 | | | |
| May 19 | 34,000 | 450 | 15,300,000 | | | | 26,000 | | 28,100,000 |
| May 26 | | | | | | | 60,000 | | |
| | | | | 2,000 | 400 | 800,000 | | | |
| | | | | 24,000 | 500 | 12,000,000 | | | |

Cost of issues for production:

| Date | Quantity Issued | Cost |
|---|---|---|
| | | ¢ |
| May 5 | 48,000 | 15,200,000 |
| May 26 | 40,000 | 19,100,000 |
| | 88,000 | 434,300,000 |

Valuation of Closing Inventory:

| Date | Quantity Outstanding | Price per Unit | Value |
|---|---|---|---|
| | | ¢ | ¢ |
| May 19 | 20,000 | 450 | 9,000,000 |

**Table 3.6** : **STORES LEDGER CARD (LIFO)**

| Date | Receipts | | | Issued | | | Balance | | |
|---|---|---|---|---|---|---|---|---|---|
| | Quantity | Unit Price ¢ | Value ¢ | Quantity | Unit Price ¢ | Value ¢ | Quantity | Unit Price ¢ | Value ¢ |
| May 2 | | | | | | | 40,000 | 300 | 12,000,000 |
| May 5 | 10,000 | 400 | 4,000,000 | | | | 50,000 | | 16,000,000 |
| May 12 | 24,000 | 500 | 12,000,000 | | | | 74,000 | | 28,000,000 |
| May 16 | | | | | | | | | |
| | | | | 24,000 | 500 | 12,000,000 | | | |
| | | | | 10,000 | 400 | 4,000,000 | | | |
| | | | | 14,000 | 300 | 4,200,000 | | | |
| | | | | 48,000 | | 20,200,000 | | | |
| | | | | | | | 26,000 | | 7,800,000 |
| | | | | | | | | 300 | |
| May 19 | 34,000 | 450 | 15,300,000 | 34,000 | | 15,300,000 | 60,000 | | 23,100,000 |

Cost of issues for production:

| Date of issue | Unit issued | Cost of issues ¢ | |
|---|---|---|---|
| May 16 | | 48,000 | 20,200,000 |
| May 26 | | 40,000 | 17,100,000 |
| | | 88,000 | 37,300,000 |

Valuation of closing inventory

May 2 20,000 @ ¢300 = ¢6,000,000

ii) JOB COST CARD FOR JOB 2499

| | FIFO | LIFO |
|---|---|---|
| | ¢ | ¢ |
| Direct materials cost | 34,300,000 | 37,300,000 |
| Direct labour cost | 10,000,000 | 10,000,000 |
| Overheads cost | | |
| (110% of direct materials) | 37,730,000 | 41,030,000 |
| Total cost | 82,030,000 | 88,030,000 |
| Profit margin | | |
| (10% of total cost | 8,203,000 | 8,833,000 |
| Price of job | 90,000,000 | 97,163,000 |

(b) Economic Order Quantity (EOQ)

$$= \sqrt{\frac{2DCo}{Cc}}$$

Where:

D = annual demand

Co = Cost of order

Cc = Carrying cost per unit

i) $EOQ = \sqrt{\frac{2\,(20{,}000)\,(30{,}000)}{400}}$

$= 1{,}732$,kg

i) Re Order Level = maximum usage X maximum lead time

$= 600 \times 3$

$= 18{,}000$kg

ii) Maximum level = ROL + EOQ (minimum usage x minimum lead time)

= (1,800 + 1,732) (400 x 1)

= 3,132 kg

iii) Minimum level

= ROL - (average usage x average lead time)

$$= \left(\frac{1{,}800 - 600}{2}\right) + 400\left(\frac{3 + 1}{2}\right)$$

= 1,800 - (500 x 2)

= 800 kg

## 3.10 PERPETUAL INVENTORY SYSTEM

This is a system of inventory control where records such as the bin card, tally card, stores ledger account, etc. are kept as a perpetual record of receipts and issues of materials. Thus the system keeps a perpetual record of all inventory transactions updated on a continuous basis.

The main objective of such a system is to know the inventory level at any particular time, facilitate regular checking of inventory level and to prevent the need for stores closure for inventory verification. The records from the perpetual inventory system should be reconciled with the result of inventory count.

## 3.11 INVENTORY TAKING

This is the physical counting of materials in the store. Inventories of raw materials, work-in-progress and finished goods are usually subject to an inventory take.

There are two basic types of inventory taking methods:

a. Annual / Periodic Inventory taking; and
b. Continuous Inventory taking

### 3.11.1 PERIODIC INVENTORY TAKING

This is a method of counting inventory where the items are counted at the end of a given period usually, annually or semi-annually.

**Merits**

a. It is less expensive to conduct an inventory count once in the whole year; and
b. The inconvenience of regular or frequent inventory count is avoided.

**Demerits**

i. It may require a disruption of operational activities at the end of the year when the inventory take is to take place;

ii. The size of the work may justify the use of a large team; some members of the team may not be familiar with the particular inventory items. This may lead to inaccuracies in the inventory count;

iii. Discrepancies which are the result of an on-going problem and which should have been noted at an earlier date may be revealed; and
iv. The deferment of the inventory taking to the end of the year may encourage theft and pilferage.

### 3.11.2 Continuous Inventory Taking

This is a method of inventory taking where inventory items are counted at frequent intervals on a random rotational basis and the results of the counts reconciled with the perpetual inventory records.

**Merits**

a. It is a deterrent to pilferage and theft;
b. The closure of stores for an annual inventory count is avoided;
c. There is ready availability of reliable inventory balances, throughout the year;
d. Discrepancies are discovered and remedied more promptly;
e. The stores system is kept constantly under review; and
f. It enhances effective inventory control.

**Demerits**

i. It is expensive to operate a continuous inventory taking system.

### 3.11.3 Effective Inventory Procedures

In order to have a successful inventory taking, the following should be considered

a) Inventory items should be arranged so as to avoid omissions and double counting;
b) Counting should be done by pairs of two persons, one person counting and the other recording;
c) The inventory take should be well organised to avoid or minimise production disruptions;
d) The counting of inventories should be done whilst production is in

progress; and

e) The final inventory sheet should be checked against the store records and discrepancies immediately investigated.

### 3.11.4 **Causes of Stock Discrepancies**

a) Incomplete entries caused by non-completion of original documents;
b) Casting errors on inventory records;
c) Over or under issues due to carelessness of the issuing officer;
d) Recording materials in the wrong inventory record;
e) Movement of inventories without proper documentation;
f) Pilferages;
g) Loss due to breakage, evaporation, etc.;
h) Inaccuracies in the inventory count; and
i) Omissions.
j) Double counting, etc.

## 3.12 **CHAPTER SUMMARY**

This chapter covers materials cost determination and control and has been treated under six sections.

a) Section one explains what constitutes materials, the functions of stores and the factors that facilitate effective materials cost control;

b) Section two concentrates on materials procurement process in organisations. This section explains the responsibilities of the procurement unit, the methods of purchasing and the various stages of the purchasing function;

c) Section three focuses on the receipts and storage of materials. Issues discussed in this section include: the stores operation process; the factors that influence the design of procedures for operating the inventory; principles of good storage and issuing, and factors that facilitate effective materials cost control;

d) Section four discusses the various methods of costing issue of inventories for production and closing inventories. Some of the methods discussed include; FIFO, LIFO, Weighted Average, Standard Price, etc.

e) Sections five and six explain the procedure for controlling inventory. This involves establishing inventory control levels such as re-order level, maximum level, minimum level and re-order quantities of inventory so that inventory cost will be minimised;

f) Some records are kept in Bin Cards, Tally Cards or Store Ledger Cards to ensure that the control levels established are maintained. To facilitate effective inventory control, it is necessary to undertake inventory counts. All of these issues have been discussed in section six.

In conclusion, material cost is usually a significant proportion of the total cost of operations of business organizations. To maximise profitability, effective ways must be adopted to control material cost to the minimum.

## MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS

1. Materials include the following EXCEPT
A. Finished goods
B. Work in progress
C. Consumable goods
D. Fixed assets
E. Raw materials

2. The work of the storekeeper does NOT include
A. Receiving inventory items
B. Issuing inventory items
C. Custody of inventory items
D. Recording of inventory items
E. Selling of inventory items

3. Purchase orders are issued by
A. Quality Control Manager
B. Procurement Manager
C. Stores Manager
D. Cost Accountant
E. Production Manager

4. When ordered materials are received, they are brought into stores via
A. Materials Requisition
B. Local Purchase Order
C. Goods Received Note
D. Tenders
E. Materials Issue Note

5. Under the First in First Out method, store issues are priced using the prices of
A. The last batches received into store
B. The first batches received into store
C. The middle batches received into store
D. The average of the first and last batches received into store
E. The next batch to be received into store

## SHORT-ANSWER QUESTIONS (SAQ)

1. The inventory valuation method that assumes that inventories are issued in reverse order of receipts is called …….

2. The level of inventory below which quantities are not expected to fall in the store is called……..
3. The optimum quantity of inventory that should be ordered from suppliers at any one time is known as …….
4. The name of the document which the storekeeper uses in recording the receipt and issue of materials is called……..
5. The system whereby bits of store items are counted at frequent intervals so that by year end all items would have been counted at least once is known as……

**SOLUTIONS**

1. D
2. E
3. B
4. C
5. A
6. Last In First Out (LIFO)
7. Minimum inventory level
8. Economic Order Quantity
9. Bin Card
10. Continuous Inventory taking

**EXAMINATION TYPE QUESTIONS**

3.1(a) One of the board members of Dunu Manufacturing Company wants to know the purchasing procedure that the company adopts to procure its raw materials. As a Cost Accountant, outline the main purchasing procedures. **(14 Marks)**

b. What are the costs associated with the following:
i. Excessive inventory holding?
ii. Stockouts? **(6 Marks)**

**(Total 20 Marks)**
ICAG, May 2001

3.2 (a) Explain briefly the following, bringing out clearly, the formulae for calculation where appropriate

i. Free inventory balance
ii. Minimum inventory balance
iii. Maximum inventory balance
iv. Re-order level
v. Average inventory level
vi. Re-order quantity /economic order quantity
vii. Lead time **(12 Marks)**

(b) Use the following information to calculate the maximum, minimum, re-order and average inventory levels for Boateng Ltd.

Boateng Ltd manufactures a special product for the domestic market. Records available at the stores department indicated the following:

| | | | |
|---|---|---|---|
| Maximum usage | - | 1,200 units per week | |
| Minimum usage | - | 500 units per week | |
| Re-order quantity | - | 1,500 units per week | |
| Delivery period | - | between 2 to 4 weeks | **(8 Marks)** |

ICAG, May 2000

3.3 (a) Distinguish between perpetual inventory system and continuous inventory taking. **(5 Marks)**

ICAG NOV. 1999

3.4 State and explain **TWO** merits of each of the following methods of pricing issues of materials from stores to production and service cost centres

(a) Weighted Average Method
(b) Standard Price Method
(c) LIFO Method
(d) Replacement Price Method
(e) FIFO Method

**(20 Marks)**
ICAG JAN. 1995

3.5 The level of stock held is controlled to enable a company operate on an economic level.

(a) Explain briefly the following inventory levels, giving the formulae of computation as appropriate:

i) minimum inventory level
ii) maximum inventory level
iii) re-order inventory level
iv) average inventory level **(12 Marks)**

(b) You are required to compute

i) re-order inventory level
ii) minimum inventory level
iii) maximum inventory level
iv) average inventory level

Using the following data: minimum usage 250 per week, maximum usage 250 per week, re-order quantity 30, unit delivery is between 4 to 6 weeks. **(8 Marks)**

**(Total 20 Marks)**
ICAG January 1994

3.6 Ebeyeye Co Ltd stocks a number of basic raw materials to meet its production demands. The following are the forecast consumptions in respect of one of the raw materials for the year 1993.

| Month | Consumption | Month | Consumption |
|---|---|---|---|
| January | 10,000 | July | 15,000 |
| February | 10,000 | August | 15,000 |
| March | 14,000 | September | 11,000 |
| April | 14,000 | October | 11,000 |
| May | 15,000 | November | 10,000 |
| June | 15,000 | December | 10,000 |

Delivery period from suppliers

Minimum 2 months
Maximum 4 months

**Required :**

(a) State the main objectives of inventory control. **(4 Marks)**

(b) Give reasons why the average inventory level ought to be maintained at the optimum level, that is, the level most favourable to a company that stocks a wide range of materials. **(6 Marks)**

(a) Calculate Ebeyeye Co Ltd's estimated average inventory level for the year from the information provided.

**(10 Marks)**
**(Total 20 Marks)**

ICAG January 1993

3.7 In manufacturing its products, a company uses three raw materials, A, B, and C in respect of which the following applies:

| Raw Material | Usage per unit of product | Price per kilo | Delivery period in months | | Re-order Level | Mini-mum level |
|---|---|---|---|---|---|---|
| | Kilo | | Min | Max | Kilos | kilos |
| A | 16 | 10 | 1 | 3 | 6750 | - |
| B | 4 | 30 | 3 | 5 | 4500 | 1300 |
| C | 6 | 15 | 2 | 4 | - | 1800 |

Monthly production varies from 175 to 225 units averaging 200 units

Re-order quantities are as follows

| Material | kilos |
|---|---|
| A | 10,000 |
| B | 5,000 |
| C | 10,000 |

**Required**

(a) Explain:

i) Minimum inventory level **(2 Marks)**
ii) Maximum inventory level **(2 Marks)**
iii) Re-order level **(2 Marks)**
(b) Calculate the quantities of the following: Minimum
i) inventory of material A **(2 Marks)**
ii) Maximum inventory of material B **(2 Marks)**
iii) Re-order level of material C **(2 Marks)**
iv) Average inventory level of material A **(2 Marks)**

(c) Material B could be the increase in 10,000 kilos lots at ¢29 per kilo: **(4 Marks)**
i) What would be the increase in the value of the average stock of B?

ii) What would be the saving in raw material cost for one unit of product B?
iii) What would be the annual saving on material costs as a percentage of the average **(**increase in investments in inventory? **(4 Marks)**

**(Total 24 Marks)**

ICAG January 1989

3.7 Describe the essential requirements of an effective material stock control system.

**(17 Marks)**

CACA Level I Costing June 1985

3.8 a. With reference to Material Control System, you are requested to:
   i. Explain continuous inventory taking
   ii. Buffer inventory

b. Explain the source and purpose of the under-mentioned documents:
   i. Goods received note;
   ii. Materials returned note
   iii. Purchase requisition

3.9 TMT Business provides the following extracts of its operations:

| | |
|---|---|
| i. Annual requirements | 12,600 units |
| ii. Inventory holding/carrying costs | ₦50 per unit of material |
| iii. Ordering cost per annum | ₦350 |
| iv. Re-Order Period: | |
| Minimum | 4 days |
| Average | 5 days |
| Maximum | 6 days |
| v. Consumption rates: | |
| Minimum consumption per day | 30 units |
| Average consumption per day | 35 units |
| Maximum consumption per day | 40 units |

**Required:**

a. What is the importance of determining maximum inventory level for the period?
b. Determine: (i) Re-order level, (ii) Minimum level, (iii) Average level; (iv) Maximum level
c. Produce EOQ graphs, for carrying cost, ordering cost, and total cost, indicating EOQ in units and in monetary cost.

3.10 Elam Ltd's inventory purchases during a recent week were as follows:

| Day | Price per unit ₦ | Units Purchased |
|---|---|---|
| 1 | 145 | 550 |
| 2 | 160 | 880 |
| 3 | 175 | 1,200 |
| 4 | 180 | 750 |
| 5 | 190 | 1,300 |

There were no inventories at the beginning of the week. 4,200 were issued to production during the week. The company updates its inventory records after every transaction.

**Required:**

a. List the benefits of FIFO methods of pricing materials issued from stores to the factory floor.
b. Produce a stores ledger to record the transactions above, using FIFO method.
c. If Elam Ltd changes to the weighted average method of inventory valuation, determine the new value of closing inventory and likely effect on profit. **20 Marks**

## ACCOUNTING FOR LABOUR

## CONTENTS

a. Explain recording and monitoring procedures, namely time-keeping and time-booking functions.
b. Explain job evaluation, merit rating, work study, methods study and work measurement.
c. State and explain the basic labour remuneration methods, including time-based systems, piecework systems, and individual and group incentives schemes.
d. Explain the treatments of overtime and idle time in labour costing.
e. Explain and prepare entries in payroll accounting.
f. Explain the causes of labour turnover and ascertain the costs of labour turnover.

**4.0 Objectives:**

After studying this chapter, a reader should be able to:

a. Explain labour as a factor of production;
b. Explain what constitutes recruitment cost;
c. Explain labour time-keeping and time-booking functions;
d. Explain job evaluation, merit rating, work study, methods study and work measurement;
e. Compute labour cost using various techniques of remuneration;
f. Explain the treatments of overtime and idle time in labour costing;
g. Compute overtime premium and explain the various treatments of overtime premium;
h. Explain group incentive schemes; Prepare entries in payroll accounting;
i. Explain the causes of labour turnover; and
j. Ascertain the costs of labour turnover.

## 4.1 INTRODUCTION

Labour is the second element of cost after materials. Labour cost represents the remuneration for employees' effort in the production process. The processes involved in controlling labour cost are complex, just like the complexities involved in the control of human beings and their behaviour. This chapter explains the processes involved in the determination and control of labour costs.

## 4.2 LABOUR ADMINISTRATION AND DOCUMENTATION

### 4.2.1 Labour: A Factor of Production.

Factors of production are resources or inputs that are used to facilitate the production of goods and services. Labour can be described as the collective efforts of employees in a production process while transforming raw materials into finished goods. Labour takes the form of employee knowledge, efforts, expertise and experience. Without the input of labour skills and expertise, production cannot take place. Labour encompasses all categories of the workforce of an organisation, spanning from the top management to the grassroots workforce. From the perspective of cost accounting, labour cost determination and control are very important because it has huge cost implications for the organisation. Labour cost consists of recruitment cost, cost of training and staff development, wages and salaries, employee bonuses, overtime premium, group incentives, etc. Therefore, if these costs are not properly controlled, the total cost of labour may not be minimised.

### 4.2.2 Labour Recruitment Process

Labour recruitment refers to the process of engaging employees to render services for an organisation. Depending on the policy of an organisation, the recruitment of new employees may be carried out internally or externally. In some organisations, both internal and external methods may be adopted. Whichever method is adopted, the following steps are imperative and must be carried out as part of the process.

a. A job analysis should be carried out to determine the roles, responsibilities and specifications required for a position.
b. If the organisation plans to adopt an internal mode of recruitment, then an advertisement must be widely circulated within the organisation and copies pasted on the company's notice boards to create awareness and allow for qualified persons within the organisation to apply for such vacant positions.

c. Where external methods are adopted, advertisements may be circulated in labour offices, newspapers or through social media platforms. Other methods could be through job fairs, outsourcing, manhunt, executive searches and so on
d. Interested applicants will respond appropriately to the company's invitation, and those found qualified will be shortlisted for the selection process.

**Selection Process**

Interviews will be conducted to evaluate all the candidates present for the interview. The method of the interview process could be structured or unstructured. A structured interview should have an agreed format with questions ordered to evaluate applicants. Using a pre-determined criterion, the most qualified or suitable candidate will be selected at the end of the process.

### 4.2.3 Labour Induction and Placement

Induction is a programme carried out to ensure that all new employees are integrated into the culture of the organisation by way of acquaintance with the values and norms in force within the system. Placement is an art of fixing the newly employed in their respective posts based on their qualifications, competencies and on the job experiences. Training may follow thereafter to ensure enhanced performance on the job.

Training may involve a consideration of the following:

a. Training needs analysis;
b. Assessing training budget;
c. Prioritising the ones considered most critical;
d. Defining skills, knowledge, abilities and objectives desired after training;
e. Considering the training method, and
f. Evaluating the training.

In summary, the total cost expended on the process of recruitment will include:

a. Advertising cost
b. Interviewing expenses
c. Cost of induction of new employees; and
d. Training expenses
e. While total labour costs will comprise total recruitment cost as stated above, plus other costs such as:
f. Wages and salaries of employees;
g. Overtime premiums;
h. Employee bonuses; and
i. Retirement payments, etc.

### 4.2.4 Labour Timing and Assessment.

After employees have been engaged and reported at the workplace, there is a need to track them as they work to assess their performance and also to make it possible to cost of the work that they do. Various organisations use different methods to time their employees. Examples of methods used include:

a. Attendance Register;
b. Clock Cards;
Attendance Boards;

c. Job Sheets; and
d. Job Books, etc.

The above records will then be used to gather pieces of evidence in respect of the following: Hours worked;

i. Time spent by each employee on a particular job or service; and
ii. Hours of idleness; etc

### 4.2.5 Labour Behaviour and Control

Labour is about human beings, and it is relatively difficult to control human behaviour. However, it must be taken by the organisation to control human behaviour. Consequently, some mechanisms have been devised to aid labour control. Some of these mechanisms include:

a. Time keeping- employees are required to record the time they report for work and departure time. Each employee will be required to record the time spent on each job or work handled.
b. Monitoring and supervision- employees are monitored and supervised by senior officers who, in addition, also observe the work attitude of the employee.
c. Evaluation- the work of the employee is evaluated by his peers, clients and management. This usually motivates staff to exhibit positive attitudes at the workplace.
d. Reward schemes, rewarding effective employees through bonuses and incentives, drive employees to work hard.

### 4.2.6 Job Evaluation

This is a technique which seeks to show the relative worth of each job to rank it against other jobs and ultimately establish the appropriate weight of remuneration to attach to the job. Job evaluation analyses the content of each job using yardsticks such as degree of responsibility, decisions involved, training and experience required, working conditions, etc., awarding points for each yardstick.

### 4.2.7 Merit Rating

Merit rating measures the jobholder's performance to determine whether the employee should be promoted, demoted or given a special reward. It also uses its yardsticks on the performance and attributes of the employee, like accuracy, initiative, level of responsiveness, willingness, etc. Most industries employ rating techniques at the end of each year to determine the progress of each employee within the organisation.

Structure.

### 4.2.8 Work Study

This is a system of increasing or maximising the productivity of an operating unit by reorganising the work of that unit. Work study is subdivided into two major methods, namely methods study and work measurement.

a. **Method study**
This is the recording and critical examination of existing methods of doing work and comparing them with proposed methods with a view to coming up with easier methods which would be more effective and cheaper in the long run.

**b. Work measurement**

As the name suggests, work measurement seeks to measure the time required for a qualified worker to complete a specific assignment at a specified level of performance.

## 4.3 LABOUR REMUNERATION METHODS

There are two basic methods of remunerating labour: a. Time-based remuneration; and
b. Output-based remuneration.

### 4.3.1 Time Rate Methods of Remuneration

Under this method, the amount earned by the employee is based on the number of hours spent at their place of work and not on the quantity of work produced. The gross wage is calculated as: Hours Worked × Rate per hour. However, when overtime is worked, the payment to the employee will also include a premium for the overtime hours.

Illustration 4.1

| | | |
|---|---|---|
| Hours worked | = | 50hours |
| Rate per Hour | = | ¢2,000 |
| Gross wage | = | ¢2000 x 50hours |
| | = | ¢100,00 |

***Advantages:***

i. It is simple to operate and easy to understand.
ii. The quality of work produced tends to be higher since the worker is not in a rush to complete a job to maximise his earnings.

***Disadvantages:***

i. There is no financial incentive to produce more than a minimum amount.
ii. There is often an incentive to produce as little as possible so that the worker can increase their wage.
iii. To monitor and check idleness, the employer will be obliged to incur supervision costs.
iv. The method is often unfair because lazy workers and hard-working staff are paid the same rates.

### 4.3.2 Piece Rate (Payment by Result) Method

Under this method, the amount earned by the employee is based on the number of units produced. Piece rates can take the form of any of the following:

(a) Straight Piece Rate
(b) Differential Piece Rate
(c) Piece Rate with guaranteed time Rate

a. **Straight Piece Rate**: Under straight piece rates, the payment to the employee is computed by multiplying the number of units produced by the rate per unit.
Straight Piece Rate = No. of units produced × Rate per unit.
The worker receives a fixed rate for each unit produced, which does not depend on the time taken to produce it. Earnings, therefore, depend on the volume of the worker's output.

Illustration 4.2
Number of units produced, Rate per Unit,
Gross wage

Example 2: Normal rate/ hour
= 1,000 unis = e16,000
= 1,000units x 16,00 = ₵16,000,000 ₵1,600

***Advantages***

1. Effort is rewarded and, in consequence, the employee is given the incentive to produce more.
2. Employees are self-motivated; hence, less supervision is required.
3. The employer benefits from a reduction in the overhead cost per unit of production.

***Disadvantages***

1. There is a danger that quality will be sacrificed, and to avoid such a situation, the employer would spend more on inspection and quality control.
2. Piece workers, after earning a certain remuneration during a week, might be satisfied and slacken their pace, arrive late or absent themselves from work. Plant will therefore be left idle and capacity will be under-utilised.
3. A considerable degree of time is involved in setting standard times and as these are subject to the agreement of trade union representatives, further time is often spent in detailed negotiation before piece rates are established.
4. If piece rates are set too high erroneously, it will be difficult to reduce rates

thereafter, and this can adversely affect the cost of products.

b. **Piece Rate with Guaranteed Day Rate**: It is a system adopted to compensate employees on account of low production, leading to earnings under the piece rate being below the normal day rate remuneration. If an employee's earnings according to the piece work are less than the normal Day Rate, he is paid the day rate instead of the Piece Rate.

Illustration 4.3

| | |
|---|---|
| Rate per hour | ℓ 15,000 |
| Cost per unit | ℓ 16,000 |
| Units produced | 8,000 units |

An organisation charges ¢15,000 per hour for a product that costs ¢16,000 per unit. Assuming 8,800 hours were spent to produce 8,000 units. You are required to calculate the piece rate using the guaranteed day rate method of remuneration.

Piece Rate Earnings = Units produced x Rate per unit 8,000 units x ¢15,000
¢120,000,000
Time Rate Earnings = Hours worked x Rate per hour
8,800 hours x ¢15,000
¢132,000,000

c. Since the guaranteed hourly rate is higher than the piece rate, the employee is paid the hourly wages of ¢132,000,000

**4.3.3 Differential Piece Rate**: Under this scheme, the piece work rate changes at different levels of efficiency or production.

Assuming the labour rates charged by XYZ Company are as follows:
¢10,000 per unit when production is below 7 units per hour
¢15,000 per unit, when production is 7 to 10 units per hour.
¢20,000 per unit when production is above 10 units per hour, etc.

The objective of this is to provide a strong incentive to reach the maximum rate of production.

**4.3.4 Premium Bonus Schemes**

Bonus schemes are intended to reward employees for their efficiency in saving costs for the organisation through the saving of time. These are therefore schemes for sharing extra profit with employees. Conceptually, the bonus can only be awarded where there has been cost savings or improved performance that leads to exceeding the profit target. However, to compute bonuses, we need to appreciate the following concepts:

Time allowed: This refers to the expected time fixed to carry out a piece of work. e.g., if the time set for one unit is 5 hours, then 100 units will be expected to be 500 hours. However, the Time allowed may not be the same as the hours worked.
Time taken: This is the number of hours actually spent in performing a piece of work. Time saved: This is the difference between the time allowed and the hours worked, when the time allowed is greater than the hours worked.

Premium bonus: This is paid when time has been saved; the magnitude of the bonus therefore depends upon the time saved.

**Types of Premium Bonus Schemes:**

a. Individual Incentive Schemes
b. Group Incentive Schemes

#### 4.3.4.1 Individual Incentive Schemes

These are bonus schemes that reward employees for their efficiency on an individual basis.
The schemes include the following:

(i) Halsey Bonus Scheme
(ii) Halsey-Weir Bonus Scheme
(iii) Rowan Bonus Scheme

**Halsey Scheme:** According to this scheme, the time saved should be apportioned equally between the employer and the employee.

Bonus = ½ × (Time Saved) × Day Rate
Note: Time allowed – Time Taken = Time Saved

**Halsey-Weir Scheme**: Under this scheme, time saved is shared in the ratio of 2:1 in favour of the employer. Thus, the employee gets a third of time saved at the specified rate per hour.

Bonus = ⅓ (Time Saved) × Day rate

**Rowan Scheme**: Under this system, the bonus awarded to the employee is based on the proportion between the time taken and the Time allowed of the Time saved.

$$\text{Bonus Awarded to Employee} = \frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{Hours Worked} \times \text{Day Rate}$$

Total Remuneration = (Time Taken x Day Rate) + Bonus Awarded to Employee
It therefore follows that if the employee saves more time, he gets a greater bonus. Example:

| | | |
|---|---|---|
| Time allowed | = | 12 hours. |
| Day rate | = | ¢18,000 |
| Time taken by A | = | 6 hours |
| Time taken by B | = | 9 hours |

**Required:**
Calculate the Bonus to be awarded to both employees

**Solution**
Bonus to be awarded under Rowan Bonus Scheme: **Employee A**

Time allowed = 12hrs

$$\text{Bonus} = \left[\frac{\text{Time Saved}}{\text{Time Allowed}}\right] \times \text{Time Taken} \times \text{Day rate}$$

Time Taken = 6hrs
Time Saved 6hrs

$$= \left[\frac{6\,hrs}{12hrs}\right] \times 6 \times ¢18{,}000$$

$$= \underline{¢54{,}000}$$

**Employee B**

Time allowed = 12hrs Bonus = $\left\lceil \frac{\textit{Time Saved}}{\textit{Time Allowed}} \right\rceil \times$ Time Taken × Day rate

Scheme: **Employee A**

Time allowed = 12hrs Bonus = $\left\lceil \frac{\textit{Time Saved}}{\textit{Time Allowed}} \right\rceil \times$ Time Taken × Day rate

Time Taken = 6hrs

Time Saved 6hrs

$$= \left\lceil \frac{6\ hrs}{12hrs} \right\rceil \times 6 \text{ x } ¢18{,}000$$

$$= ¢54{,}000$$

**Employee B**

Time allowed = 12hrs Bonus = $\left\lceil \frac{\textit{Time Saved}}{} \right\rceil \times$ Time Taken × Day rate

$$\left\lfloor \frac{\text{Time Allowed}}{\text{Time Taken}} \right\rfloor = \left\lfloor \frac{12-9}{12} \right\rfloor \times 9 \times ¢18{,}000$$

Time Taken = 9hrs

Time Saved = 3 hrs

$$= \left\lfloor \frac{3}{12} \right\rfloor \times 9 \times ¢18{,}000$$

$$= \underline{¢40{,}500}$$

Time Taken = 9hrs $= \frac{\lceil 12-9 \rceil}{\lfloor 12 \rfloor} \times 9 \times ¢18{,}000$

Time Saved = 3 hrs

$= \frac{\lceil 3 \rceil}{\lfloor 12 \rfloor} \times 9 \times ¢18{,}000$

$= \underline{¢40{,}500}$

Many business organisations determine their bonuses through negotiations with employee groups. However, the factors that influence the size of the bonus paid to employees include the following:

- Time saved by employees
- Cost saved by employees
- Improved productivity
- The number of super profits made by the business organisation
- The achievements of other budgetary targets, etc.
- Market demand for the product

#### 4.3.4.2 Advantages and Disadvantages of Individual Incentive Schemes

**Advantages of Individual Incentive Schemes**

- Individual employees are motivated to be more efficient and productive
- It may generate a competitive spirit among employees
- Employees' morale is raised since individual effort is rewarded
- Ultimately, both the employee and the business organisation obtain enhanced benefits

**Disadvantages of Individual Incentive Schemes**

- Employees may compromise on quality to increase their bonus earnings
- Excessive competition can bring about unhealthy rivalry
- The determination of standard performance levels determine efficiency levels may give rise to conflict in an organisation
- It is relatively more difficult and expensive to operate an individual incentive scheme compared to a group incentive scheme.

### 4.4 Overtime Remuneration Schemes

Overtime is the time spent beyond the normal working hours or days. Overtime wage rates are expressed as time plus a fraction or multiples of time, e.g.

i. Time and a half
ii. Time and one-third
iii. Double time
iv. Time and one-fifth, etc.

**a.** **Meaning of Time:** Time refers to the basic rate, e.g. if the normal rate of pay is ¢500 an hour, then the time is ¢500. Meaning of Additional Rate The additional rate is called the overtime premium, and the amount involved is arrived at by multiplying the description by the basic rate, e.g. the overtime rate is time and a half, the basic rate is ¢500. The premium shall be ¢250 = ½ × ¢500.

b. **Overtime Premium**: It is the portion of the overtime pay over and above the basic rate of pay. Overtime premium is treated as indirect wages. The only time it is treated as direct wages is when overtime is worked according to the customer's request to complete their order within a specified period.

## 4.5 Idle (Waiting) Time

Idle or waiting time is defined in the CIMA terminology as "the period of time for which a workstation is available for production but is not utilised due to shortage of tooling, material and operators". From the definition, idle time is the time lost for which there is no production. In most cases, employees received payments for the lost production created because of idle time.

**Idle time can be classified into:**

1. Normal idle time; and
2. Abnormal idle time

Normal idle time is uncontrollable, while abnormal idle time is controllable. Therefore, Management should ensure that all facilities are in good condition and materials readily available for production.

Operation costs associated with normal idle time are charged to work in progress, for which customers are to bear the burden. Ordinary, costs associated with abnormal idle time are to be written off to profit and loss account or income statement, but some organisations treat them as part of indirect costs incurred

**Illustration 4.2**

Jobs are issued to operative X to make 189 units and to operative Y to make 204 units for which a time allowance of 20 standard minutes and 15 standard minutes per unit respectively is credited. For every hour saved, bonus is paid at 50% of the basic rate which is ¢200 per hour for both employees.

The basic working week is 42 hours. Hours in excess are paid at time and half. X completes his units in 45 hours and Y his in 39 hours (but works a full week). Because of defective material, 6 of X's units and 4 of Y's units are subsequently scrapped although all units produced are paid for.

You are required to calculate for each of X and Y:

(a) the amount of bonus payable
(b) the total gross wage payable.
(c) the wages cost per good unit made.

Solution

| | X | Y |
|---|---|---|
| Time allowed | $\frac{20}{60} \times 189$ | $\frac{15}{60} \times 204$ |
| | = 63hrs | = 51hrs |
| Time Taken | 45 hours | 39 hours |
| Time Saved | 18 hours | 12 hours |

| | | | |
|---|---|---|---|
| (a) Bonus (½ x Time saved x Day rate) | ½ ¢200 = ¢1800 | | ½ × 12 × ¢200 = ¢1200 |
| Basic wages (42 hrs. × ¢200) | *X* ¢8,400 | | *Y* (*b*) ¢8,400 |
| Over time (3 hrs. × 1.5 × ¢200) | 900 | | Nil |
| Bonus | 1.800 | | 1,200 |
| | 11,100 | | 9,600 |
| (c) | 183 units | (204-4) | 200 units |
| Good units made (189-6) | | | |
| Gross wage | 11,100 | (3 x 200) | ¢9,600 |
| Less non-productive pay | - | | 600 |
| | 11,100 | | 9000 |
| Wages cost/good unit | 11,100 / 183 = ¢9,600 | 9,600 / 200 = ¢48 | |

**Illustration 4.3**

Kokonsekyi Ltd. presents the following information which relates to a week's work of 3 of its employees

| Employee | Mumu | Opi | Adam |
|---|---|---|---|
| Work issued (units) | 400 | 600 | 120 |
| Time allowed (hrs. per unit) | 0.2 | 0.1 | 0.4 |

Output Rejected (units) 60 90 40

The basic working week is 40 hours; the first 6hours overtime are paid time plus ½ and the next 10hours at time plus ¾. Hours worked by Mumu, Opi and Adam were 52, 45 and 40 respectively. Adam spent 8 hours on indirect work. These hours being included in the 40 hours worked by him. The three employees' basic hourly rates of pay were as follows: Mumu ¢30, Opi ¢40 and Adam ¢25.

Bonus is paid at ½ of the basic rate for all time saved. The high rate of rejection was considered abnormal due to faulty materials, and it was agreed to credit all output for bonus purposes.

**Required:**

You are required to present in a tabular form for each employee:

(a) number of bonus hours worked 3 marks
(b) basic wages including overtime 4 marks
(c) amount of bonus earned 4 marks
(d) gross wages 3 marks
(e) direct wages cost per unit accepted when overtime is worked:
(i) regularly throughout the year, as company policy, due to labour shortages and
(ii) specifically at the customer's request to expedite delivery.

ICAG Jan. 1992

Solution

| | Mumu | Opi | Adam |
|---|---|---|---|
| Time allowed | (400 x 0.2) | (600 x 0.1) | (120 x 0.4) |
| | = 80hrs | = 60hrs | = 48hrs |
| Time allowed | 52hrs | 45hrs (40-8) | 32hrs |
| Time saved | 28hrs | 15hrs | 16hrs |

(b)

Hours worked × rate/hours 52hrs × ¢30 = ¢1,560 45hrs × ¢40 = ¢1,800 40hrs × ¢25 =¢1,00

Overtime premium: 6hrs×½ 30 = 90 5hrs x ½ x ¢40 = 100 132

6hrs×¢¾× ¢30 = 1650

Basic wages (including overtime ¢ 1900 ¢1135

(c) Bonus ½ × ¢30 × 28 hrs. ½ × ¢40 × 15 hrs. ½ × ¢25 × 16 hrs.

= ¢420 = ¢300 = ¢200

(d) Gross wages:

Basic wages 1,785 1,900 1,000

Bonus 420 300 200

¢2 205 ¢2,200 ¢1 200

(e) Direct labour for good, unit accepted:

Good units 400-60 600-90 120-40

= 340 units = 510 units = 80 units

(i) 156 ₵4.59 1800 ₵3.53 800/80 ₵10

(ii) 0 ₵4.85 (32x25) ₵3.75 935/80 ₵11.69

510

340 1900/510 (32x25)

165

0

340

**Summary of answers**

| | Mumu | Opi | Adam |
|---|---|---|---|
| (a) Bonus hours | 28hours | 15hours | 16hours |
| (b) Basic wages | ¢1,650 | ¢1,900 | ¢1,135 |
| (c)Bonus earnings | 420 | 300 | 200 |
| | 2,070 | | 1,335 |
| (d) Gross wages | | 2,200 | 1,135(e) |
| Direct wages per unit | - | | |
| (i) Regular overtime | 4.59 | 3.53 | 10 |
| (ii) Requested overtime | 5.25 | 3.75 | 10 |

**Illustration 4.4**

A Company's basic wage rate is ¢0.45 per hour, and its overtime rates are:

Evenings - 1 1/3
Weekends - double time

During the previous year, the following hours were worked:

| | |
|---|---|
| Normal time | 440,000 clock hours |
| Time plus 1/3 | 40,000 clock hours |
| Double time | 20,000 clock hours |

The following times have been worked on the stated jobs.

| | Job X | Job Y | Job Z Clock |
|---|---|---|---|
| | Clock Hours | Clock Hours | Clock Hours |
| Normal Time | 60,000 | 10,000 | 8,000 |
| Evening overtime | 600 | 1200 | 2100 |
| Weekend overtime | 200 | 100 | 600 |

You are required to calculate the labour cost chargeable to each job in each of the

The following circumstances:

(a) Where overtime is worked regularly throughout the year as company policy due to a labour shortage.
(b) Where Overtime is worked specifically at the customer's request to expedite delivery.

**Solution**

| | Job X | Job Y | Job Z |
|---|---|---|---|
| (a) Hours worked | 6,800hrs | 11,300hrs | 10,700 |
| x Hourly Rates | x0.45 | x0.45 | x0.45 |
| | =¢3,060 | =¢5,085 | = ¢4,815 |
| (b) 6,800hrs x¢0.45 | | 11,300hrs x¢0.45 | 10,700 x¢0.45 |
| =3,060 | | = 5,085 | = 4,815 |

Overtime premium:

| | | | |
|---|---|---|---|
| Evening overtime (1/3) (600 x 1/3 x0.45) | 90 | (1200 x 1/3 x 0.45) 180 | (400x 1/3 x0.45)315 |
| Weekend overtime (1) (200x0.45x1) | 90 | (100x1x0.45) 45 | (60x1x0.45) 270 |
| | ℓ3,240 | ℓ5,310 | ℓ5,400 |

**Group Incentive Schemes**

These are bonuses awarded to a team of employees rather than individual employees. The incentives are enjoyed by every member of the team based on an agreed-upon formula of sharing.

### 4.6. Advantages and Disadvantages of Group Incentive Schemes

***Advantages of Group Incentive Schemes***

- It enhances team spirit among employees and organisational cohesiveness ·
- Quality of output is not unduly compromised
- Compared to individual incentive schemes, it is relatively easy and less expensive to administer
- It avoids unhealthy competitive rivalry among employees

***Disadvantages of Group Incentive Schemes***

- Lazy team members are rewarded as hard-working group members.
- This does not t o motivate individual hard work.

### 4.7 Direct and Indirect Cost of Labour

As already discussed, labour cost is either direct or indirect. The direct labour cost is the Labour cost incurred on employees who are engaged in directly transforming the raw materials into finished goods. It must be noted that it is only the basic wages paid to direct workers that constitute direct labour cost. Policy-related costs incurred on direct workers is not direct labour but rather indirect labour. Examples of these policy-related costs include:

- Workers' compensation premiums paid to insurance companies
- Employer's social security fund contribution
- Bonuses paid to employees

- Overtime premium paid to employees where the overtime is worked regularly as company policy, etc.

Wages cost incurred on indirect workers are indirect wages.

**Illustration 4.5**

Mame Coins employs on her farm 120 workers as direct labour and 15 workers as indirect labour. The farm remunerates its labour as follows:

a. Direct labour is paid regularly based on units of output at the rate of ¢12,250 per unit
b. Indirect labour is paid regularly based on hours worked at the rate of ¢14,000 per hour
c. Overtime premium is paid to all factory workers for the hours worked at the following rates

   i. ¢26,000 per hour for direct labour
   ii. ¢28,000 per hour for indirect labour

d. The firm makes deductions and contributions to the social security scheme. The employer and employee contribute 10% and 5%, respectively, of gross pay

e. Each worker contributes dues to the local workers' union at the following rates:

   i. ¢50,000 per direct labour month
   ii. ¢80,000 per indirect labour per month.

   The union dues are deducted at source

f. The firm, in addition, operates a workmen's compensation insurance scheme with SIC for all factory workers. The monthly premium is ¢7,500,000.00 approximately divisible between direct and indirect labour in the ratio of 4:1

g. PAYE TAX is appropriately 8% of the monthly Gross Income

During the month of June 1999, the farm produced 30,000 units of products during regular working time of 8 hours a day for 22 days a month. The farm worked overtime of 6 hours a day for 4 days during the month and produced 3,200 more units of products.

You are required to determine:

i. the cost to Mame Coins of farm labour for the month of June 1999 distinguishing between direct and indirect labour costs
ii. the total net wages paid to the workers for June 1999

**( 20 marks)**

ICAG NOV. 1999

**Solution**

**Analysis of wages**

| | Direct Workers | Indirect Workers |
|---|---|---|
| No of units produced in normal time | 30,000 units | - |
| Overtime production | 3,200 units | - |
| Total units produced | 33,200 units | - |
| Hours worked regular time | 120x176hrs | 45x176hrs |
| | =21,120hrs | =2,640hrs |

Overtime hours: 24 hrs×120 = 2880hr

25hr

s x15 = 360hrs Wages rate

¢12.

250unit

¢14,

000/hrs.

Overtime rate

2,880hrs. @ ¢26,000

Overti

me pay = ¢74.880,000 Overtime piece rate

ℓ74,880,000

3,200

**Solution**

Overtime

premium = ¢23,400

¢

2

8

,

0

0

0

h

Direct

workers 3320

units @12,250

Indirect workers 3000hrs

14,000

Basic wages

| | | | |
|---|---|---|---|
| Overtime premium | | | |
| Direct workers (3200x11,150) | 35,680 | - | 35,680 |
| Indirect workers (360hrs x 14,000) | | 5,040 | 5,040 |
| Premium for workman's compensation | 6,000 | 1,500 | 7,500 |
| Employer SSF 10% of basic wages | | | |
| Direct workers | 40,670 | - | 40,670 |
| Indirect works | | 4,200 | 4,200 |
| | 489,050 | 52,740 | 541,790 |

**DIRECT WORKERS AND INDIRECT WORKERS PAYROLL**

| | ¢ | ¢ |
|---|---|---|
| Basic wages | 406,700 | 42,000 |
| SSF contribution 5% | 20,335 | 2,100 |
| | 386.365 | 39,900 |
| Other allowances: | | |
| Overtime premium | 35,680 | 5,040 |
| Taxable premium | 422,045 | 44,940 |
| PAYE 8% | (33,763) | (3,595.2) |
| Local union | (6,000) | (1,200) |
| New wages | 382,281.4 | 40,144.8 |
| Net wage per employee | 382,218.4 | 40,144.8 |
| | 120 | 15 |
| | = | 3185.68 |
| | 2 | 2676.3 |

## 4.8 Labour Cost Minimisation Techniques

Some of the techniques include the following:
Effective labour monitoring and control
Reduction of labour turnover
Elimination or reduction of labour-related fraud

## 4.9 Labour Turnover

a. **Meaning**

This is a term which signifies percentage change in labour force during a given period. It can be measured by using the following formula:

$$\frac{\textit{Number of employees leaving during a period}}{\textit{Average number of employees for the same period}} \times 100$$

The percentage calculated is referred to as the labour turnover ratio (or rate) The details below relate to the mouth of March

| | |
|---|---|
| Total number of employees at the beginning of month | 201 |
| Number of employees who started during the month | 3 |
| Number of employees leaving during the month | 5 |

Calculate the labour turnover rate

$$\text{Labour turnover rate} = \frac{\textit{Number of employees leaving during}}{\textit{Average number of employees}} \times 100$$

$$= \frac{5}{\frac{1}{2}(201+199)} \times 100$$

$$= \frac{5}{200} \times 100$$

$$= \underline{2.50\%}$$

Note: Average number of employees

$= {}^{1}/_{2}$ [ Number of employees at beginning of period + Number of employees at end of period ]

To obtain the maximum benefit from the calculation, the rate of labour turnover should be compared with the rate for previous periods and, if available, the rate for other businesses in the area and the industry as a whole. If the number of leavers is high relative to the total number of employees, a high ratio will emerge. An increase in the number of employees leaving or a reduction in the total workforce will cause an increase in the rate compared with previous periods. The effect of a high rate is reflected in loss of output, lowering of morale and higher cost. Loss of output occurs because of:

(a) The gap between the person leaving and his replacement;
(b) The length of time taken to train a new employee to the level of efficiency of the previous employee;
(c) The reduced effort given by an employee during the days or weeks immediately to the date of departure.

Note that two other methods of computing labour turnover are:

Flux Method:

$$\frac{\text{number of workers joining the organisation plus a number of workers leaving during the period}}{\text{Average number of employees during the same period}}$$

Separation Method:

$$\frac{\text{Number of workers replaced during the period}}{\text{Average number of employees during the same period}}$$

**a. Cost of Labour Turnover**

In addition to the increase in the cost per unit from reduced production and low morale, the following costs are likely to be higher when the rate of labour turnover is on the increase:

(i) Advertising for personnel and interviewing expenses
(ii) Re-imbursement of removal and settling in expenses removal of furniture to new house and subsistence allowance between the date of commencement and date of moving.
(iii) Training, including the new employee's wages during the training period, the wages and salaries of instructors, materials used in the training process.
(iv) Scrap and defective work during initial stages
(v) Machine break-down
(vi) Pension scheme administration, etc.

**Causes of Labour Turnover**

Although Labour turnover cannot be eliminated completely, the problem can be tackled by an understanding of the events and circumstances by which it is caused. Why do people leave? The following are some of the possible reasons:

(i) Dissatisfaction with the job, wages;
(ii) Hours of work or working conditions;
(iii) Discontent due to the relationship with the employee's supervisors and or colleagues;
(iv) Lack of promotion opportunities;
(v) Personal matters, e.g. ill health, marriage, pregnancy, moving to a new area; and
(vi) Sometimes employees are discharged due to redundancy, incompetence, lateness, absenteeism and for disciplinary reasons.

**Reduction in the Labour Turnover Rate**

To reduce the rate of labour turnover, the following should be considered:

(i) Regular statistics should be provided, analysing labour turnover by cause and indicating whether the cause was avoidable or unavoidable and whether the person left voluntarily or was dismissed; we should however note that there are some dangers in such an analysis because employees who leave do not always give the true reasons for leaving.

(ii) Seek ways in which the selection of applicants can be improved to prevent situations arising where employees are discontented or unsuitable.

(iii) Ensure that the labour requirements are properly planned to avoid redundancy

(iv) Consider the introduction of a high wages plan or some other form of incentives.
(v) See to it that the working environment is congenial.

(vi) Consider whether the transfer of a dissatisfied employee to another department will remove the cause of the dissatisfaction.

(vii) Hold annual medical check-ups to prevent ill-health at a later date.

(viii) Consider whether the fringe benefits are competitive, e.g. pension scheme, subsidised meals, sports and social facilities.

(ix) Develop better human relationships between management and workers

As it is important for an organisationto keep its skilled and experienced employees, a target labour turnover rate could be set with the personnel manager being responsible if the rate is higher than target.

**Internal Check and Payroll Fraud**

There is ample scope for perpetration of fraud at some stage during the preparation and payment of wages. For this reason, some form of controls must exist. Controls must exist to prevent fraud or to detect it in its early stages. The dangers to be avoided are:

- The payment of wages to employees who do not exist – (ghost workers),
- The payment to an employee of an amount which is more than he is entitled to receive.

**Precautions to reduce payroll fraud**

1. Segregating the work so that the payroll procedures are not the responsibility

   of only one or two persons e.g. the clerk who prepared the wage sheet should not also make-up or distribute the wage packets.
2. Adopting adequate authorisation procedures:
   - Piece work tickets should be signed by an inspector and the employee's foreman;
   - Only overtime properly authorized should be paid;
   - Changes in wage rates and salaries should only be processed if the personnel manager completes and signs the appropriate forms.
3. Proper Supervision: The distribution of the pay envelopes must be supervised by a senior official and preferably by one who can identify the recipients.
4. Instituting suitable procedures, e.g. a procedure should be laid down for dealing with unclaimed wages.
5. Creating a programme for internal checks. As part of the programme, the personnel officer might periodically check the payroll to ensure that payments are not being made to fictitious employees.
6. If the organisation is sufficiently large, the internal audit department could be very much involved in the creating and working of the internal check programme.

## 4.10 PAYROLL PREPARATION AND ACCOUNTING

### 4.10.1 Types of Payrolls for Labour

The payroll is the process of determining the net wage earnings of each employee in the organisation. The payroll could either be computerised or done manually. With the manual payroll, the earnings of each employee are computed by human effort without the use of a computer. The computerised payroll uses computer software and inputs the basic variables for each employee such as:

4.10.1.1 hours worked
4.10.1.2 overtime hours
4.10.1.3 units produced
4.10.1.4 wage rate per unit or hour
4.10.1.5 statutory deductions
4.10.1.6 voluntary deductions, etc.

into the computer and the software processes the data to arrive at the net earnings of each employee as well as the total amount of various deductions.

### 4.10.2 Labour Payroll Preparation

In preparing the payroll, the following procedure is followed:

4.10.2.1 Determine the basic wages of each employee. This is done by multiplying the hours worked or units produced by the agreed wage rate

4.10.2.2 Add all other allowances to the basic wages to obtain the gross wages

4.10.2.3 Calculate the various deductions to be made. These are of two types: statutory and voluntary deductions. The statutory deductions are required to be deducted by law and so the amount of deduction is determined by the law. These deductions include the following:

4.10.2.3.1 Social Security contributions
4.10.2.3.2 Employee Personal Income Tax (P.A.Y.E.)
4.10.2.3.3 National Health Insurance, etc.

Examples of voluntary deductions include:

4.10.2.3.4 Trade Union contributions
4.10.2.3.5 Provident fund contribution
4.10.2.3.6 Employee welfare contribution, etc.

When the total deductions are taken from the gross wages, then we obtain the net wages of each employee.

**Illustration 4.6**

Clay Products Limited manufactures and sells a variety of clay products. Management of the company is proposing to introduce an incentive scheme into its factory, but is undecided on what kind of scheme to introduce.

Below is an extract of payroll information concerning four of the company's employees for July 2005

| Name of employee | Michael | Gerald | Cynthia | William |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Actual hours worked | 152 hours | 144 hours | 160hours | 136 hours |
| Wage rate per hour ¢ | 45,000 | 30,000 | 37,500 | 54,000 |

Output produced – units

| | | | | |
|---|---|---|---|---|
| Product A | 168 | 480 | - | 480 |
| Product B | 288 | 304 | - | 1,080 |
| Product C | 368 | - | 200 | - |

The standard time per unit of each product is:

| Product | Minutes Allowed |
|---|---|
| A | 6 |
| B | 9 |
| C | 15 |

Each minutes earned is valued at ¢750 for piecework calculations

**Required:**

a. Calculate the earnings of each employee using the following methods:

i. Guaranteed hourly rates (basic pay)

ii. Piece work, but earnings guaranteed at 75% of basic pay

iii. premium bonus in which the employee receives (2/3) two thirds of time saved in addition to hourly pay. **(14 marks)**

b. State three (3) advantages and three (3) disadvantages of individual incentive schemes **(6 Marks)**

ICAG NOV. 2005

**Solution**

Payroll of Clay Products Ltd

i. Guaranteed hourly rates method

| Name of Employment | Hrs. Worked | Wage Rate per Hour ¢ | Basic Wage ¢'000 |
|---|---|---|---|
| Michael | 152hrs. | 45,000 | 6,840 |
| Gerald | 144hrs. | 30,000 | 4,320 |
| Cynthia | 160hrs | 37,500 | 6,000 |

| | | | |
|---|---|---|---|
| William | 136 hrs | 54,000 | 7,344 |
| | | | 24,504 |

ii. Piece work method
Rate per unit of output produced

| Procedure | A | B | C |
|---|---|---|---|
| (a) minutes allowed | 6 | 9 | 15 |
| (b) value of a minute ¢ | 750 | 750 | 750 |
| Rate per unit (a) + (b) = | ¢4,500 | ¢6,750 | ¢11,250 |
| | Product A | Product B | Product C |
| i. rate per unit = | ¢4,500 | ¢6,750 | ¢11,250 |
| ii. units produced (in units) | | | |
| Michael | 168 | 288 | 368 |
| Gerald | 480 | 304 | - |
| Cynthia | - | - | 200 |
| William | 480 | 1,080 | - |

Wages computation

| Michael | | ¢ |
|---|---|---|
| Product A | 168 @ 4500 = | 756,000 |
| Product B | 288 @ 6750 = | 1,944,000 |
| Product C | 268 @ 11250 = | 4,140,000 |
| | | 6,840,000 |

| Gerald | | |
|---|---|---|
| Product A | 480 @ 4500 = | 2,160,000 |
| Product B | 304 @ 6750 = | 2,052,000 |
| | | 4,212,000 |

Since 75% of his basic pay which is ¢3,240,000 is less than the piece rate wage he will be entitled to the piece rate wage at 4,212,000.

Cynthia
Product C 200 @ ¢11,250 = ¢2,250,000
75% of her basic hourly wages is ¢4,500,000
Therefore, she will be entitled to ¢4,500,000 and not ¢2,250,000

William

| | | |
|---|---|---|
| Product A | 480 @ 4500 = | 2,160,000 |
| Product B | 1,080 @ 6750 = | 7,290,000 |
| | | 9,450,000 |

iii. Bonus premiums

| Time allowed for actual production | Time taken | Time saved | Bonus premium |
|---|---|---|---|

Michael
A 6x168 x 1/60 = 16.8hrs
B 9x288x1/60 = 43.2hrs
C 15x268x1/60 = 92.0hrs

| 152hrs | 152hrs | Nil | None |
|---|---|---|---|

Therefore his wages = ¢6,840,000

Gerald
A 6x480 x 1/10 = 48hrs
B 9x304x1/10 = 45.6hrs
C - -

| Total | 93.6hrs | 144hrs | Nil | None |
|---|---|---|---|---|

Therefore wages = ¢4,3200,000

Cynthia

Product C 15 x 200 x 1/40

= 50hrs 160hrs Nil None

Therefore wages = ¢6,000,000

William

| | | | | |
|---|---|---|---|---|
| A 6x480 x 1/60 | = | 48hrs | | |
| B 9x1080x1/0 | = | 112hrs | | |
| Total | | 210hrs | 136hrs | 74hrs |

Therefore, William will be entitled to bonus premium

| | ¢ |
|---|---|
| His hourly basic pay | = 7,344,000 |
| Bonus premium | |
| 2/3 of 74hrs @ 54,000 | = 2,664,000 |
| | 10,008,000 |

4.10.3 **The Differences between Labour Cost Accounting and Payroll Accounting**

Labour cost accounting relates to the determination of the cost of labour chargeable to various jobs, customers, clients and overhead accounts. Under labour cost accounting, the objective is to ascertain the labour cost that can be charged to products and services. In this regard, the following are considered as Labour cost:

- Wages and salaries of employees
- Insurance for workmen's compensation
- Employee bonuses
- Employee share schemes, etc.

Payroll accounting relates to the process of computing the amount of earnings of employees as well as the various payments made for and on behalf of employees. Examples of payment on behalf of employees include the following:

- Employer's Social Security Contribution for the employee ·
- Employee Social Security Contributions
- Personal Income Tax of employees
- Welfare deductions for employees, etc.

Under payroll accounting, labour costs such as workmen's compensation, employee insurance, etc., are not considered. It is only amounts payable to or on behalf of employees that are considered.

## 4.11 CHAPTER SUMMARY

In this chapter, we have discussed labour cost determination and control under three sections. Section one dealt with the preliminary issues of labour; including: labour

As a factor of production; labour recruitment, induction and placement; labour behaviour and control.

Section two focused on the computation of labour cost, explaining the two main methods of labour remuneration: time-based methods and piece rate methods. The concept of overtime was also explained as well as the treatment of overtime premium when overtime is regularly worked and when overtime is requested by a customer. Bonuses and incentives were also discussed. We explained the difference between group incentive schemes and individual incentive schemes as well as the merits and demerits of each of these. Section two concluded with a discussion on labour cost minimization minimisation techniques. To minimise labour cost, labour turnover rates should be reduced, payroll fraud eliminated and employees effectively monitored and supervised.

The last section of this chapter covered payroll preparation and administration. Here we explained manual payrolls and computerized payrolls and illustrated how the payroll is prepared. We also explained the difference between labour cost accounting and payroll cost accounting.

In conclusion, labour cost is an important element of cost and forms a significant part of the total cost of most organisations, especially service organisations. Its accurate determination and effective control is therefore critical to the success of organisations.

### MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS

1. Under labour incentive schemes, bonus is paid
   - A. Every December
   - B. Each time the company received a large order
   - C. To very good employees
   - D. Anytime there is surplus money in the treasury
   - E. Over and above the basic pay to reward extra time worked or time saved

2. Overtime is
   - A. Work done over a period of time
   - B. Time spent in calling over production figures
   - C. Time spent by the employee working beyond the normal working hours
   - D. Time spent by the employee in the changing room
   - E. Time wasted when the production machines are idle

3. PAYE is an acronym for
   - A. Pay As You Engage
   - B. Pay According to Your Expectation
   - C. Pay All Your Employees
   - D. Payment At Year End
   - E. Pay As You Earn

4. Ghost workers are
   A. Former employees who are now dead
   B. Employees who were involved in accident whilst working
   C. Employees always on sick leave
   D. Workers who do not exist but in whose names salaries are being paid
   E. Those who work in the cemetery

5. Gross wages are calculated by adding
   A. All allowances to basic wages
   B. All deductions to basic wages
   C. Pension deduction to PAYE
   D. Bonuses and allowances together
   E. Twelve months' wages together
6. Time allowed minus time taken equals……
7. Labour costs incurred on employees engaged in directly transforming the raw materials into finished goods are referred to as …….
8. The extent to which employees leave an organization is known as ……

9. The product of hours worked and wage rate per hour is …….

10. The card issued in the name of each employee, which is inserted into an electronic recording machine to capture time spent at work by the employee, is called...……

**SOLUTION**

1. E
2. C
3. E
4. D
5. A
6. Time saved
7. Direct Labour Costs
8. Labour Turnover
9. Basic Wage
10. Clock Card

**EXAMINATION TYPE QUESTIONS**

**4.1** Cee ltd is a waste collection company engaged in the collection of household refuse in the Gonor District. The District is divided into three areas made up of a total of 1,000 households.

Each area enjoys 2 days of refuse collection during the 6-day working week of 48 hours.
During April 2000, the company employed 2 drivers and 4 labourers, further grouped into 2 Gangs, Gang A and Gang B. A gang is made up of 1 driver and 2 labourers. A driver is paid ¢2,500 per hour and a labourer ¢3,125 per hour.

A group bonus scheme is in operation, which requires that hours over the normal working hours are shared equally among members of a gang and paid at a rate of ¢10,000 per driver and ¢12,000 per labourer.

During April 2000, gang A worked 219 hours and gang B worked 228 hours.

**Required:**

(a) Calculate the total cost of providing this service in the district for April 2003. **(8 Marks)**

(b) Calculate the net pay due to the workers for April 2003. **(8 Marks)**

(c) The company charges the households at cost plus 20%. How much should Cee Ltd charge each household for the month of April 2003? **(4 Marks)**

**Assume**

1. An income tax rate of 10% of total earnings
2. Statutory SSF deductions of 12% for employer and 5% for employee
3. 4 weeks in the month of April

(**Total 20 Marks)**

ICAG NOV., 2003

**4.2**

An item passes through 5 successive hand operations in sequence as follows:

| Operation | Time per Article (minutes) | Grace of Employee | Wage rate per hour ¢ |
|---|---|---|---|
| 1 | 12 | A | 50p |
| 2 | 20 | B | 55p |
| 3 | 30 | C | 60p |
| 4 | 10 | D | 70p |
| 5 | 15 | E | 80p |

An Hour week is worked and a production target of 720 dozen items has been set. You are required to calculate:

a. the number of workers required;
b. the weekly cost of each operation;
c. the total labour cost per week.

**4.3.** Kofi, Yaw and Kwesi are three craft men employed by Handcraft Ltd. on the carving of two models of traditional sculptures for the local market known as "SS 77' and 'SS 88". Each craft man completes 8 'SS 77" sculptures and 10 'SS 88' sculptures per week. Production is then inspected for defective work. The company's remuneration scheme provides for employees to work a 40-hour week for which Kofi is paid ¢48 per hour, Yaw ¢54 per hour and Kwesi ¢60 per hour. Time worked in excess of 40 hours in any week is paid for as follows:

| | | |
|---|---|---|
| The first4 hours | - | Time and a third |
| Additional hours | - | Time and a half |

In addition, a bonus of 25% of time saved is paid on work completed during the week. The bonus hourly rate for all workers is ¢50. No bonus payment is made on defective work. The time allowance in hours per completed sculpture passing inspection is as follows:

| | Craft man | | |
|---|---|---|---|
| | Kofi | Yam | Kwesi |
| 'SS 77' model | 5 | 4 | 3 |
| 'SS 88' model | 3 | 5 | 2 |

The following facts relate to week No. 34

(a) Hours worked: Kofi 44; Yaw 46; Kwesi 40.

(b) Hours worked during week No. 33 on uncompleted sculpture brought forward to week No. 34:

Kofi 8; Yaw 7; and Kwesi 5.

(c) Hours worked on uncompleted sculptures carried forward to week No. 35:

Kofi 7; Yaw 8 and Kwesi 3

(d) Defective work

| | Kofi | Yam | Kwesi |
|---|---|---|---|
| 'SS 77' Sculptures | 1 | 2 | - |
| 'SS 88' sculpture | - | 1 | 1 |

You are required to calculate the total wages earned by each craftsman for week No. 34, showing separately:

(i) The standard hours allowed, excluding time for defective work;

(ii) Bonus hours and bonus payable

(iii) Overtime payable

(iv) Basic wage payable **(20 Marks)**

ICAG Jan., 1989

4.4 End Times Co. ltd manufactures two different types of products, namely Miaweni and Yirewoho. Below is a summary of the payroll data of the company's production department for the month of March 2006

| | Direct workers | Indirect workers |
|---|---|---|
| Hours worked: | | |
| Ordinary time, | 3600 hrs | 800hrs |
| Overtime | 650 hrs | 80 hrs |
| Wage rate per hour | ¢15000 | ¢ 20000 |

The company pays overtime at a time and a half. Overtime is generally worked to meet budgeted production targets.
Analysis of the time used by the direct workers on the two products reveals the following:

| | |
|---|---|
| Miaweni | 2,400 hours |
| Yirewoho | 1,150 hours |
| Idle time | 700 hours |

**Required -**

a. Prepare the following accounts:
   (i) Wages control account
   (ii) work-in progress control account
   (iii) production overhead control account

**(12 Marks)**

b. Explain the term labour turnover cost and state **FOUR** examples of such cost **(3 Marks)**

c. Outline **FIVE** causes of labour turnover **(5 Marks)**

**(Total 20 Marks)**

How can the Cost Accountant help to control labour cost in an organisation and what are some of the problems to be faced in labour cost control?

**(15 Marks)**
ACCA Costing

4.6 Explain the differences between labour cost control and labour cost reduction and outline five practical ways of achieving labour cost control as well as labour cost reduction. **(20 Marks)**

4.7 What is the difference between individual incentive scheme and group incentive scheme? Outline **THREE** advantages and **THREE** disadvantages for each of the **TWO** types of Incentive schemes. **(20 Marks)**

4.8 (a) Explain the differences between labour cost accounting and payroll accounting (**6 Marks)**
(b) List examples of labour turnover cost (**7 marks)**
(c) Explain seven (7) ways of avoiding or reducing labour turnover. **(7 Marks)**

**(Total 20 Marks)**

4.9 a. Write formulae for remuneration payable using: (i) Time rate system; and (ii) Payment by result

b. PUCT provides you with the extracted details of its employees' labour records as under:

| **Data** | Name of employee | Sanity | Moses |
|---|---|---|---|
| | Units produced; Standard time allowed | 90 | 80 |
| | (in minutes per unit produced) | 30 | 30 |
| | Actual time worked in hours | 35 | 40 |
| | Rate per hour | ₦200 | ₦250 |
| | Rate per unit | ₦66.7 | ₦68 |

**Required:**

1. Determine the remuneration payment using (i) the Time rate system, (ii) Payment by result; and (iii) Bonus Scheme is based on the product of time saved and hourly rate.
2. Which of the two employees is efficient?

## CHAPTER FIVE

## ELEMENTS OF COST: OVERHEADS

### CHAPTER CONTENTS

a. Definition, classifications and analysis
b. Allocation and apportionment
c. Absorption - bases and calculation of overhead absorption rates
d. Treatment of under- and over-absorption of overhead
e. Activity Based Costing

### 5.0 Objectives:

After studying this chapter, readers should be able to:

a. define overheads and identify the various ways of classifying overheads
b. distinguish between overheads allocation, overheads apportionment and overheads absorption
c. use the overheads analysis sheet to allocate and apportion overheads to cost centres
d. calculate overheads absorption rates using six methods
e. explain why predetermined overheads absorption rates are often used
f. explain the reasons for the preference of departmental overhead absorption rates to blanket or company-wide overhead absorption rates
g. calculate and explain the alternative treatments of overhead under-absorbed and overheads over-absorbed
h. make accounting entries with respect to overheads

### 5.1 INTRODUCTION

Overheads are the aggregate of indirect materials, indirect labour and indirect expenses. Overheads are a very important issue in cost accounting because of a number of reasons, namely:

i. it is made up of a number of cost elements; materials; labour and expenses therefore, its determination and control is complex and more challenging.

In many organizations, the overheads component of total cost is often significant and therefore requires careful analysis, monitoring and control to minimise total cost and thus maximise profitability.

## 5.2 DEFINITION AND CLASSIFICATION OF OVERHEADS

Overhead cost consists of indirect material cost, indirect labour cost and indirect expenses. Examples include:

a. Cost of cleaning materials
b. Cost of stationery
c. Cost of consumable materials
d. Supervision cost
e. Bonus payable to employees
f. Salaries of indirect workers
g. Lighting and heating
h. Rent and rates
i. Insurance premium
j. Depreciation of fixed assets.

The objective of overhead cost determination is to:

a. enable overhead cost to be absorbed by products and
b. provide useful information for management decision making and cost control decisions

## 5.3 TYPES AND CLASSIFICATION OF OVERHEADS

Overheads can be classified variously as follows:

a. production overheads
b. administrative overheads
c. marketing, selling and distribution overheads
d. research and development overheads

Let us explain each of them.

a. **Production Overheads**

These are the indirect costs of manufacturing a cost unit. It comprises of indirect materials consumed in the factory, indirect factory wages and other indirect expenses incurred in connection with production.

b. **Administrative Overheads**

These are the costs of formulating policy, directing and controlling operations not related directly to production, selling, distribution or research and development.

c. **Marketing, Selling and Distribution Overheads**

These are marketing costs incurred in securing orders, publicising and presenting to customers the products of the undertaking in suitably attractive forms at acceptable prices and delivery of the goods to customers. It also includes the costs of after sales services or processes.

d. **Research and Development Overheads**

These are the costs of seeking new ideas, materials, methods of production and improved products and the development and design of such ideas so that they can be applied to formal production.

Overheads can be analysed as consisting of:

i. Indirect materials cost
ii. Indirect labour cost and
iii. Indirect expenses.

Overheads can also be analysed either as

- Fixed overheads or
- Variable overheads

These forms of analysis have already been discussed in chapter one under the classification of overheads.

In order to charge overhead cost to products and services, a system of overhead allocation, overhead apportionment and overhead absorption is adopted. This is because; overheads by their nature cannot be assigned to any specific cost unit. Overheads must first be assigned to cost centres before they can be absorbed by products.

### 5.3.1 Overhead Allocation and Apportionment Methods

a. **Overhead Allocation**

This is the assignment of overheads to cost centres directly without sharing. Overhead cost that is specifically incurred in respect of a particular cost centre is wholly assigned to that cost centre; this is referred to as overheads allocation.

b. **Overhead Apportionment**

This is the sharing of overhead cost incurred in respect of a number of cost centres to the centres involved using a fair and equitable basis of apportionment. There are some overhead costs that are incurred for a number of cost centres and at times even for the whole organization. Such overheads cannot be allocated but must be apportioned. The basis of apportionment should be fair and equitable. Examples of bases of apportionment commonly used are:

i. Space occupied by cost centre
ii. Number of employees
iii. Value of plant
iv. Number of materials requisitioned
v. Kilowatt hours of energy used
vi. Horsepower energy used etc.

Where the cost centres involved are production and service cost centres, then the overheads relating to the service cost centres must be re-apportioned to the production cost centre.

One of the following methods can be used for the re-apportionment of service cost centre overheads to production cost centres where service cost centres serve one another. That is where there is reciprocal service.

.

**Illustration 5.1**

Apple Ltd. makes wooden crates which are sold to brewers and soft drinks bottling companies. The production work involves three production departments, Sawing, Assembling and Finishing. There are also two service departments, Maintenance and Materials Handling.

During the year ended 31st December 2025, 40,000 crates were made:

| | Sawing ¢ | Assembly ¢ | Finishing ¢ |
|---|---|---|---|
| Costs incurred: | | | |
| Materials issued | 800,000 | 600,000 | 100,000 |
| Direct Wages | 300,000 | 150,000 | 250,000 |
| Overheads | 120,000 | 80,000 | 30,000 |

Materials handling wages totalled ¢21,000

Maintenance wages totalled ¢45,000

Consumable stores totalled ¢15,000 (maintenance)

The departments' benefits from the service departments are as follows:

| | Sawing % | Assembly % | Finishing % | Materials Handling % |
|---|---|---|---|---|
| Maintenance | 30 | 40 | 20 | 10 |
| Materials Handling | 50 | 20 | 30 | |

**Required:**

(a) Prepare a Statement showing the overheads allotted to each production department **(6 Marks)**

(b) Calculate the unit cost of a wooden crate. **(9 Marks)**

**(Total 15 Marks)**

ICAG JULY 1993

**SOLUTION**

**OVERHEAD ANALYSIS SHEET**

| | SAWING | ASSY. | FNISHG | MTCE | MAT. HANDL. |
|---|---|---|---|---|---|
| | ¢ | ¢ | ¢ | ¢ | ¢ |
| Overheads | 120,000 | 80,000 | 30,000 | 45,000 | 21,000 |
| Consumables | | | | 15,000 | |
| | 120,000 | 80,000 | 30,000 | 60,000 | 21,000 |
| Apportion Maintenance | 18,000 | 24,000 | 12,000 | (60,000) | 6,000 |
| Material Handling | 13,500 | 5,400 | 8,100 | - | (27,000) |
| 151,500 109,400 | 50,100 | NIL | NIL | | |

| SAWING | ASSEMBLY | FINISHING | TOTAL | COST PER UNIT |
|---|---|---|---|---|
| Material Cost | 800,000 | 600,000 | 100,000 | 1,500,000 | 37.5 |
| | ¢ | ¢ | ¢ | ¢ | ¢ |
| Direct wages | 300,000 | 150,000 | 250,000 | 700,000 | 17.5 |
| Overheads (a) | 151,500 | 109,400 | 50,100 | 311,000 | 7.78 |
| | 1,251,500 | 859,400 | 400,100 | 2,511,000 | 62.78 |

Unit cost per crate = ¢62.78

### 5.3.2 The Concept of Overhead Absorption

This is the process of assigning overhead costs to products or services produced. Overheads are absorbed into products by first calculating the overhead absorption rate and then

applying the calculated overhead absorption rate to determine the overhead absorbed by each cost unit.

### 5.3.3 Methods of overheads analysis.

The ascertainment of overhead cost per cost unit involves a more complex procedure than the ascertainment of direct expenses. This is because overhead cost consists of indirect material, indirect labour and indirect expenses.

First of all, the overhead is not only one type of cost, and secondly, overheads cost cannot be attributed directly to a cost unit.

To determine the overhead cost per unit of any product or service therefore, the following process is followed:

a. The organization is divided into cost centres which represent areas of the business, items of equipment or persons with respect to which cost can be gathered and related to cost unit. There are two types of such cost centres
   i. production cost centres
   ii. service cost centres

b. Allocate to the various cost centres, the cost incurred for each of them that can be wholly assigned to the cost centre without sharing or apportionment.

c. Those overhead costs that are incurred for more than one cost centre, should be apportioned among the beneficiary cost centres. The apportionment should be done using a fair, equitable and reasonable basis of apportionment. Some bases that can be used include:

| | Basis | Type of overhead |
|---|---|---|
| i. | Floor space or area occupied | Rent |
| | | Rates |
| | | Heating and lighting |
| | | Insurance of buildings, etc. |
| ii. | Kilowatt hours | Electricity |
| | | Power, etc. |
| iii. | Value of Plant and Machinery | Depreciation of Plant & Machinery |
| | | Insurance of plant |
| | | Plant maintenance, etc. |
| iv. | Number of employees | Canteen expenses |
| | | Supervisors' wages |
| | | Personnel dept expenses |
| v. | Number of requisitions | Stores expenses |
| | | Indirect material expenses, etc. |

At this stage, the overhead cost for each cost centre can be determined

d. Re-apportion the overhead cost of the service cost centres to the production cost centres.
The reason for this re-apportionment is that cost units are not produced in the service cost centres and so it will not be possible for the cost units to absorb the overhead costs of the service cost centres unless and until such have been reapportioned to the production cost Centres.

When re-apportioning service cost centre overheads it is important to observe whether or not the service cost centres serve each other. That is, whether or not they provide reciprocal services. If they don't provide reciprocal services, then the following procedure should be followed:

i. First re-apportion the service cost centre that serves other service cost centres, beginning with the service cost centre that serves the largest number of cost centres.

ii. Repeat the first process until all service cost centres overheads have been re-apportioned to production cost centres. Note that under these circumstances once a service cost centre's overheads have been re-apportioned, that centre does not receive any re-apportioned overheads.

Where the service cost centres provide reciprocal services, it means one service cost centre serves another and receives services from that other. In such a situation the following methods can be used for the re-apportionment:

i. elimination method
ii. continuous allotment method
iii. Simultaneous equation method

**Illustration 5.2**

Lamina Manufacturing Company has four production departments and three service departments. Indirect labour and other indirect costs for a typical month have been allocated as shown below:

**Lamina Manufacturing Company**

| | Production Departments | | | | Service Departments | | |
|---|---|---|---|---|---|---|---|
| Maintenance | Grinding | Blending | Firing | Polishing | Personnel | Adminis-tration | Mainten-ance |
| | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
| Indirect labour | 4,600 | 3,300 | 5,400 | 2,900 | 700 | 1,800 | 800 |
| Other Indirect Costs | 1,400 | 1,200 | 2,800 | 1,600 | 500 | 300 | 1,200 |

The service departments' costs are allocated as follows:

| | Grinding | Blending | Firing | Polishing | Adminis-tration | Maintenance |
|---|---|---|---|---|---|---|
| Personnel (%) | 15 | 25 | 30 | 20 | 5 | 5 |
| Administration (%) | 10 | 30 | 40 | 15 | - | 5 |
| Maintenance (%) | 15 | 30 | 40 | 5 | 10 | - |

In the Grinding and Firing Departments, an overhead rate per machine hour is used; whereas in the Blending and Polishing Departments, an overhead rate per direct labour hour is used.

Machine hours are budgeted as 620 in the Grinding Department and 520 in the Firing Department. Direct labour hours are budgeted as 1,050 in the Blending Department and 450 in the Polishing Department.

**Required:**

(a) Determine the total overheads for each of the production cost centres.

(b) Calculate the overhead recovery rates for each of the production departments

**Note:** We shall use this example to illustrate each of the methods.
For the purpose of computing the overhead absorption rates, we shall use the production cost centre overheads determined through the elimination method.

5.3.4 **Elimination method:**

In this method, once a service cost centre's overheads have been re-apportioned, that service cost centre is eliminated from further apportionments

Using the above illustrative example, we will have the solution below.

OVERHEADS ANAYSIS SHEET (ELIMINATION METHOD)

| | | Production cost centres | | | | Service cost centres | | |
|---|---|---|---|---|---|---|---|---|
| Overheads | Basis | Grinding | Blending | Firing | Polish-ing | Personnel | Admini stration | Mainte nance |
| | | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
| Indirect labour | Allocated | 4600 | 3300 | 5400 | 2900 | 700 | 1800 | 800 |
| Other indirect Cost | Allocated | 1400 | 1200 | 2800 | 1600 | 500 | 300 | 1200 |
| Total | | 6000 | 4500 | 8200 | 4500 | 1200 | 2100 | 2000 |
| re-apportionment of service cost: Personnel | 15:25:30: 20:5:5 | 180 | 300 | 360 | 240 | (1200) | 60 | 60 |
| Maintenance | 15:30:40: 5:10 | 309 | 618 | 824 | 103 | - | 206 | (2060) |
| Administration | 10:30:40: 15 | 249 | 747 | 996 | 374 | - | (2366) | - |
| Overheads production cost centres | | 6,738 | 6,165 | 10,380 | 5,217 | Nil | Nil | Nil |

**Please observe the following points:**

| Overheads | Basis | Production cost centres | | | Service cost centres | | | |
|---|---|---|---|---|---|---|---|---|
| | | Grinding | Blending | Firing | Polish-ing | Personnel | Admini stration | Mainte nance |

i. the service cost centre that does not receive any service is first re-apportioned

ii. the next service cost centre to be re-apportioned is the service cost centre that receives the least portion of service from other service cost centres

iii. once a service cost centre's overheads have been re-apportioned, that cost centre does receive any further re-apportionment

iv. this process is continued until all service cost centre overheads have been re apportioned to production cost centres

### 5.3.5 Continuous allotment:

Each of the service cost centre overheads will be continuously re-allocated until the amount to be re-apportioned becomes insignificant, then re-apportionment is ignored.

Again, if we use the illustrative example we will have the solution below.

**Overhead Analysis Sheet**

**Please observe the following points:**

| | | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
|---|---|---|---|---|---|---|---|---|
| Indirect labour | Allocation | 4600 | 3300 | 5400 | 2900 | 700 | 1800 | 800 |
| Other indirect | cost | | | | | | | |
| | allocation | 1400 | 1200 | 2800 | 1600 | 500 | 300 | 1200 |
| total | | | | | | | | |
| Re-apportionment of service cost: personnel | | 6000 | 4500 | 8200 | 4500 | 1200 | 2 100 | 2000 |
| Maintenance | 15:25:30: 20:5:5 | 180 | 300 | 360 | 240 | (1200) | 60 | 60 |
| Administration | 15:30:40: 5:10 | 309 | 618 | 824 | 103 | - | 206 | (2060) |
| Maintenance | 10:30:40: 15:5 | 237 | 710 | 947 | 354 | - | (2366) | 118 |
| Administration | 15:30:40: 5:10 | 18 | 35 | 47 | 6 | - | 12 | (118) |
| Overheads of production cost centres | 10:30:40: 15 | 1 | 4 | 5 | 2 | - | (12) | - Nil |
| | | 6,745 | 6,167 | 10,383 | 5,205 | Nil | Nil | |

The service cost centres that serve each other are continuously re-apportioned to each other until the figure becomes so small that we will ignore the fact of the reciprocal service

5.3.6 **Simultaneous Equations:**

This method uses equations. The amount to be charged to each service cost centre is considered an unknown variable and worked for using the concept of simultaneous equations. If we use the above illustrative example, we will have the solution below.

**SOLUTION**

Under this method, the total allotment of overheads to a particular service cost centre is the overheads allocated direct to that cost centre plus the amount of overheads allotted to it from other service cost centres.

Using the illustrative example let us denote the service cost centres as

follows: P = total personnel department overheads

A = total administrative departments overhead

M = total maintenance department overheads

Now, P = 1200

A = 2100 + 0.05p +0.10m

But we know P to be1,200

Thus A= 2100 +0.05(1200) + 0.10m

A = 2160 +0.10m - (1)

M = 2000 + 0.05p +0.05 A

= 2000 + 0.05(1200) +0.05 A

M = 2060 + 0.05 A - (2)

By re-arrangement, equation 2 will be:

M - 0.05A = 2060

- 0.05A = 2060 M

$$A = \frac{2060}{-0.05} - \frac{1\,M}{-0.05}$$

A = - 41,200 + 20M (3)

Deduct equation (1) from (3)

A = - 41,200 + 20M -

A = 2160 + 0.10M

0 = - 43,360 + 19.90M

- 19.90M = - 43,360

$$M = \frac{-43{,}360}{-19.90}$$

= 2179

Substitute M = 2179 into - (1)

A = 2160 + 0. 10(2179) = 2378

Therefore

P = 1200

M = 2179

A = 2378

| | | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
|---|---|---|---|---|---|---|---|---|
| Indirect labour | Allocated | 4,600 | 3,300 | 5,400 | 2,900 | 700 | 1,800 | 800 |
| Other indirect cost | Allocated | 1,400 | 1,200 | 2,800 | 1,600 | 500 | 300 | 1,200 |
| Total | | 6,000 | 4,500 | 8,200 | 4,500 | 1,200 | 2,100 | 2,000 |
| Overheads re-apportionment: | | | | | | | | |
| Personnel | 15:25:30:20:5 | 180 | 300 | 360 | 240 | (1,200) | 60 | 60 |
| Administration | 15:25:30:20:5:5 | 238 | 713 | 951 | 357 | - | (2,378) | 119 |
| Maintenance | 10:30:40:15:5 | 327 | 654 | 872 | 108 | - | 218 | (2,179) |
| Production cost Centre overheads | | 6,745 | 6,167 | 10,383 | 5,205 | Nil | Nil | Nil |

At the end of all of these stages, the total overheads cost for each production cost centre would have been determined. The next stage then is to absorb the overheads to cost units using any of the overheads absorption methods.

### 5.3.7 Overheads Absorption Methods

#### a. The Concept of Overhead Absorption.

This is the process of assigning overhead costs to products or services produced.

Overheads are absorbed into products by following the process below:

i. Calculate the overhead absorption rates

ii. Apply the calculated overhead absorption rate to determine the overhead absorbed

b. **Methods of Overhead Absorption.**

Generally, overhead absorption rate (OAR) is computed as OAR

$$= \frac{\text{Budgeted overheads}}{\text{Budgeted activity level}}$$

There are different factors that could possibly be used as activity level. These include the following:

i. direct labour hours
ii. machine hours
iii. units of production
iv. direct material cost
v. direct labour cost
vi. prime cost, etc.

i. Direct labour hours

This method assumes that direct labour hours are the most significant factor that influences the amount of overhead incurred in the production of products and services. Direct labour hours are therefore used as basis for the absorption.

$$\text{OAR} = \frac{\text{budgeted overhead}}{\text{Budgeted direct labour hours}}$$

$$= \text{OAR per direct labour hour}$$

This method is suitable where the production technology is labour intensive and wage rates are stable.

From our illustrative example, we were told that overheads from the Blending and Polishing departments are absorbed using Direct Labour Hours.

Overheads for these two departments are:

Blending Department = ¢6,165,000

Polishing Department = ¢5,217,000 and the budgeted direct labour hours for these two departments are 1050 hours and 450 hours respectively.

The Overhead Absorption Rates will be computed as follows:

| | BLENDING | POLISHING |
|---|---|---|
| Budgeted Overheads | ¢6,165,000 | ¢5,217,000 |
| Budgeted Direct Labour Hours | 1,050 hours | 450hours |
| Therefore O.A.R. | ¢5.87 Per DLH | ¢11.5 Per DLH |

Where DLH is Direct Labour Hour.

ii. Machine hours

Where the production process is highly mechanised and is not labour intensive, the use of labour hours may be inappropriate. Under such circumstances, machine hours may be preferred as activity base for overhead absorption.

$$OAR = \frac{\text{Budgeted overheads}}{\text{Budgeted machine hours}}$$

= OAR per machine hour.

From our illustrative example, we were told that overheads from the Grinding and Firing departments are absorbed using Machine Hours.

Overheads for these two departments are:

Grinding Department = ¢6,738,000

Firing Department = ¢10,380,000 and the budgeted machine hours for these two departments are 620 hours and 520 hours respectively.

The Overhead Absorption Rates will thus be computed as follows:

| | GRINDING | FIRING |
|---|---|---|
| Budgeted Overheads | ¢ 6,738,000 | ¢10,380,000 |
| Budgeted Machine Hours | 620 hours | 520 hours |
| Therefore O.A.R. | ¢10.87 per MH | ¢19.96 per MH |

Where MH is Machine Hour.

iii. Units of production

Where the production process turns out uniform products in mass quantities, units of production could be used as basis for overheads absorption. Where the units are not uniform however, it will be inappropriate to use units of production as a basis for overhead absorption.

$$\text{OAR} = \frac{\text{Budgeted overheads}}{\text{Budgeted production units}}$$

= OAR per unit of output produced

iv. Direct material cost percentage

Overheads are absorbed as a percentage of direct materials cost.

This method is used when there are stable and uniform materials prices. Where material prices are not uniform, this method will not yield good results

$$\text{OAR} = \frac{\text{Budgeted overheads}}{\text{Budgeted direct materials cost}} \times 100$$

= OAR as a percentage of direct material cost

v. Direct labour cost percentage

Overheads are absorbed as a percentage of direct labour cost. Where intensive labour technology is used and wage rates are stable and uniform, this method may be used

$$\text{OAR} = \frac{\text{Budgeted overheads}}{\text{Budgeted direct labour cost}} \times 100$$

= OAR as a percentage of direct labour cost

vi Prime cost percentage

Prime cost as we have already explained is the aggregate of direct materials cost, direct labour cost and direct expenses. Overheads can therefore be absorbed as a percentage of prime cost. Where there is relative stability of rates and prices, this method could be used.

$$\text{OAR} = \frac{\text{Budgeted overheads}}{\text{Budgeted price cost}} \times 100$$

= OAR as a percentage of prime cost

## 5.4 Calculation of Overhead Absorption Rates

We shall use the illustrative example below to demonstrate further, how overhead absorption rates are calculated and absorbed.

**Illustration 5.3**

(a) Distinguish between Overhead Absorption and Overhead Apportionment (6 Marks)

(b) Enyo Manufacturing Company Ltd. produces three products- Frytol, Brilliant and Pepsodent. The cost estimates of each of the products are given below:

| | Frytol | Brilliant | Pepsodent |
|---|---|---|---|
| Direct Materials | ¢ 720 | ¢1,280 | ¢1,800 |
| Direct Labour: | | | |
| Production Dept. at ¢60/hr. | 2 hrs. | 2.5 hrs. | 2 hrs. |
| Finishing Dept. at ¢40/hr. | 2 hrs. | 2.5 hrs. | 1 hr. |
| Variable Overheads | ¢80 | ¢30 | ¢40 |

Fixed overhead cost for the following year is estimated at ¢6 million and planned production is:

| | |
|---|---|
| Frytol | 10,000 units |
| Brilliant | 20,000 units |
| Pepsodent | 40,000 units |

The directors are considering alternative methods of absorbing fixed overheads into product costs and you have been asked, as the Cost Accountant, to calculate the rates to be applied for the following alternatives:

i, Direct Labour Costs (7 Marks)

ii, Percentage of Total Variable Cost (7 Marks)

**(20 Marks)**

ICAG MAY 2000

**SOLUTION**

Please refer to sections 5.2.1 a and b for the answer to (a)

(b)

| | | Production department | | Finishing department | |
|---|---|---|---|---|---|
| Direct labour hour budgeted | | | | | |
| Frytol | 2 x 10,000 | = 20,000 hrs. | | 2 x 10,000 | = 20,000 hrs. |
| Brilliant | 2.5 x 20,000 | = 50,000 hrs. | | 2.5 x 20,000 | = 50,000 hrs. |
| Pepsodent | 2 x 40,000 | = 80,000 hrs. | | 1 x 40,000 | = 40,000 hrs. |
| | 150,000hrs | | | | 110,000hrs |
| Budgeted wage rate per hour | | ¢60 | | ¢ 40 | |
| Budgeted direct labour cost | | 150,000hrs @ 60 | | 110,000hrs@40 | |
| **= ¢9,000,000** | | | | **= ¢4,400,000** | |
| Total direct labour cost | =9,000,000 + 4,400,000 | | = | ¢13,400,000 | |

| Budgeted material cost: | | | ¢ |
|---|---|---|---|
| Frytol | 720 x 10,000 | = | 7,200,000 |
| Brilliant | 1,280 x 20,000 | = | 25,600,000 |
| Pepsodent | 1,800 x 40,000 | = | 72,000,000 |
| | | | 104,800,000 |
| Budgeted labour cost | | | 13,400,000 |
| | | | |
| Budgeted variable overhead | | | |
| Frytol | 80 x 10,000 | = | 800,000 |
| Brilliant | 30 x 20,000 | = | 600,000 |
| Pepsodent | 40 x 40,000 | = | 1,600,000 |
| Budgeted total variable cost | | | 121,200,000 |

Overheads Absorption Rate (OAR) = $\frac{\text{Budgeted overhead}}{\text{Budgeted activity level}}$

(i) O.A.R. (Direct Labour cost percentage) = $\frac{\text{Budgeted Overheads}}{\text{Budgeted Direct Labour cost}}$

= $\frac{¢600,000}{13,400,000} \times 100$

= 4.48% of direct labour cost

(ii) O.A.R. (Variable cost percentage) = $\frac{\text{Budgeted Overheads}}{\text{Budgeted Variable cost}}$

= $\frac{¢6,000,000}{121,200,000} \times 100$

= 5% of total variable cost.

## 5.5 Why Predetermined Rates Are Used

a. Actual overheads are not known until after the end of a given period. If budgets are not used product cost cannot be quickly determined. The alternative will be to calculate the overhead recovery rate as frequently as possible. This could however be cumbersome and inconvenient.

b. Actual overheads are substantially influenced by the general price level and actual activity levels are subject to wide fluctuations. The combined effect of these two is large fluctuations in the absorption rates if based on actual overhead cost and actual activity level. To reduce the effects of such fluctuations on overhead recovery, it is considered better to use budgeted overheads and budgeted activity for overhead absorption since these reflect standards.

## 5.6 Blanket Overhead Absorption Rates Versus Cost Centre Overhead Absorption Rates.

Overhead recovery rates are usually calculated for each production cost centre. Where an overhead recovery rate is in respect of the whole factory it is termed a blanket overhead recovery rate or factory wide overhead recovery rate.

Blanket overhead recovery rates are not very appropriate because of the following reasons:

a. The factory consists of different production cost centres and products may consume cost centre overheads in different proportions.

b. Different activity bases drive cost for different products in different proportions.

## 5.7 Determination and Treatment of Overheads Over Absorbed or Overheads Under Absorbed.

The overhead cost absorbed to a cost unit is computed as:

**Overhead absorption rate x actual quantity of activity base.**

Having regard to the way overheads are absorbed, it is unlikely for the actual overhead incurred to be the same as the overheads absorbed. This thus creates a situation of over absorbed overheads or under absorbed overheads.

Treatment of under or over absorption of overheads.

a. Adjust the under or over recovery to cost units produced during the period.
   - This may not be worthwhile and such historical information is of no use to management

b. Carry the under or over recovery to future accounting periods
   -This results in a distortion of performance figures.

c. Treat the under or over recovery as a period cost by writing it off in the Profit and Loss account.
   - This is the most preferred treatment.

The treatment of overheads could be done using

a. Absorption costing system where both fixed and variable overheads are considered and charged to products.

b. Marginal costing system where only variable overheads is considered and absorbed into products and fixed overheads treated as a period cost and charged to Profit & Loss account.

**Illustration 5.4**

(a) The use of pre-determined rates for overhead absorption may lead to under-absorption or over-absorption of overheads.

You are required to discuss any three methods for disposing of overhead discrepancy.

**(5 Marks)**

(b) Agape Co. Ltd. operates two production departments (machining and assembling) and one service department (which is responsible for maintenance). The data below was extracted from the books of the company on 31st July,1998.

| Hours Spent | Machining | Assembling | Maintenance |
|---|---|---|---|
| Direct labour hours | | 600 | |
| Machine hours | 300 | | |
| Expenses | ¢000 | ¢000 | ¢000 |
| Indirect labour | 200 | 250 | 120 |
| Indirect materials | 420 | 450 | 500 |
| Machine handling | - | - | 80 |
| Housekeeping | 50 | 70 | 100 |
| Air Conditioning | 30 | 30 | 50 |
| Rent and rates | 160 | 120 | 150 |
| Supervision | 240 | 180 | 200 |

Expenses of the maintenance centre are subsequently re-apportioned to the two production centres in proportion to the total expenses incurred by those departments during the month. Predetermined overhead rates were established as follows:

| | Machining | Assembling |
|---|---|---|
| | ¢000 | ¢000 |
| Budgeted Monthly overhead | 2,000 | 1,200 |
| Planned machine hours | 500 | - |
| Planned direct labour hours | - | 400 |

You are required to:

a. calculate the overhead absorption rates which were in operation during July;

b. prepare a statement showing the actual overhead borne by the production department for July;

c. state the extent to which overhead was under- absorbed or over-absorbed by the production departments during the month of July.

ICAG NOV. 1998

**SOLUTION**

Please for the answer to (a) of this illustrative example, refer to section 3.6.

(b)

Overheads Absorption Rate (OAR) = Budgeted overhead / Budgeted activity level

| | Machinery | Assembly |
|---|---|---|
| Budgeted overheads ¢ | 2,000,000 | 1,200,000 |
| Budgeted machine hours | 500,000 | |
| Budgeted direct labour | - | 400,000 |
| OAR ¢ | 2,000,000 / 500,000 | ¢ 1,200,000 / 400,000 |
| | ¢4 per machine hour | ¢3 per direct labour hour |

Overheads analysis sheet

| Overhead | Basis | Production cost centre | | Service cost Centre |
|---|---|---|---|---|
| | | Machinery | Assembly | Maintenance |
| Indirect labour | Allocated | 200 | 250 | 120 |
| Indirect Material | Allocated | 420 | 450 | 500 |
| Material handing | Allocated | - | - | 80 |
| Housekeeping | Allocated | 50 | 70 | 100 |
| Air conditioning | Allocated | 30 | 30 | 50 |
| s Rent & Rate | Allocated | 160 | 120 | 150 |
| Supervision | Allocated | 240 | 180 | 200 |
| | | 1100 | 1100 | 1200 |
| Maintenance | 1,100:1,100 | 600 | 600 | (1200) |
| | | 1700 | 1700 | Nil |

Overheads absorbed = OAR x actual activity

| | Machining | Assembly |
|---|---|---|
| OAR | ¢4 per machine hour | ¢3 per direct labour hour |
| Actual machine hours | 300,000 hrs. | |
| Actual direct hours | | 600,000 hrs. |
| Overheads Absorbed | 4 x 300,000 | 3 x 600,000 |
| | 1,200,000 | 1,800,000 |
| Overheads absorbed | 1,200,000 | 1,800,000 |
| Overheads incurred | 1,700,000 | 1,700,000 |
| Overheads under absorbed | 500,000 | - |
| Overheads over absorbed | | 100,000 |

## 5.8 Activity-Based Costing

In a conventional costing of indirect cost, overhead is absorbed to different products through a single base such as units of production, labour hours etc. This method of overhead absorption is suitable for a small or medium firm that is not having complex operational activities.

However, in a large firm that is highly automated, a single base is not suitable for recovery overheads simply because different activities are associated with incurring indirect costs. In such a complex automated firm, activity-based costing is used for absorbing overhead to products.

**Definition:**

**Activity-Based Costing** is a costing approach that focuses on how costs of resources utilised are traced to activities that cause them to incur and then to absorb overhead to products or services based on the activities utilisation.

Thus, ABC identifies factors that cause or drive costs and termed those factors as **cost drivers.**

**Cost Driver** is defined as "any factor that causes a change in the cost of an activity (CIMA).

Examples of Cost Drivers are:

- Number of materials handling or issued
- Number of machine set-up
- Number of machine hours
- Number of inspector orders

### 5.8.1 Stages in ABC

Two stages are involved in how overheads are absorbed into products through ABC. They include:

1. **Stage One:** Grouping of overheads into pool in accordance with identifying activities that drive them
2. **Stage Two:** Determination of how much overhead should be charged

The first stage involves identification of cost driver that allows cost to be incurred. The steps required in this first stage include:

i. Step 1: Identify and classify activities;

ii Step 2: Selection of cost driver bases such as number of machine set-up etc;
Step 3: Assigning of cost to each activity or cost pools; and
Step 4: Ascertainment of activity cost per cost driver base. The formula is:

$$\text{Cost Driver Rate} = \frac{\text{Activity cost for each cost pool}}{\text{Quantity of cost driver available}}$$

Stage Two: This stage computes amount of overheads to be recovered i.e assuming 100 units of product Lare to be manufactured. If total overhead is determined as:

| | | | |
|---|---|---|---|
| Machine set up: | 20 set up x ₦40 | = | 800 |
| Materials handling: | 10 issues- 10 x ₦20 | = | 200 |
| Total | | | 1,000 |

Overhead rate per unit of output of product L

= ₦1,000 ÷ 100 = ₦100

**Illustration 5.5**

MMT manufactures two products. The company provides you with the following details relating to 2015 financial year

1. Units Produced:
   Product AB - 150,000 units
   Product CD - 200,000 units

2. Cost drivers identified

| Activity | Activity Cost Pool ₦ | |
|---|---|---|
| Machine set up | 300,000 | 60 set up |
| Materials issued | 200,000 | 90 requisition |
| Quality control | 100,000 | 40 inspection |

3. Cost drivers identified

| Activity | PRODUCT AB | CD |
|---|---|---|
| Machine set up | 10 | 10 |
| Materials issued | 20 | 10 |
| Quality control | 10 | 5 |

Determine overhead absorption rate through activity based costing

**Illustration 5.6**

MMT manufactures two products. The company provides you with the following details relating to 2015 financial year

1. Units Produced:
   Product AB - 150,000 units
   Product CD - 200,000 units

2. Cost drivers identified

| Activity | Activity Cost Pool ₦ | |
|---|---|---|
| Machine set up | 300,000 | 60 set up |
| Materials issued | 200,000 | 90 requisition |
| Quality control | 100,000 | 40 inspection |

3. Cost drivers identified

| Activity | PRODUCT AB | CD |
|---|---|---|
| Machine set up | 10 | 10 |
| Materials issued | 20 | 10 |
| Quality control | 10 | 5 |

Determine overhead absorption rate through activity based costing

**SOLUTION**

Unit cost of each cost driver:

Activity cost for each pool
Total quantity per pool

| | | | |
|---|---|---|---|
| Machine set up: | ₦300,000 ÷ 60 | = | ₦5,000 per set up |
| Material issued: | ₦200,000 ÷ 80 | = | ₦2,500 per issue |
| Quality Control | ₦100,000 ÷ 40 | = | ₦2,500 per inspection |

Amount of cost driver to charge to each product:

| **PRODUCT** | | **AB** | | **CD** |
|---|---|---|---|---|
| | | ₦ | | ₦ |
| Machine set up | 5,000 x 10 = | 50,000 | 5,000 x 10 = | 50,000 |
| Materials issued | 2,500 x 20 = | 50,000 | 2,500 x 20 = | 25,000 |
| Quality control | 2,500 x 10 = | 25,000 | 2,500 x 5 = | 12,500 |
| | | **125,000** | | **87,500** |

Total amount of Overhead charged:

| | ₦ |
|---|---|
| Product AB | 125,000 |
| Product CD | 87,500 |
| | **212,500** |

## 5.9 CHAPTER SUMMARY

In this chapter, we have discussed accounting for overheads in three sections. The first section addressed the introductory issues in overheads; explaining the meaning of overheads, the classification of overheads and the treatment of overheads.

The second section discussed the processes of overhead allocation, overhead apportionment and overhead absorption.

The final section of the chapter explained the concept of overhead absorption; explaining six methods that could be used to absorb overheads, the reasons for the preference of pre-determined overhead absorption rates to actual overhead absorption rates and the treatment of overheads under absorbed or overheads over absorbed.

Overheads are a very important concept in cost accounting and it is very important that you thoroughly understand it because you encounter overheads concepts in subsequent topics as you progress.

**MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS**

1. Overheads are the aggregates of
   A. Indirect materials, indirect labour and indirect expenses
   B. Expenses incurred over the normal expense heads
   C. Expenses incurred by the owners of the business
   D. All uncontrollable expenses
   E. All unauthorized expenses

2. Distinguish between overhead allocation and overhead apportionment
   A. Allocation is used in government whilst apportionment is used in the private sector
   B. Allocation is assigned direct to cost centres whilst apportionment is shared to a number of cost centres
   C. Allocation is made scientifically whilst apportionment is done arbitrarily
   D. Allocation benefits only managers whilst apportionment benefits junior staff
   E. Both terms mean the same thing

3. What is the usual basis of apportioning factory rent?
   A. Number of employees
   B. Number of machines
   C. Floor space occupied
   D. Kilowatt of energy used
   E. Number of pillars

4. What distinguishes a profit centre from a cost centre?
   A. Direct Cost
   B. Indirect Cost
   C. Profit
   D. Loss
   E. Revenue

5. What does OAR stand for in overheads accounting?
   A. Organisational Accounting Returns
   B. Overheads Accounting Rate

C. Overheads Accounting Returns

D. Overheads Absorption Rate

E. Organisational Absorption Rate

6. Where the actual overheads for a period exceeds the predetermined overheads, we have a case of ……..

7. A system of continuously reallocating each service cost centre overhead until the amounts become insignificant is known as …….

8. Overheads covering cost of securing orders, publicizing and presenting to customers the products of a company in proper forms at acceptable prices and delivery of the goods to customers are generally known as ……

9. Where the production process is highly mechanized making the use of labour hours inappropriate for overhead absorption, what alternative basis may be used?

10. Which is the best way to treat over- and under-absorbed overheads for a period?

**SOLUTION**

1. A
2. B
3. C
4. E
5. D
6. under-absorption
7. continuous allotment
8. marketing, selling and distribution overheads
9. machine hour
10. Write off to Profit & Loss Account

## EXAMINATION TYPE QUESTIONS

5.1

The following information is available for a firm producing and selling a single product:

| **Budgeted costs (at normal activity):** | **¢000** |
|---|---|
| Direct material and labour | 264,000 |
| Variable production overhead | 48,000 |
| Fixed production overhead | 144,000 |
| Variable selling and administration overhead | 24,000 |
| Fixed selling and administration overhead | 96,000 |

The overhead absorption rates are based upon normal activity of 240,000 units per period. During the period just ended 260,000 units of product were produced and 230,000 units were sold at ¢3,000 per unit. At the beginning of the period 40,000 units were in stock. These were valued at the budgeted costs shown above. Actual costs incurred were as per the budget.

**Required:**

a. Calculate the fixed production overhead absorbed during the period and the extent of any over or under absorption. For both of these calculations you should use absorption costing.

b. Calculate profits for the period using absorption costing and marginal costing respectively.

c. Reconcile the profit figures which you calculated in (b) above.

d. State the situations in which the profit figures calculated under both absorption costing and marginal costing would be the same. **(20 Marks)**

ICAG MAY 2000

**5.2**

Atta Kwesi Manufacturing Company has three departments X, Y and Z. The company uses a single production overheads rate for overhead absorption purposes, the rate expressed as a percentage of direct wages.

A change in policy in favour of departmental absorption rates has been contemplated on the grounds that these rates can be more precise.

The following information relates to the factory's budgeted and actual costs:

Budgeted:

| | Direct Wages ¢000 | Direct Labour hrs. '000 | Machine Hrs '000 | Production Overheads '000 |
|---|---|---|---|---|
| Dept. X | 250 | 100 | 400 | 1,200 |
| Dept. Y | 1,000 | 500 | 100 | 300 |
| Dept. Z | 250 | 250 | - | 750 |
| Total | 1,500 | 850 | 500 | 2,250 |

| Actual | | | | | | |
|---|---|---|---|---|---|---|
| Dept. X | 300 | 120 | | 450 | | 1,300 |
| Dept. Y | 800 | 450 | | 140 | | 280 |
| Dept. Z | 300 | 300 | | - | | - |
| | 1,400 | 870 | | 590 | | 1,580 |

A Job No. FX18 has been completed currently and the actual information on this job is as follows:

| | Direct Materials | Direct Wages | Direct Labour hrs. | Machine Hrs. |
|---|---|---|---|---|
| Dept X | 1,200 | 1,000 | 200 | 400 |
| Dept Y | 600 | 600 | 400 | 100 |
| Dept. Z | 100 | 100 | 100 | - |

The pricing policy of the company is to make a gross mark-up of 331/3 on completion of jobs, which is expected to cover other business expenses plus net profit.

**Required:**

(a) Determine the overhead absorption rate which has been in use.

(b) Determine the production overhead to be absorbed by Job FX18 using the absorption rate determined in (a) above and determine the price of Job FX 18.

(c) Considering the new policy, determine the relevant overhead absorption rates, on a departmental basis, that can ensure precise cost computations of jobs. Give your reasons for each rate chosen.

(d) With the new rates in (c) above, determine the new departmental overheads and the subsequent total production overheads to be absorbed by Job FX 18 as well as the price of Job FX 18 with this new information.

**(25 Marks)**

ICAG MAY 1997

Q.1

**5.3**

A manufacturing company that produces in batches is preparing a budget for the next financial period. There are two production departments - Moulding and Finishing. The following information for the coming year has been compiled:

| | Moulding | Finishing |
|---|---|---|
| | (¢'000) | (¢'000) |
| Direct Materials | 640,000 (56,000 hours) | 120,000 (14,000 hours) |
| Direct Labour | 224,000 | 44,800 |
| Overhead (apportioned) | 313,600 | 62,720 |
| Total | ¢ 1,177,600 | ¢227,520 |

The following forecast has been made:

(i) In the Moulding Dept. 90% of the labour time will be spent in actual production, and 10% of the time will be used for adjusting machines before actual production operation.

In the Finishing Dept. 80% of the labour time will be for actual production and 20% for preparing machines for actual operation.

(ii) 2,000 batches of work will be produced. The work passes through the two stages of production - Moulding and Finishing before reaching completion. There are two ways by which labour employed in preparing machines could be treated:

(i) as part of direct labour cost, or

(ii) as an addition to the overhead apportioned to each department.

**Required:**

Calculate for each Department:

(a) Overhead absorption rates on a percentage of prime cost bases, assuming labour spent in preparing machines as part of:

(i) direct labour cost

(ii) overhead cost

Note: The prime cost percentage should be calculated to the nearest whole number

(12 marks)

(b) The total production cost of each of the following three batches:

| | Direct Labour hours (Kilograms) | | Direct Materials | | Price of Materials per kilo |
|---|---|---|---|---|---|
| Batch No. | Moulding | Finishing | Moulding | Finishing | |
| C061 | 35 | 22 | 250 | 101 | ¢ 800 |
| L024 | 38 | 29 | 411 | 139 | ¢ 690 |
| M115 | 19 | 12 | 89 | 26 | ¢ 1,100 |

Your calculation should be based on the overhead rates calculated in (a) above:

(i) with preparation of machines time as direct labour cost,

(ii) with preparation of machines time as overhead cost

**(12Marks)**

**(Total 24 Marks)**

**ICAG** JAN 1996

**5.4**

Larson Co. Ltd., a jobbing engineering firm, has in its single factory three production departments (moulding, assembly and finishing) supported by two service departments boiler and maintenance. The Company's production costs for the past month are:

| | ¢ | ¢ |
|---|---|---|
| Prime Cost | | |
| Direct Materials | 360,000 | |
| Direct Wages | 300,000 | 660,000 |
| Indirect Materials | | |
| Dept. moulding | 11,240 | |
| Dept. assembly | 17,890 | |
| Dept. finishing | 6,390 | |
| Dept. boiler | 11,000 | |
| Dept. maintenance | 5,000 | 51,520 |
| Supervision and indirect wages: | | |
| Dept. moulding | 13,000 | |
| Dept. assembly | 16,800 | |
| Dept. finishing | 13,600 | |
| Dept. boiler | 12,420 | |
| Dept. maintenance | 10,380 | 6,200 |
| General production overheads: | | |
| Factory rent and rates | 25,200 | |
| Insurance – machinery | 9,900 | |
| - building | 7,700 | |
| Depreciation – machinery | 18,000 | |
| - building | 7,280 | |
| Administrative Costs of wages dept. | 16,800 | 84,880 |
| Total Factory Cost | | 882,600 |

Further information is available

| | Moulding | Assembly | Finishing | Boiler | Maintenance |
|---|---|---|---|---|---|
| Area (Sq. metre) | 600 | 400 | 200 | 100 | 100 |
| Direct labour (hours)15,000 | 15,000 | 10,000 | - | - | |
| Direct labour cost (¢)45,000 | 30,000 | 25,000 | - | - | |
| Machine hours | 20,000 | 30,000 | 40,000 | - | |
| Net book value of plant (¢) | 30,000 | 60,000 | 90,000 | 120,000 | nil |
| Number of employees | 75 | 75 | 50 | 15 | 25 |
| Machine maintenance hrs. | 156 | 312 | 468 | 104 | |

The factory manager estimates that 10% of total maintenance costs relates to the boiler and the balance to the three production departments (the maintenance department uses none of the power produced by the boiler).

The most important factor in determining the requirement for maintenance is total machine hours, all machines used in the factory can be taken to be of approximately equal value.

The company sets its selling prices on the following basis:

(1) Cost of production, plus

(2) 25% (uplift to cover normal administrative and selling cost; plus

(3) A profit margin of 10% of the final selling price.

It is the policy of the Company to absorb overhead into production on the basis of Direct Labour Hours.

Two jobs X and Y are currently being priced. Details are:

| | Job X | | Job Y | |
|---|---|---|---|---|
| Direct materials | ¢483.00 | | ¢361.00 | |
| (a) Direct labour: | (10 hrs.) | ¢ 30.00 | (15 hrs.) | ¢ 45.00 |
| Dept. Moulding | (20 hrs.) | ¢ 40.00 | (2hrs.) | ¢ 4.00 |
| Dept. assembly | ( 8 hrs.) | ¢ 20.00 | (12 hrs.) | ¢ 30.00 |
| Dept. finishing | (5hrs.) | | (8 hrs.) | |
| (b) Machine hours: | (7 hrs.) | | (1 hr.) | |
| Dept. moulding | (1 hr.) | | (5 hrs.) | |
| Dept. assembly | | | | |
| Dept. finishing | | | | |

**You are required to:**

(a) Calculate overhead absorption rates for the three production departments: (20 Marks)

(b) Calculate the prices to be charged for jobs X and Y. (8 Marks)

Answer to 2 decimal places **(Total 28 Marks)**

1CAG JAN. 1988

5.5

(a) Explain why predetermined overhead absorption rates are preferred to overhead absorption rates calculated from factual information after the end of a financial period.

(b) The production overhead absorption rates of factories X and Y are calculated using similar methods. However, the rate used by X is lower than that used by factory Y. Both factories produce the same type of product. You are required to discuss whether, or not, this can be taken to be a sign that factory X is more efficient than factory Y.

**(20 Marks)**

CIMA Cost Accounting 1
Nov. 1979

5.6

(a) Explain the differences that exist between the following terms as used in cost accounting:

(i) Allocation of overheads
(ii) Apportionment of overheads
(iii) Absorption of overheads **(12 Marks)**

(b) Explain the reasons for overhead under absorbed and overhead over absorbed.

**(8 Marks)**

**(Total 20 Marks)**

5.7 Specify and explain the factors to be considered in determining whether to utilise a single factory wide recovery rate for all production overheads or a separate rate for each cost centre, production or service department. **(12 Marks)**

(a) Describe THREE methods of determining overhead recovery rates and specify the circumstances under which each method is superior to the other methods mentioned

(8 Marks)

**(Total 20 Marks)**

CACA Management Accounting June 1981

5.8

(a) Outline the procedures and information required in order to determine a set of predetermined production overhead absorption rates for a company manufacturing a range of different products in a factory containing a number of production departments and several service departments.

(b) Critically examine the purpose of calculating overhead absorption rates.

**(20 Marks)**

ACCA Costing

**5.9 OVERHEAD ABSORPTION RATE**

a. Distinguish between Blanket rate and Cost Centre (multiple) rate, used for absorbing overheads

b. The following data were provided by XYZ Ltd:

| | |
|---|---|
| Budgeted labour hours | 8,500 |
| Budgeted Overheads | ₦148,750 |
| Actual labour hours | 7,928 |
| Actual overheads | ₦146,200 |

**Required:**

i. What is the labour hour overhead absorption rate?
ii. What is the amount of overhead under/over absorbed if any?
iii. Produce overhead control account for the transaction above
iv. Give the major reason for the under/over absorption of overhead in (ii) above

**(12½ Marks)**

**CHAPTER SIX**

**INTERLOCKING AND INTEGRATED ACCOUNTS CHAPTER**

**CONTENTS**

(a) Explain bookkeeping entries and ledger for interlocking systems
(b) Reconcile financial and cost accounting profits
(c) Explain and illustrate an integrated accounting system

6.0 **Objectives**: After studying this chapter, readers should be able to:

(a) post cost accounting transactions into the appropriate books of account;
(b) apply double-entry principles in cost accounting;
(c) differentiate between "interlocking accounts" and"integrated accounts";
(d) Reconcile the financial accounting profit and the cost accounting profit;
(e) design accounts code in a summarised and suitable form for effective management of assets; and
(f) explain the application of computer-assisted techniques.

## 6.1 Introduction

Bookkeeping is identifying, recording, and summarising transactions in the appropriate books of accounts. The balances in the accounts, to be obtained at the end of an accounting period, are to be extracted to the trial balance, which is just a list of balances of both debit and credit. The system of bookkeeping is not restricted to Financial Accounting; it applies to Cost Accounting too. Both financial and cost accounting data are to be properly kept and processed to develop and produce the required information for decision making.

The size of a business normally determines the complexity of the cost records. Small businesses keep simple records not linked to the financial accounts, while larger businesses maintain a full set of cost accounts, which provide the details behind the totals shown in the financial accounts. In some firms, cost accounts are kept independent of financial accounts, and periodic reconciliation will be necessary. In other firms, the cost accounts form part of the formal accounting system, and no reconciliation is necessary. The latter method is more satisfactory because it avoids duplication of effort and reconciliation problems. The two methods will be explained in this chapter.

This chapter focuses attention on the double-entry aspect of cost accounting and the reconciliation needed between cost accounts and financial accounts when the two are maintained separately. The reconciliation of accounting and costing systems will be required where a firm adopts an interlocking system. Where the integral system is maintained, both the financial and cost accounting systems would be combined to generate a single profit.

## 6.2 INTERLOCKING ACCOUNTING SYSTEM AND COST LEDGER

The above-named accounting system uses separate cost accounts, which are periodically reconciled with the financial accounts.

The cost accountant uses a great deal of financial information, which is also available to the financial accountant, e.g., purchase daybook, expenses daybook, payroll, main journal, sales daybook, etc.

The structure of the interlocking cost accounting system in any organisation will be determined by the volume of its business transactions. The cost accounting records usually kept will include those of cost elements, cost centres (production, marketing and general administration), job cost (production, repairs, capital) and inventory (i.e., raw materials, components or spare parts,

finished goods). The interlocking of the two systems is carried out by the use of the following control accounts:

(a) Cost ledger control account in the financial ledger
(b) Financial ledger control account in the cost ledger.

## 6.3 Cost Ledger

In an interlocking cost accounting system, the cost ledger is the principal ledger. Generally, the cost ledger will contain the following:

(a) Control accounts for stores, work-in-progress, finished goods and overhead. Each control account will be supported by a subsidiary ledger to provide the necessary details for accounting control and the preparation of information.
(b) Accounts for wages, sales, cost of sales, together with any other account needed in the cost book-keeping system, e.g., Notional charges, Suspense accounts.
(c) Cost Ledger Contra Account (General Ledger Adjustment Account or Cost Ledger Control Account). This account is needed to maintain the double-entry principle in the cost ledger. It contains postings which affect accounts outside the costing system,

e.g., Debtors, Creditors or Cash Accounts of Real and Personal accounting transactions.

The focal point of the cost accounting system is the work-in-progress ledger. In an organisation that operates and uses specific order costing, the ledger kept by such an organisation will contain a separate account for each job. Each account will be debited with the cost of material used, wages allocated and production overhead absorbed. When the jobs are completed, the cost of completed jobs will be transferred to the Finished Goods Ledger. When the job is delivered to the customer, the cost of sales will be debited with the Job Cost and Sales credited with the selling price. Thus, a (costing) profit and loss statement can be prepared and analysed according to each completed job. The balances on the control accounts represent the cost value of stores, work-in-progress and finished goods.

## 6.4 Relevant Cost Accounting entries under the interlocking system up to WIP Control Account.

1. **Raw Material Incurred**:
   Debit Raw Material Control A/c Or Stores Ledger Control Account
   Credit Cost Ledger Control Account/General Ledger Adjustment Account/Cost Ledger Control Account

2. **Materials returned to Suppliers**:
   Debit Cost Ledger Control Account or General Ledger Adjustment Account
   Credit Raw Materials Control Account or Stores Ledger Control Account

3. **Material Issued**:
   Debit Work-in-Progress Control Account, WIP (Direct) Debit Production Overhead Control Account, POCA (Indirect Material issued to factory)
   Credit Raw Materials Control Account

4. **Wages Incurred**:
   Debit Wages Control Account, WCA
   Credit Cost Ledger Control Account

5. Wages Allocated:
   Debit Work-in-Progress Control A/C (Direct)
   Debit Production Overhead Control A/c (Indirect to factory)
   Credit Cost Ledger Control Account

6. Production Overhead Absorbed:
   Debit WIP Control Account
   Credit Production Overhead Control Account

   Note: An Adjustment entry is made for the under- or overabsorption of overhead

7. Output Achieved:
   Debit Finished Goods Accounts
   Credit Work-in-Progress Control Account

**Illustration 6.1**

Below are incomplete accounts in the cost ledger of a manufacturing business for a period in which final accounts are to be prepared:

**Raw materials Control Account/ Store Ledger Control A/C**

| | ₦ | | ₦ |
|---|---|---|---|
| Bal. b/f | 8,000 | Work – in – progress | 9,000 |
| Cost ledger contra | 110,000 | | |
| | 118,000 | | |

**Work-in-Progress Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Bal. b/f | 7,000 | | |
| Raw materials control | 90,000 | | |

**Finished Goods Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Bal. b/f | 30,000 | | |

**Production Overhead Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Cost Ledger Control | 117,000 | | |

**Production Wage Control Account**

| ₦ | | ₦ |
|---|---|---|
| Cost Ledger contra | 46,000 | |

**Marketing and Admin Control Account**

| ₦ | | ₦ |
|---|---|---|
| Cost Ledger contra | 50000 | |

**Cost of Sales**

| ₦ | ₦ |
|---|---|

**Sales**

| | |
|---|---|
| P & L Acct. | 300,000 |

**Cost Ledger Contra**

| | ₦ | ₦ |
|---|---|---|
| Bal. b/f | 110,000 | |
| Raw materials control | 46,000 | |
| Production OH control | 117,000 | |
| Prod. Wage control Marketing & Admin. OH | 50,000 | |

You are given the following additional information:

(a) The closing inventory balances are:

₦

| | |
|---|---|
| Raw materials control | 18,000 |
| Work-in-Progress control | 68,000 |
| Finished Goods Control | 45,000 |

(b) The ratio of direct wages incurred to indirect wages is 11:2

(c) Marketing and administration overhead is absorbed by jobs based on 25% of the cost of goods sold.

(d) Production overhead is absorbed based on N3.50 per direct labour hour. During the period total direct labour hours worked is 22,000.

**Required**

(a) Identify the transactions indicated by the accounting entries numbered (i) to (vi)

(b) Complete the cost accounts for the period

(c) Compute the cost accounting profit for the period

(d) Prepare a trial balance as at the end of the period

(Adapted from ACCA).

**Solution 6.1**

(a) The transactions indicated by the accounting entries (i-viii) are as follows.

(i) Issue of indirect materials to production

(ii) Direct wages allocated

(iii) Production overhead absorbed

(iv) Finished goods produced during the period transferred from WIPCA

(v) Cost of goods sold during the period

(vi) Indirect wages allocated

(vii) Marketing and administration overhead absorbed

(viii) Value of sales during the period.

**Workings**

(a) The total production wages incurred= ₦117,000, and the ratio of direct wages to indirect wages is 11:2.

Therefore:

$$\text{Direct wages (ii)} = \frac{11}{13} \times 117{,}000 = \text{N}99{,}000$$

$$\text{Indirect wages (vi)} = \frac{2}{13} \times 117{,}000 = \text{N}18{,}000$$

(b) Production overhead absorbed (iii) = 22,000 x N3.50 = ₦77,000

(c) Indirect materials issued to production (i)

| | ₦ | ₦ |
|---|---|---|
| Total raw materials available | | 118,000 |
| Less: Direct materials issued | 90,000 | |
| Closing inventory of raw materials | 18,000 | 108,000 |
| Materials | | 10,000 |

The values of other missing figures can be determined in the same way as indirect materials issued to production as follows:

(d) Value of finished goods produced during the period (iv) =

₦(7,000+ 90,000 + 99,000 + 77,000 - 68,000) = ₦205,000

(e) Cost of goods sold (v)
= ₦(30,000+ 205,000 - 45,000) = ₦190,000

(f) Marketing & Administration overhead absorbed (vii) = 25% of cost of goods sold
= 25% of ₦190,000
= ₦47,500

(g) Value of opening balance of cost ledger contra account (ix) = Net sum of all the opening balances, i.e.

| | Dr | Cr | |
|---|---|---|---|
| | ₦ | | ₦ |
| Raw materials control | 8,000 | | |
| WIP control | 7,000 | | |
| Finished goods control | 30,000 | Cost Ledger Contra | 45,000 |
| | 45,000 | | 45,000 |

(b) Completion of the accounts:

**Raw Materials Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Bal b/f | 8,000 | Work-in-progress | 90,000 |
| Cost Ledger Contra | 110,000 | Production OH (i) | 10,000 |
| | | Bal. c/d | 18,000 |
| | 118,000 | | 118,000 |
| Bal. b/d | 18,000 | | |

**Work-in-Progress Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Bal b/f | 7,000 | Finished goods (iv) | 205,000 |
| Raw materials control | 90,000 | | |
| Wage control (ii) | 99,000 | Bal. c/d | 68,000 |
| Production OH control (iii) | 77,000 | | |
| | 273,000 | | |
| Bal. b/d | 68,000 | | 273,000 |

**Finished Goods Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Bal b/f | 30,000 | Cost of sales (v) | 190,000 |
| W.I.P. control (vi) | 205,000 | Bal. c/d | 45,000 |
| | 235,000 | | 235,000 |
| Bal. b/d | 45,000 | | |

**Production Overhead Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Raw materials control | 10,000 | W.I.P Control | 77,000 |
| Wages control | 18,000 | | |
| Cost Ledge Contra | 46,000 | | |
| Profit & Loss (Over absorbed overhead) | 3,000 | | |
| | 77,000 | | 77,000 |

**Production Wages Control Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Cost Ledger control | 117,000 | W.I.P Control | 99,000 |
| | | Production Wages (a) | 18,000 |
| | 117,000 | | 117,000 |

**Marketing and Admin. Overhead Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Cost Ledger contra | 50,000 | Cost of sales (vii) | 47,500 |
| Profit and Loss | | (under absorbed OH) | 2,500 |
| | 50,000 | | 50,000 |

| | | | |
|---|---|---|---|
| Bal. b/d | | | |

**Cost of Sales Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Finished goods control | 190,000 | Profit & Loss | 237,500 |
| Marketing & Admin. OH (vi) | 47,500 | | |
| | 237,500 | | 237,500 |

**Sales Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Profit & Loss | 300,000 | Cost Ledger contra (viii) | 300,000 |

**Profit and Loss Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Cost of Sales | 237,500 | Sales | 300,000 |
| Marketing & Admin. OH | 2,500 | Production OH control | 3,000 |
| Net Profit (transferred to cost ledger contra) | 63,000 | | |
| | 303,000 | | 303,000 |

**Cost Ledger Contra Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Sales | 300,000 | Balance b/f | 45,000 |
| Balance c/d | 131,000 | Raw materials control | 110,000 |
| | | Production OH control | 46,000 |
| | | Production wages control | 117,000 |
| Profit and Loss | 63,000 | Marketing & Admin. OH | 50,000 |
| | 431,000 | | 431,000 |
| | | Balance b/d | 131,000 |

**Trial Balance as at the end of the Period**

| | Dr | Cr |
|---|---|---|
| | ₦ | ₦ |
| Raw materials control | 18,000 | |
| Work-in-Progress control | 68,000 | |
| Finished goods control | 45,000 | |
| Cost Ledger contra | | |
| | | 131,000 |
| | 131,000 | 131,000 |

**NB**: The term 'control' account implies that a subsidiary ledger is kept. Entries to control accounts represent summary postings.

## 6.5 RECONCILIATION OF COSTING AND FINANCIAL ACCOUNTING PROFITS

The profit disclosed by the Costing Profit and Loss Account may differ from that shown in the financial Profit and Loss Account where interlocking system is adopted and it is necessary to reconcile them. The difference may be due to the following reason:

### 6.5.1 Appropriations of Profit not dealt with in Cost Accounts

(a) Corporate tax
(b) Transfers to general reserve or any other fund of accumulated profits, e.g. dividend equalization reserve.
(c) Charitable donations, where no direct benefit is derived therefrom by the company's employees.
(d) Dividends paid on the share capital of the company
(e) Additional provisions for depreciation of buildings, plants, etc., and for bad debts.
(f) Amounts written off on goodwill, preliminary expenses, underwriting commission, debenture discounts, expenses of capital issues, etc.

(g) Appropriations to sinking funds for the purpose of providing for the repayment of loans or debentures.

### 6.5.2 Financial Matters that are outside the scope of Cost Accounts

(a) Interest received on bank deposits and investments.
(b) Rents receivable. If, however, rents are received from a portion of the business premises which has been sublet, the profit on rental only will be excluded from cost records. The proportion which represents recovery of rent paid will be deducted from the total rent to ascertain the net cost of the premises will be included in cost records.
(c) Profits or losses on sales of investments, buildings and other fixed assets; capital expenditure charged specifically to revenue.
(d) Interest on bank loans, mortgages, debentures and other borrowed money, if interest on capital is ignored in the cost records.
(e) Damages payable according to law.
(f) Penalties payable for late completion of contracts.
(g) Losses due to scrapping of machinery before the date estimated when depreciation and obsolescence were provided for.

### 6.5.3 Items whose Costing treatment differs from those adopted in the Financial Accounts

**(a)** Under and over recoveries of overheads due to deliberate omission from the cost accounts, e.g. selling costs, or to approximation in the basis of recovery or to fluctuations in output.

**(b)** Differences in the valuation of inventory and work-in-progress. Inventories are frequently valued in the cost accounts at a figure which includes an appropriate allotment of overhead, which from the point of view of ascertaining the profit on manufacture, is perfectly justifiable. It may be thought prudent when preparing the financial accounts to value inventories at prime costs only.

**(c)** Depreciation in the cost accounts may in certain circumstances be charged on the basis of production units or hours, while in the financial accounts it is usually provided for in the form of a charge for the period, irrespective of the actual level of production output.

**(d)** Abnormal losses in production and storage, and abnormal idle time. In the financial accounts these losses will be "merged" in the appropriate accounts, i.e. the figure of sales will be exclusive of finished products wasted, and the debits for materials and wages will be inclusive of any abnormal losses of material or time. In the cost accounts, such abnormal losses may be ignored as outside the scope of manufacturing costs.

The reconciliation of financial accounts and cost accounts is carried out using a memorandum reconciliation account in order to show why the respective profit figures of the two accounts are different.

**Illustration 6.2**

NUASA Limited has separate financial and cost accounting systems. Extracts from both sets of accounts for the year ended 31 December 2021 were as follows

**(a) Financial Accounts**

| | Opening Inventory | Closing Inventory |
|---|---|---|
| | ₦ | ₦ |
| Inventory valuations: | | |
| Raw materials | 34,000 | 36,000 |
| Work-in-progress | 7,000 | 5,000 |
| Finished goods | 16,000 | 20,500 |
| | | ₦ |
| Debenture interest | | 5,000 |
| Interest received | | 750 |
| Discounts allowed | | 2,000 |
| Discounts received | | 1,000 |
| Net Profit (before taxation) | | 17,500 |

**Cost Accounts**

| | Opening Inventory | Closing Inventory |
|---|---|---|
| ₦ | | ₦ |
| Inventory valuations: | | |
| Raw materials | | |
| Work-in-progress | 33,000 | 34,500 |
| Finished goods | 8,000 | 7,000 |
| | 17,000 | 20,500 |

**Required:**

Reconcile the profit as reported in the financial accounts with the cost accounts, and calculate the cost accounting profit for the ***period.***

**Solution 6.2**

**Memorandum Reconciliation Account**

| | ₦ | ₦ | | ₦ | ₦ |
|---|---|---|---|---|---|
| Profit as per financial a/cs | | 17,500 | * profit as per cost a/cs | | 21,750 |
| Debenture interest | 5,000 | | Items not credited | | |
| Discount allowed | 2,000 | | Discount received | 1,000 | |
| Raw material opening | 1,000 | | Interest received | 750 | |
| W.I.P closing | 2,000 | 10,000 | Raw mat. Closing | 1,500 | |
| | | | W.I.P closing | 1,000 | |
| | | | Finished Goods: | | |
| | | | - Opening | 1,000 | |
| | | | - Closing | 500 | 5,750 |
| | | 27,500 | | | 27,500 |

The sum of ₦21,750 stands for the difference between the two sides of the account as well as the Cost Accounting profit.

## 6.6 INTEGRATED ACCOUNTING SYSTEM

With respect respect o this accounting system, the concept of separate Profit and Loss Accounts for financial and cost purposes is discarded in favour of a unified set of accounts which serves both financial and management purposes. Such a system of accounting is referred to as an "integrated" or "integral" system. The terminology (CIMA) defines integrated accounts as "set of accounting records that integrates both financial and cost accounts using a common input and data for all accounting purposes." The system is designed on the following principles:

(a) The degree of integration must be determined. Some undertakings find it satisfactory merely to integrate up to the stage of prime cost, leaving overhead allotment to be dealt with in memorandum form only.

(b) Dependent upon the degree of integration decided upon, the normal classification of expenditure by nature is discarded. In some systems, a single control account is opened into which all expenditure and revenue which is the subject of costing treatment is posted. Such a control account operates in the same way as a debtor's control account, i.e., it contains a summary of transactions posted to the cost ledger

in detail. It is, however, more usual for separate control accounts to be opened in the formal records in the same way as control accounts were opened within the cost ledger for stores, Work-in-Progress, Overhead, etc., under a separate accounting system.

(c) Full details of items posted to the control accounts are supplied to the cost department at conventional intervals, accompanied by prelists or registers of amounts to ensure completeness.

(d) The information is then dealt with by the cost department following the system of costing in force and with the following objectives in mind:

(i) To provide the necessary costing data.

(ii) To form the basis for journal entries so that the control accounts can be cleared to suitable revenue accounts, culminating in a profit and loss account.

(e) The amount of detail which is recorded in the ledger itself depends upon circumstances, but is usually kept to a minimum, with full information regarding each department or process being contained in subsidiary ledgers (based of course upon exactly the same data). Postings to subsidiary ledgers are sometimes referred to as third entries to emphasize that they are not part of the double entry of the accounting system.

Application of the above principles will facilitate preparation of operating statements (in the form required) from data which are subject to routine accounting control. The conventional analysis of expenditure required for statutory purposes and for external reporting can be accommodated by modifying the classification procedures within the subsidiary ledgers.

The integrated accounting system saves administrative effort since only one set of accounts is kept, instead of two separate systems of cost and financial accounting. In this regard the possible confusion arising from having two sets of accounts is eliminated. The main problem about the integrated accounting system is that one set of accounts is expected to fulfil two different purposes, which are:

(i) Stewardship of the business and external reporting

(ii) Provision of information to aid management decision and control of operations.

At times these different purposes may conflict. For example, the valuation of inventories will conform to the requirement of International Accounting Standards, whereas the cost accountant might prefer to value closing inventories at say, marginal cost or replacement cost.

**Entries for Integrated System**

1. Materials Purchased:
   Debit Raw Materials/Stores Ledger Control A/c
   Credit Bank/Creditors for Materials
2. Materials returned to Suppliers:
   Debit Creditors for Materials
   Credit Raw Materials Control Account

3. Materials Issued:
   Debit Work-in-Progress Control Account (Direct)

   Debit Production Overhead Control Account (Indirect Factory) Credit Raw Materials Control Account

4. Wages Incurred:
   Debit Wages Control Account, WCA
   Credit Wages Payable Account
5. Wages Allocated:
   Debit Work-In-Progress Control Account
   Credit Production Overhead Control Account
   Credit Wages Control Account
6. Production Overhead Incurred:
   Debit Production Overhead Control Account, POCA
   Credit Bank/Creditors for **Overheads**
7. Absorption of Overhead:
   Debit WIP Control Account
   Credit Production Overhead Control Account
   Note: Adjust for under/over absorption
8. Output Achieved:
   Debit Finished Goods Account
   Credit Work-In-Progress Account

**Illustration 6.3**

Gwammaja Limited manufactures a range of products which are sold through a network of wholesalers and dealers. A set of integrated accounts is kept, and for the year 2021, the following information is relevant:

(a) Production overhead is absorbed into the cost of products based on a budgeted rate of 80% of direct labour cost.
(b) Finished inventories are valued at factory cost.

(c)

| | 31 March 2021 ₦ | 30 April 2021 ₦ |
|---|---|---|
| Raw materials inventory, | 8,600 | 7,580 |
| Work-in-Progress, | 2,800 | 2,375 |
| Finished goods inventory, | 5,250 | 6,045 |
| Debtors for goods sold | 4,600 | 5,570 |
| Creditors for raw materials | 3,800 | 4,710 |
| Fixed assets at net book value | 3,000 | 3,000 |

(d) Bank transactions for April 2021 were as follows:

| | ₦ |
|---|---|
| Bank balance at 31st March | 750 |
| Receipts from debtors | 13,780 |
| Payments made: | |
| Direct labour | 3,200 |
| Creditors for raw materials | 4,480 |
| Production overhead | 2,700 |
| Administration overhead | 350 |
| Selling & Distribution overhead | 1,150 |

**Required:**

(a) Use the information provided above to write up the following control accounts:
- (i) Raw materials inventory
- (ii) Work-in-Progress
- (iii) Finished Goods
- (iv Production Overhead

(b) Prepare the following statements:
- (i) A Profit and Loss Account for April, 2021
- (ii) A balance sheet as at 30th April, 2021

**Solution 6.3**

**Raw Materials Inventory**

| | ₦ | | ₦ |
|---|---|---|---|
| Opening balance | 8,600 | Work-in-progress | 6,410 |
| Credit (W1) | 5,390 | (Balance figure) | |
| | | Balance c/d | 7,580 |
| | | | 13,990 |
| | 13,990 | | |
| Balance b/d | 7,580 | | |

**Work in Progress**

| | ₦ | | ₦ |
|---|---|---|---|
| Opening balance | 2,800 | Finished goods | 12,595 |
| Raw materials inventory | 6,410 | (Balancing figure) | |
| Direct wages | 3,200 | | |
| Production OH (W2) | 2,560 | Balance c/d | 2,375 |
| | 14,970 | | 14,970 |

| | | | |
|---|---|---|---|
| Balance b/d | 2,375 | | |

**Finished Goods**

| | ₦ | | ₦ |
|---|---|---|---|
| Opening Bal. | 5,250 | P & L. A/c (cost of sales) | |
| W.I. P | 12,595 | (Balancing figure) | 11,800 |
| | | Balance c/d | 6,045 |
| | 17,845 | | 17,845 |

**Production Overhead**

| | ₦ | | ₦ |
|---|---|---|---|
| Cash | 2,700 | Work-in-progress | 2.560 |
| | | Profit & Loss account (under-absorbed) | 140 |
| 2,700 | | | 2,700 |

(b) (i) **Income Statement for April, 2021**

| | ₦ | ₦ |
|---|---|---|
| Sales (W3) | | 14,750 |
| Production cost of sales | | 11,800 |
| Gross profit | | 2,950 |
| Less: Administration Overhead | 350 | |
| Sales & Distribution overhead | 1,150 | |
| Under-absorbed production overhead | 140 | 1,640 |
| Profit | | 1,310 |

**(ii) Position Statement as at 30th April, 2021**

| | ₦ | ₦ |
|---|---|---|

| | | |
|---|---|---|
| Fixed assets (net book value) | | 3,000 |
| **Current assets**: | | |
| Raw materials | 7,580 | |
| Work-in-progress | 2,375 | |
| Finished goods | 6,045 | |
| Debtors | 5,570 | |
| Cash (W4) | 2,650 | |
| | 24,220 | |
| Less: Current Liabilities | | |
| | | 19,510 |
| | | 22,510 |
| Long-term capital N21,200 (balancing figure) + 1,310) | | 21,200 |
| Profit | | 1,310 |

**Workings**

**Creditors for Raw Materials**

| | ₦ | | ₦ |
|---|---|---|---|
| Cash | 4,480 | Opening balances | 3,800 |
| Ba; c/d | 4,710 | Raw materials purchases | |

| | | | |
|---|---|---|---|
| | | (Balancing figure) | 5,390 |
| | 9,199 | | 9,199 |

(2) Direct labour (80% of Direct. Labour) ₦3,200
Production overhead absorbed (80%) ₦2,560

**Debtors**

| | ₦ | | ₦ |
|---|---|---|---|
| Opening balance | 4,600 | Cash | 13,780 |
| Sales (bal.. figure) | 14,750 | Balance c/f | 5,570 |
| | | | 19,350 |
| | 19,350 | | |

**Cash**

| | ₦ | | ₦ |
|---|---|---|---|
| Opening bal. | 750 | Creditors for raw materials | 4,480 |
| Debtors | 13,780 | Direct labour | 3,200 |
| | | Production overhead | 2,700 |
| | | Admin. Overhead | 350 |
| | | Sales & Distribution OH | 1,150 |
| | | Balance c/d | 2,650 |
| | 14,530 | | 14,530 |

## 6.7 ACCOUNTS CLASSIFICATION AND CODES

In an organisation of any size, the cost and financial data will be so numerous that details are summarised under suitable headings before management can process and assimilate the data for various decision making. For example, many types of indirect materials would be used in the factory and in reporting consumption. Typical headings under which data are summarised will include:

a. Cleaning materials
b. Lubrication
c. Fuel
d. Protective clothing
e. Stationery
f. Small tools
g. Packing material
h. Foodstuffs
i. Replacement parts
j. Miscellaneous

By using such groups, the number of accounts can be reduced and the groups can be used for budgeting purposes, etc. In order to assemble data under these headings, it is vital to use code numbers or letters and to prepare an accounts manual which contains, among other things, a list of all current codes with explanatory notes on their use. A good manual will be one which is comprehensive, unambiguous and well indexed, so that invoices, requisitions, time-sheets, etc. can be coded quickly and accurately.

The code structure adopted can be prepared in several ways, but it is important that it follows a logical sequence, i.e., all common items will be pulled together, in such a manner that it will be flexible enough to accommodate additions and that it is arranged in a way that aids memory. If the system is so complex that a code in excess of 6 digits must be used, then it is preferable to adopt a composite code suitably structured like 024-824-03. A simple accounts code showing some of the fixed asset accounts that are likely to be encountered **will include the following:**

0000 - 0099 Land
0100 - 0299 Buildings
0300 - 0399 Machine tools
0400 - 0499 Containers and tanks
0500 - 0599 Vehicles
0600 - 0699 Handling equipment
0700 - 0799 Heating plant
0800 - 0899 Tools and patterns
0900 - 0999 Miscellaneous fixed assets, patents and goodwill

The same coding system may be followed when designing accounts code for current assets, liabilities, cost centre, expenditure and miscellaneous accounts.

## 6.8 COMPUTER ASSISTED COSTING TECHNIQUES

In today's dynamic business environment, it is doubtful if the cost accountant can be effective without the use of computer techniques. Aside from the volume of transactions and the speed of processing required, there is the need for systems that simplify decisions by offering a wide range of options for users of cost accounting information.

First among the computer assisted costing techniques are transaction processing systems. These are systems designed to handle large volumes of monotonous transactions from various points. They are coordinated by a central server which processes and harmonizes such mass of data. Usually, such systems have set functions and their formats cannot be easily amended without the intervention of data processing experts such as system analysts and programmers.

Secondly, some systems are designed to support decision making. They are generally very user-friendly and interactive, offering a variety of tools that users can employ to perform required tasks more effectively unlike the mass data processing systems.

Decision Support Systems have the advantages of:

(a) guiding the manager through a process of using available data to quickly arrive at useful decisions. They provide the support but still leave the manager to take the decisions because they do not offer pre-determined solutions.
(b) interaction which enhances effective problem solving through wide options provided.
(c) not having a structured solution. The manager must determine the direction of decisions to be reached seeing that the system guides him towards possible solution and their implications.

Decision Support Systems include;

i. Spreadsheets
ii. Forecasting
iii. Regression Analysis
iv. Non-linear and Linear Programming
v. Modelling and simulation
vi. Statistical Analysis
vii. Financial modelling
viii. Sensitivity and Risk Analysis
ix. Activity Cost Management

All the systems cited above come with application packages which are flexible and cheap to use (does not require the employment of specialist or system analysts who are expensive to maintain) and can be readily customized to suit the user's requirements. This flexibility includes transactions types, data layouts, reports formats and frequency. Most packages are designed to deal with a single specific application, but by far the most versatile and widely applied package is the electronic spreadsheet. The most useful electronic spreadsheet at the disposal of cost accountants is the Microsoft excel. The spreadsheet is capable of storing both alphabetical and numeric data which are amenable to easy manipulations through simple commands. They are arranged in a grid of columns and rows. With the data safely entered in the cells, a change in a simple command is automatically effected in the mass of data throughout the worksheets that are linked together. Spreadsheets can be used for monthly expenditure reports, cash budgets, standard cost records and profitability reports amongst others. The data in Microsoft Excel is also capable of being exported into packages like Power Point and others with capabilities for pictorial presentation, thus enhancing the use of computer systems in decision making.

## 6.9 CHAPTER SUMMARY

In this chapter, efforts were made to explain the double-entry aspects of cost accounting, and the reconciliation of the cost accounts with financial accounts. It has been made clear that cost accounts and financial accounts might be reconciled through the use of "interlocking accounts" or "integrated accounts". We have also shown how the two different accounts are reconciled with the use of memorandum reconciliation account. Finally, we have described how accounts code should be used to summarize cost and financial data before data could be processed and assimilated by the management for relevant decisions.

It is clear from this chapter that double entry system of book keeping is not restricted to financial accounting. The system is also very applicable to cost accounting and management accounting. It all depends on the type of organization and its commitment to proper record keeping, transparency and accountability.

**MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS**

1. An interlocking accounting system has
   A. Various ledgers relating to one another
   B. A separate set of financial ledgers and a separate set of costing ledgers
   C. Accounting entries locked up in the ledger
   D. Different ledger for direct expenses and indirect expenses
   E. A single set of ledgers serving a dual purpose
2. A trial balance is
   A. The set of accounting records presented at court cases
   B. An account prepared to determine a company's profitability
   C. List of account balances both debit and credit sides balance
   D. The balance of the cost ledger control account
   E. A statement of balances on bank accounts

3. What is a notional charge?
   A. A charge on the assets of the company
   B. A government charge to be paid by the company
   C. Charges introduced to reduce tax liability
   D. Charges which though not payable are meant to reflect the normal costs of running the business
   E. Expected income which may be recognized in the accounts

4. In reconciliation of profits disclosed by interlocking accounts, what are purely financial matters?
   A. Items involving cash transactions
   B. Balance sheet items
   C. Matters relating to the banks
   D. Salaries and wages paid to casual workers
   E. Financial matters outside the scope of production

5. What is a control account?
   A. An account in the main ledger summarizing the subsidiary ledger accounts
   B. The account maintained by the Financial Controller
   C. An account maintained by the Cost Controller
   D. A separate account for monitoring factory performance
   E. A secret account for monitoring factory performance
6. A system where a set of accounts is kept for both financial and costing transactions is known as ……
7. ….
8. In the accounts manual, processing of large mass of data under different accounts heads are made possible through the use of …….
9. Raw materials issued to production but yet to reach the finished stage at period end is to be found in which account?

10. The conflict between the profit figures arrived at under the financial account and cost account ledger is settled by way of a ……..

**SOLUTIONS TO MULTIPLE CHOICE AND SHORTANSWER QUESTIONS**

1. B
2. C
3. D
4. E
5. A
6. Integrated Accounting Systems
7. Accounts codes
8. Work in progress account
9. Memorandum Reconciliation Statement

**EXAMINATION TYPE QUESTIONS**

1. What do you understand by the term "Integrated Accounting and what advantages does it offer compared to separate systems of cost accounting and financial accounting?
2. The growth in the volume of data handled and current technological advances in data processing equipment continues to emphasize the need for a satisfactory system of codification of accounts. In determining such a code, what principles should be observed?
3. You have been appointed as adviser to a trade association of toy manufacturers and are aske d to prepare a report on implementing uniform costing systems.
   List **TWO** aspects of overhead costing to be examined and explain how uniformity may be achieved in respect of each.
4. The following trial balance results from entries in the cost ledger of a jobbing manufacturer:

| | Dr ₦ | Cr ₦ |
|---|---|---|
| (a) Cost Ledger Contra Account | | 11,590 |
| (b) Stores Ledger Account | 3,790 | |
| (c) Work-in-Progress Account | 5,430 | |
| (d) Finished Goods Account | 2,350 | |
| (e) Factory Overhead Account | 50 | |
| (f) Administration cost Account | | 30 |
| | 11,620 | 11,620 |

Explain what each balance represents and the transaction(s) out of which it has arisen.

5. The following Trial Balance was extracted from the books of Easy Going Limited on 1 January, 2010:

| | ₦ | ₦ |
|---|---|---|
| Raw materials | 18,000 | |
| Work-in-progress | 27,000 | |
| Finished goods | | 9,000 |
| General ledger control a/c | | 54,000 |
| | 54,000 | 54,000 |

The company keeps separate cost accounting records. Transactions for January 2010 are given below:

| | ₦ |
|---|---|
| Raw materials purchased | 59,000 |
| Direct wages | 40,000 |
| | 7,200 |
| Indirect wages | 6,800 |
| Administration salaries | 4,400 |
| Selling and Distribution expenses | 7,500 |
| | 4,000 |
| Administration expenses | 33,100 |
| Production expenses | 2,070 |
| | 1,000 |
| Stores issued: - Production | |
| Factory maintenance | 14,000 |
| Administration | |
| | 15,000 |
| | 11,000 |
| Production overhead absorbed | 78,000 |
| Production overhead absorbed | 80,000 |
| by finished goods | 120,000 |
| Selling and distribution Overhead absorbed by sales | |
| Factory cost of finished goods | |
| Cost of finished goods sold | |
| Sales | |

You are required to record the transactions for the month and prepare a Costing Profit and Loss Account and a Trial Balance as at the end of the month (ICAN, May 1995 Adopted)

6. Explain how the results shown by the Cost Accounts and those shown by the Financial Accounts may be reconciled.

(b) According to the Cost Accounts of Gwammaja PLC, profit for the year was ₦48,390. The Financial Accounts disclosed the following position:

**Manufacturing Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Raw materials opening inventory | 1,900 | Transfer to finished inventory account | 111,400 |
| Purchases (less returns) | 54,900 | | |
| | 56,800 | | |
| Less: Closing inventory | 1,800 | | |
| | 55,000 | | |
| Direct labour | 35,500 | | |
| Factory overhead | 21,400 | | |
| | 111,900 | | |
| Work-in-progress: | | | |
| Opening 8,400 | | | |
| Closing 8,900 | 500 | | |
| | 111,400 | | 14,530 |

**Finished Inventory Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Opening inventory | 11,600 | Cost of Sales transfer. | |
| Transfer from manuf. a/c | 111,400 | to trading A/c | 110,700 |
| | | Closing Inventory | 12,300 |
| | 123,000 | | 123,000 |

**Trading Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Factory cost of sales | Sales | | 184,500 |
| transferred from inventory a/c | 110,700 | | |
| gross profit c/d | 73,800 | | |
| | 184,500 | | 123,000 |

**Profit and Loss Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Administration expenses | 15,600 | Gross profit b/d | 73,800 |
| Distribution expenses | 10,182 | Discount received | 1,800 |
| Discounts allowed | 1,511 | Bank interest received | 37 |
| Debenture interest | 850 | Dividend received | 300 |
| Fines | 500 | | |
| Losses of a non-trading nature | 350 | | |
| Net profit | 46,950 | | |
| | 76,943 | | 75,943 |

Inventory valuations in the Cost Accounts were as follows:

| | Opening Balance ₦ | Closing Balance ₦ |
|---|---|---|
| Raw materials | 1,969 | 1,850 |
| work-in-progress | 8,280 | 8,730 |
| Finished inventory | 11,396 | 12,810 |

Depreciation amounting to ₦6,146 was charged in the cost accounts, whereas Factory Overhead in the Financial Accounts included ₦5,873 for this expense. The profit shown in the Cost Accounts has been arrived at before charging the notional rent of ₦1,500. You are required to present a logical statement to reconcile the two profit figures.

7 Indabawa Limited, a company engaged in the manufacture of specialised marine engines, operates a historic job cost accounting system which is not integrated with the financial accounts. At the beginning of May 2021, the opening balances in the cost ledger were:

| | ₦ |
|---|---|
| Stores Ledger Control Account | 85,400 |
| Work-in-progress Control Account | 167,350 |
| Finished Goods, Control Account | 49,250 |
| Cost Ledger Control Account | 302,000 |

During the month, the following transactions took place:

| | ₦ |
|---|---|
| Materials: Purchases | 42,700 |
| | 63,400 |
| Issues – to general maintenance | 1,450 |
| - to the construction of manufacturing | |

| | |
|---|---|
| - to the construction of manufacturing equipment | 7,650<br>124,000 |
| Factory wages: Total gross wages paid | |

₦125,000 of the above gross wages were incurred on the construction of manufacturing equipment, ₦35,750 were indirect wages, and the balance was direct.

**Production overheads:**
Actual amount incurred, excluding items shown above, was ₦152,350; ₦30,000 was absorbed by the manufacturing equipment under construction and under-absorbed overheads written off at the end of the month amounted to ₦7,550.

**Royalty Payments:**
One of the engines produced is manufactured under licence.
₦2,150 will be paid to the inventor for the month's production of that particular engine.

| | |
|---|---|
| Selling overheads: | ₦22,000 |
| Sales | ₦410,000 |

The company's gross profit margin is 25% on factory cost.

At the end of May, the inventory of Work-in-Progress had increased by ₦12,000. The manufacturing equipment under construction was completed within the month and transferred out of the cost ledger at the end of the month.

**Required:**

Prepare the relevant control accounts, costing profit and loss account, and any other accounts you consider necessary to record the above transactions in the cost ledger for May 2021. (ACCA, Costing adapted)

6.8 Chiroki and Ibro Nig. Ltd. maintained separate cost and financial accounts, and the costing profit for 2020 differed to that revealed in the financial accounts, which was shown as ₦50,000.

i. The following information is available:

| | Cost Account ₦ | Financial Accounts ₦ |
|---|---|---|
| Opening inventory of raw materials, | 5,000 | 5,500 |
| Closing inventory of raw materials, | 4,000 | 5,300 |
| Opening inventory of finished goods | 12,000 | 15,000 |
| Closing inventory of finished goods | 14,000 | 16,000 |

(ii) The company received a dividend of N1,000

(iii) A machine with a Net Book Value of ₦10,000 was sold during the year for ₦8,000

(iv) The company charged 10% interest on its opening capital employed of ₦80,000 to its process costs.

**Required:**

(i) Determine the profit figure shown in the cost accounts.

(ii) Explain the advantages a business might obtain from the integration of cost and financial accounting data.

**INTEGRAL**

6.9 You are provided with the extracted operational data of Tamsat Limited:

Opening Trial Balance as at 1st January, 2022

| ₦ | ₦ |
|---|---|
| Debit: Work-in-progress | 25,000 |
| Store ledger control A/c | 10,000 |
| Finished goods control A/c | 20,000 |
| Fixed assets (Non-current assets) | 80,000 |
| Credit: Creditors for Material | 30,000 |
| Prov. For depreciation on F.A | 10,000 |

| Transactions for the period of ended 31st January | ₦ | ₦ |
|---|---|---|
| Materials purchased | | 70,000 |
| Normal loss on materials | | 3,000 |
| Direct materials issued | | 50,000 |
| Indirect materials issued to factory | | 6,000 |
| Wages allocated as indirect to factory | | |
| Wages allocated as indirect to factory | | |
| Total wages incurred | 35,000 | |
| Production overhead incurred | 10,000 | 45,000 |
| Production overhead absorbed | | 12,000 |
| Output for the period | | 24,000 |
| | | 100,000 |

**Required:**

a. Using integrated account, produce the following ledgers: (i) Stores Ledger (ii) Wages Control Account (iii) Production Overhead Control Account (iv) Work-in-progress control account; and (v) finished goods account.

b. Prepare the closing trial balance

(12½ Marks)

**INTERLOCKING**

6.10 The trial balance of Wusu on 31st March, 2021 is provided below:

| | ₦ | ₦ |
|---|---|---|
| Stores Ledger Control Account | 20,000 | |
| Work-in-progress control account | 30,000 | |
| Finished goods control account | 10,000 | |
| Cost ledger contra or general ledger adjustment account | | 60,000 |
| | 60,000 | 60,000 |

The extracted details for April 2021 transactions are:

| | ₦ | ₦ |
|---|---|---|
| Purchase of Stores | | 70,000 |
| Returned to suppliers | | 15,000 |
| Normal loss in stores | | 1,000 |
| Direct materials issues to production | | 50,000 |
| Stores issued to maintenance | | 5,000 |
| Stores: Wages allocated | 25,000 | |
| Indirect factory wages | 8,000 | |
| Production Overhead incurred | | 33,000 |
| Production Overhead recovered | | 9,000 |
| Output for the period | | 22,000 |
| | | 120,000 |

**Required:**

a. Prepare the following ledgers:
   i. Cost ledger contra Account
   ii. Store/raw materials ledger control account
   iii. Wages control account
   iv. Production overhead control account
   v. Work-in-Progress Control Account
   vi. Finished Goods Account
   vii. Closing Trial Balance at 30th Aril 2021

**(12½ Marks)**

**RECONCILIATION**

6.11 a. Dambat Limited is a medium size firm that has been in existence for the past five years. The firm uses interlocking system in its Cost Accounting Department. It has been observed that the profit disclosed in the costing department books differs from profit disclosed in the financial reports generated in the Financial Account Department.

**Required: a.**

i. Why are profits disclosed in cost accounting department different from those of financial accounting department?
ii. List four (4) items that are recorded in the financial books only
iii. List two (2) items that are considered in the cost accounting books only.

b. Given below are extracted data from both financial and costing books:

**Financial Books:**

| | ₦ |
|---|---|
| Opening inventory on raw materials | 6,000 |
| Closing inventory on raw materials | 7,000 |

**Costing Books:**

| | |
|---|---|
| Opening inventory on raw materials | 5,000 |
| Closing inventory on raw materials | 12,000 |

**Other details:**

| | |
|---|---|
| Interest received on investment | 2,000 |
| Profit on sale of fixed asset (non-current asset) | 1,000 |
| Profit disclosed in financial books | 77,000 |
| Profit disclosed in costing books | 80,000 |

You are required to reconcile the differences in the profits disclosed in both books

## CHAPTER SEVEN

## COSTING METHODS: SPECIFIC ORDER COSTING

**CONTENTS**


a. Introduction
b. Specific Order Costing methods:
   i. Job;
   ii. Batch;
   iii. Contract costing; and
   iv. Service costing


### 7.0 Objectives:

After studying this chapter, readers should be able to:

a. appreciate what order costing is all about and its difference with process costing.
b understand various methods of absorbing general overhead expenses by jobs, batches, contracts, services or products.
c. appreciate the important features of batch, service and contract costing.
d. understand how to account for the value of a product, service, batch or contract.
e. understand how to account for the total cost of an order, for effective pricing decision purpose.

### 7.1 INTRODUCTION

### 7.2 Specific Order Costing Methods

#### 7.2.1 The Concept of Order Costing

This is a situation where expenses are linked to the cost of products or services ordered from the supplier. The ordering expenses are to be fully accounted for in the course of costing the product or service. Examples of ordering items are job, contract, or batch. Thus, job costing, contract costing and batch costing are specific order costing methods.

Order costing is a technique used by organisations which produce specific jobs, batches, contracts or orders that can be identified throughout the various stages of production. A job cost system, therefore, enables an organization to determine the total cost incurred in the course of performing a job. The system is normally used by organisations that are involved in construction (bridges, roads, etc.),

building (of houses, offices, etc.), machine tool manufacturing, printing (of books, journals, newsletters, etc.), and general engineering where each product is produced to the specification of the customer.

Order costing is also widely used for costing of batches of similar articles such as shoes, bags, shirts, etc. Order costing is a basic costing procedure which may be used in conjunction with other costing techniques.

The main purpose of order is to determine the profit or loss realised on each order. This means that total cost per order has to be estimated as accurately as possible for a good price to be charged on the order, if the organisation is to make reasonable profit. Order costing assists organisations (in the private or public sectors) to distinguish between profitable and unprofitable orders and to make comparison with past estimates in order to assess productivity and profitability levels. The technique is also used as a basis of pricing government contracts and other contracts, where "cost plus" basis (that is at a price based on an agreed cost plus agreed target profit) of pricing is in use.

### 7.2.3 **Job Order Costing Procedure**

Costing a job, batches of product, or contract, involves aggregation of all the cost elements necessary for the final product. For factory working process, adequate production control records must be maintained. Each job must be given an order number which must be cited or identified at all stages of production.

Where an order for a very large quantity of output is placed, it should be divided into separate batches so that job costing can be made without waiting for the completion of the whole work. The batches are then to be numbered along the line of the number given to the large quantity order. For example, where the order number is 30, the separate batches of the job could be 30.1, 30.2, 30.3, etc. for the purpose of distinguishing between them.

Each order is to be put into production control records and a job cost sheet or job ledger account prepared. On this account will be recorded the cost of material used, the wages paid to labour and the machine time taken. The departmental general overhead cost is to be apportioned for absorption by each of the jobs, batches or contracts performed by the department.

### 7.2.4 **Overhead Cost Absorption Methods in Job Costing**

In costing a job, the total of the direct material cost, direct wages, and direct expense of the job must be taken. Overhead is then applied as a percentage of direct material cost, percentage of wages, percentage of prime cost, rate per unit or rate per hour as required. The direct materials, wages and expenses, and the absorbed overhead are totalled to give the production cost of the job. Each job must absorb, in addition to production overhead, part of the administration, selling and distribution overheads, unless they are regarded as direct charges to the Profit and Loss Account of the period. The addition of selling, administrative and distribution overhead costs to production cost gives the total cost of the job.

The final effect in accounting for overhead is to charge overhead costs to cost units. The charging of overheads of a production cost centre to all the units passing through the centre is termed Absorption.

In order to absorb overhead, it is necessary to develop a basis upon which overhead costs can be charged to cost unit. In general, the following arbitrary bases, which have been discussed earlier (in chapter 4), are applied:

1. Percentage Rate on Direct Labour Cost, Direct material Cost and Prime Cost.
2. Cost unit Rate
3. Time Rate: Machine Hour Rate and Direct Labour Hour Rate.

We shall illustrate some of these bases of overhead absorption, as follows:

(a) **Percentage Rate on Direct Wages:** This basis (method) is about the overhead rate determinable by dividing the total overhead cost attributable to a cost centre by the total direct wages incurred and expressing the same as a percentage.

$$\text{Overhead Rate} = \frac{\text{Total overhead charges}}{\text{Total Direct Labour Cost}} \times 100$$

This method is suitable where the overhead cost is lower than the direct wages; where wages are major portion of total costs; where similar types of labour and rates of pay are employed; and where relatively little machines are in use.

**Illustration 7.1**

Star Modern Shoemakers Company Limited, Wambai, Kano, found that their production costs for the 2025 accounting year had been as follows:

| | |
|---|---|
| Direct wages | ₦8,000,000 |
| Direct material cost | ₦5,000,000 |
| Direct expenses | ₦1,500,000 |
| Production overhead cost | ₦2,400,000 |
| Total shoe manufactured | 160,000 units |
| Machine time used | 10,000 hours |
| Direct Labour time used | 40,000 hours |

500 pairs of shoes were produced during the first two weeks of the year as a single job, which had a direct wages cost of ₦8,000, direct material cost of ₦5,000 and direct expenses of ₦1,500. The company apportions fixed overhead cost using the percentage rate on direct wages and charges administrative expenses and selling expenses at the rate of 15% and 10% respectively, of production cost.

Required: Determine the following:

(a) Production cost
(b) Total cost
(c) Cost per pairs of shoe

**Solution**

**Star Modern Shoemakers Company Ltd, Wambai, Kano**

**Computation of Production Cost & Total Cost**

| | | |
|---|---|---|
| Direct wages | 8,000 | |
| Direct material cost | 5,000 | |
| Direct expenses | 1,500 | |
| Prime cost | 14,500 | |
| Production Overhead | 2,400 | |
| Production Cost | 16,900 | (a) |
| Admin. Expenses | 2,535 | |
| Selling expenses | 1,690 | |
| Total Cost | 21,125 | (b) |

(c) Cost per pair of shoes = 21,125/ 500 = ₦42.25

Workings

1. % Rate on Direct wages = ₦2,400,000 x 100 = 30% ₦8,000,000
2. Production Overhead cost on the job = 30% of 8,000 = ₦2,400
3. Administrative expenses = 15% of 16,900 = ₦2335
4. Selling expenses = 10% of 16,900 = ₦1690

**(b) Percentage Rate on Direct Material Cost**

This is determinable by dividing the total overhead cost by the total direct material cost and expressing the figure in percentage.

Overhead Rate = Total Overhead Charges x 100
Total Direct Material cost

**Illustration 7.2**

Assume that the Shoe Manufacturing Company in illustration 6.1 apportions fixed overhead cost using the percentage rate on direct material cost, and all other figures remain the same, determine i - iii.

Workings:

1. % Rate on Direct Material Cost = ₦2,400,000 x 100 = 80%
   ₦3,000,000
2. Production Overhead on the job = 80% of ₦ 5,000 = ₦4,000
3. Administrative expenses = 15% of ₦18,500 = ₦2,775
4. Selling expenses = 10% of 18,500 = ₦1,850

Computation of Production Cost & Total Cost

| | |
|---|---|
| ₦ Direct wages | 8,000 |
| Direct material cost | 5,000 |
| Direct expenses | 1,500 |
| Prime cost | 14,500 |
| Production overhead | 4,000 |
| Production Cost | 18,500 (i) |

| | |
|---|---|
| Administrative expenses | 2,775 |
| Selling expenses | 1,850 |
| Total Costs | 23,125 (ii) |

(iii) Cost per pair of Shoes = ₦23,125 ÷ 500 = ₦46.25

The percentage rate on direct material cost method is more appropriate where direct material cost forms a major part of total cost for a job, contract or batch and where overhead cost tends to relate to direct material cost. The method is also equitable where only one type of product is produced and, therefore, the material cost charged to each cost unit is identical.

**Note:**

A Job Card is opened for each job, contract or batch that is undertaken by the company. The format of a job card varies from one organization to the other but essentially it should contain certain information as contained in a typical job.

It should be appreciated that the Job Cost Card is supplementary to the cost book-keeping system, that is, it exists independently of the cost ledger system, for up-to-date control of costs. This is why in most jobbing concerns, such as a Motor Garage, the job card is kept in the shop floor while costs (expenses) are recorded in the cost ledger via the normal source documents like Material Requisition Note, Wages Analysis Journals and Expenses Analysis Books.

### 7.2.5 Batch Costing

Batch Costing arises where a customer orders a quantity of identical products at the same time. The job consists of a number of units of similar products covered by a single job number: for example, a batch of 1000 copies of a book printed by a printer. The batch cost must be divided by the total quantity to give a cost per unit.

Peculiar features of Batch Costing System:

- ***(a) Set-Up Costs:*** This is the cost of setting up production facilities, for example, the Engineering time, Factory consumables to set up moulds, etc.
- ***(b) Unit Cost:*** Average cost per unit will tend to decrease as the batch size increases due

to some fixed costs (such as set up costs mentioned above) which will remain the same irrespective of the batch size.

***(c)*** ***Batches of Dissimilar Products*****:** Sometimes batches of different products are brought together because a common operation can be performed on them so as to save set-up time on the machine. The common cost of this Joint Operation becomes a problem to apportion. However, it should be realized that common costs should be apportioned on equitable basis.

**Illustration 7.3**

Ayingba Press Limited was asked to quote for supplying 5,000 and 25,000 booklets. The company normally expects a profit of 10% on sales. Costs were recognized as follows:

| | | | |
|---|---|---|---|
| Paper and other materials, | | - | ₦ |
| (per 1,000 copies) | | 30.00 | |
| Wages (per 1,000 copies) | | 20.00 | |
| Layout cost | | 500.00 | |
| Fixed overhead | | 200.00 | |
| Variable overhead= | 12% of wages | | |

Draft a cost computation, showing minimum selling prices that might be quoted per 1,000 copies each of the supplies.

Although this question is asking for the selling price to quote, we shall also use it to demonstrate how cost per batch and how cost per unit is obtained in addition to solving the simple arithmetical exercise of calculating selling price.

**Solution**

Calculation of Batch Cost

As earlier on mentioned, this is just the accumulation of all the expenses incurred in producing the batch of similar products; in this case, the booklets:

| | 1000 Copies ₦ | 5000 copies ₦ | 25000 copies ₦ |
|---|---|---|---|
| Setup Cost or | 500 | 500 | 500 |
| Lay-out Cost | | | |
| Material | | | |
| Paper | 30 | 150 | 750 |
| Labour | | | |
| Wages | 20 | 100 | 500 |
| Overhead | | | |
| Fixed Overhead | 200 | 200 | 200 |
| Variable Overhead | 2.4 | 12 | 60 |
| Total Cost | 752.40 | 962 | 2,010 |
| Profit Margin | 83.60 | 107 | 223 |
| Selling Price | 836 | 1069 | 2,233 |

It should be noted that Total Cost was obtained by simply aggregating the expenses, set-up cost, and material, labour and overhead cost. If cost per unit is desired, it will be obtained by simply averaging the total cost per batch over the units that made up the batch size. At 5,000 units, total cost was ₦962: cost per unit, therefore, will be ₦962/5,000, i.e. 19.24 kobo. At 25,000 copies, total was ₦2,010; cost per copy therefore is 8.04 kobo. This proves a common saying that unit cost decreases as the batch size increases. Selling price per copy will also be calculated in the same way.

## 7.3 Contract Costing and Accounting

Contract Costing is a form of order costing. Perhaps the following features would explain how contract costing differs from job order costing. The peculiar features of contract costing system are:

(a) The jobs are usually of long-term duration, often more than one accounting period.

(b) The contracts are often carried out in the contractee's premises, that is, site-based.

(c) There may be sub-contractors where there are some special jobs to be done such as plumbing and electrical installation.

- There is often an architect engaged by the contractee who supervises the work and issues a valuation certificate upon which payment would be made to the contractor. This is otherwise known as work certified or architect's valuation.

It should be appreciated by the reader that the problem with contract costing is not the ascertainment of cost. Cost is ascertained by simply aggregating all the expenses relating to each contract. A contract account is opened for each contract wherein the expenses will be recorded as they are incurred.

The major problem with contract costing is the calculation of profit (if any) that can be taken on an on-going contract which has reached an advanced stage, but has not been completed. This major problem (accounting for profit on uncompleted contracts) should, therefore, be given proper attention.

Contracts can be classified into three (3) different categories, according to their stages of completion, as follows:

(a) Contracts at the early stage.
(b) Contracts that have reached an advanced stage but for which cost to complete cannot be reasonably estimated.
(c) Contracts that are nearing completion, that is, have reached an advanced stage and for which cost is somehow complete and can be reasonably estimated.

For any of these types of contract, the accountant is advised to first open the contract account and debit thereto all relevant expenses and apportionments. This will give the up- to- date cost of the contract, and then move to the next stage of considering whether profit can be taken at the given stage of the contract. It is at this stage that the three classifications given above become important.

A Notional profit is usually first obtained before a calculation of profit is taken. The procedure for calculating Notional Profit and profit to be taken is discussed according to the stage of the contract.

### (a) Contract at the Early Stage

Where the contract has just been started and not in any way nearing completion, the expenses debited to the contract account will simply be carried forward as cost of the contract or the cost of the Work-In-Progress, just in the same manner as Job Costing.

(b) **Fairly Advanced Stage**

At this stage of a contract's life, profit may be recognized on the efforts on the contract to date, although clear estimate of the cost to completion may not be available. The profit taken is usually limited by the excess of the architect's value of work certified over the contractor's cost to date (notional profit) adjusted for the cash actually received. This approach accords with the prudence concept.

Where a contract is at this stage, the profit recognized is usually computed as follows: Step 1: Calculate the Notional Profit as follows:

| | |
|---|---|
| Value of work certified | XX |
| Less: Cost of work certified | (X) |
| Notional Profit | XX |

It should be noted that cost of work certified might not be the same with the cost to date as cost may be incurred on work which, for one reason or the other, the architect is yet to certify.

Step 2: Calculate the Profit taken as follows:

2/3 x Notional Profit x Cash received

Value of work certified

Readers should note that the proportion of notional profit to be taken will be as determined by the contract paper or, in the case of students, by the examiner.

(c) **Nearing Completion Stage**

At this point, reliable estimates could be made on the cost of completed contract, thus the total contract cost is in sight. Profits are taken on the contracts at this stage, using a fairly different basis compared to those discussed above. Here the profit taken is with reference to the estimated total profit of the contract over its life, while adjusting for whatever profit that has been recognized previously.

The calculation of profit taken follows these steps: Step 1: **Calculate the Estimated Total Profit (ETP).**

| | N |
|---|---|
| Total Contract Price | |
| Less: Contract Cost: | |
| Cost to date | X |
| Cost to complete | X |
| Estimated total cost | |
| Estimated total profit | |

Step 2: **Calculate Profit Realised to Date.**

This is the lower of:

(i) Value of work certified x ETP Estimated Total Cost (ETC) and

(ii) Cost to date x ETP Estimated total cost

Step 3: **Profit taken in Current Period**

| | ₦ |
|---|---|
| Profit realized to date | X |
| Profit recognized in prior periods | (X) |
| Profit taken in current period | XX |

Where it is expected that the contract as a whole will result in a loss for the contractor, the loss must be written off to the maximum possible extent immediately. If the contract is unprofitable, it shouldn't have been accepted in the first place. The foreseeable loss should be written off in the period in which it is recognized, in line with the prudence principle.

**Final Step**

The final step is to provide the value at which a contract should be reflected in the balance sheet. This should not pose a big problem as what is required is to deduct payments received from the contractee from the value of the asset created so far by the contract. This is the amount to which the contractor remains entitled and is to be shown as contract value receivable in the balance sheet.

The calculation is to be done as follows:

| | ₦ |
|---|---|
| Total Cost incurred to date | X |
| Profit taken/realised on contract | X |
| **Total value of the contract** | **X** |
| Less cash received from contractee | (X) |
| Contract value for balance sheet | X |

## Service Costing

Service costing, also known as operation costing, is a method of cost ascertainment applied in organisations which provide services as opposed to products. Examples are transport companies, electricity companies, hotels, hospitals, schools and cinema houses.

### Characteristics of Service Costing

- Services provided are standardised, i.e. similar type of services are provided to all customers
- Services are provided on an uninterrupted and regular basis
- Huge capital is invested in fixed assets, e.g. transport vehicles, hotels buildings and hospital equipment.
- Some portion of the operating cost of the organisations is fixed while some other part is variable.

FORMAT OF OPERATION COST SHEET OF A TRANSPORT COMPANY

| Vehicle No: | | Period : |
|---|---|---|
| Cost Unit: | | No. of Units: |
| | TOTAL ₦ | Per Km. ₦ |
| **Fixed Costs (Standing Charges)** | | |
| Motor Park Dues | | |
| Vehicle Licencing and Insurance | | |
| Manager's Salary | | |
| Administrative Charges | | |
| Finance Charges | | |
| Driver's Salaries | | |
| TOTAL FIXED COSTS | | |
| **Variable (Running Costs)** | | |
| Depreciation | | |
| Fuel and Oil | | |
| Spare Parts, Service and Repairs | | |
| TOTAL VARIABLE COSTS | | |
| GRAND TOTAL | | |

The total costs are aggregated as per the format above and the cost per vehicle per kilometre can be ascertained

### Service-based Cost Models in a Digital Economy

The rise of the digital economy has significantly altered how businesses set prices for goods and services. Traditional cost-plus pricing models are giving way to dynamic, data-

driven strategies that leverage real-time information. Companies now employ sophisticated algorithms and artificial intelligence to optimize pricing based on consumer demand, competitor actions, and market conditions. This shift presents both opportunities and challenges for businesses and consumers alike.

Microeconomic principles, particularly demand and supply dynamics, play a crucial role in digital pricing strategies. Businesses aim to set prices that maximize profits while considering consumer preferences, willingness to pay, and market competition. The elasticity of demand—a measure of how sensitive consumers are to price changes—guides firms in adjusting their pricing strategies to optimize revenue.

One of the most prominent pricing strategies in the digital economy is dynamic pricing, where prices fluctuate based on factors such as demand, time, and consumer behavior. Online retailers, airlines, and ride-sharing companies use algorithms to adjust prices in real-time. For example, Amazon frequently changes product prices based on competitors' pricing and customer demand, while Uber's surge pricing adjusts fares based on rider demand and driver availability.

The freemium model is widely used in digital services, where basic services are provided for free, while premium features require payment. Platforms like Spotify, LinkedIn, and many mobile applications rely on this strategy to attract a large user base before converting free users into paying customers. This model leverages the concept of consumer surplus, where users perceive value in premium services and are willing to pay for additional features.

Subscription models have gained popularity in the digital economy, particularly in industries such as entertainment (Netflix, Disney+), software (Microsoft 365, Adobe), and e-commerce (Amazon Prime). This pricing strategy provides businesses with a steady revenue stream while offering consumers cost-effective access to services. Subscription pricing also reduces customer acquisition costs, as retaining existing customers is generally cheaper than acquiring new ones.

Price discrimination involves charging different prices to different consumer segments based on their willingness to pay. In the digital economy, firms use data analytics to implement personalized pricing strategies. Airlines, hotels, and e-commerce platforms analyse browsing history, location, and purchasing patterns to offer customised prices. While this strategy enhances revenue, it raises ethical concerns regarding fairness and consumer privacy.

Digital businesses frequently use product bundling, where multiple products or services are offered at a discounted price. Software companies, for example, bundle productivity tools into a single subscription plan. Another effective approach is psychological pricing, where firms set prices slightly below a round number (e.g., $9.99 instead of $10) to influence consumer perception and increase sales.

Market structure influences pricing strategies in the digital economy. In highly competitive markets (e.g., e-commerce), firms often engage in penetration pricing, setting

initial prices low to gain market share before raising them later. Conversely, monopolistic digital platforms (e.g., Google Ads, Apple App Store) may leverage premium pricing due to their dominant market position and unique value propositions.

While advanced pricing strategies benefit businesses, they also raise ethical concerns. Algorithmic pricing can lead to price gouging, where firms exploit high demand to impose excessive prices, as seen in emergency situations or scarce product availability.

Additionally, excessive reliance on consumer data for pricing personalization raises concerns about privacy and potential discrimination. Regulatory bodies are increasingly scrutinising these practices to ensure fair competition and consumer protection

**Cost for digital products**

Digital products are intangible products you sell online.

For example:

- Online courses
- E-books
- Apps
- Digital art
- Stock photos/videos
- Software templates

Because these are online assets with no raw materials, storage, or shipping costs, digital pricing models are completely different from physical products.

The costs for digital products can vary widely due to the following factors:

- Production Costs. These include expenses for software subscriptions, freelancers, and other resources required to create the product.
- Market Research. Understanding the market and competition can help in setting a competitive price
- Profit Margins. Digital products often have high margins due to low overhead costs, but initial startup costs should be considered.
- Value Perception. The price should reflect the value that customers see in the product, which can influence how much they are willing to pay.

- Pricing Strategies. Different strategies can be employed, such as cost-plus pricing or value-based pricing, to determine the best price point.

**Subscription-based Costing Models**

In today's fast-evolving digital landscape, subscription pricing models have become a cornerstone for organisations intending to build recurring revenue streams and foster long-term customer relationships. From SaaS companies to e-commerce brands, the subscription economy is booming. It is therefore very critical to carefully select the best pricing models from the various ones being paraded

Subscription pricing models are recurring payment structures where customers pay a regular fee—typically monthly or annually—to access a product or service. Unlike one-time purchases, subscriptions emphasize continuous value delivery, making them ideal for businesses aiming to secure predictable revenue and enhance customer retention.

The flexibility of subscription models allows companies to cater to diverse industries, including software, media streaming, fitness, and even physical goods.

**Top Subscription Models**

1. Flat-Rate Pricing. A single fixed price grants unlimited access to all the features of the product, e.g. Netflix

2. Tiered Pricing. Multiple pricing tiers with varying features or usage limits, e.g. based on team size and functionality

3. Usage-Based Pricing. Customers pay based on how much they use the service

4. Freemium Model. A free basic plan with premium upgrades available for a fee

5. Per-User Pricing. Charges are based on the number of users accessing the service.

6. Hybrid Pricing. Combines elements of the models above.

## 7.4 CHAPTER SUMMARY

In this chapter, you have acquired the knowledge of job costing, batch costing, service costing and contract costing, which were discussed under specific order and non-specific order costing methods. It has been made clear that job costing is quite different from process costing. Efforts have been made to simplify some of the methods of absorbing general overhead expense for the determination of total cost of an order for the supply of a product or service, which may come in the form of a job, a batch or a contract. These methods are basically classified into three: (i) Percentage rate on direct cost (material, labour or prime cost); (ii) Cost per unit rate and (iii) Time rate (machine hour or direct labour hour rate). The outlines and steps for job, batch, service and contract costings were all presented and discussed.

It is hoped that readers would be able to show clear understanding of the following, anywhere, anytime:

- Meaning of order costing.
- Differences between job order costing and process costing.
- Different methods of absorbing overhead costs by jobs.
- Job Costing procedure as guide to effective pricing decision.
- Batch costing principles and process.
- Service costing, including costing of professional services.
- Steps to be followed in contract costing and accounting.

## MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS

1. Distinguish between a Job Card and a Job Cost Card
   - A. Job card is kept for employees whilst Job Cost Card is kept for assets
   - B. Job card is kept on the shop floor whilst Job Cost Card in the cost office
   - C. Job card is for engineering jobs whilst Job Cost Card is for financial jobs
   - D. Job card is for jobs whilst Job Cost Card is for batches
   - E. Job card is given to the customer whilst Job Cost Card is kept by the company

2. What are set-up costs?

   A. Cost of buying equipment sets
   B. Cost of constructing the factory
   C. Cost of setting up production facilities for a particular job or production
   D. Cost of laying ambush for dishonest employees
   E. Cost of assembling factory operatives in readiness for production

3. What is mark-up?

   A. A percentage of total costs added as profit to determine the selling price
   B. The distinctive mark on packages to avoid mix-up
   C. The total selling price of an order
   D. The total cost of executing a Job
   E. The amount of bonus to be given to those who worked on an order

4. What is Notional profit on contracts?

   A. The total profit of the contractor for the years
   B. The amount of profit to be paid to the nation's coffers
   C. The profit on the mobilization fee received on a contract
   D. The amount of profit brought forward from previous years
   E. The portion of profit to be recognized on an on-going contract at year end

5. "Value of work certified" in contract means……..

   A. Value of materials used in the contract
   B. Total cost of executing a contract
   C. Total progress payments received on a contract
   D. The aggregate total value of architect's certificates issued so far
   E. Total value of items sub-contracted on the main contract

6. How is a job identified among other jobs being carried out simultaneously in the accounts?

7. What document gives the value of contract completed on a progressive basis?

8. Large jobs usually of long-term duration, usually extending beyond the company's financial year-end, and carried out on the customer's premises are known as ……...

9. Where a company decided to recognize total provisional profit on a contract almost fully completed at its year end, what is the name given to the amount added to cost to date to arrive at estimated total cost?

10. An amount, usually a small percentage of the total contract sum, withheld by the contractee, for almost a year after the end of a contract as insurance against possible faults discovered later is known as……..

**SOLUTION**

1. B
2. C
3. A
4. E
5. D
6. Job Number or Order Number
7. Architect's certificate
8. Contracts
9. Cost to completion
10. Retention money or fee

**EXAMINATION TYPE QUESTIONS**

1. Discuss the concept of Job Order Costing as clearly as possible.
2. Briefly explain any TWO methods of absorbing overhead costs by jobs.
3. Write short notes on each of the following:
   (i) Overhead absorption
   (ii) Cost Plus basis of pricing
   (iii) Transport service costing
   (iv) Batch costing
4. Mr. Hankaka, a shoe manufacturer, found that his production costs for an accounting period had been as follows:

| | |
|---|---|
| Direct wages | ₦800,000 |
| Direct materials | ₦300,000 |
| Direct expenses | ₦100,000 |
| Prime cost | ₦1,200,000 |
| Production overhead | ₦240,000 |
| Factory Cost | ₦1,440,00 |

| | | |
|---|---|---|
| Total shoes manufactured | = | 160,000 units |
| Total machine hours used | = | 10,000 hours |
| Total Direct labour hours used | = | 40,000 hours |

A job performed had a direct wages cost of ₦800, direct material cost of ₦500 and direct expenses of ₦150.

**Required:**

(a) Find the total cost of the job, using:

(i) Direct Wages percentage rate (ii) Direct material percentage rate.

(b) If the job was that of producing 120 units of shoes, what would be the total cost?

5. Hantsi Limited operates a factory consisting of two departments: Assembly and Galvanizing. All products pass through the 2 departments. The following information in respect of the two departments was given:

| | Assembly | Galvanizing |
|---|---|---|
| Materials | ₦40,000 | ₦12,000 |
| Direct wages | ₦20,000 | ₦36,000 |
| Production OH | ₦16,000 | ₦24,000 |
| Direct labour hours | 38,000 | 24,000 |
| Machine hours | 66,000 | 38,000 |

One of the jobs completed in the factory was No. 707 and the following data relate to the job.

| | Assembly | Galvanizing |
|---|---|---|
| Materials | ₦400 | ₦80 |
| Direct wages | ₦300 | ₦150 |
| Direct labour hours | 800 | 300 |
| Machine hours | 700 | 800 |

**Required:**

(a) Use 3 different methods by which production overheads could be recovered and prepare the job cost card in respect of Job No. 707, using 50% of production overheads to recover administrative overheads and 25% of production overheads to recover selling and distribution

overheads.

(b) Show the total proceeds on the job if target profits is put at 45% of total cost.

6. Shahada Ltd operates a factory consisting of two (2) departments: Washing and Cleaning. All activities pass through the two (2) departments:

| | Washing | Cleaning |
|---|---|---|
| Materials | ₦60,000 | ₦25,000 |
| Direct wages | ₦30,000 | ₦40,000 |
| Production OH | ₦20,000 | ₦32,000 |
| Direct labour hours | 40,000 | 25,000 |
| Machine hours | 50,000 | 43,000 |

One of the jobs completed in the factory was Job No. 807 and the following data relate to the Job:

| | Washing | Cleaning |
|---|---|---|
| Materials | ₦500 | ₦120 |
| Direct wages | ₦450 | ₦200 |
| Direct labour hours | 900 | 450 |
| Machine hours | 200 | 150 |

**Required:** Prepare Job Cost Card in respect of Job No. 807 using:

(i) Direct material percentage rate; and

(ii) Direct labour hour rate of recovering production.

You are to use 30% of Production Overhead of the Job to recover administrative overhead and 35% of the production overhead to recover selling and distribution overhead. (BUK, 1993).

7. The following figures have been extracted from the books of Ray soap Manufacturing Company. All jobs pass through the company's two departments:

| | Working Department | Finishing Department |
|---|---|---|
| Materials used | ₦6,000 | ₦500 |
| Direct Labour cost | ₦3,000 | ₦1,500 |
| Factory overheads | ₦1,800 | ₦1,200 |
| Direct labour hours | 12,000 | 5,000 |
| Machine hours | 10,000 | 2,000 |

The following information relates to Job S.10.

| | Working Department | Finishing Department |
|---|---|---|
| Materials used | ₦20 | ₦10 |
| Direct Labour | ₦65 | ₦25 |
| Direct Labour hours | 265 (hrs.) | 70 (hrs.) |
| Machine hours | 255 (hrs.) | 25 (hrs.) |

**You are required:**

(a) To use FOUR methods of absorbing factory overhead to show overhead absorbed in each department.

(b) To prepare statements showing the different cost results for Job S.10 under any two of the methods mentioned above (HUK Poly, 1997).

8. Discuss the procedures involved in normal service costing.

9. Discuss the necessary features of a good batch costing system.

Write an article suitable for publication in an Accounting Students Journal/ Newsletter on the topic: "Peculiarities of contract costing and accounting"

A transport company runs a fleet of six buses between two towns 75 kilometres apart. Each bus has a seating capacity of 40 passengers. The following details are available for the month of June 2025:

| | ₦ |
|---|---|
| Wages of drivers, conductors and cleaners | 360,000 |
| Salaries of office staff | 150,000 |
| Fuel and Oils | 1,350,000 |
| Repairs and maintenance | 280,000 |
| Vehicle licencing, Insurance and other road levies | 240,000 |
| Depreciation of buses | 390,000 |
| Finance charges (Interest on loan) | 300,000 |

On the average, the buses are 80% full during the trips, and they operate on 25 days during the month, doing a round trip each day.

Compute the cost per passenger/kilometre (12 marks)

# CONTINUOUS OPERATION COSTING:
# PROCESS COSTING AND JOINT PRODUCTS COSTING

**CHAPTER CONTENTS**

a. Continuous Operation costing:
   i. Process costing including determination of work–in–progress, equivalent units, Treatment of Losses, Gains and Accounting for Scrap
   ii. Accounting treatment of By-Products and Joint Products
   iii. Output costing

b. Non-specific order costing Methods - service costing.

## 8.0 OBJECTIVES:

After studying this chapter, the reader should be able to:

a. Explain the meaning and importance of Process Costing.
b. Explain the areas where Process Costing is applied.
c. Determine 'equivalent units' and their meaning;
d. Treat process losses and wastages;
e. Describe the valuation process in Process Costing.
f. Determine completed product cost;
g. Explain joint costs and joint products;
h. Understand various methods of apportioning joint costs to joint products;
i. Explain the accounting treatment of by-products; and
j. Determine the total costs of producing joint products.

## 8.1 INTRODUCTION TO CONTINUOUS OPERATION COSTING

Continuous operation costing is another method of costing. Unlike specific order costing, which is applicable where work is undertaken to the customer's specification, continuous operation costing is applicable in industries where there are repetitive operations. In the manufacturing industries, continuous operation is applicable in (a) process costing and (b) output or operation costing. In organisations that render services, continuous operation costing is termed as operating or service costing.

### 8.1.1 PRINCIPLES OF PROCESS COSTING

Process Costing is a method used in a situation where production follows a series of sequential processes. The method is used to ascertain the cost of a product or service at each stage.

Stage of production, manufacture or process.

It is generally applied in particular industries where continuous mass production is possible. Because of the continuous nature of the process and the uniformity of the output, it is not possible or necessary to identify a particular unit of output with a time of manufacture. In process costing, the cost of any particular unit must be taken as the average cost of manufacture over a period. This can be complicated by the necessity to apportion costs between completed output and unfinished production at the end of the period.

Wastage must also be accounted for. In this costing system, it is the average cost incurred that concerns management, as managerial pricing decisions would be taken on that basis.

In this section of the chapter, we shall learn about the application areas of process costing, the treatment of losses and wastages, the concept of equivalent units and the determination process of cost of completed products.

### 8.1.2 **Application Areas of Process Costing**

Process costing is used in a variety of industries, including food processing, paper milling, chemical and drug manufacturing, oil refining, soap making, textiles, box-making, paint and ink manufacturing, brewery, flour milling, bottling and canning, biscuit products, meat products, sugar making, etc. It is probably the most widely used cost accounting system everywhere in the world.

Although details will vary from one business concern to another, there are common features in most process costing systems that should be taken note of. These are:

(a) Expenditure for each cost centre is collected and, at the end of the accounting period, the cost of the completed units is then transferred into an inventory account or to a further process cost centre.

(b) Accurate records are required of units produced and part-produced, as well as the cost incurred by the cost centres.

(c) The cost unit chosen should be relevant to the organisation.

(d) The cost of the output of one process is the raw material input cost of the next process stage. The cost incurred in a process cost centre could include, therefore, costs transferred from a previous process plus the raw materials, labour and overhead costs relevant to the cost centre.

(e) Wastage due to scrap, chemical reaction or evaporation is unavoidable. The operation or manufacturing should, however, be in such a way that wastage can be reduced to the barest Minimum.

Either the main product or a by-product of the production process may require further processing before reaching a marketable state.

### 8.1.3. **Distinction between Process and Order Costing**

Order costing and process costing are two extreme systems of product costing. The distinction between the two centres mainly on how product costing is accomplished. Unlike process costing, which deals with broad averages and great masses of like units, the essential feature of order costing is the attempt to apply costs to specific jobs, batches or contracts, which may consist of either a single physical unit or a few like units.

The most important point is that product costing is an averaging process. The unit cost used for products is the result of taking some accumulated cost that has been allocated to production departments and dividing it by the level of production (activity). The basic distinction between order costing and process costing is the size of the denominator. For order costing, the denominator is small (for example, one bridge constructed or one hundred copies of a book, etc.). For process costing, the denominator is large (for example, thousands of litres of kerosene, thousands of bags of cement, etc.).

As indicated earlier, order costing is used in industries like construction, printing, furniture-making, aircraft, machinery, shoe-making, etc., where tailor-made or unique goods are produced. Process costing, as we would learn in chapter 7, is mostly found in such industries as chemicals, oil, textile, plastic, paints, flour milling, canneries, rubber, food processing, glass-making, mining, cement, meat processing, etc., where there is mass production of like units which usually pass continuously, through a series of uniform production steps called processes.

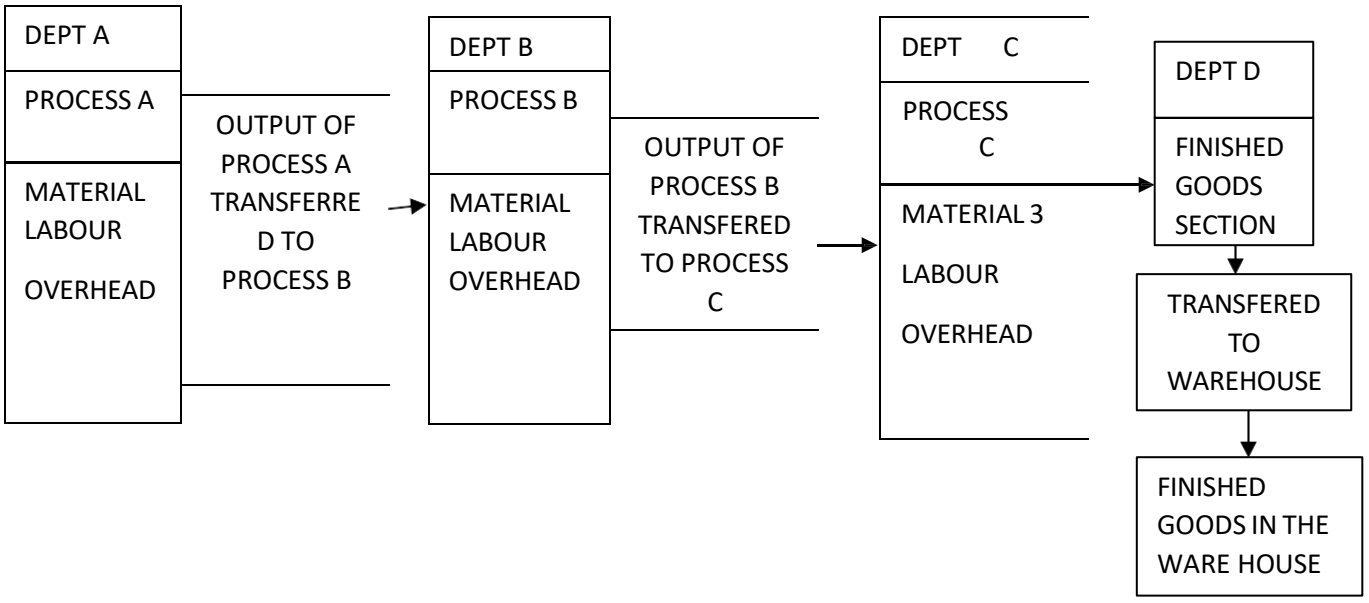
### 8.1.4 Production Stages in Process Costing

It is noticed from the definition of process costing that production follows a series of sequential processes and that the output of one process automatically becomes an input of the intermediate product for the next process stage.

Generally, an intermediate product is further processed by adding further material and incurring labour and overhead until another output is obtained. This process is continuous.

Until the finished goods are obtained in the finished section and transferred to the warehouse. At each process, the unit cost is arrived at by dividing the total cost of the output achieved by the total units produced. This stage-by-stage is indicated in Fig. 7.1

**Fig. 8.1: Production Stages in Process Costing**



8.1.5 **Treatment of Process Losses and Wastages**

In many industries, the amount of the process output will be less than the amount of the materials input. Such shortages are known as process losses or wastages, which may arise due to a variety of factors such as evaporation, scrap, shrinkage, unavoidable handling, breakages, etc.

If the losses are by normal practice, they are known as normal process losses. But where losses are above reasonable and acceptable levels, they are known as abnormal losses, and as such, they should be charged to an appropriate account pending investigation.

Normal process wastages are unavoidable losses arising from the nature of the production process, and, so, it is logical and equitable that the cost of such losses is included as part of the cost of good production. This is because the attainment of good units necessitates normal wastages. Since the wastage arises under efficient operating conditions, it can be estimated with some degree of accuracy.

Abnormal process wastages are those above the level deemed normal in the production process. Abnormal wastage cannot be predicted and may be due to special circumstances.

Such as plant breakdown, inefficient working, or unexpected defects in materials. Abnormal wastage is the difference between actual spoilage in the period and the normal (estimated) wastage. Abnormal gain is where the actual wastage is less than the normal wastage.

The cost of abnormal wastage is to be charged to the profit and loss account, unlike the cost of normal wastage, which is to be part of the good products' total cost. Process account is to be credited as abnormal loss account is debited. The abnormal loss account is then to be closed to the profit and loss account.

Abnormal gain realised is to be credited to the abnormal gain account, as the process account is debited. The abnormal gain account is to be closed to the credit of the profit and loss account.

**Illustration 8.1: Accounting for Process Wastages**

TMR produces 100 units from the input of 120 units from material that is ₦27, 000 and conversion cost of N13,000. Normal process wastage was determined in advance as 5% of input. The output is to be sold at ₦150 per kg. Wastages is valued as ₦50=00 per kg.

**Required to determine:**

a. Cost per unit of the expected output,
b. Production cost of the output achieved,
c. Cost of abnormal loss or abnormal gain obtained

**Solution 8.1**

| | Units produced (kg) | Cost |
|---|---|---|
| Input: Material | 120 | 27,000 |
| Conversion | | 13,000 |
| | | 40,000 |
| Less Normal wastage (5%) | 6 | 300 |

| | | |
|---|---|---|
| Expected output | 114 | 39,700 |
| Output | 100 | 34,825 |
| Abnormal Loss | 14 | 4875 |

a. Cost per kg of Expected output = ₦39,700 / 114 = ₦348.246

b. Production cost of the output achieved: 100 units x ₦348.246
= ₦34,824.6

c. Cost of abnormal loss: 14 units x ₦348.246 = N4875.44

**Illustration 8.2**
Using the information given in illustration 8.1, prepare the Process account and the abnormal loss account.

**Solution 8.2**

**Process Account**

| | Kg | Value ₦ | | Kg | Value ₦ |
|---|---|---|---|---|---|
| Material | 120 | 27,000 | Good production | 100 | 34,825 |
| Conversion | | 13,000 | Normal wastage | 6 | 300 |
| | | | Abnormal wastage | 11 4 | 4,,875 |
| | 100 | 40,000 | | 120 | 40,000 |

**Abnormal wastage Account**

| | ₦ | | ₦ |
|---|---|---|---|
| Process A/c | 4,875 | Profit & Loss | 4 8 7 5 |

**Illustration 8.3: On product flowing**

A product passes through three distinct processes (A, B, and C) to completion. During the period 15 May, 1,000 litres were produced. The following information was obtained:

| | Process A | Process B | Process C |
|---|---|---|---|
| | ₦ | ₦ | ₦ |
| Materials cost | 40,000 | 15,000 | 5,000 |
| Labour cost | 20,000 | 25,000 | 15,000 |
| Direct OH expenses | 5,000 | 3,000 | 3,000 |

Indirect overhead expenses for the period were ₦30,000 apportioned to the processes on the basis of wages. There was no work-in-process at the beginning or end of the period.

**Required:** Calculate the cost of output to be transferred to finished goods inventory and the cost per litre.

**Solution 8.3**

Process A

| Details | Cost/litre ₦ | Total ₦ | Details | Cost/litre | Total ₦ |
|---|---|---|---|---|---|
| Materials | 40 | 40,000 | Transferred-in to | | |
| Labour | 20 | 20,000 | Process B | 75 | 75,000 |
| Dir. Exps. | 5 | 5,000 | | | |
| Indir. Exps. | 10 | 10,000 | | | |
| | 75 | 75,000 | | 75 | 75,000 |

Process B

| Details | Cost/litre ₦ | Total ₦ | Details | Cost/litre | Total ₦ |
|---|---|---|---|---|---|
| Process A | 75 | 75,000 | | | |
| Material | 15 | 15,000 | Transferred-in to | | |
| | 130.50 | 130,500 | | 130.50 | 130,500 |

Process C

| Details | Cost/litre ₦ | Total ₦ | Details | Cost/litre ₦ | Total ₦ |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Process B | 130.50 | 130,500 | | | |
| Materials | 5.00 | 5,000 | Output to finished | | |
| Labour | 15.00 | 15,000 | goods inventory | 161.00 | 161,000 |
| Dir. Exps. | 3.00 | 3,000 | | | |
| dir. Exps. | 7.50 | 7,500 | | | |
| | 161.00 | 161,000 | | 161.00 | 161,000 |

**Note:** Indirect expenses were apportioned as follows:

1. Process A $= \frac{20,000}{60,000} \times 30,000 = ₦10,000$

   Process B $= \frac{25,000}{60,000} \times 30,000 = ₦12,500$

Process C $= \frac{15,000}{60,000} \times 30,000 = ₦7,500$

2 The cost per litre of the process product is ₦161 and so the selling price must be higher than that amount if the business is to make any positive profit.

3 Indirect expenses include all expenses that cannot be directly traced to the productive process and, so, they include general administrative, selling and distributive costs.

8.1.6 **Determination of Equivalent Units**

At the end of a given period, in the course of the production process, it is virtually certain that some items will only be partly completed (work-in-progress). Some of the cost for the period, therefore, is attributable to these partly completed units as well as to those that are fully completed. To spread the costs equitably over part-finished and fully completed units, the concept of 'equivalent units' is used.

For the calculation of costs, the number of equivalent units is the number of equivalent fully completed units which the partly completed units represent. For example, in a given period, production was 3,000 completed units, and 1,600

Partly completed tasks were deemed to be 60% complete.

Total equivalent production = completed units plus equivalent units produced in WI

= 3,000 + (60% of 1600)

= 3,000 + 960

= 3960 units

The total costs for the period would be spread over the total equivalent production as follows: Cost per unit = Total Costs.

Total equivalent production (units)

In calculating equivalent units, it is more desirable to consider the percentage completion of each of the cost elements: material, labour and overhead. Here, each cost element must be treated separately, and then the costs per unit of each element are added to give the cost of a complete unit.

**Illustration 8.4: Equivalent Units Determination**

The production and cost data of Elsemco Shoemakers for January 1999 were as follows:

| | N |
|---|---|
| Materials | 422,400 |
| Labour | 395,600 |
| Overhead | 225,000 |
| Total cost | 1,043,000 |

Production was 8,000 fully completed units and 2,000 partly completed. The percentage completion of the 2,000 units' work-in-progress was:

| | |
|---|---|
| Material | 80% |
| Labout | 60% |
| Overhead | 50% |

**Required:** Find the value of completed production and the value of work-in-progress.

**Solution 8.4**

| Cost Elements | Equiv. Units in WIP | Fully compl. Units | Total prod. | Total Cost | Cost/unit |
|---|---|---|---|---|---|
| ₦ | | | | ₦ | |
| Material | 80% of 2000 = 1600 | 8000 | 9600 | 422,400 | 44 |
| Labour | 60% of 2000 = 1200 | 8000 | 9200 | 395,600 | 43 |
| Overhead | 50% of 2000 = 1000 | 8000 | 9000 | 225,000 | 25 |
| | | | | 1,043,000 | 112 |

Value of completed units = ₦112 x 8,000 = N896,000
Value of WIP= TC - Value of completed units
= 1,043,000 - 896,000 =
₦147,000

To check the value of WIP, the cost of each cost element is to be multiplied by the number of equivalent units of production in WIP related to each cost element.

| Element | No of equiv. units in WIP | Cost/unit ₦ | Value of WIP ₦ |
|---|---|---|---|
| Material | 1,600 | 44 | 70,400 |
| Labour | 1,200 | 43 | 51,600 |
| Overhead | 1,000 | 25 | 25,000 |
| | | | 147,000 |

**Process Account**

| Element | Unit Cost ₦ | Total ₦ | Element | Unit ₦ | Total Cost ₦ |
|---|---|---|---|---|---|
| Material | 10,000 | 422,400 | | | |
| Labour | | 395,600 | Goods transferred | | |
| Overhead | | 225,000 | to the next stage | 8,000 | 896,000 |

| | | | | | |
|---|---|---|---|---|---|
| | | | WIP c/d | 2,000 | 147,000 |
| | 10,000 | 1,043,000 | | 10,000 | 1,043,000 |
| WIP b/d | 2,000 | 147,000 | | | |

It is important to remind the reader that the output of one process level forms the input material to the next process level. The full cost of the completed units transferred forms the input material cost of the subsequent process, and by its nature, must be 100% complete. Material introduced is an extra material required by the process and should always be shown separately.

If there are partly completed units at the end of one period, there will be opening WIP at the beginning of the next period. The values of the cost elements of the brought forward WIP are normally known and are to be added to the costs incurred during the period.

### 8.1.7 **Process and Cost of Production Report**

Several stages are passed through in the valuation necessary for process costing for cost statement. First, the physical flow of the units of production must be calculated having regards to the total number of units to be accounted for, regardless of the degree of completion.

Secondly, the equivalent units involved in the physical flow are to be calculated. In this respect, it is often necessary to divide the flow into its material cost element and conversion cost element, as the degree of completion may well vary between them.
Thirdly, having already established the physical units to be accounted for using the first two stages, the total equivalent units and the current equivalent units involved are to be calculated. These are to be accounted for in respect of the cost elements (transferred-in cost, material cost and conversion cost).

Fourthly, the unit costs are to be calculated, paying regards to the inventory valuation method assumed (FIFO, WAP, LIFO, etc.). Fifthly and finally, the total cost of the transferred-out products and work-in-progress are to be calculated, ensuring that all costs are accounted for.

The cost of production report is to show the number of units of output to be accounted for.

The total equivalent units of completed output, the cost statement showing the impact of all the cost elements and the cost of completed units as well as that of the work-in-process at the end of the reporting period.

In the illustration that follows, two methods of inventory valuation, FIFO and WAP, would be used and two processes of production are assumed.

**Illustration 8.5: Cost of Production Report**

Within the production department of Savannah Sugar Company Limited, there are two processes which produce the finished product. Raw materials are introduced initially at the commencement of Process 1 and further raw materials are added at the end of process

2. Conversion costs accrue uniformly throughout both processes. The flow of the product is continuous; the completed output of process 1 passes immediately into process 2 and the completed output of process 2 passes immediately into the finished goods warehouse.

The following information is available for the month of June:

**Process 1**

| | | |
|---|---|---|
| Opening WIP | | 35,000 units |
| Materials | | ₦210,000 |
| Conversion | (2/5 complete) | ₦52,500 |
| Completion of units in June | | 168,000 |
| Units commenced in June | | 140,000 |
| Units | | |
| Closing WIP | | |
| (1/2 complete as to conversion) | | 7,000 units |
| Material introduced in June | | ₦770,000 |
| Conversion cost added in June | | ₦630,000 |

**Process 2**

| | |
|---|---|
| Opening WIP units | 42,000 |
| Material from Process 1 | ₦343,000 |
| Conversion (2/3 complete) | ₦392,000 |
| Completion of Units in June | 154,000 |

units

| | |
|---|---|
| Units commenced in June | - Closing WIP: |
| (3/8 complete as to conversion) | 56,000 units |
| | |
| Material introduced in June | ₦462,000 |
| Conversion costs added in June | ₦2,205,000 |

**Required:**

Prepare the cost of production report of Savannah Sugar Company Limited for the Month of June, using each of the WAP and FIFO methods, and showing clearly the cost of finished products and WIP at the end of the period.

**Solution 8.5**

*Tutorial Note:* The units to be accounted for, total equivalent units and current equivalent units are to be determined before going to the cost statement, using each of the two inventory valuation methods. The heading of the report should be well expressed.

**Cost of Production Report of Savannah Sugar Company Limited for June, using Weighted Average Price (WAP) Method**

**Process 1**

Physical flow of units of material:

Units

| | |
|---|---|
| WIP (beginning) | 35,000 |
| Material introduced | 140,000 |
| Total units to be accounted for | 175,000 |

| Units Accounted for: | | Equivalent Units: Material | Equivalent Units: Conversion |
|---|---|---|---|
| Units completed & Transferred out: | 168,000 | 168,000 | 168,000 |
| WIP (ending): | 7,000 | 7,000(100%) | 3,500 (1/2 of 7,000) |
| Total units accounted for | 175,000 | | |
| Total equivalent units (TEU) | | 175,000 | 171,500 |
| Less WIP (beginning) | 35,000 | | 14,000 |
| Current Equivalent Units (CEU) | | 140,000 | 157,500 |

**Note:**

1. Conversion WIP ending = ½ x 7,000 = 3,500 units
2. Conversion WIP beginning = 2/5 x 35,000 = 14,000 units

**Cost Statement**

| Cost Element | Cost of WIP | Current Cost (beginning) | TC | TEU | Cost/unit |
|---|---|---|---|---|---|
| | ₦ | ₦ | ₦ | ₦ | ₦ |
| Material | 210,000 | 770,000 | 980,000 | 175,000 | 5.60 |
| Conversion | 52,500 | 630,000 | 682,500 | 171,500 | 3.98 |
| | 262,500 | 1,400,000 | 1,662,500 | | 9.58 |

Cost of Units completed and transferred out = 168,000 x ₦9.58 = ₦1,609,440

Cost of WIP(Ending)

| | | | |
|---|---|---|---|
| Material: | 7,000 x 1 x ₦5.6 | = | ₦39,200 |
| Conversion: | 7,000 x ½ x ₦3.98 | = | ₦13,930 |
| | | | ₦53,130 |

Another way (which is easier) of determining the cost of WIP ending is to find the difference between the total cost and the cost of the completed units.

Cost of WIP (end) = TC - Cost of completed units

= ₦1,662,500 - ₦1,609,440

= ₦53,060

**Note:** The difference of N70 is due to the approximation made to two decimal places.

**Cost of Production Report Using First-In-First-Out (FIFO) Method**

**Process 1**

| Current costs: | | C.E. Units | Unit's cost |
|---|---|---|---|
| Material | ₦770,000 | 140,000 | ₦5.50 |
| Conversion | 630,000 | 157,500 | 4.00 |

Cost of Closing WIP:

| | | |
|---|---|---|
| Material: 7,000 x 1 x ₦5.5 | = | ₦38,500 |
| Conversion: 7,000 x ½ x ₦4.0 | = | 14,000 |
| | | 52,500 |

Units completed and transferred out = 168,000 units
Cost of the completed units = TC - cost of closing WIP
= ₦1,662,500 - ₦52,500
= ₦1,610,000

**Process 2, using the WAP method**
Physical flow of units of material:

| | |
|---|---|
| WIP (beginning) | 42,000 |
| Units transferred in | 168,000 |
| Units to be accounted for | 210,000 |

| Units Accounted for: | Transferred Material | Equivalent Units | | Conversion |
|---|---|---|---|---|
| | 154,000 | 154,000 | 154,000 | 154,000 |
| WIP (ending): | 56,000 | 56,000 | 0 | 21,000 |
| Total units accounted for | 210,000 | | | |
| Total equivalent units | | 210,000 | 154,000 | 175,000 |
| Less WIP (beginning) | | 42,000 | 0 | 28,000 |
| Current Equivalent units (CEU) | | 168,000 | 154,000 | 147,000 |

**Cost Statement**

| Cost Element | Cost of WIP (beginning) | Current Cost | TC | TEU | Cost/unit |
|---|---|---|---|---|---|
| ₦ | | ₦ | ₦ | ₦ | ₦ |
| Transferred in | 343,000 | 1,609,440 | 1,952,440 | 210,000 | 9.2973 |
| Material | 0 | 462,000 | 462,000 | 154,000 | 3.0000 |
| Conversion | 392,000 | 2,205,000 | 2,597,000 | 175,000 | 14.8400 |
| | 735,000 | 4,276,440 | 5,011,440 | TC/unit | = 27.1373 |

Cost of complete units = ₦154,000 x ₦27.1373 = ₦4,179,144.20

Cost of WIP(Ending)

| | | | |
|---|---|---|---|
| Transferred-in | 56,000 x 1 x ₦9.2973 | = | ₦520648.80 |
| Material: | 56,000 x 0 x ₦3 | = | ₦0 |
| Conversion: | 56,000 x 3/8 x ₦14.84 | = | 311640.00 |
| | | | ₦832,288.80 |

Another Way:

Cost of Ending WIP TC- Cost of completed units

= 5,011,440 - 4,179,144.20

= ₦832,295.80

Note that the difference of ₦7 is due to the approximation made to four decimal places.

**Process 2: Using the FIFO Method**

| Current Costs: | | Current Equiv. Units | Unit Cost |
|---|---|---|---|
| ₦ Transferred in | ₦1,609,440 | 168,000 | 9.58 |
| Material | 462,000 | 154,000 | 3.00 |
| Conversion | 2,205,000 | 147,000 | 15.00 |
| TC/C.E.U. | | | ₦27.58 |

Cost of Closing WIP:

| | | | |
|---|---|---|---|
| Transferred in: | 56,000 x 1 x 9.58 | = | ₦536,480 |
| Material: | 56,000 x 0 x ₦3 | = | ₦0 |
| Conversion | 56,000 x 3/8 x 15 | = | ₦315,000 |
| | | | ₦851,480 |

Units completed and transferred-out = 154,000 units.
Cost of completed units = TC - Cost of ending WIP
= 5,011,440 - 851,480
= ₦4,159,960

FIFO: An Improved Method

| | ₦ | |
|---|---|---|
| Opening WIP to be out first | = 735,000. | Completion |
| of opening WIP: | | |
| Material: 42,000 x ₦3 = | | 126,000 |
| Conversion: 42,000 x 1/3 x 15 = | | 210,000 |
| Cost of units started & completed in June: | | |
| 154,000 42,000 = 112,000 x ₦27.58 = | | 3,088,960 |
| | | 4,159,960 |

## 8.2 Joint Products and By-Products

### 8.2.1 The Concept of Joint Products

The process costing principles discussed in Paragraph 7.1 are shown to be all about determining the cost of processing some inputs that yield the same types of products. At the end of the processing activities, only one type of product would result from the processed raw material.

However, it is not always that we have only one type of product from a processing operation. It is possible for a single raw material to yield two or more products simultaneously when processed. Such products are known as Joint Products. For example, when crude oil (a single raw material) is processed or refined, petrol, kerosene, gas, etc., could be obtained from it.

The cost of processing a production input (raw material) that would amount to joint products is known as the Joint Cost. The joint cost is to be restricted to the split-off point (the point after which each joint product would incur separate processing cost). Joint cost is not to be traced to any particular product but rather to all the joint products as a group. There are many ways of apportioning joint cost to joint products for financial accounting purposes. These would be discussed later.

In practice, it is normal to identify one product out of the joint products as the principal product and the rest to be treated as joint products or even by-products. In the example above, it is clear that petrol is the main product to be identified as crude oil is processed. Pairs of shoes could be the main products as leather is processed, while bags, wallets, etc., could be seen as joint or by-products.

One way of differentiating between by-product and joint product is to consider their cost of production or sales value. A product that costs 10% to 15% of the main product cost should be treated as a by-product. Any product that costs 15% to 40% of the main product cost is a joint product. Any product that costs above 40% of the identified main product cost should also be treated as a main product.

As a result of price changes, therefore, a by-product can become a joint-product or even a main product and vice versa.

### 8.2.2 Accounting Treatment of By- Products

A by-product is a secondary product arising as a result of a processing activity aimed at producing a certain main product. The market value of a by-product less the processing cost after the split off point is usually negligible, compared to the total market value of all the joint products or the market value of the main product.

The usual treatment of by-product is to deduct its Net Realizable Value (NRV) from the total Joint Cost (JC) and then divide the net joint cost among the joint or main products. The NRV of the by-product is the difference between its market value and its separate processing cost.

**Illustration 8.6: Treatment of by- products**

Wambai Shoemakers have a process that yields two main products: A and B and a by-product C at a total cost of ₦3,000,000. There are 1,000 units of C requiring no further processing and each can be sold at ₦60 with negligible market cost. The two main products take equal share of joint cost.

**Required:** What should be the share of Product A from the Joint Cost?

**Solution 8.6**

The total market value of Product C = 100 x ₦60 = ₦60,000. This is the NRV, since its market cost is negligible.

Net Joint Cost = ₦3,000,000 - ₦60,000 = ₦2,940,000

Share of Product A = ₦2,940,000 = ₦1,470,000

2

**Note:** It can be concluded that in deducting the NRV of by-product C from the Joint Cost, we are in effect, assigning to the by-product a joint cost which is equal to its NRV.

### 8.2.3 Methods of Apportionment of Joint Cost

There are three popular methods of sharing joint costs among the joint (or main) products. These are the Physical Units Bases, Sales Value (at the point of separation) and Net Realisation Basis. These methods are briefly discussed below:

(a) **Physical Units Basis**

Under this method, the joint cost is shared among the joint products on the basis of the quantities of physical units, provided all the products are measured by a common unit of measurement, such as kilograms or litres. The problem with this method is that consideration is not given to price, and so, it does not consider the value of the products. Usually, the value of products is the most important factor to be considered.

**Illustration 8.7**

Anadariya Company Ltd., Tiga has a processing system that produces three products: K, S and T with 5,000 kg, 3,000 kg and 2,000 kg, respectively, in a year. The total cost incurred up to the split-off point in the year 2000 was ₦1,000,000. Use the physical units basis to share the joint cost among the three products. Calculate also their unit cost.

**Solution 8.7**

1. The Ratio:

K = 5,000 x 100 = 50%
10,000

S = 3,000 x 100 = 30%
10,000

T = 2,000 x 100 = 20%
10,000

Share of joint cost:

K = 50% of ₦1,000,000 = ₦500,000

S = 30% of ₦1,000,000 = ₦300,000

T = 20% of ₦1,000,000 = ₦200,000

2. Unit Cost based on the share of joint cost:

K = ₦500,000 = ₦100/unit
5,000

S = ₦300,000 = ₦100/unit
3,000

T = ₦200,000 = ₦100/unit
2,000

(b) **Sales Value (at the Point of Separation)**

Under this method, the joint cost is shared among the joint products on the basis of their sales value before further processing. At the split-off point, market value can be estimated per unit of each of the joint products. It is the ratios of the sales value of the joint products that are to be used as the basis of apportioning the joint cost.

The problems with this method are twofold: One, a product may have zero value at the point of separation but significant value with little processing cost after the split-off point. Secondly, a product may have high selling price at the split-off point and hence high sales value but may involve large selling and distribution cost (advert, carriage, etc.).

**Illustration 8.8**

Assuming that Anadariya Company Ltd has estimated the following selling prices for its three products at the point of separation:

K = ₦400/unit

S = ₦440/unit

T = ₦340/unit

Use the Sales Value method to apportion the joint cost and determine the unit cost of each of the three products.

**Solution 8.8**

1.

| Product | | Ratio | Unit Share of JC | SP/Unit | Sales Value |
|---|---|---|---|---|---|
| | | ₦ | ₦ | | ₦ |
| K | 5000 | 400 | 2,000,000 | 50% | 500,000 |
| S | 3000 | 440 | 1,320,000 | 33% | 330,000 |
| T | 2000 | 340 | 680,000 | 17% | 170,000 |
| | | | 4,000,000 | | 1,000,000 |

2. Unit cost based on the share of joint cost:

K = ₦500,000 / 5000 = ₦100/Unit
S = ₦330,000 / 3000 = ₦110/Unit
T = ₦170,000 / 2000 = ₦85/Unit

(c) **Net Realisable Value Basis**

This method is used only when further processing is necessary or when high marketing and distribution costs are involved. Net realisation is sales value less the incremental cost of further processing after the split-off point and any other cost necessary for the selling or distribution of the product.

**Illustration 8.9**

Assuming that the sales values in illustration 7.9 are market prices after further processing and that separate processing and marketing costs are as follows:

K = ₦200,000
S = ₦300,000
T = ₦160,000

Determine the share of the joint cost to the three (3) products. Also show the unit cost of each of the three products.

**Solution 8.9**

1.

| Product | Units | SP/Unit | Sales Value | SPC | NRV | Share of JC |
|---|---|---|---|---|---|---|
| | | ₦ | ₦ | ₦ | ₦ | ₦ |
| K | 5000 | 400 | 2,000,000 | 200,000 | 1,800,000 | 538,922 |
| S | 3000 | 440 | 1,320,000 | 300,000 | 1,020,000 | 305,389 |
| T | 2000 | 340 | 680,000 | 160,000 | 520,000 | 155,689 |
| Total | 10,000 | | 4,000,000 | | 3,340,000 | 1,000,000 |

**Note:**

(a) Net Realisable Value (NRV) = Sales Value Less separate processing costs (SPC)
(b) The total of the NRV of all the joint products is obtained, and the joint cost is shared in proportion to the NRV of each product.
(c) This method is the 'best' as it considers the quantity (units) produced of all the joint products, their sales values and their further processing costs.

2. Unit cost based on the share of joint cost:

K = ₦538,922 / 5,000 = ₦108/unit

S = ₦305,389 / 3,000 = ₦102/unit

T = ₦155,689 / 2,000 = ₦78/unit

8.2.4 **Total Cost Determination Using NRV Method**

The Total Cost of a joint product is given by its share of the joint cost plus its further processing and marketing costs. To arrive at a product's total cost per unit, its total cost is divided by the units produced.

Using illustration 7.10, the total cost per unit could be determined for each of the three products as follows:

K = ₦538,922 + ₦200,000 = ₦738,922 = ₦147.78
5,000 5,000

S = ₦305,389 + ₦300,000 = ₦605,389 = ₦201.80
3,000 3,000

T = ₦155,689 + ₦160,000 = ₦315,689 = ₦157.84
2,000 2,000

If there are closing inventory of Product K (900 units), S (500 units) and T (400 units), the value of closing inventory for reflection in the balance sheet could be determined as follows:

| | | ₦ | |
|---|---|---|---|
| K | = | 900 x ₦147.78 = | 133,002 |
| S | = | 500 x ₦201.80 = | 100,900 |
| T | = | 400 x ₦157.84 = | 63,136 |
| | | | 297,038 |

**Note:**
It should be understood that profit is always the difference between total revenue (sales value) and total cost. That economic principle is very much applicable in joint-product costing.

## 8.3 Non- Specific Order Costing Methods

### 8.3.1 Service Costing

This is the application of costing principles to the costing of service operations as opposed to the manufacture of goods. Hotels, transport, airlines, and school businesses are examples of areas of application. It may also be applied to a service cost centre in a manufacturing concern. It may be decided to treat the haulage department of a manufacturing and trading concern as a service cost centre after determining whether it is worthwhile to run the delivery of goods to customers internally or to sub-contract it to a Haulage Company. To take such a decision, information is required on how much it costs to run it in-house.

The principles of cost ascertainment remain the same in that all expenses incurred in the process of providing services are appropriately classified and aggregated and then averaged over the total number of units of that service to obtain cost per unit.

The manner in which costs are classified would vary from one industry to the other. The following formats for a transport and canteen businesses are illustrated:

**Format of a Cost Statement for a Transport Business:**
**Fixed or Standing Charges:**

| | | |
|---|---|---|
| Motor Insurance | | X |
| Motor Licence | X | |
| Driver's Remuneration | | X |
| General Administration | | X |
| | | X |
| **Running Costs** | | |
| Fuel | | X |
| Servicing | | X |
| Lubricants | | X |
| Repairs and Maintenance | | X |
| | | X |
| Total Cost | XX | |

Fixed or Standing Charges represent those expenses which are incurred irrespective of the distance covered.

| | | CANTEEN COST | | |
|---|---|---|---|---|
| | ACTUAL | | BUDGET | |
| **Wages and Salaries** | X | X | X | X |
| Supervisor<br>Cooks<br>Counter helpers<br>**Provisions**<br>Meat<br>Fish and Poultry<br>Vegetable and Fruit<br>Eggs<br>Coffee<br>**Miscellaneous**<br>Maintenance<br>Crockery and Glassware<br>Insurance<br>Table line<br>Rent<br>Steer<br>**Deduct Income from**<br>Sales<br>Company Subsidy | | | | |

**(a) Cost Unit in the Service Industry**

The major problem with service costing is the determination of the cost unit to use in measuring the output of the service. For example, for a transport business, service could be measured in terms of kilometres travelled or the weight of goods carried, or a combination of both.

Other cost units in the service industry are as follows:

| **Service** | **Cost units** |
|---|---|
| Hotel | Occupied room-day |
| Restaurant | Meal served |
| College | Full-time equivalent student |
| Electricity | Kilowatt-hour |
| Hospital | Patient-day |
| Transport | Miles travelled |

(b) **Costing of Professional Services**

In most professional offices, such as those of Architects, Accountants, Tax Practitioners and Medical Doctors, where services are rendered and classified as belonging to the service industry, it may be appropriate to apply service costing principles. This is because the services rendered by professional offices usually differ from one client to another, as opposed to the service organisations that render standardised services. In such a case, the job costing method is used in the costing of such professional services.

## 8.4 CHAPTER SUMMARY

This chapter has introduced the reader to the meaning of Process Costing, its application areas, and how it can be put to use for proper accountability. The characteristics of Process Costing, how products flow in the course of processing, the equivalent units of production to be transferred to the next stage of production, accounting for wastages/losses, and the valuation process for the cost of production report have all been discussed. Process costing, which is arguably the most widely used costing in the world, has been given the coverage it deserves, hoping that readers would find the contents of the chapter so simple that process costing would be 'counted in' as one of their simplest accounting topics.

Detailed discussion was also made on joint and by-product costing, where three different methods of apportioning joint cost to joint products were explained. These methods are the Physical Units' basis, Sales Value basis and the Net Realisable Value basis. By-products and their accounting treatment have also been discussed in the chapter.

## MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS

1. In process costing, what does “Abnormal Loss” mean?
   - A. Inventory items stolen during production
   - B. Inventory items given out as gifts before full production
   - C. The greater of the present production's loss over the previous production
   - D. The greater of the actual process loss over what is deemed to be normal in the industry
   - E. Losses in production due to abnormal behaviour of workers

2. Joint products are the name given to two or more
   A. The main products arising from the same process
   B. dissimilar products manufactured from the same factory
   C. complementary products that must be sold together
   D. Raw materials mixed to manufacture a single item
   E. products which work alike but are from different manufacturers

3. What is scrap?
   A. Metal residue of a production process
   B. Residues of little or no value obtained during production
   C. Failed production items
   D. Cleaning materials
   E. Packaging materials

4. Explain "Net Realisable Value" in joint product costing
   A. Sales value less the cost of further processing before the point of separation
   B. Market value of each of the joint products
   C. Sales value less the cost of further processing, selling and distribution after the split-off point
   D. Sales value less cost of production up to the point of separation
   E. Sales value less commission

5. Which of these statements is correct as far as process costing is concerned?
   A. Each process's output can be sold without further processing
   B. Only the materials introduced in process 1 are sufficient for the whole production
   C. Each manufacturing process has a normal gain
   D. Process costing is not significantly different from contract accounting
   E. The output of one process forms the input of another process

6. In process costing, partly completed units are calculated to represent fully completed units based on the percentage of completion. Such representation is known as ……
7. Where actual process wastage is less than what is regarded as normal in the industry, the difference is called……...
8. A product of lesser value incidentally produced during the manufacture of another product is known as …….
9. The point during the production process where two or more joint products assume their identities is called ……...
10. Where is the total credit on the Abnormal Gains Account transferred to at the end of the accounting period?

**SOLUTION**

1. D
2. A
3. B
4. C
5. E
6. Equivalent units

7. Abnormal Gain
8. By-product
9. Split-off point or separation point
10. Profit & Loss Account

**EXAMINATION TYPE QUESTIONS**

1 "Process Costing is a cost accounting system used to determine the cost of production in a series of sequential processes". Discuss this comment, giving diagrammatic example(s).

2. Briefly discuss the characteristics of a good process costing system.

3. Write short notes on each of the following terms:
   (a) Product flow
   (b) Work-in-progress
   (c) Equivalent units
   (d) Transferred in costs

4. Differentiate between the following:
   (a) Actual spoilage and normal spoilage
   (b) Conversion cost and overhead cost
   (c) Material cost and transferred-in cost
   (d) WAP and FIFO methods in process costing

5. A process has a normal wastage of 4%, which has a resale value of ₦50 per kg. Find the cost per kg of good production, if the material cost is ₦320,000 and the conversion cost is ₦240,000 for producing 500 kg.

6. ABC uses process costing for its operations. You are given the following information by the officers of the company for the year ended December 31, 2008:

| | |
|---|---|
| Work-in-Process (1January, 2008) | 4,800 units |
| Units started during the year | 19,200 units |
| Abnormal spoilage | 1,392 units |
| Units completed | 17,280 units |
| Work-in-Process (31December, 2008) | 3,600 units |
| Stages of Completion of Work-in-Process | |
| W.I.P. January 1, 2008: Material | 100% |
| Conversion | 80% |
| W.I.P. Dec. 31, 2008: Material | 100% |
| Conversion | ⅔ |
| *Costs* | |
| WIP 1January, 2008 | ₦35,520 |
| Material: ₦18,000; Conversion: | ₦17,520 |

| | |
|---|---|
| Costs incurred during the year<br>(Material: ₦73,200; Conversion: ₦96,480) | ₦169,680 |

**Required:**
Using the weighted average pricing method, draw up a cost of production report showing the cost of completed units, abnormal spoilage and the cost of the inventory (WIP) at 31 December, 2008. Also, show the relevant computations in detail. Spoilage is detected at the end of operations.

7. BCB Company has the following data for October: Department

| | A | B |
|---|---|---|
| Beginning Inventory in process | 2,000 units | 2,000 units |
| Transferred in cost | - | ₦4,600 |
| Cost of materials added last month | ₦2,000 (100%) | ₦3,200 (80%) |
| Conversion costs were added last month. | ₦900 (60%) | ₦800 (40%) |
| Units put in process this month | 30,000 | 29,000 |
| Cost of materials added this month | ₦30,968 | ₦56,580 |
| Conversion costs added this month. | ₦41,800 | ₦29,803 |
| Units completed and transferred | 29,000 | 30,000 |
| Ending inventory in process: | 3,000 units | 1,000 units |
| Material content | 90% | 50% |
| Conversion costs | 50% | 30% |

**Required:**

a. Using weighted average costing method, prepare a production report for the month of October for each department. Show the cost of the ending work-in-progress as well as the cost of goods completed and transferred.
b. Using the FIFO costing method, repeat the requirements of (a) above

8. Differentiate clearly between joint-product, by-product and main-product in a joint costing system.
9. (a) Discuss briefly, the three methods of apportioning joint costs to joint-products.
   (b) Explain how do a profit or loss on a joint-product is arrived at?

10. (a) Briefly describe the Accounting Treatment of By-product.

(b) A Process has 2 products A and B with a joint cost of ₦900. After the split-off point, Product Ahas 1,000 units with a selling price of ₦2 per unit and a separate processing cost of ₦600. Product B has 500 units with a selling price of ₦1 per unit and a separate processing cost of ₦100.

**Required:** Calculate the total cost of each product using:
a) Physical unit basis
b) Net Realisation method

11. A firm manufactures two products, X and Y, from the same material. In Department I, the material is split into X and Y. X requires no further processing, while Y must be processed further in Department 2. The following costs pertain to November.

| *Dept.* | *Direct mat.* | *Direct labour* | *Factory OH* | *Total* |
|---|---|---|---|---|
| | ₦ | ₦ | ₦ | ₦ |
| 1 | 60,000 | 15,000 | 10,000 | 85,000 |
| 2 | - | 3,000 | 12,000 | 15,000 |
| | 60,000 | 18,000 | 22,000 | 100,000 |

| Product *Sold* | Kilograms | Kilograms *produced* | Sales for *the period* |
|---|---|---|---|
| X | 30,000 | 45,000 | ₦60,000 |
| Y | 20,000 | 20,000 | ₦95,000 |

There were no finished goods at the beginning of the month.

**Required:**
Using the net realisation method to apportion the joint cost, calculate the total cost of Producing X and Y, also during the month.

Assume that selling price of X will remain constant in the immediate future.

12. A firm operates a process which produces four refined products A, B, C and D. The joint costs of operating the process in the month of December were:

| | ₦ |
|---|---|
| Raw materials | 15,000 |
| Labour | 4,800 |
| Overhead | 1,200 |
| Total | 21,000 |

Output for the period was:
A 14,000kg
B 12,000 kg
C 8,000 kg
D 2,000 kg

Work-in-progress at the beginning and end is equal to zero. Also equal to zero was finished output at the beginning of the month. Finished output at the end of the period:

A 600 kg
B 400 kg
C 1,600 kg
D 100 kg

The price received per kilogram of refined product during the period was:

A ₦0.20
B ₦0.45
C ₦0.60
D ₦1.00

The price is expected to be maintained shortly.

**Required:**

Calculate the cost of the finished units using as your basis of calculation:

(a) Physical units
(b) Sales value

13. JAY Chemicals Ltd manufactures a range of industrial chemicals. Basic materials are put through an initial process, which produces the ingredients for three products X, Y, Z. The ingredients for each product then undergo separate further processing to form the finished products:

The following is the budget for year:

Joint Cost apportioned: ₦

| Total | | X | Y | Z |
|---|---|---|---|---|
| Sales (kilos) | | 6,000 | 6,000 | 6,000 |
| Sales value | ₦540,000 | ₦180,000 | ₦240,000 | ₦120,000 |
| Materials | 120,000 | 40,000 | 60,000 | 20,000 |
| Labour | 80,000 | 30,000 | 40,000 | 10,000 |
| Overheads | 100,000 | 30,000 | 50,000 | 20,000 |
| | ₦300,000 | ₦100,000 | ₦150,000 | ₦50,000 |

Separate further:
Processing cost (₦)

| | | | |
|---|---|---|---|
| Material | 10,000 | 20,000 | 10,000 |
| Labour | 10,000 | 10,000 | 10,000 |
| Overhead | 20,000 | 20,000 | 20,000 |
| Total | 40,000 | 50,000 | 40,000 |
| Profit/Loss | 40,000 | 40,000 | 30,000 |

The overhead costs are fixed, except for the separate processing where the variable portion is estimated at ₦6,000 for X, ₦12,000 for Yand ₦3,000 for Z.

The company can increase its production by 25% of the present level without incurring additional fixed costs, and market research has shown that to sell more than the budgeted quantities in the present market, the price would have to be reduced to the following levels:

X ₦28; Y ₦35; and Z ₦18

Although the products are jointly produced, sales of the products are independent of each other, thus, an increase in the quantity sold of a product would not affect the sales of the other products.

A customer from a neighbouring country has offered to purchase the following quantities:

| | | |
|---|---|---|
| X | 1,500 kilos | @ ₦27 per kilo Y |
| | 500 kilos | @ ₦37 per kilo Z |
| | 1,000 kilos | @ ₦18 per kilo |

Based on the budget for the next year, direct labour will be working at full capacity. To meet the order, the direct labour force will have to work overtime to be paid at five and a half. Evaluate whether the company should accept the order. *(ICAN Foundation, May 1997)*

14. Okoro Ltd is attempting to decide the sales price for two products 'wids' and 'leads'. The products are both made by the same workforce and in the same department. ₦30,000 direct labour hours are budgeted for the year. The budgeted fixed costs are ₦30,000, and it is expected that the department will operate at full capacity. Variable costs per unit are as follows:

| | Wids | Leads |
|---|---|---|
| Materials | 4 | 4 |
| Labour (2 hours) | 2 | 9 |
| Expenses (1 machine hr.) | 6 | 2 |
| (3 hours) | 12 | 15 |

Expected sales demand is estimated roughly to be 7,500 'wids' and 5,000 'leads'.
What would the unit sales prices be to give a profit of 20% on full cost if overheads are absorbed?
On a direct labour hour basis?
On a machine hour basis? *(ICAN, PE I November)*

**SIMPLE PROCESS**

**15**

a. Summarise what you understand by abnormal loss and as well show how it is treated in Process Costing.
b. Given below relates to the data for **Process I** during the month of January 2017 as extracted from the books of MMM Ltd.

| | | |
|---|---|---|
| 2nd Jan, 2017: | Input | 40,000 units |
| | Opening WIP | Nil Value |
| Process Costs: | | ₦ |

| | |
|---|---|
| Material | 1,140,000 |
| Labour | 633,600 |
| Overhead | 528,000 |
| 31st Jan, 2017: | Units |
| Output | 32,000 |
| Closing WIP | 8,000 |

Degree of completion of Clowing WIP:

| | |
|---|---|
| Materials | 75% |
| Labour | 40% |
| Overhead | 40% |

**Determine:**

i. Equivalent Production Units
ii. Cost per unit of equivalent production
iii. Statement of evaluation
iv. Process I Account

(12½ Marks)

16 Tamco Limited manufactures TACO product. The product passes through two distinct Processes before it is turned out as a finished product. Extracted details are provided below:

**i. Input Details:**

| **Items** | **L** ₦ | **M** ₦ | **Total** ₦ |
|---|---|---|---|
| Direct materials | 400,000 | 158,000 | 558,000 |
| Direct wages | 200,000 | 80,000 | 280,000 |
| Production overhead | 110,000 | 90,000 | 200,000 |

20,000 units @₦3 were introduced to Process L. The output of Process L passes to Process M, where finished goods are obtained. The finished goods are then forwarded to the warehouse.

**ii. Output, normal loss details and total scrap realised in each process**

| | **Actual Output** | **% of normal loss on input** | **Total scrap to realise** |
|---|---|---|---|
| | **Unit** | **Unit** | ₦ |
| 2Process L | 18,900 | 5% | 10,000 |
| Process M | 18,000 | 5% | 6,700 |

**Required to prepare:**

a. Process accounts; and
b. Process losses and process gain account

(12½ Marks)

**CHAPTER NINE**

**OTHER COSTING TECHNIQUES AND DECISION MAKING TOOLS**

**CHAPTER CONTENTS**

Decision making:

a. Use of Marginal Costing and Absorption Costing Techniques in the preparation of Income Statements.
b. Concept of Relevant Cost
c. Short Term decisions: Make or buy, products mix, special order pricing, shut down problems with simple illustration, etc.

## 9.0 OBJECTIVES:

After studying this chapter, readers should be able to:

a. Explain the concepts and differences between marginal and absorption costing;
b. Explain the equation, advantages and limitations of marginal costing;
c. Discuss the importance of contribution margin in managerial decision making;
d. Use marginal costing as a technique for short-term tactical planning and decision-making; and
e. Make decisions on whether to make or buy a component, accept or reject an order, among others.

## 9.1 Marginal Costing Versus Absorption Costing

### 9.1.1. Introduction

In taking short term, medium term or long term decisions, management of organisations in both the public and private sectors of the economy must adopt some tools that would aid better decision making for the achievement of organisational goals. In Accounting, there are many tools to be used in guiding effective decision making by management. Two of the most popular decision making techniques are marginal costing and absorption costing.

The two techniques are expected to be appropriately used in short term tactical managerial decision making exercise that would amount to efficient management of resources for the production of. income, profit and wealth. The decision to be taken would normally be about the future, which is full of risks and uncertainties. This calls for a lot of care and attention when using either of the two techniques.

This section of the chapter explains the concepts of marginal costing and absorption costing; the differences between marginal and absorption costing; and the environments for the application of marginal and absorption costing.

### 9.1.2. The Concepts of Marginal Costing and Absorption Costing

Marginal costing has been described by different names in different continents. The term 'Marginal costing' is common in the United Kingdom (UK) and other countries of the European continent, while the expression 'Direct Costing' or 'Variable Costing' is preferred in the United States (US). The technique has generated strong views both for and against, with the result that it has become a subject of lively raging controversy during recent times.

'Marginal Cost', is derived from the word 'margin', which is a well-known concept in economic theories. Thus, quite in tune with the economic connotation of the term, it is described in simple words as the cost which arises from the production of additional increments of output and it does not arise if the additional increments are not produced. This shows that marginal cost is the cost of additional unit of product or service produced in any production process.

From this point of view, marginal costs will be synonymous with variable costs, i.e. prime costs and variable overheads in the short run, but in a way, would also include fixed costs in planning production activities over a long period of time involving an increase in the productive capacity of the business. Thus, marginal costs are related to change in output under a particular circumstance of a case. Marginal costing therefore is about costing an additional product or service on its merit without relating it to the general cost being incurred by the producer in the course of the production process.

Marginal costing is not a system of costing in the sense in which other systems of costing, like process or job costing are, but it has been designed simply as an approach to the presentation of accounting information meaningful to management from the viewpoint of adjudging the profitability of an enterprise by carefully studying the impact of the entire range of costs according to their respective nature.

The concept of marginal costing is a formal recognition of ideas underlying flexible budgets, break-even analysis and/or Cost-Volume-Profit relationships. It is an application of these relationships which involves a change in the conventional treatment of fixed overheads in relation to income determination.

The concept of marginal costing is based on the important distinction between product costs and period costs, the former being related to the volume of output and the latter to the period of time rather than the volume of production. Marginal costing regards as products costs, only those manufacturing costs which have a tendency to vary directly with the volume of output. This is in complete contrast to the conventional system of costing under which all manufacturing costs - fixed as well as variable are treated as product costs.

Absorption costing, on the other hand, is the technique of costing that is used in preparing income statements to ascertain the result of operations of private and public sector organisations. The term absorption is about absorbing the general overhead (fixed) costs being shared to all the units of goods or services produced to ascertain total cost of production per unit rather than total variable (marginal) cost per unit.

Absorption costing, therefore, is about costing a product or service with due regards to all the cost elements involved in the production process and, so, appropriate apportionment is to be made for indirect costs to be incorporated into the total cost for each unit of goods or services produced.

### 9.1.3. Major Differences Between Marginal Costing and Absorption Costing

Marginal costing, as explained above, is a technique of costing that considers fixed costs as a period cost and so, irrelevant when taking decision on the total cost of a product to be compared against its benefit for the determination of its profitability/viability. It is conceptually about variable or direct costing. Absorption costing, on the other hand, is about appreciating all cost elements as essential for production and sales and, so, all of them are to be captured when determining total cost at all levels of production up to sales.

As the two techniques are used in preparing income statements, some important differences could be noticed when the income statements are compared. The most fundamental difference is in the treatment of fixed overhead production cost. While absorption costing accepts the overhead cost as part of the cost of goods sold, marginal costing treats it as a period cost which must be incurred whether or not there is production and, so, it is not to be part of the cost to be aggregated for a product or service.

The second difference is in the determination of cost of closing inventory, where absorption costing recognises total cost per unit to be multiplied by the closing units to get the value of closing inventory while marginal costing recognizes only the total variable cost per unit for that purpose as all fixed costs (production, administrative and selling) must have been covered by contribution margin.

The third difference is in the initial profit shown under each of the two techniques. While absorption costing income statement first shows gross profit before net profit and so on, marginal costing income statement first shows contribution margin, which is the difference between sales and total variable costs. This shows that except where all the units produced are sold, the two income statements would not amount to the same level of profit at the initial stage.

Absorption costing is used in preparing income statements at the end of an accounting period by all the three forms of businesses (sole proprietorship, partnership and company) in an economy and various governmental establishments. The method is, therefore, better used to aid decision on performance evaluation in respect of result of operations at the end of an accounting period. Marginal costing technique is only relevant for short term tactical managerial decision making, focusing the future happenings of businesses. The areas of application of marginal costing would be discussed in this chapter.

Students would learn more about the differences and the environments for application in the subsequent section and subsections of the chapter.

### 9.1.1 Basic Equation, Advantages and Limitations of Marginal Costing

#### (a) Basic Equation of Marginal Costing

The technique of marginal costing hinges on the contribution made by a product towards fixed costs and profit of the undertaking.

Contribution can be looked at from two angles - either as the total of fixed costs and profit or as the difference between sales and variable costs.

For the sake of convenience, contribution can be stated by way of equation as: Sales - variable costs = Contribution.

This is sometimes

shortened as: S - V = C

It is also true that: Contribution - Fixed Cost = Profit

i.e. C - FC = P

The formula has been so framed that, if some of the above four factors are known, the remaining one can be easily found out. From this point of view, the equation is of fundamental importance.

Moreover, its significance also lies in the fact that a formula for calculating sales at break-

even point has been derived from the basic equation and the formula is:

$$\text{Sales at break-even point} = \frac{\text{Fixed Cost x Sales}}{\text{Sales - Variable costs}}$$

(b) **Advantages of Marginal Costing**

I. ***Constant in nature:*** Marginal costs remain the same per unit of output irrespective of the volume of production;

ii. ***Facilitating cost control:*** The clear-cut division of costs into their fixed and variable components paves the way for a better cost control through flexible budgeting which is based on this important distinction;

iii. ***Simplicity of overhead treatment:*** Marginal costing does away with the need for allocation, apportionment and absorption of fixed overheads thereby removing an important source of accounting complications by way of under-absorbed or over-absorbed overheads;

iv. ***Basis for pricing and tendering:*** Marginal costing furnishes a better and more logical basis for the fixation of sales prices as well as in tendering for contracts when business is at a low ebb;

v. ***Aid to Profit planning:*** The technique of marginal costing enables data to be presented to management in a manner as to show cost-volume profit relationships; and

vi. ***Realistic valuation:*** Elimination of fixed overheads from the cost of production means that finished goods and work-in-progress are valued at their marginal cost and, therefore, the valuation is more realistic and uniform as compared to the one when they are valued at their total cost.

(c) **Limitations of Marginal Costing**

i. ***Difficulty in analysis:*** Considerable difficulty is always experienced in analysing overheads into their fixed and variable components;

ii. ***Lop-sided Emphasis:*** Marginal costing has a tendency to attach more importance to the selling function which has the effect of relegating the production function to a comparatively unimportant position. However, the efficiency of a business is to be judged by taking together its selling as well as production functions into account;

iii. ***Difficulty in application:*** The technique of marginal costing cannot be adequately applied in the case of industries in which, according to the nature of business, large inventories have to be kept in the form of work-in-progress;

iv. ***Limited Scope:*** As marginal costing distinguishes between the treatment of fixed and variable components of costs, it is difficult to adopt the technique in capital- intensive industries where fixed costs are very large; and

v. ***Inappropriate basis for pricing:*** Selling price cannot reasonably be fixed on the basis of contribution alone.

In the light of these advantages and disadvantages, marginal costing may be considered to be a very useful technique from the point of view of management but it must be applied with a full awareness of its limitations as well as of the circumstances in which it can be fruitfully used.

## 9.2 PREPARATION OFINCOME STATEMENTS

### 9.2.1 Income Determination: Under Absorption and Marginal Costing

According to traditional costing system, fixed costs of production are assigned to products to be subsequently released by way of expenses as part of cost of goods sold or are carried forward as part of the cost of inventory. Such an approach to the treatment of fixed costs has brought into attention various methods of allocation of overheads to different departments on an equitable basis and their proper apportionment to units produced. However, the various methods devised fail to give precise results and sometimes even lead to absurd situations.

Marginal costing removes all the difficulties involved in the allocation, apportionment and recovery of fixed costs; it is able to accomplish this by excluding fixed costs from product costs and by covering them off entirely using contribution margin. Consequently, when the volume of output differs from the volume of sales, the net income reported under marginal costing will differ from the net profit reported under absorption costing.

**Illustration 9.1: (Comparative Income Statements)**

Green-Green Limited furnishes the following details for the year ended 31 December 2025 for preparing the income statement of the year:

| | | |
|---|---|---|
| Sales | 1,000 units @ | ₦10 per unit |
| Manufacturing costs | | ₦2,200 |

| | | |
|---|---|---|
| Variable manufacturing cost | 1,100 units @ ₦6 per unit | |
| Inventory at close | 100 units | |
| Fixed selling & admin. Expenses | ₦500 | |
| Variable selling & admin expenses | ₦400 | |

**Solution 9.1**

**Green-Green Limited**

**Income Statement for the year ended 31st December, 1992**

(a) **Under Absorption Costing**

| ₦ | | ₦ |
|---|---|---|
| Sales 1,000 units @ ₦10 each | | 10,000 |
| Less: Cost of sales: | | |
| Variable manufacturing cost: | | |
| 1,100 units @ ₦6 each | 6,600 | |
| Fixed manufacturing costs | 2,200 | 8,800 |
| Les: Inventory at close: | | |
| 100 units @ ₦8 each | 800 | 8,000 |
| Gross Margin or profit | | 2,000 |
| Less: Total selling & admin. exps. | | 900 |
| Net Income | | 1,100 |

**Workings:**

The Cost of inventory at close, as well as cost of each unit of inventory at close, has been calculated as follows:

| | | |
|---|---|---|
| Units manufactured | 1,100 | |
| Units of inventory at close | 100 | |
| Ratio of closing inventory to total production | = 100 / 1,100 | = 1/11 |

Cost of Inventory at close = $\frac{1}{11}$ x ₦8,800 = ₦800

Thus, the cost of each unit of inventory at close

= ₦800 / 100 units = ₦8 per unit

(b) **Under Marginal Costing**

| | ₦ | ₦ |
|---|---|---|
| Sales: 1,000 units @ N10 each | | 10,000 |
| Less: variable cost of sales: | | |
| Variable manufacturing cost | | |
| 1,100 units @ N6 each | 6,600 | |
| Less: Inventory at close: | | |
| 100 units @ N6 each | 600 | 6,000 |
| Variable Gross Margin | | 4,000 |
| Less: Variable selling & Admin. exps. | | 400 |
| Operating contribution margin | | 3,600 |
| Less: Fixed Costs: | | |
| Fixed manufacturing costs | 2,200 | |
| Fixed selling & admin. exps. | 500 | 2,700 |
| Net Income | | 900 |

**Workings**

The cost of inventory at close, as well as the cost of each unit of inventory at close, has been calculated as follows:Ratio of closing inventory to total production

= 100 / 1,100 = 1/11

Cost of Inventory at close = $\frac{1}{11}$ x ₦6,600 = ₦600

Thus, the cost of each unit of inventory at close

= $\frac{₦600}{100 \text{ units}}$ = ₦6 per unit

**Note:**
The difference of ₦200 in the net income calculated under the two methods is due to the difference between the costs of closing inventory which, under absorption costing, is ₦800 and, under marginal costing, is ₦600. This is the result of holding back ₦200 out of the total fixed manufacturing cost of ₦2,200 as cost of inventory under absorption costing, whereas

₦200 is released immediately as period charge under marginal costing.

A perusal of the absorption costing statement and marginal costing statement shows that the former gives more information than the latter. Although, this might be correct, the information furnished by the absorption costing statement cannot be as useful as the one given by marginal costing because the conventional costing statement rarely classifies costs into fixed and variable components.

### 9.2.2 **Role of Contribution**

Contribution is of vital importance for the system of marginal costing. The rationale of contribution lies in the fact that, where a business manufactures more than one product, the profit realized on individual products cannot be possibly calculated due to the problem of apportionment of fixed costs to different products which is done away with under marginal costing. Therefore, some methods are required for the treatment of fixed costs and marginal costing and answer to the challenge is 'contribution'.

Contribution is the difference between sales and the variable cost of sales and is, therefore, sometimes referred to as a 'gross margin'. It is visualized as some sort of a 'fund' or 'pool' out of which all fixed costs, irrespective of their nature, are to be met and to which each product has to contribute its share. The difference between contribution and fixed costs is either profit or loss as the case may be.

The concept of contribution is useful in the fixation of selling prices, determination of break-even point, selection of product mix for profit maximization, make or buy decisions, acceptance or rejection of special orders and ascertainment of the profitability of products, departments, etc.

**Illustration 9.2: Contribution Margin**

The following information has been gathered from the accounts of Prudent Engineering Limited,

organized on departmental basis:

**Departments**

| | X | Y | Z |
|---|---|---|---|
| | ₦ | ₦ | ₦ |
| Variable costs | 90,000 | 100,000 | 150,000 |
| Fixed costs: | | | |
| (apportioned on the basis of sales) | 20,000 | 30,000 | 50,000 |
| Total Costs | 110,000 | 130,000 | 200,000 |
| Sales | 100,000 | 150,000 | 250,000 |
| Profit (Loss) | (10,000) | 20,000 | 50,000 |

With the object of doing away with the 'X' department incurring a loss, the management asks for your opinion on:

a. The closure of Department X on the basis of the above information.

b. The comparative profitability of different departments if specific fixed costs are ascertained to be ₦5,000 for Department X, ₦55,000 for Department Y and ₦30,000 for Department 'Z' the remaining ₦10,000 being general fixed costs.

**Required:**

Prepare appropriate statements so as to help management in arriving at a decision on the above points. Also give your comments, explaining the position presented in the statements.

**Solution 9.2**

i. **All Fixed Costs regarded as general fixed costs:**

**Marginal Cost Statement**

**Departments**

| | X | Y | Z | Total |
|---|---|---|---|---|
| | ₦ | ₦ | ₦ | ₦ |
| Sales | 100,000 | 150,000 | 250,000 | 500,000 |
| Less: Var. Cost | 90,000 | 100,000 | 150,000 | 340,000 |
| Contribution Mar. | 10,000 | 50,000 | 100,000 | 160,000 |
| Less: Fixed Costs | | | | 100,000 |
| Profit | | | | 60,000 |

**Comment**

The loss of ₦10,000 shown by Department X has been due to the arbitrary basis of the apportionment of fixed overheads to different departments.

The fact has been underlined by the marginal cost statement which does not attempt to allocate fixed costs to the various departments with the result that, in the event of the closure of Department X, the total contribution as well as profit would be reduced by ₦10,000. Therefore, the company would not stand to gain anything by closing Department X; rather, it would lose under such an eventuality.

ii. **On the basis of specific fixed costs**

*Statement of Comparative Profitability Departments*

| | *X* | *Y* | *Z* | *Total* |
|---|---|---|---|---|
| | ₦ | ₦ | ₦ | ₦ |
| Sales | 100,000 | 150,000 | 250,000 | 500,000 |
| Less: Var. cost | 90,000 | 100,000 | 150,000 | 340,000 |
| Contribution | 10,000 | 50,000 | 100,000 | 160,000 |
| Less: Specific fixed Costs | 5,000 | 55,000 | 30,000 | 90,000 |
| Net contribution | 5,000 | (5,000) | 70,000 | 70,000 |
| Less: General Fixed costs | | | | 10,000 |
| Profit | | | | 60,000 |

**Comment**

Although, the second statement shows the same profit of N60,000 as is shown by the first statement, it tells an entirely different story.

From the above statement, it is Department Y which incurs a loss of N5,000 instead of Department X. The position has been brought about by the availability of fixed costs specifically assigned to different departments. It also highlights the anomalies created by the apportionment of fixed costs on some arbitrary basis as according to sales in this example.

It is also evident that if Department Y is closed, the total contribution as well as profit would increase by ₦5,000. Advice on these lines would be tendered only if either the working of Department Y cannot be improved so as to show a profit, or at least to break-even, or the price of its products cannot be increased. It is so because, generally, it is better from the point of view of business to offer a wider range of choices to its customers.

The position on the closure of Department Y can be shown as follows:

**Statement of Comparative Profitability**

Departments

| *X* | | *Z* | *Total* |
|---|---|---|---|
| | ₦ | ₦ | ₦ |
| Sales | 100,000 | 250,000 | 350,000 |
| Less: Var. Costs | 90,000 | 150,000 | 240,000 |
| Contribution | 10,000 | 100,000 | 110,000 |
| Less: Specific fixed costs | 5,000 | 30,000 | 35,000 |
| Net Contribution | 5,000 | 70,000 | 75,000 |
| Less: Gen. Fixed cost | | | 10,000 |
| Profit | | | 65,000 |

Thus, profit increases by ₦5,000 on the closure of Department Y, the amount being exactly equal to the loss incurred by that Department.

### 9.3 Relevant Cost and Relevant Revenue

Environment affected by limiting factors often requires consideration of relevant costs and relevant revenues in taking short term decisions. For instance, decisions involving application of marginal costing such as special order pricing, make or buy decision, etc. often make use of relevant costing. The CIMA Official Terminology defines Relevant Costs/Relevant Revenues as "Costs and revenues appropriate to a specific management decision". It further expresses that these costs and revenues are represented by future cash flows whose magnitude will vary depending upon the outcome of management decision made. If inventory is used, the relevant cost, used in the determination of the profitability of the transaction, would be the cost of replacing the inventory, not its original purchasing price which is a sunk cost."

The three major characteristics of relevant costs are:

- They are future cost/future revenues
- They are differential costs/revenues that differs between alternative decisions
- This involves cash flows

Thus, to consider a cost or revenue as relevant, it must satisfy the three major characteristics listed above. The implication, which is also contained in the definition above, is that whenever a decision is to be taken involving application of marginal costing, past costs, known as historical costs are sunk cost.

Some of the items of relevant costs include:

1. **Notional Imputed Costs:** They are costs not considered in financial accounting because they do not involve actual cost spending but useful in decision making. The terminology defines it as "cost used in product evaluation, decision making and performance measurement to reflect the use of resources that have no actual (observable) cost" e.g notional interest for internally generated fund or notional rent for use of space.
2. **Opportunity Costs:** They are value of benefit forgone in favour of taking alternative course of action. Others are
3. Out of Pocket Cost
4. Replacement Cost
5. Differential Costs that are more than zero value

**The Concept of Sunk Costs**

These costs are past or historical costs that are not useful in taking economic decisions. The terminology defines sunk cost as "cost that has been irreversibly incurred or committed and cannot therefore be considered relevant to a decision". Sunk cost may also be termed as irrecoverable cost.

A past or historical cost can be considered relevant if only such item has a replacement or market value or can be put into an alternative use; otherwise the value is zero in decision making.

**Relevant costs of elements of cost**

1. **Relevant Cost of Material:** Where materials in stores will be used to implement decision, relevant cost is NIL value. However, if materials in stores have replacement or market value or can be put into an alternative use, then, the values involved are relevant.
2. **Relevant Cost of Labour:** Fixed salaries of workers are irrelevant, only the associate opportunity cost of fixed salary earner is relevant.

**Illustration**

A firm received a special order. The order will utilise material worth of ₦10,000 in stores. The material has no market value and cannot be put into an alternative use. The order will require additional material CD to procured for ₦30,000. Factory workers will be paid a total of ₦50,000.

The order will also make use of an idle plant that has a written down value of ₦200,000. Depreciation is calculated at ₦5,000 per month. The order will take one month to complete.

**Required:** Determine the minimum price to charge.

**Computation of Minimum Price**

| | | ₦ |
|---|---|---|
| Existing Material in Stores | | NIL |
| Material CD | 30,000 | |
| Factory workers 'wages | 50,000 | |

| | | |
|---|---|---|
| Manager's salary | | NIL |
| Plant depreciation | | NIL |
| Minimum Price | **80,000** | |

**Bases for Computation:**

- Existing materials in stores are sunk cost
- Manager's salary considered as irrelevant/sunk
- Plant depreciation on an idle plant – sunk cost
- Relevant cost includes material CD value and factory workers' wages.

## 9.4 PRACTICAL APPLICATIONS OF MARGINAL COSTING

### 9.4.1 Make or Buy Decisions

This type of situation arises when a manufacturer is faced with the decision as to whether to

(a) Manufacture one of its components in-house or

(b) Buy such components from an outside supplier.

The decision maker will be interested in the difference between the supplier's quotation and the cost of producing in-house. The cost of in-house production of the components is made up of:

i. The incremental cost of production; and

ii. Any opportunity cost that may arise from producing in-house.

The following qualitative factors will also be considered in deciding whether or not to produce in-house:

- The quality of the product that will be bought from outside;
- The reliability of the outside supplier;
- The effect of future market prices;
- Ability of management to cope with the present and future production capacity;
- The possible adverse effects of revealing trade secrets; and
- The possible problems of transport and handling costs.

### 9.4.2 Decision Criteria

a. If the outside supplier's quotation is greater than the total variable cost of producing the component in-house, then the component should be manufactured in- house.

b. If the total variable cost of in-house production is higher than the quotation of the outside supplier, then it will be quite discrete for the management to purchase the component from the supplier.

**Illustration 9. 3: Make or Buy Decision**

Nagarta Nig. PLC is considering an opportunity to subcontract the production of a certain component part of its product to another manufacturer. The subcontractor has offered to produce the part for ₦1.80 per piece. The management accountant makes a study of the savings of cost resulting from subcontracting as follows:

a. The standard unit cost of material is reduced by 50k

b. The direct labour cost per unit is reduced by 25k

c. The reduction in variable overhead is estimated at ₦1.00 per unit

It is believed by the plant superintendent that supervision costs to the extent of ₦2,000 per year can be saved if the job is subcontracted and that the cost of special tools will be reduced by ₦500 per year.

The other relevant cost data in connection with the manufacture of this product are as follows:

| | |
|---|---|
| Units sold | 14,000 |
| | ₦ |
| Net sales - 14,000 @ ₦25 each | 350,000 |
| Standard cost of goods sold: | |
| Direct materials | 70,000 |
| Direct labour | 28,000 |
| Factory overheads | 140,000 |
| Total Standard cost of goods sold | 238,000 |
| Standard Gross Margin | 112,000 |
| Less: Unabsorbed factory overheads | 36,000 |
| Actual Gross margin | 76,000 |
| Less: Selling and distr. exp. | 100,000 |
| Net Loss | 24,000 |
| | ======= |

The break-up of factory overheads into their fixed and variable components at normal capacity level of 20,000 units is as follows:

| | Per Unit | Total |
|---|---|---|
| | ₦ | ₦ |
| Variable | 4 | 80,000 |
| Fixed | 6 | 120,000 |

| | |
|---|---|
| 10 | 200,000 |
| === | ===== |

The company is working at less than normal capacity which is manufacture of 20,000 units of this product.

You are required to advise the management on whether to subcontract the production of the component or to produce it internally.

**Solution 9.3**

Solution to this problem may best be presented in form of differential cost analysis as follows:

**Effects of Make or Buy Decision:**

| Make | | | Difference | Buy |
|---|---|---|---|---|
| | ₦ | | ₦ | ₦ |
| Net Sales | 350,000 | | 350,000 | |
| Dir. Materials | 70,000 | (14,000 x ₦0.50), i.e. | -7000 | 63,000 |
| Purchased parts | | - (14,000 x ₦1.80), i.e. | +25,200 | +25,200 |
| Direct labour | 28,000 | (14,000 x ₦0.25),i.e. | -3,500 | 24,500 |
| Var. overheads | 56,000 | (14,000 x ₦1),i.e. | -14,000 | 42,000 |
| Fixed overheads | 120,000 | (2,000 + ₦500),i.e. | -2,500 | 117,500 |
| Total manuf. Cost | 274,000 | | -1,800 | 272,200 |
| Selling & Admin. | 100,000 | | - | 100,000 |
| | 374,000 | | -1,800 | 372,200 |
| Net Loss | (24,000) | | -1,800 | (22,200) |

It is to be noted that, no change in volume or selling price is assumed so that revenue is not a relevant factor. The benefit lies purely in cost saving which at this level of volume is ₦1,800.

Despite the fact that the company is operating at less than normal capacity, the small monetary advantage offered by this opportunity may not be considered sufficient to offset the ill-effects of laying off more of the labour force due to the non-manufacture of the component. In spite of the impact of non-quantitative factors, the differential cost analysis remains at the core of make or buy decision.

**Illustration 9.4 : Further Exampleon Make or Buy Decision**

A.B. Limited makes four components, W, X, Y and Z for which costs in the following year are expected to be:

| | W 1,000 | X 2,000 | Y 4,000 | Z 3,000 |
|---|---|---|---|---|
| Production (in units) | | | | |
| Unit marginal costs: Direct materials | ₦ 4 | ₦ 5 | ₦ 2 | ₦ 4 |
| Direct labour | 8 | 9 | 4 | 6 |
| Variable prod. Overheads | 2 | 3 | 1 | 2 |
| | 14 | 17 | 7 | 12 |

Directly attributable fixed costs per annum and committed fixed costs: ₦

| | |
|---|---|
| Incurred as a direct consequence of making W | 1,000 |
| Incurred as a direct consequence of making X | 5,000 |
| Incurred as a direct consequence of making Y | 6,000 |
| Incurred as a direct consequence of making Z | 8,000 |
| Other fixed costs (committed) | 30,000 |
| | 50,000 |

Asub-contractor has offered to supply units of W, X, Yand Z for ₦12, ₦21, ₦10 and ₦14, respectively.

Should A.B. Limited make or buy the components?

**Solution 9.4**

a) The relevant costs are the differential costs between making and buying, and they consist of differences in unit variable costs plus differences in directly attributable fixed costs. Sub-contracting will result in some fixed costs savings;

| | W | X | Y | Z |
|---|---|---|---|---|
| | ₦ | ₦ | ₦ | ₦ |
| Unit var. cost of making | 14 | 17 | 7 | 12 |
| Unit var. cost of buying | 12 | 21 | 10 | 14 |
| | (2) | 4 | 3 | 2 |
| Annual requirements (units) | 1,000 | 2,000 | 4,000 | 3,000 |
| | ₦ | ₦ | ₦ | ₦ |
| Extra var. cost of buying (per annum) | (2,000) | 8,000 | 12,000 | 6,000 |
| Fixed cost saved by buying | 1,000 | 5,000 | 6,000 | 8,000 |
| | (3,000) | 3,000 | 6,000 | (2,000) |

b) The company would save ₦3,000 per annum by subcontracting component W (where the purchase cost would be less than the marginal cost per unit to make internally) and would save ₦2,000 per annum by subcontracting component Z (because of the saving in fixed costs of ₦8,000);

c) In this example, relevant costs are the variable costs of in-house manufacture, the variable costs of sub-contracted units, and the saving in fixed costs;

d) Other important considerations would be:
   (a) If components W and Z are sub-contracted, the company will have spare capacity. The question now is how such spare capacity can be profitably used?
   (b) Would the sub-contractor be reliable with delivery times, and would he supply components of the same quality as those manufactured internally?
   (c) Does the company wish to be flexible and maintain better control over operations by making everything itself?
   (d) Whether the estimates of fixed costs savings are reliable?

### 9.4.3 **Acceptance/Rejection of Special Order**

This type of situation arises when a company receives an order from a customer at a price lower than its normal selling price. The company, if working below capacity, may be advised to accept the offer after taking into consideration the marginal cost of its production.

Therefore, if the answers to the following questions are in the affirmative, the special price quoted by the customers should be accepted:

(a) Does the price quoted by the special customer cover the marginal cost of production?
(b) Does the company have excess capacity?

**Illustration 9.5: Acceptance or Rejection**

Shukura Nigeria Limited manufactures a special product for ladies called 'the slimming stick'. A stick sells for 20k. Current output is 400,000 sticks which represents 80% level of activity. A customer Gyatin Stores Limited, recently placed an order at 13k per slimming stick. The total cost for the period was ₦56,000 of which N16,000 were fixed costs. This represents a total cost of 14k per slimming stick.

**Required:**

(a) Based on the above information, advise Shukura Nigeria Limited whether or not to accept the offer.
(b) What other factors may be taken into consideration in taking such a decision?

**Solution 9.5**

(a)

| (i) Present position | ₦ |
|---|---|
| Sales (400,000 units) | 80,000 |
| Less: marginal cost (400,000 x10k) | 40,000 |
| Contribution | 40,000 |
| Less: Fixed costs | 16,000 |
| Net Profit | 24,000 |

(ii) If the order from Gyatin Stores is accepted: ₦

| | |
|---|---|
| Sales (100,000 x 13k) | 13,000 |
| Less: marginal cost (100,000x10k) | 10,000 |
| Contribution | 3,000 |

**Recommendation**

The Order from Gyatin Stores should be accepted, since it will increase profit by ₦3,000.

(b) Other factors to consider include:

i. Will the acceptance of one order at a lower than market price make other customers to ask for price reductions?

ii. Is the so called special price the most profitable way of utilizing the excess capacity?

iii. If accepted, will the special order not block the acceptance of offers which may be at true market price? and

iv. Will fixed cost remain constant?

### 9.4.4 Marginal Costing and Pricing Decision

The determination of prices of products manufactured or services rendered, by a business is often considered to be a difficult problem generally faced by management of an enterprise. However, the basic problem involved in pricing is the matching of demand and supply.

The marginal costing technique offers a simple as well as clear portrayal of the relationship between specific products costs and the different possible selling prices being considered. This is due to the fact that contribution margin is unaffected by the allocation of indirect costs.

**Illustration 9.6: Marginal Costing and Pricing**

M & K Industries Limited produces and markets industrial containers and packing cases. Due to competition, the company proposes to reduce its selling price.

If the present level of profit is to be maintained, indicate the number of units to be sold if the proposed reduction in selling prices is 5%, 10% and 15%.

The following information is available:

| | | ₦ |
|---|---|---|
| Present sales turnover (30,000 units) | | 300,000 |
| Variable cost (30,000 units) | 180,000 | |
| Fixed cost | 70,000 | 250,000 |
| Net profit | | 50,000 |

**Solution 9.6**

Marginal Cost Statement

| | **Present Price** ` ₦ | **price reduction of 5%** ₦ | **Price Reduction of 10%** ₦ | **Price Reduction of 15%** ₦ |
|---|---|---|---|---|
| Sales | 300,000 | 285,000 | 270,000 | 255,000 |
| Less: variable costs | 180,000 | 180,000 | 180,000 | 180,000 |
| Contribution | 120,000 | 105,000 | 90,000 | 75,000 |
| Less: Fixed costs | 70,000 | 70,000 | 70,000 | 70,000 |
| Net Profit | 50,000 | 35,000 | 20,000 | 5,000 |
| Contribution per unit per unit | ₦4.00 | ₦3.50 | ₦3.00 | ₦2.50 |

As profit is to be maintained at the present level of ₦50,000, the contribution has to be ₦120,000 as at present when the total sale proceeds from 30,000 units are ₦300,000.

The number of units required to be sold at different levels of price reduction would be calculated as follows:

Total Contribution Required

Contribution per unit

Therefore:

At 5% Reduction: ₦120,000 = 34,286 units approximately

₦3.50

At 10% Reduction: ₦120,000 = 40,000 units

₦3

At 15% Reduction: ₦120,000 = 48,000 units

₦2.50

**Verification:**

The result obtained at 10% price reduction can be verified as follows:

₦

| | |
|---|---|
| Sales (40,000 x 270,000/30,000) | 360,000 |
| Less: variable costs (40,000 x 180,000) | |
| 30,000 | 240,000 |
| Contribution | 120,000 |
| Less: Fixed Costs | 70,000 |
| Profit | 50,000 |

The other results can also be similarly verified.

### 9.4.5 Marginal Costing and Profit Planning

Marginal costing, through the calculation of contribution ratio, enables the planning of future operations in such a way as to attain either maximum profit or to maintain a specified level of profit. Thus, it is helpful in profit planning.

**Illustration 9.7: Profit Planning**

A toy manufacturer earns an average net profit of ₦3 per piece in a selling price of ₦15 by producing and selling 60,000 pieces at 60% of the potential capacity. Composition of his cost of sales is:

| | |
|---|---|
| Direct materials | ₦4.00 |
| Direct wages | ₦1.00 |
| Works overhead | ₦6.00 (50% fixed) |
| Sales overhead | ₦1.00 (25% variable) |

During the current year, he intends to produce the same number but anticipates that:

(a) his fixed costs will go up by 10%

(b) rates of direct labour will go up by 20%

(c) rates of direct material will increase by 5%

(d) selling price cannot be increased.

Under these circumstances, he obtains an order for a further 20% of his capacity.

What minimum price will you recommend for accepting the order to ensure the manufacturer earns an overall profit of ₦180,500.

**Solution 9.7**

**Marginal costing statement for current year**

**(prior to acceptance of 20% excess order)**

| | Per Piece | Total Amount |
|---|---|---|
| | ₦ | ₦ |
| Variable costs: | | |
| Direct material | 4.20 | |
| Direct labour | 1.20 | |
| Var. works overhead | 3.00 | |
| Var. sales overhead | 0.25 | |
| | 8.65 | 519,000 |
| Sales value | 15.00 | 900,000 |
| Contribution | 6.35 | 381,000 |

Fixed Costs:

₦

| | | | |
|---|---|---|---|
| Works overhead | 180,000 | | |
| Add: 10% | 18,000 | 198,000 | |
| Sales overhead | 45,000 | | |
| Add: 10% | 4,500 | 49,500 | 247,500 |
| Profit | | | 133,500 |

Planned profit = ₦180,500

increased in profit = ₦180,500 - ₦133,500 = ₦47,000

The minimum price for 20,000 toys (order for 20% capacity) can be worked out as thus:

₦

| | | |
|---|---|---|
| Variable cost @ ₦8.65 | 173,000 | |
| Add: Increase in profit | 47,000 | 220,000 |

Minimum sales price per unit = ₦220,000 / 20,000 units

= ₦11

## 9.5 CHAPTER SUMMARY

In this chapter, efforts have been made to differentiate clearly between marginal costing and absorption costing, two techniques that guide decision making based on income statements prepared using them. Greater emphasis has been put on marginal costing, being the technique that is purely for managerial decision making. Its advantages and limitations were highlighted for better appreciation and some of the popular areas of its application for short term tactical managerial decision making were discussed and illustrated.

It is expected that the set objectives of the chapter have been achieved by now and that the reader is in a position to educate beginners on:

- The concepts of marginal and absorption costing;
- The role and importance of contribution margin in managerial decision making;
- The advantages and limitations of marginal costing; and
- The various application areas of marginal costing.

## MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS

1. What is marginal cost?
   A. Cost incurred when increasing profit margin
   B. Total cost of manufacture
   C. Cost of producing an additional unit
   D. Cost of manufacturing a new product
   E. Aggregate cost of manufacturing diverse products

2. What are period costs?
   A. Costs incurred during a stated period of production
   B. Costs incurred during certain periods of the year
   C. Costs which serves to enhance profitability
   D. Costs of maintaining equipment during a period
   E. Costs incurred during a period irrespective of production volume

3. How does marginal costing differ from absorption costing in inventory valuation?
   A. Marginal costing is only concerned with margins
   B. Absorption costing uses LIFO
   C. Absorption costing recognizes slow-moving inventory
   D. Marginal costing uses total variable costs, writing off fixed assets
   E. Marginal costing uses FIFO

4. What is contribution?
   A. Total sales minus total variable costs
   B. Total sales plus total variable costs
   C. Total variable costs plus total fixed costs
   D. Total fixed costs minus profit
   E. Total sales plus total fixed costs

5. What is limiting factor?
   A. The extent to which the company may manufacture its products
   B. The factor of production which is in short supply
   C. The limit to which government may allow the company to manufacture
   D. The factor of production which is most expensive
   E. The factor of production which is easily obtainable

6. Under absorption costing, closing inventory include a portion of the period's…….
7. In most decision making scenarios, marginal costing advises that the relevant cost is the ……
8. Under marginal costing, sales minus variable costs equals ……

9. What is the major cause of differences in profit figures derived from profitability statements using absorption costing and marginal costing?

10. Based purely on financial information, a company should manufacture more of the products giving the highest ……. towards the absorption of its fixed assets and ultimately earning profit.

**SOLUTION TO MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS**

1. C
2. E
3. D
4. A
5. B
6. Fixed costs
7. Variable cost (marginal cost)
8. Fixed Costs plus profit or Contribution
9. Value of closing inventory/Treatment of fixed costs
10. Contribution

**EXAMINATION TYPE QUESTIONS**

1. Dala Equipment Limited, manufactures four components, the cost particulars of which are given below:

**Components**

| | A | B | C | D |
|---|---|---|---|---|
| Elements of cost: | ₦ | ₦ | ₦ | ₦ |
| Direct material | 80 | 100 | 100 | 120 |
| Direct labour | 20 | 25 | 25 | 30 |
| Variable overhead | 10 | 12 | 15 | 10 |
| Fixed overhead | 15 | 23 | 20 | 20 |
| | 125 | 160 | 160 | 180 |
| Output per machine hour (units) | 4 | 2 | 3 | 3 |

The key factor is shortage of machine capacity.

You are required to advise management as to whether they should continue to produce all or some of the components (which are used in its main product) or they should buy them from a supplier who has quoted the following prices:

A - ₦115; B - ₦175; C - ₦135; D - ₦185

2. Three different product lines are sold by Indabawa Mills. Product A contributes 20% of its revenue to fixed costs and profits, product B contributes 10% and product C contributes 60%.

The products sell for the following prices:

Product A ₦8 per unit

Product B ₦2 per unit

Product C ₦3 per unit

The company earned a net profit of ₦50,000 before income taxes last year by selling 50,000 units of product A, 150,000 units of product B and 60,000 units of product C.

Muhammed Ibrahim, who is the sales manager believes that the profit picture can be improved by eliminating product B and concentrating the sale effort on products A and C. He sees an opportunity to increase the sales of product A to 70,000 units but admits that product C will probably sell at the same amount next year.

Prepare a projected income statement using Ibrahim's assumption. Do you agree with him? Explain what would cause the profit to increase or decrease as the case might be.

3. The management of Wali and Gwangwazo Ltd, is considering the proposal to discontinue the manufacture of product X out of the list of its products - X, Y and Z. The details of which are given below:

| | **X** | | **Y** | **Z** |
|---|---|---|---|---|
| Capacity in actual production | | | | |
| | 20% | | 40% | 40% |
| Units manufactured | 4,000 | | 10,000 | 12,000 |
| Costs per unit | | ₦ | ₦ | ₦ |
| Materials | | 25 | 20 | 40 |
| Labour | | 15 | 10 | 20 |
| Fixed overhead | 4 | | 4 | 5 |
| Var. overhead | 4 | | 3 | 5 |
| | | 48 | 37 | 70 |
| Profit/loss per unit | | (4) | 13 | 10 |
| | | 52 | 50 | 80 |

Product X having persistently shown a loss for a number of years due to the saturation of demand and the future prospects of the other two products being bright as a result their having been accepted as components by a newly established machine-building complex in the vicinity, the production capacity released by the discontinuation of product X is sought to be transferred equally to products Yand Z.

Moreover, the nature of product X has been such that the transfer of production capacity engaged therein to the other two products would bring about an increase of 30% and 50% in the number of units to be manufactured of products Y and Z respectively, than those involved in the capacity transferred to them under the proposal without, of course, involving any change in total fixed costs from those when all the three products were manufactured.

The anticipated increases in the structure of costs and the selling price are as thus:

| | Y | Z |
|---|---|---|
| Materials | 5% | 5% |
| Labour | 10% | 10% |
| Selling price | 10% | 5% |

You are required to:

(a) prepare a statement of projected profitability; and

(b) advise management as to whether the proposal may be accepted for implementation.

4. The management of Jigawa Limited is considering next year's production and purchase budgets. One of the components produced by the company which is incorporated into another product before being sold has a budgeted manufacturing cost as follows:

| | ₦ |
|---|---|
| Direct material | 14.00 |
| Direct labour (4 hours at ₦3) | 12.00 |
| Var. overhead (4 hours @ ₦2) | 8.00 |
| Fixed overhead (4 hours @ ₦5) | 20.00 |
| | 54.00 |

Ball Limited has offered to supply the above component at a guaranteed price of ₦50 per unit.

**Required:**

(a) Considering cost criterion only, advise management as to whether the above component should be purchased from Ball Limited.

(b) Explain how your above advice would be affected by the situation shown below:

As a result of government legislation, if Jigawa Limited continues to manufacture this component, the company will incur additional inspection and testing expenses of ₦56,000 per annum, which are not included in the above budgeted manufacturing costs.

(c) The production director of Jigawa Limited recently said "we must continue to manufacture the component as only one year ago we purchased some special grinding equipment to be used exclusively by this component. The equipment cost ₦100,000, it cannot be resold or used elsewhere and if we cease production of this component, we will have to write off the written down value (book value) which is ₦ 80,000.

Draft a reply to the production director commenting on his statement.

5(a) Koki Limited which manufactures part M - 6 for use in its production cycle has the following costs per unit for production of 25,000 units.

| | ₦ |
|---|---|
| Direct materials | 5 |
| Direct labour | 20 |
| Manufacturing overhead | 24 |
| | 49 |

It has been established that 66 2/3% of the overhead is fixed. Sahara Limited has offered to sell 25,000 units of part M - 6 to Koki Ltd, for ₦45 per unit. If Koki Ltd., accepts Sahara's offer, some of the facilities presently used to manufacture M - 6 could be rented to a third party at an annual rent of ₦65,000.

Additionally, ₦6 per unit of the fixed overhead costs applied to part M - 6 would be totally eliminated. The managing director of Koki Ltd, has called on you to advise on whether he should accept Sahara's offer.

(b) State factors other than the relevant costs above which would influence your decision to accept or reject Sahara's offer.

6. Aminu Motors Limited specializes in the manufacture of sport cars making some of the components which are required and buying others. Alparts Limited offers to supply a component currently made by Aminu Motors Ltd. at a price of ₦7. The costs incurred by Aminu Motors in making the part are as follows:

| | ₦ |
|---|---|
| Variable costs | 4 |
| Traceable fixed costs | 2 |
| Allocated fixed costs | 3 |
| | 9 |

Monthly output is 5,000 units.

**Required**:
Present information to management on the assumption that the production capacity realised as a result of accepting the order will remain idle.

7(a) What benefits are to be gained from marginal costing? Are there any pitfalls in the application of marginal costs? Discuss these matters critically.

(b) Give a brief account of a practical application of marginal costing which you consider sound from a policy viewpoint.

8. Tapgun Industries Limited manufactures paints in 4-litre cans. The selling price of the product is N280 with standard cost of N180 made up as follows:

| | |
|---|---|
| Direct materials | ₦100 |
| Direct labour | ₦20 |
| Variable overheads | ₦30 |
| Fixed overhead | ₦30 |

In the months of April and May 1999, the company's production and sales reports were as follows:

| | April | May |
|---|---|---|
| Production (qty) | 10,000 | 10,000 |
| Sales (qty) | 8,000 | 12,000 |

You are required to:

(a) Prepare the company's operating statements for the two months using:
  (i) the marginal costing technique; and
  (ii) the absorption costing technique.

(b) Comment on the results of the two techniques with particular reference to difference in the inventory valuation; and the operating profit.

*(ICAN Intermediate, Nov. 1999)*

## CHAPTER TEN

## COST-VOLUME-PROFIT ANALYSIS

a. The concept of Cost-Volume-Profit Analysis, C-V-P Analysis
b. Break-Even-Point
c. Margin of Safety
d. Graphical presentation of C-V-P Analysis

### 10.0 Objectives

After studying this chapter, readers should be able to:

- explain the meaning and importance of C-V-P Analysis;
- explain the concept of break-even analysis and computation of Break-Even-Point; and P/V ratio or contribution margin ratio
- explain the meaning of margin of safety and use a formula to determine the margin of safety;
- present a graphical approach for C-V-P analysis

### 10.1 Cost-Volume-Profit Analysis

#### 10.1.1 The Concept of Cost-Volume-Profit

Cost-Volume-Profit (CVP) analysis involves the analysis of how total costs, total revenues and total profits are related to sales volume and is, therefore, concerned with predicting the effects of changes in costs and sales volume on profit. It is an analysis that guides decision making in respect of going into a production process, expanding an existing production level or diversifying into unusual areas of operations.

CVP analysis seeks to find answers to the following questions:

a. What would be the cost of production under different circumstances?
b. What has to be the volume of production?
c. How many units of each product can be manufactured to earn the target profit?
d. What is the difference between the selling price and the cost of production?
e. What are the effects of a percentage increase or decrease in cost on production volume and profit?

If carefully used, the technique might be helpful in:

i. Budgeting Process. The volume of sales required to make a profit and the "safety margin" for profits in the budget can be measured.
ii. Pricing decision and sales volume decisions;
iii. sales mix decision, i.e. in deciding what proportions of each product should be sold;

iv. decisions affecting the cost structure and production capacity of the company.

### 10.1.2 Importance of Cost-Volume-Profit Relationship

An understanding of the inter-relationship between these three forces, and of the probable effect that any change in sales volume would have on the business, is extremely helpful to management in a broad variety of problems involving planning and control. The relationship between cost, volume and profit makes up the profit structure of an enterprise. It is only through the knowledge and intelligent use of such information that the prediction of the probable effect of any number of contemplated actions is made possible. This makes Cost-Volume-Profit relationship analysis an important tool for budgeting and profit planning.

The data used in the review of such relationships may come from several sources and may differ considerably in adaptability and usefulness. In companies where a rather complete sales analysis is made and flexible budgets and standard costs are available, the records would provide the necessary information in a readily usable form. Costs, in all probability, would have been segregated into the fixed and variable elements. In case such sources are not available, the conventional historical records might be utilised. Quite a detailed analysis may be necessary to isolate the effects of changes in volume, selling prices and variable costs.

However, if cost control has been weak, the relationship between volume and cost will be difficult to trace, and the margin of error will be high. Thus, the accuracy of results would depend greatly on the reliability of the data and the validity of assumptions. Very often, for investment and credit purposes, published financial statements are used as source data in studying the effects of volume on the business.

## 10.2 The concept of Break-even Analysis

The C-V-P analyses short-term profit planning that establishes the relationship among costs, volume and profit. Even though profit determination is important, it is equally necessary to find out production capacity utilisation that enables a firm to break even to enable the firm to determine the viability of its business. The C-V-P Analysis is broad, while the narrow concept of C-V-P analysis is referred to as Break-even point analysis.

The Break-Even Point is the level of production activity at which there is neither profit nor loss. At the break-even point, BEP, the total cost incurred for a product equates total revenue realised for selling the product.

Thus, at BEP:

Total Cost = Total Revenue

i.e. TC = TR **or** TR – TC = Zero Profit

where TC = Total Fixed Cost + Total Variable Cost.

After the BEP, fixed cost is no longer incurred. The cost incurred after BEP is limited to variable costs.

### 10.2.1 Assumptions of Break-even Analysis

Break-even analysis and Cost-Volume-Profit analysis are based upon certain assumed conditions that are rarely found in practice. Some of these basic assumptions are as follows:

a. The principle of cost variability is valid.
b. Costs can be resolved into their fixed and variable components.
c. Total fixed costs remain constant throughout the relevant range.
d. Variable costs per unit vary proportionally with volume.
e. Selling price does not change as volume changes.
f. There is only one product or, in the case of multiple products, the sale mix remains constant.
g. There will be no change in the general price level.
h. Productivity per worker remains mostly unchanged.
i. There is synchronisation between production and sales, i.e., no inventory
j. Revenue and costs are being compared with a common activity base, e.g., sales value of production or units produced.
k. The efficiency of the plant can be predicted.

A change in any one of the above factors will alter the break-even point so that profits are affected by changes in factors other than volume. Thus, the break-even chart must be interpreted in the light of the limitations of underlying assumptions, especially concerning price and sales mix factors.

### 10.2.2 Computation of Break-Even-Point, BEP

BEP can be computed through different mathematical approaches. However, the two most popular mathematical approaches are: (i) the equation technique or (ii) the contribution margin technique.

(i) *The equation technique:* This technique uses the following formula, which also expresses the relationship of the items of the income statement.

Sales at BEP = Variable Expenses + Fixed Expenses + Zero Profit
**or** TR = TC + Zero Profit.

This simple equation may be adapted to any break-even or profit estimate situation.

(ii) *Contribution margin or marginal income technique: This* is based on the concept of marginal costing. Contribution margin is the difference between sales and variable expenses. Where the break-even point is desired, sales and expenses are analysed as thus:

Unit selling price - Unit variable expenses = Unit contribution.
And, Contribution – Fixed Cost = Profit

This unit contribution margin is divided by total fixed expenses to determine the number of units which have to be sold to break even.

i.e. $\frac{\text{Fixed Expenses}}{\text{Contribution per unit}}$

These two techniques can be illustrated by an example.

**Illustration 10.1: Mathematical Approach**

Alhaji Gazali plans to sell toy rockets at the Kano International Trade Fair. He may purchase these rockets at ₦5 each, with the privilege of returning all unsold rockets. The booth rent at the fair is ₦2,000 payable in advance. The rockets will be sold at ₦9 each.

Determine the number of rockets which must be sold to break even, as well as the number of rockets to be sold to yield a 20 per cent operating margin on sales.

(a) **Equation Technique**

Sales = Variable costs + Fixed Costs + Profit.

Assuming that A is the number of units to be sold to break-even, the values in the above formula can be substituted as follows:

9A = 5A+ ₦2,000 + 0
9A- 5A = ₦2,000
4A = ₦2,000
A = ₦2,000/4
A = 500 units

(b) **Contribution Margin Technique**

Here, BEP is determined as:

FC÷ Contribution per unit

From the illustration, FC = ₦2,000

Unit contribution = Unit sales price – Unit variable cost i.e.

Unit contribution = N9 – ₦5 = ₦4 per unit

Then, BEP in the unit =

$$\frac{\text{Fixed Costs}}{\text{Contribution per unit}}$$

= N2000 ÷ 4 or 500 units

**Determination of production volume or production units that will realise a 20% profit margin:**

Sales = Variable costs + Fixed costs + Profit

Assuming that X is the number of units to be sold to yield the desired profit, the values in the above formula can be substituted as:

9X = 5 X + ₦2,000 + .2(9 X)

9X = 5 X + ₦2,000 + 1.8 X

9 X - 1.8 X – 5 X = ₦2,000

2.2 X = ₦2,000

X = 909.09 units approx. Or X =

910 units.

Other simple formulae that can be used to determine the cost-volume-profit relationship are as follows:

i. Break-even point (in ₦)

$$= \frac{\text{Fixed Cost}}{\text{Contribution Margin Ratio, CMR}}$$

OR C/S Ratio

Note that C/S Ratio is also termed as Profit-Volume Ratio, i.e P/V Ratio

Where:

$$\text{C/S ratio} = \frac{\text{Contribution per unit}}{\text{Sales price per unit}} \times 100$$

ii. Level of sales to result in target profit (in units)

$$= \frac{\text{Fixed Costs + Target Profit}}{\text{Contribution per unit}}$$

Note: This formula was applied above to solve for the production volume required to earn 20% profit margin.

iii. Level of sales to result in target profit (in ₦ or in monetary value)

$$= \frac{\text{Fixed Costs + Target Profit}}{\text{Contribution per unit}} \times \text{sales price/unit}$$

Or

$$\frac{\text{FC + Target Profit}}{\text{C/S Ratio or P/V Ratio}}$$

iv. Level of sales to result in target profit after tax (in units)

$$\text{Target Profit after Tax} = \frac{\text{Fixed costs + Profit (1- tax rate)}}{\text{Contribution per unit}}$$

Note that the Profit specified in the above equation is termed as Profit before tax

**NB:** The above formulae apply only to a single-product firm or one with a constant mix of sales. In case of a multi-product firm, the break-even point is calculated as follows:

$$\text{Break-even point} = \text{Total } \frac{\text{Fixed Cost for all the products}}{\text{Weighted C/S Ratio } \textbf{or} \text{ P/V Ratio}}$$

Where Weighted P/V Ratio =
Total contribution realised for all products ÷ Total sales revenue earned on all the products sold.

### 10.2.3 **Profit-Volume Ratio, P/V Ratio**

Profit Volume Ratio, P/V Ratio or Contribution Margin Ratio, CMR, establishes the relationship between contribution and sales revenue. It determines the profitability of a product. It can be calculated in different ways below:

1. $\text{P/V Ratio} = \frac{\text{Unit Contribution}}{\text{Unit Sales Price}}$, i.e $\frac{S - V}{S}$

2. $\text{P/V Ratio} = \frac{\text{Total Contribution}}{\text{Total Sales Revenue}} = \frac{TSR - TVC}{TSR}$

Where: S indicates unit contribution,
V indicates unit variable cost
TSR indicates total sales revenue,
TVC indicates total variable cost

3. $\text{P/V Ratio} = \frac{FC + Profit}{\text{Total Sales}}$

**Illustration 10.2: Another Mathematical presentation**
Time Dey Nigeria Limited manufactures and sells a unique product, the selling price of which is ₦20.

The summarised profit and loss statement for last year is as follows:

| | ₦ | ₦ |
|---|---|---|
| Sales | | 800,000 |
| Direct materials | 120,000 | |
| Direct wages | 160,000 | |
| Variable production overhead | 80,000 | |
| Fixed production overhead | 100,000 | |
| Fixed Administration overhead | 75,000 | |
| Fixed Selling and Distr. Overhead | 60,000 | 595,000 |
| Net Profit before tax | | 205,000 |
| Less: provision for taxation (40%) | | 82,000 |
| Net Profit after tax | | 123,000 |

**Required:**

a. Calculate the break-even point for last year.

b. What do you understand by the terms 'profit volume ratio' and 'margin of safety'. Illustrate using last year's result.

c. Determine the number of units to sell in the current year to achieve an after-tax profit of

₦150,000

d. Calculate the sales value required to achieve a net profit before tax of 15% of total revenue.

e. Assuming no change in unit selling price and cost structure, calculate the percentage increase in sales volume required in the current year to produce a profit before tax 20% higher than last year's results.

f. Calculate the selling price per unit that the company must charge in the current year to cover a potential increase of 12% in variable production costs this current year and still maintain last year's contribution margin ratio.

g. Determine the volume of sales (in ₦) that the company must achieve in the current year to maintain the same net profit of last year, if the selling price remains at ₦20 and variable cost per unit increases by 12%.

h. Recalculate last year's result if salesmen commission of 10% is introduced, selling price is reduced by 13% and volume increases by 30%.

**Solution 10.2**

i. (a) Break-even point (in ₦)

B. E point = Fixed Costs x selling price/unit
Contribution/unit

= ₦235,000 x ₦20
11

= ₦427,273

Note that P/V Ratio = Unit or Total Contribution
Unit or Total Sales Price

Or 11 ÷ 20

(b) Break-even point (in units)

= Fixed costs
Contribution/unit

= N 235,000
11

= 21,364 units

**Workings:**

1. Calculation of fixed costs:

| | ₦ |
|---|---|
| Production overhead | 100,000 |
| Administration overhead | 75,000 |
| Selling & Distr. Overhead | 60,000 |
| | 235,000 |

2. Units of Production = N800,000 ÷ Unit S.P
= N800,000 ÷ 20 = 40,000 units

3. Unit VC = Total VC ÷ 40,000 units
= N360,000 ÷ 40,000 units = ₦9 per unit

4. Contribution per unit:

Selling price per unit - Variable cost per unit

= N20 - N9 = ₦11

ii. (a) A profit volume (P/V) ratio indicates the relationship between contribution and revenue. It is otherwise referred to as the contribution margin ratio (CMR) or the contribution sales ratio (CSR).

It is calculated in the following ways:

1. P/V ratio = [ SP– VC] ÷ SP
2. P/V ratio = CM ÷ SP
3. P/V ratio = $\frac{TS - TVC}{TS}$
4. P/V ratio = [ Total Contribution] ÷ Total sales
5. P/V ratio = [ FC + Profit ] ÷ Total Sales

Calculation of P/V ratio:

P/V ratio = $\frac{CM}{SP} = \frac{₦11}{₦20} = 0.55$

**Interpretation**

The P/V ratio of 0.55 means that for every one ₦ sale generated, 55 kobo profit is generated.

10.2.4 **The Concept of Margin of Safety**

**Margin of Safety, M/S**

Margin of safety is expressed as the amount by which forecast revenue exceeds or falls short of that required to break even (CIMA). It is the difference between the forecast sales at the profit level and those of sales at the break-even point. It is expressed mathematically as:

i. M/S = Sf - Sb (in monetary value)

Where:

$S_f$ = forecast sales

$S_b$ = sales at the Break-even point

ii. M/S Ratio:

$$\left\{\frac{S_f - S_b}{S_f}\right\} \times 100$$

**Calculation of Margin of Safety from Illustration 9.2**

| | | ₦ | Units |
|---|---|---|---|
| Sales | (B) | 800,000 | 40,000 |
| Break-even point | | 427,273 | 21,364 |
| Margin of safety | (A) | 372,727 | 18,636 |
| in percentage for | **or** | | |
| M/S Ratio: (A/B x 100%) | | 47% | 47% |

$M/S = S_f - S_b$ = ₦800,000 - 427.273 = ₦372,727

The margin of safety of 47% means that our sales must fall by more than 47% before we sustain aloss.

The level of sales required to achieve a target profit (after tax) of ₦150,000 ( in units). The answer is given by the Target profit

$$= \frac{\text{Fixed cost} + 150{,}000\ (1-\text{rate})}{\text{Unit Contribution}}$$

$$= \frac{₦235{,}000 + 150{,}000\ (1-0.4)}{11}$$

$$= \underline{44{,}091}\ \text{units}$$

iv. Level of sales to result in net profit before tax of 15% of total revenue.

$$Y = \frac{\text{Fixed Cost + Target profit}}{\text{Contribution margin ratio}}$$

$$Y = \frac{235{,}000 + 0.15Y}{0.55}$$

$0.55Y = ₦235{,}000 + 0.15Y$

$0.55Y - 0.15Y = ₦235{,}000$

$0.40Y = ₦235{,}000$

$$Y = \frac{₦235{,}000}{0.4}$$

= **₦587,500**

**Workings:**

Contribution margin ratio is determined as follows:

$$= \frac{\text{Selling price - Variable cost}}{\text{Selling price}}$$

$$= \frac{[20 - 9]}{20}$$

Or $11 \div 20 = 0.55$

The level of sales required to result in the target profit.

$$= \frac{\text{Fixed costs + Target profit}}{\text{Contribution per unit}}$$

$$= \frac{₦235{,}000 + (120\% \text{ of } 205{,}000)}{11}$$

$$= \frac{₦235{,}000 + ₦246{,}000}{11}$$

$$= \underline{\underline{43{,}727 \text{ units}}}$$

Percentage increase:

$$= \frac{43{,}727 \text{units} - 40{,}000 \text{ units}}{40{,}000 \text{ units}} \times 100\%$$

$$= \underline{\underline{9.32\%}}$$

vi. Contribution margin ratio of last year = Contribution margin ratio of this year

$$0.55 = \frac{\text{Revised SP- Revised VC per unit}}{\text{Revised SP.}}$$

$$0.55 = \frac{RSP - RVC}{RSP}$$

$$0.55 = RSP - \frac{(9 \times 1.12)}{RSP}$$

$$0.55 = \frac{RSP - 10.08}{RSP}$$

$$0.55 \, RSP = RSP = 10.08$$

$$RSP - 0.55 \, RSP = 10.08$$

$$0.45 \, RSP = 10.08$$

$$RSP = 10.08 \div 0.45$$

$$RSP = ₦22.4$$

$$\text{Confirmation P/V} = \frac{RSP - RVC}{RSP} = \frac{22.4 - 20.08}{22.4}$$

$$= 12.32 \div 22.4 = 0.55$$

Note:

Variable Production Cost

| | |
|---|---|
| DM | 120,000 |
| DW | 160,000 |
| VPO | 80,000 |
| Variable Production Cost | 360,000 |
| Unit VPC | 360,000 ÷ 40,000 = ₦9 per unit |

vii. Sales value required to achieve a target profit of ₦205,000

$$= \frac{FC + TP}{\text{Contribution/unit}} \times \text{Unit selling price}$$

$$= \frac{₦235,000 + 205,000}{9.92} \times 20$$

= ₦887,097

**Workings:**

Contribution per unit is determined as follows:

= SP - RVC/U
= ₦20 - (₦9 x 1.12)
= ₦20 - ₦10.08 = ₦9.9

viii. Revised Income Statement

| | ₦ | ₦ |
|---|---|---|
| Sales ((₦20 x 0.87) x (40,000 x1.30)) | | 904,800 |
| Less: Variable Cost (52,000 units x 9) | 468,000 | |
| Salesmen comm. (10% of 904,800) | 90,480 | 558,480 |
| Total Contribution | | 346,320 |
| Less: Fixed Costs | | 235,000 |
| Net Profit before tax | | 111,320 |
| Less: 40% tax | | 44,528 |
| Net Profit after tax | | 66,792 |

10.2.5 **C.V.P Graphs**

In addition to determining break-even value and P/V Value mathematically, some business organisations usually at the same time to enable managers to appraise the business profitability at a glance.

Thus, it removes the danger accompanying many accounting reports, a danger that the report would get bogged down with unnecessary details in such a way that managers may never come to grips with the heart of the matter.

Generally, C-V-P graphs can be split into different component graphs. However, this study pack only covers two types of C-V-P graphs, which include:

i. Break-Even Point Graph, and

ii. Profit-Volume, P/V Graph

### 10.2.5(a) The Break-Even Point Graph

A break-even chart shows the level at which the total cost incurred for a product equates total revenue realised. The break-even point is obtained where the total cost graph intersects with the total revenue graph. The costs and revenues are indicated in the Y-axis, while output in units is indicated in the X-axis.

The total cost line is obtained from:

Total Fixed Cost + Variable Cost at different production units

While:

The total revenue line is obtained from:

Total revenue at each level of output, which is the product of output multiplied by the unit selling price.

**Illustration 10.3: Graphical demonstration**

A company makes and sells a single product. The variable cost of production is ₦3 per unit, and the variable cost of selling is ₦1 per unit. Fixed costs total ₦6,000, and the unit selling price is ₦6. The company budgets to make and sell 3,600 units in the next years.

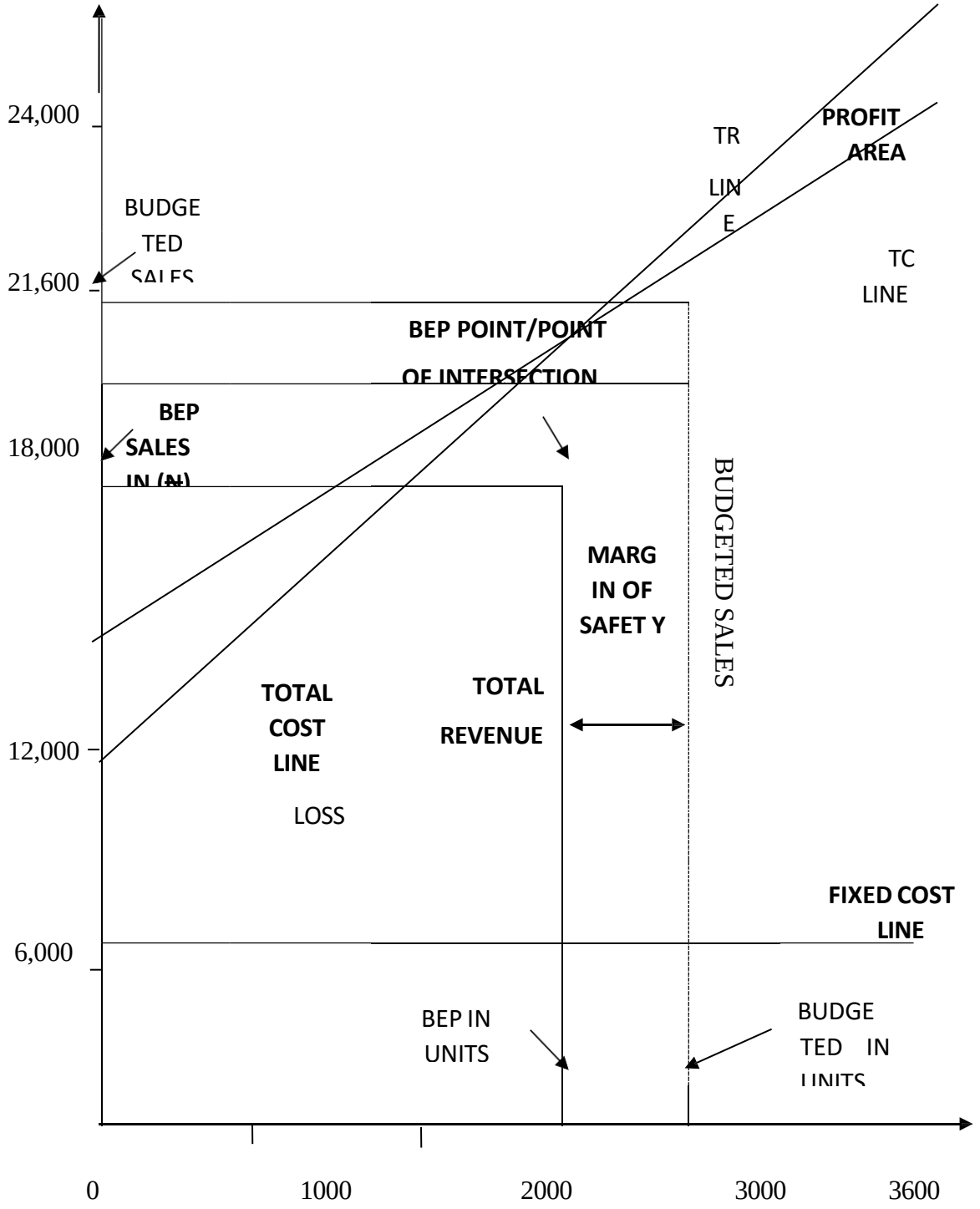
**Required**: A break-even chart and later to plot (i)Profit-Volume graph, with each graph showing the expected amount of output and sales required to break even, and the safety margin in the budget.

**Solution 10.3 (i) Break-Even Chart**

The break-even chart shows:

- Fixed cost line of ₦6,000,
- Total costs line, with variable cost added to fixed cost line, and
- Total revenue line

**Figure 10.1: Break-Even Chart**




Note:

i. At BEP, TC Line = TR Line = Point of intersection
= 3,000 units sold for ₦18,000
At BEP, Profit = Zero

ii. M/S from.The graph:

| | Revenue in: | |
|---|---|---|
| | ₦ | Units |
| Budgeted | 21,600 | 3,600 |
| Break-Even Point | 18,000 | 3,000 |
| Margin of Safety | 3,600 | 600 |

Thus, the sales revenue can be fall by ₦3,600 sales revenue or by 600 units and the firm can still break even with zero profit.

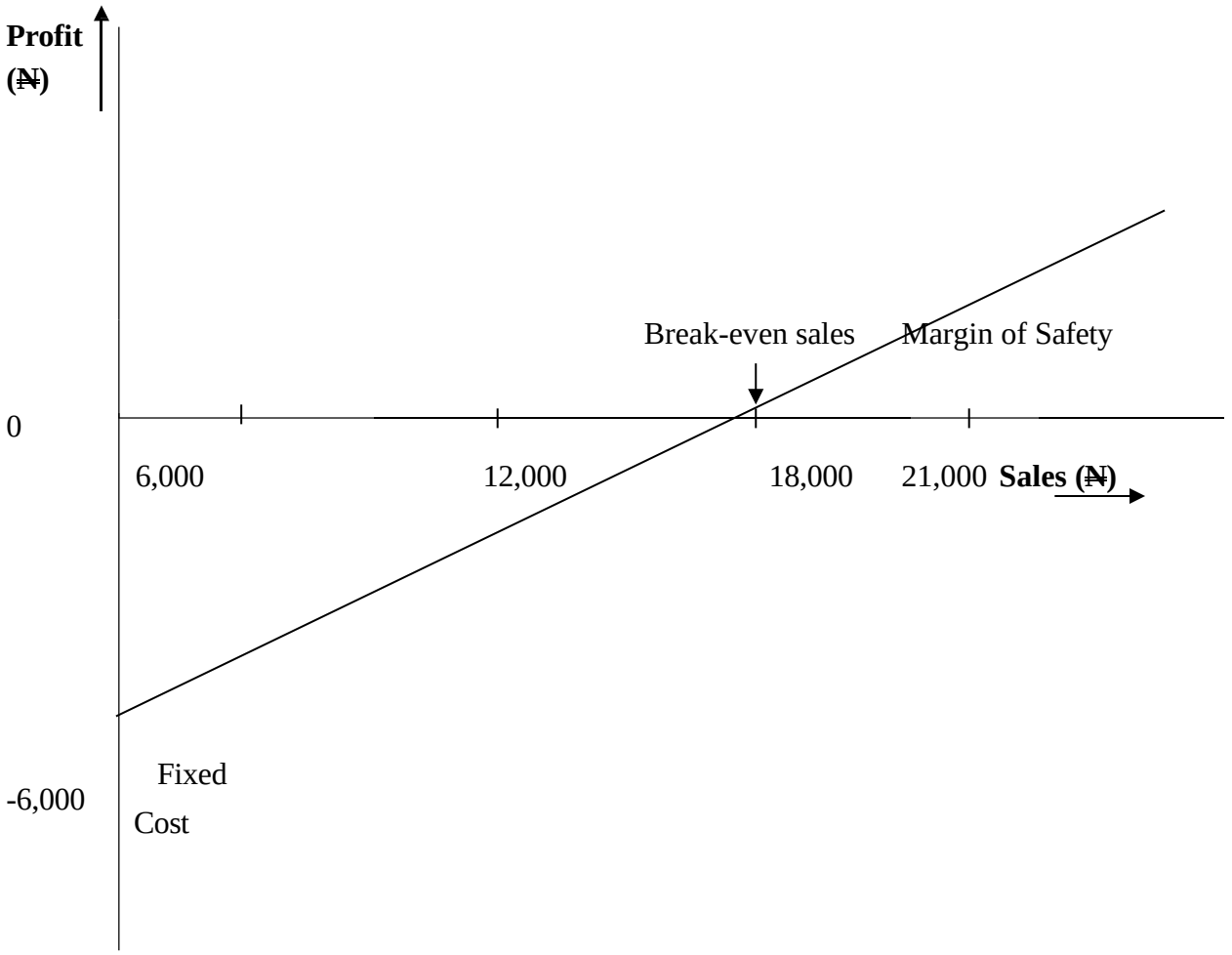
### 10.2.5(b) The Profit-Volume Chart, P/V Chart

The P/V graph establishes relationship that exists between sales revenue and profit at different level of activities.

In plotting P/V graph:

i. Determine break even sales in monetary value, having zero profit at x-axis.
ii. Determine the loss incurred when there is no production and no sales. This is equal to the total fixed cost.
iii. Join the two points with a straight line.

**Figure 10.2: P/V Chart**




**From the graph:**

i. Break-even sales at zero profit = ₦18,000

ii. Margin of Safety = N21,600 – ₦18,000 = ₦3,600 i.e $S_f - S_b$
or 16.67% of budgeted revenue

### 10.2.6 Limitations of Break-Even Analysis

The limitations of this concept stem from its numerous underlying assumptions.

a) The postulation that fixed costs will remain constant at all activity levels is a faulty one. This is because fixed costs are likely to vary at different activity levels, thus giving preference to the use of a stepped fixed cost as the most accurate representation of break-even analysis

b) The outcomes of the analysis can only be counted on within the activity level, where all related costs can be correctly determined.

c) The cost-volume-profit analysis represents short-term relationships, and for this reason it is considered inappropriate in long-term relationships.

d) The postulations that variable cost and sales will be linear appear incorrect, seeing that the effect of extra discount, overtime payments, learning curve, special price contracts, and other similar matters present variable cost and revenue as a curve rather than a straight line

e) Break-even analysis assumes a perfect knowledge of cost and revenue function, thereby ignoring risk and uncertainty, which in practice remains a crucial factor in the economic decision-making process.

f) The concept of CVP analysis relies heavily on cost behavior and classification. This implies that activity level determines the changes in revenue and costs, this assumption sounds so good theoretically. There are numerous factors, in practice, that will influence changes in cost and revenue in addition to their activity levels.

g) Revenue and variable cost do vary with the level of activity nonetheless the reaction of individual cost components such as material cost, labour cost, etc are not the same

h) CVP analysis unrealistically assumes a single product /constant product mix/ constant rate of markup on marginal cost. This is considered an oversimplification of the concept.

i) The concept relies on a wrong assumption that a perfect market exists and that the firm is a price taker

## 10.3 CHAPTER SUMMARY

In this chapter efforts have been made to focus on cost-volume profit analysis in decision making. The chapter has also discussed the concept of break-even point, the P/V ratio and margin of safety. At the same time, the chapter illustrated the determination of BEP points, P/V ratio and margin of safety through mathematical formulae, and finally plotted break-even point chart as well as profit-volume chart.

**MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS**

1. Which one of these is NOT an assumption of Break-even analysis?
   A. Fixed costs remain constant
   B. Variable costs vary proportionally with volume
   C. There is no synchronization between production and sales
   D. Costs can be resolved into their fixed and variable components
   E. Selling price does not change as volume changes

2. What is margin of safety?
   A. The amount by which actual or budgeted output/sales may fall short of the budget without incurring a loss
   B. The amount by which budgeted output/sales may fall short of the actual without incurring a loss
   C. The point where actual output/sales agree with the budget
   D. The area immediately preceding the break-even point
   E. The area where fixed assets are equal to variable costs

3. What happens at the area before the break-even point?
   A. No profit, no loss
   B. Marginal profit
   C. Huge profit
   D. Loss
   E. No revenue

4. The point at which total revenues equal total costs is known as ……..
5. The number of units of products to be sold to be able to break-even is given by the formula .....
6. Break-even point in Naira can be calculated by multiplying the break-even point in units with ……

**SOLUTION**

1. C
2. A
3. D
4. Break-even point
5. $\frac{\textit{Fixed Costs}}{\textit{Contribution Per Unit}}$
6. Unit selling price

## EXAMINATION TYPE QUESTIONS

9.1 An electrical goods manufacturing company made a 10% profit on sales of ₦ 1 million in the last trading year. The composition of its costs was direct labour 25%, direct materials 60% and fixed overhead 15%.

The general manager has drawn your attention to the fact that, although the sales were just below forecast, the profit was very much lower than he had expected.

Your initial investigation shows that the significant difference appears to be caused by the direct labour costs. The company uses the marginal cost accounting principle to price its products. In all price quotations the direct labour was treated as a variable cost directly related to volume of output. However, the review indicates that the direct labour cost showed little change when output decreased for any reason.

You are required to:

(a) state whether you agree that direct labour costs should be treated as wholly variable with output.

(b) calculate, using the last year's results, the sales volume at break-even point and the margin of safety when the direct labour costs are treated as:

(i) wholly variable

(ii) fixed.

9.2 The following figures relate to Prudent Nigeria Ltd which manufactures varied range of products:

| Year | Total Sales | Total Costs |
|---|---|---|
| | ₦ | ₦ |
| 1 | 39,000 | 34,800 |
| 2 | 43,000 | 37,600 |

**Required:**

Assuming stability in prices, with variable costs carefully controlled to reflect pre-determined relationships, and an unvarying figure for fixed costs.

(i) Calculate:

(a) The fixed cost

(b) The profit/volume ratio

(c) The break-even point

(d) the margin of safety for years 1 and 2

(ii) How might the profit/volume ratio be improved, apart from increasing prices or reducing costs? 9.3

(a) Cost-Volume-Profit (C-V-P) analysis is a technique available to management to understand better the inter-relationship of several factors which affect a firm's profit. As with many such techniques, the accountant over simplifies the real world by making certain assumptions. Mention five major assumptions underlying C-V-P analysis

(b) Yakasai Toys Company's tentative budget for product Q for 2004 is as follows:

| | ₦ | ₦ |
|---|---|---|
| Sales (2,500) @ ₦40 | | 100,000 |
| Manufacturing cost of goods sold: | | |
| Direct labour | 15,000 | |
| Direct materials | 14,000 | |
| Variable factory O.H. | 10,000 | |
| Fixed factory O.H. | 5,000 | |
| Total manufacturing costs | | 44,000 |
| Gross Profit | | 56,000 |
| Selling expenses: | | |
| Variable | 6,000 | |
| Fixed | 10,000 | |
| Admin. expenses: | | |
| Variable | 5,000 | |
| Fixed | 10,000 | |
| Total selling & Admin. Exps. | | 31,000 |
| Operating Income | | 25,000 |

**Required:**

(i) How many units of product Q would have to be sold to break-even?

(ii) What would the operating income be if projected sales increased by 30%?

9.4 Universal Cloth Manufacturing Limited has recently rented a new factory and intends to produce a newly developed man-made fibre "Texlon". The company previously had two other large mills. The combined installed capacity of the three mills will enable the company to provide a wide range of products to its numerous customers.

It has been planned to produce "Texlon" in the new mill, first on a single shift and six months later, a double shift will be introduced. Three months thereafter, a treble shift becomes operational. The production, cost and selling price per unit are as follows:

| | Single Shift | Double Shift | Treble Shift |
|---|---|---|---|
| Production/ week (units) up to | 1,500 | 1500-2,500 | 2501-4,000 |
| Selling price/unit | ₦5 | ₦5 | ₦5 |
| Variable cost/unit | ₦3.00 | ₦3.20 | ₦3.50 |
| Fixed cost/week | ₦2,000 | ₦2,520 | ₦3,030 |

The managing director has requested the following information:

i. The break-even units per week of each shift.

ii. What is the total unit cost for each shift at break-even and at full capacity?

iii. What is the profit/loss for each shift at 50 per-cent of the installed capacity (or increased capacity)?

| | |
|---|---|
| Single shift | 750 units |
| Double shifts | 2,000 units |
| Treble shifts | 3,250 units |

iv. What is the profit per week at full capacity for each of the three shifts and the unit variable and fixed cost?

9.5(a) Explain the term 'break-even point'. Give an example illustrating your answer. Also mention the types of problems which an accountant can expect to solve with the help of such analysis.

(b) Draw a break-even chart with a few illustrative figures.

(c) Explain the Cost-Volume-Profit relationship. How would a change in the selling price affect it?

9.6 You own the Finance Manager of a manufacturing company based in Gusau. In the past year, the company has been facing very difficult times and incurring trading losses, due to unfavourable fiscal policy and mismanagement. Your new Managing Director, a young intelligent man who believes that all problems have a solution, has just returned from a workshop in the UK on Finance for Non-Finance Managers. He was told in the workshop that one of the main benefits of the Profit/Volume Ratio is that it could be used to calculate the effect of changes in volume or costs on profit. The MD is convinced that this concept will be very helpful for the dynamic marketing strategy which the company is presently considering.

The following details are available:

₦

| | |
|---|---|
| Selling price per carton of product | 1,200 |
| Variable cost per carton | 1,000 |
| Variable distribution expenses per carton | 50 |
| Annual fixed expenses: | |
| Storage | 200,000 |
| Wages | 100,000 |

An Exco meeting has been scheduled for next week, and you have been requested by your MD to consider the following scenarios and provide computations on your spreadsheet showing the company's position under each scenario. Please remember to consider each case independently.

(i) The Marketing Director thinks that by spending ₦20,000 monthly on advertising, he can achieve annual sales of 5,000 cartons. What will the profit be under this scheme?

(ii) The wages are paid to the Marketing Field Force that sells the products. They have now offered to forgo these wages in exchange for a ₦25 commission for every carton sold as an alternative to the Marketing Director's proposal. If this proposal is accepted, what would be the annual break-even point in sales revenue and in unit sales?

(iii) In the company plan, the target profit is ₦600,000. How many cases must be sold to achieve this figure (ignore the Marketing Director's proposal):

(a) Assuming the Marketing Field Force received their ₦100,000 salaries?

(b) Assuming the Marketing Field Force is paid commission?

(iv) What level of sales produces a profit figure which is the same whether the Marketing Department is paid by salary or by commission? (Ignore the Marketing Director's proposal)

*(ICAN Intermediate, Nov., 2000)*

9.7 ABC Ltd manufactures one product only, which it sells at ₦20 per unit. The existing plant has a maximum capacity of 20,000 units per annum, at which level net profit is ₦1.50 per unit and the profit/volume ratio is 20%. A new plant is to be purchased, having a maximum capacity of 30,000 units per annum, but this will result in fixed costs being increased by ₦15,000 per annum. Variable costs will be reduced by ₦4 per unit, and to achieve an increase in sales, the selling price is also to be reduced by ₦4 per unit.

You are required to:

(a) Explain what you understand by the terms profit/volume ratio and margin of safety.

(b) Calculate the revised P/V ratio as a result of the purchase of the new plant.

(c) Calculate the number of units which would be required to be produced using the new plant to give 50% increase in profit compared with the maximum production using the old plant.

(d) Calculate the margin of safety if the actual level of sales is 25,000 units, assuming the new plant is purchased. *(PE I - May, 2001)*

9.8 Briefly describe the following forms of pricing:

(i) Total cost-plus profit percentage

(ii) Marginal cost-plus contribution percentage

(iii) Break-even pricing

(iv) Negotiated pricing

(v) Competitors.

9.9 **Electrical Appliances Ltd records and sells video films at ₦30 each. The result for 1993 and 1994 are summarised below:**

| Sales | Units | Profit |
|---|---|---|
| 1993 | 6,800 | ₦23,800 |
| 1994 | 8,400 | ₦39,400 |

In 1993 and 1994, the company operated at an average of 80% of the available capacity.

In 1995, the company had budgeted to operate at 100% capacity. In 1996, the company proposed to acquire additional equipment worth ₦150,000. The equipment will have an effective life of 10 years with no residual value. With the introduction of the new equipment, the company proposes to operate at 150% of its present capacity. Variable costs will decrease by 10%, and the selling price will be reduced by 5% to enable all units produced to be sold. Using the above information, you are required to calculate:

(a) The break-even point of the company for 1995

(b) The forecast break-even point of the company for 1996

(c) The forecast profit of the company for 1995 and 1996; and

(d) The expected rate of return on the additional capital employed in 1996.

*(ICAN P E II, May/June, 1995)*

9.10 Tom Limited manufactures a single product. The unit selling price is fixed at ₦40.00 per unit.

Other data for the 2017 budgeted year are as follows:

| | ₦ |
|---|---|
| Sales | 600,000,000 |
| Variable Cost: | |
| Production Cost: ₦16 per unit | |
| Others ₦8 per unit | |
| Total Fixed Cost | 40,000,000 |

Management is of the view that the budget preparation is realistic, provided all the departments co-operate with each other. The budget for 2017 is 80% of the 2016 budget because of the recession in the economy.

**Required:**

**a.**

ai. Define the principal budget factor.

aii. What is the principal budget factor for 2017

a. Produce a Profit-Volume graph for the 2017 budget financial year

b. From your graph, determine:

i. Break-Even Point

ii P/V Ratio

iii. Margin of Safety

iv. Profit when sales are ₦160,000,000

(12½ Marks)

## CHAPTER 11

## SIMPLE INVESTMENT APPRAISAL TECHNIQUES

**CONTENTS**

a. Capital Budgets and Decision Making; and
b. Capital Budgeting Methods

### 11.0 Objectives:

After studying this chapter, readers should be able to:

a) appreciate the concept of capital budgeting;
b) explain the nature and role of capital budgeting decisions;
c) identify and discuss the traditional methods of appraising investment and one of the Discounted Cash Flow (DCF) techniques (NPV); and
d) appreciate the necessity of capital budgeting in long-term decision making.

### 11.1 INTRODUCTION

#### 11.1.1 The Concept of Capital Budgeting

Capital budgeting is the planning of expenditure whose returns extend beyond one year; it involves deciding whether or not to commit resources to a project whose benefits would be spread over several periods. It evaluates proposed capital expenditures and their funding. The primary task in capital budgeting is to correlate benefits with costs in a rational manner, aligning with the profit-maximising goal of the enterprise. Capital budgeting decisions are among the most critical managerial choices, necessitating estimation and forecasting that demand a high level of intellectual capacity for economic analysis.

Capital budgeting entails making judgments on projects with long gestation periods. These projects may consist of tangible or intangible assets whose advantages will persist beyond the fiscal year. The choice of projects to finance for long-term benefits is solely a managerial decision.

#### 11.1.2 The Objectives of Capital Budgeting

When expenditures are incurred and benefits persist over an extended duration, several significant consequences arise, which provide the basis for capital planning.

Initially, by undertaking a capital investment, the decision-maker dedicates resources to the future, thereby diminishing their financial flexibility. Consequently, the acquisition of an asset with a decade-long economic lifespan necessitates an extended period of anticipation before the outcomes materialise. Once the benefits commence, the organisation will be substantially recompensed for the initial investment.

Secondly, asset expansion correlates with future sales, and the economic lifespan of the acquired asset serves as an indirect projection of sales throughout its economic duration. Inaccurate forecasting could lead to either over-investment or under-investment in fixed assets. A flawed assessment of asset requirements may have significant repercussions for the firm. Excessive investment in fixed assets may impose unnecessary financial burdens, while insufficient expenditure could hinder productive operations due to inadequate capacity.

Thirdly, effective capital planning would enhance the timing of asset acquisition and elevate the quality of assets procured. This is due to the nature of demand for and supply of capital goods. The former does not arise until sales impinge on productive capacity, and such situations occur only intermittently; on the other hand, production of capital goods involves a relatively long period, so that the business would ordinarily have to wait for about a year or so before new capital goods become available. Matching the need for capital goods with their availability is one of the functions of capital budgeting.

Finally, asset expansion requires substantial funds, which might not be immediately and automatically available. Therefore, a determined effort would have to be made to procure them. Naturally, a capital expenditure programme of a sizeable amount would entail finance arrangements well in advance to ensure their availability at the right time.

Thus, the size of funds involved in asset expansion and the expenditures are designed to be recovered in the future, which is distant and seemingly imperceptible, making capital budgeting one of the most critical, delicate and sometimes complex managerial decisions. This is so because, so many variables, e.g., changes in the quality of the product, changes in the quantity of output, changes in the quality and quantity of direct labour, changes in the amount and cost of scrap, changes in the maintenance expenses, down-time, safety, flexibility, etc., are involved in capital budgeting decisions.

Capital budgeting decision is not a routine clerical task to be performed on a mechanical basis, though electronic equipment has facilitated the same to some extent. It requires continuous monitoring and evaluation of a conglomeration of factors by engineers, cost analysts, economists and other individuals in an undertaking who are competent to make such an evaluation.

### 11.1.3 Data for Capital Expenditure Decisions

The type of costs required in capital investment decisions differs from those required for accounting purposes because only future costs are relevant for such decisions. Recorded costs may be useful in investment decisions, but only to the extent that they furnish a starting point for future cost projections.

All estimated costs pertinent to the project under consideration should be included; any expected savings in material costs, particularly those arising from an expected reduction in scrap, should be reflected; prospective changes in direct labour, materials, handling, inspection, etc., should be reckoned; anticipated increases or decreases in specific overhead costs, such as taxes, insurance, power, maintenance, repairs, supplies, etc., must be considered.

The use of the plant overhead rate should be avoided. Very often, new equipment is more automatic than the old, and a different cost pattern will emerge. Thus, there is a reduction in supervision, overtime premiums and other overhead costs, which tend to vary with direct labour, while there is an increase in repairs, maintenance, power and other costs varying with machinery usage.

### 11.1.4 Opportunity and Interest Costs

Opportunity cost plays an important role in capital budgeting decisions. It represents the loss of alternative income as a consequence of the actions adopted. For example, in an expansion project, the economic rather than the book value of the space required for expansion should be taken into account against a proposed investment. In a replacement decision, the realisable value of the existing asset should be treated as a reduction of the cost of replacement.

Accounting reports and statements typically give recognition to contractual interest but ignore imputed interest on capital. While the inclusion of interest is indispensable in investment studies, the determination of an appropriate rate presents difficulties. Interest is sometimes misunderstood as return on investment, which consists of two elements: interest and profit.

The former represents the cost of money, while the latter is the reward for risk and uncertainty. Interest cost constitutes the minimum acceptance criterion for capital investment projects undertaken for profit. A firm must at least recover its costs before it can realise a profit on its investment. On the other hand, the minimum acceptance criterion that can be considered as a reward for risk and uncertainty varies with the nature of the assumed risk.

Depreciation is another cost whose treatment in capital investment analysis differs from other cost accounting reports and analyses. In calculations designed to reveal the desirability of replacing existing machinery and equipment, depreciation of the unabsorbed book value of an existing asset is a sunk cost and is not relevant except for its tax effects. It is only the economic value of the assets that has relevance in replacement decisions.

In capital project decisions, differential revenue, where measurable, cannot be ignored. There are two facets to this - the potentiality or capacity of the asset under consideration and the marketability of increased output.

## 11.2 CAPITAL BUDGETING METHODS

In most business firms, there are more proposals for capital projects than the firm is able and willing to finance. Some proposals are good while others are bad. A screening process has to be devised for finding out the real content of such proposals; methods of differentiating between them have to be developed, and, for this purpose, a ranking procedure has to be evolved.
Essentially, the ranking procedure envisaged should relate a stream of future earnings to the cost of obtaining those earnings. Among the many methods of ranking investment proposals, the

The following are very widely used;

a) Payback Period method;
(b) Accounting Rate of Return method;
(c.) Net Present Value method;
(d) Profitability Index method; and
(e) Internal Rate of Return method.

### 11.2.1 Pay Back Period (PBP) Method

The payback, sometimes called pay-out or payoff method, represents the years required to return the original investments by savings before depreciation, but after the payment of taxes. It is about the time it takes a project to recoup its investment. Thus, the method attempts to measure the period it takes for the original cost of a project to be recovered from the project's earnings. If an investment of ₦100,000 earns ₦50,000 cash proceeds in each of its first two years, its PBP is two years.

$$\text{Pay Back Period} = \frac{\text{Cost of investment (Initial Outlay)}}{\text{Annual Cash Flow (Annual Cash Proceeds/ Savings)}}$$

$$= \frac{₦\ 100,000}{₦\ 50,000}$$

$$= 2 \text{ Years}$$

The available investments are then ranked according to the length of their payback periods, so that an investment with a payback of two years is bound to be considered more desirable than an investment with a payback of three years.

**Illustration 11.1**

Use the information provided in the table below to determine the most viable project between A& B, if N10,000 is to be spent on either of the projects.

| **Savings Before Depreciation Year** | **A** | **B** |
|---|---|---|
| 1 | ₦5,000 | ₦1,000 |
| 2 | 4,000 | 2,000 |
| 3 | 3,000 | 3,000 |
| 4 | 1,000 | 4,000 |
| 5 | - | 5,000 |
| 6 | - | 6,000 |

**Solution 11.1** **PROJECT A** **PROJECT B**

| Year | Cash flow ₦ | Cumulative Cash Flow ₦ | Cash flow ₦ | Cumulative Cash Flow ₦ |
|---|---|---|---|---|
| 0 | -10,000 | -10,000 | -10,000 | -10,000 |
| 1 | 1,000 | -9,000 | 5,000 | -5,000 |
| 2 | 2,000 | -7,000 | 4,000 | -1,000 |
| 3 | 3,000 | -4,000 | 3,000 | 2,000 |
| 4 | 4,000 | 0 | 1,000 | 3,000 |
| 5 | 5,000 | 5,000 | | |
| 6 | 6,000 | 11,000 | | |

PBP (A) = 4Years PBP (B)= 2 Years + 1000/3000
2. 33 Years

From the solution provided above, it is clear that project B would require less than three years to recover an original investment of ₦10,000 (its PBP is precisely 2.33 years) while project A would need a payback period of four years. From this point of view, Project B would be preferred over Project A.

**Illustration 11.2**

Koki Nig. Plc. is considering the purchase of a new machine which would carry out some operations that are presently performed by manual labour. The two alternative models under consideration are "Damsel" and "Shylock".

| | Machine 'Damsel' | Machine 'Shylock' |
|---|---|---|
| Cost of Machine | ₦300,000 | ₦ 500,000 |
| Estimated life (in years) | 10 | 12 |
| Estimated savings in scrap per annum | ₦20,000 | ₦30,000 |
| Additional cost of supervision per annum | ₦24,000 | ₦32,000 |
| Additional cost of maintenance per annum | ₦14,000 | ₦22,000 |
| Cost of indirect material per annum | ₦12,000 | ₦16,000 |

The following information, from which a profitability statement is to be prepared for submission to the Board of Directors, is available:

Estimated savings in wages:

| | | |
|---|---|---|
| Wages per worker per annum | ₦1,200 | ₦1,200 |
| workers not required | 150 | 200 |

**Required:**
The rate of taxation may be regarded as 50 per cent of profits. Which model can be recommended for purchase? Give reasons for your answer.
**Solution**

**Statement of Profitability**

| | **Machine 'Damsel'** | | **Machine ' Shylock'** | |
|---|---|---|---|---|
| | ₦ | ₦ | ₦ | ₦ |
| Estimate savings per year: | | | | |
| Scrap | 20,000 | | 30,000 | |
| Wages | 180,000 | 200,000 | 240,000 | 270,000 |
| Less: Estimated additional Costs per year: | | | | |
| - indirect materials | 12,000 | | 16,000 | |
| - Cost of supervision | 24,000 | | 32,000 | |
| - Cost of maintenance | 14,000 | 50,000 | 22,000 | 70,000 |
| Savings before dept. | | 150,000 | | 200,000 |
| Less depreciation | | 30,000 | | 41,667 |
| Savings after depreciation | | 120,000 | | 158,333 |
| less tax @ 50 percent | | 60,000 | | 79,167 |
| Net savings per year | | 60,000 | | 79,166 |
| Annual cash flow: | | | | |
| Net savings | 60,000 | | 79,166 | |
| Depreciation | 30,000 | 90,000 | 41,667 | 120,833 |

Based on the profitability statement and according to the payback period method, the board may be advised on the following lines:

| | Machine 'Damsel' | Machine 'Shylock' |
|---|---|---|
| Payback period (before tax) | 2 years | 2 1/2 years |
| Payback period (after tax) | 3.3 years | 4 years |

Thus, machine 'Damsel' should be recommended for purchase, because it has lower payback period as compared with machine 'shylock'.

Additional statistics about profitability beyond the payback period could support the information given and the conclusion reached.

| | Machine 'Damsel' | Machine 'Shylock' |
|---|---|---|
| Before tax | ₦ 960,000 | ₦ 1,266,664 |
| After tax | ₦ 720,000 | ₦ 966,664 |

Note: Approximations have been made wherever necessary.

Working notes

(i) Calculation of the payback period: Payback period = Cost of machine / Savings per annum

Therefore

| | **Machine Damsel** ₦ | | **Machine Shylock** ₦ | |
|---|---|---|---|---|
| Before | 300,000 | 2 Years | 500,000 | 2 ½ Years |
| Depreciation | 150,000 | | 200,000 | |
| After Tax | 300,000 | 3.3 Years | 500,000 | 4 Years Approx. |
| | 90,000 | | 120,833 | |

### 11.2.2 Advantages of Payback Period Method

a) It is easy to understand and simple to operate;
b) The short-term approach typified by the PBP method reduces the possibility of loss through obsolescence;
c) A firm which is short of cash must necessarily place a premium on the quick return of its funds; and
d) The payback period method uses project cash flows rather than accounting profit. This qualifies it to be more precise and objective than the accounting rate of return.

### 11.2.3 Disadvantages of the Payback Method

a) It ignores cash flows beyond the payback period, so that, if a project is one with a longer gestation period, the use of the method would lead to a wrong selection of investments. Such investments are characteristically involved in long-range planning, i.e., developing a new product, developing a new market, etc., requiring initial investments of a huge amount which do not yield their highest returns until several years later; and

b) The payback period method for ranking investment proposals fails to take into account the period over which an investment is likely to yield savings. In a way, it ignores the interest factor, which is a very important consideration in making sound investment decisions.

Despite its weaknesses, the payback period method is the most popular appraisal technique in practice. In developing economies, foreign investors, and even local investors, are more at home with this method than with any other method of investment appraisal.

### 11.3 Accounting Rate of Return Method

The Accounting Rate of Return method requires the addition of all the earnings after depreciation and dividing them by the project's economic life. When this figure of average earnings over the period is obtained, it is divided by the capital invested. It is, therefore, given by average return divided by average investment.

Capital invested may mean the initial capital investment or the average of the capital invested over the life of the project. It may also mean initial investment plus salvage value divided by 2.

**Illustration 11.3**

Gidauniya Nigeria Limited is considering two projects, each with an initial investment of ₦1,000. Project A has a life of four years, while Project B has a life of six years. The depreciation each year for projects A and B is ₦250 and ₦167, respectively. The profits realised by the projects are estimated to be as follows:

**Table 11.3**

| | **Project A** | | | **Project B** | | |
|---|---|---|---|---|---|---|
| | Earnings ₦ | Depr | Earnings after depr. ₦ | Earnings ₦ | Depr ₦ | Earnings after Depr. ₦ |
| 1 | 500 | 250 | 250 | 100 | 167 | (67) |
| 2 | 400 | 250 | 150 | 200 | 167 | 33 |
| 3 | 300 | 250 | 50 | 300 | 167 | 133 |
| 4 | 100 | 250 | (150) | 400 | 167 | 233 |
| 5 | - | - | - | 500 | 167 | 333 |
| 6 | - | - | | 600 | 167 | 433 |
| | | | 300 | | | 1,098 |

**Required:**

a. Calculate the accounting rate of return (ARR) on:

(i) Initial capital; and

(ii) Average capital

**Solution**

(i) Accounting Rate of Return on Initial Capital:

| | **Project A** | | **Project B** | |
|---|---|---|---|---|
| Average profits = | ₦300 | | ₦1,098 | |
| | 4 | 6 | | |
| | = N 75 p.a. | | =₦ 183 p.a. | |
| ARR = | 75 x 100% | | 183 x 100% | |
| | 1,000 | | 1,000 | |
| = | 7.5% | | = | 18.3% |
| | === | | | === |

(ii) Accounting Rate of Return on Average Capital

| | | |
|---|---|---|
| Average capital = | ₦1,000 + 2 | ₦1000 + 0 2 |
| = | ₦500 | = N 500 |
| ARR = | 75 x 100% 500 | 183 x 100% 500 |
| =15% | = 37% | (approx.) |

**In all these cases, project B will be selected in preference to project A because the accounting rate of return from the former is higher. It might be of interest to note that the use of the payback method would give priority to project A over project B**

**11.3.1 Advantages of ARR**

a) It is very easy to calculate;
b) It takes savings into account, over the entire economic life of the project; and
c) It embodies the concept of 'net earnings' (after allowing for depreciation), which is vital in the appraisal of projects.

**11.3.2 Disadvantages of ARR**

a) The method suffers from the same fundamental weakness as the payback method, i.e. it ignores the fact that cash flows and outflows occur at different time intervals;
b) ARR uses the concept of accounting profit, which is quite subjective as compared with cash flows generated by the project; and
c) The method has different variants, each of which produces a different rate of return for any one proposal. This situation arises due to the diverse concepts of investment as well as earnings.

**11.4 Discounted Cash Flow (DCF) Methods**

Discounted cash flow methods use cash flows and automatically make allowance for the time value of money. As previously mentioned, accounting profits are based on conventions and are less objective than cash flows, so cash flows are preferred for decision making.

There is general acceptance that any serious investment appraisal must consider the time value of money. Some monies arising at different times are not directly comparable. They must be converted to equivalent values at some common date, using a discounting factor.

As investments are essentially outlays of funds in anticipation of future returns, the presence of time as a factor in investment is fundamental rather than incidental. Time is always crucial for the investor, so that a sum received today is worth more than the same sum to be received tomorrow. Thus, in evaluating investment projects, it is important to consider the timing of returns on investments. In a way, time is the dimension through which the monetary variables involved in investments - the capital outlays and subsequent receipts - must be related.
The discounted cash flow technique considers the net cash flow as representing the recovery of original investments, plus a return on capital invested.
There are three main DCF methods, viz: Net Present Value (NPV), Profitability Index (PI), and Internal Rate of Return (IRR), however, only the NPV is treated in this study guide as other methods are outside the coverage of the syllabus at this level.

**11.5 Net Present Value (NPV) Method**
This method generally establishes a 'target' - a minimum rate below which a proposal would be rejected by management as undesirable in the light of the profit goals and above which it would be considered favourable. It seeks to determine whether the present value of estimated future cash inflows at management's desired rate of return will be greater or less than the cost of the proposal. Thus, cash inflow, initial investment and desired rate of return are given in this method, while the present value of cash inflows and its deviation from initial investment are to be determined.

The profitability of a project is ascertained by comparing the amount which meets the stated condition (i.e., pre-determined minimum rate) with the actual amount of investment required by the contemplated project. If a project's earnings could return with interest at the market rate, an amount greater than that required for investment in the project, the project is adjudged profitable.

The NPV method involves calculating the present values of expected cash inflows and outflows through the process of discounting and establishing whether, in total, the present value of cash inflows is higher than the present value of cash outflows.

The decision rule is to accept a project which has a positive NPV

**Illustration 11.4**
Alhaji Aminu Tijjani is considering buying a machine which will improve his cash flows by ₦60,000 per annum for the next five years, at the end of which the machine will be worn out and of no value. The machine, which will cost ₦150,000 and be bought for cash, has an estimated working capital of ₦10,000. Assume that the cost of the capital to the company is 15%.

You are required to calculate the Net Present Value of this project and advice Alhaji Aminu Tijjani.

| Year | Cash Flow ₦ | Discount Factor@15 % | Present Value ₦ |
|---|---|---|---|
| 0 | -160,000 | 1 | -160,000 |
| 1 | 60,000 | 0.8696 | 52,174 |
| 2 | 60,000 | 0.7561 | 45,369 |
| 3 | 60,000 | 0.6575 | 39,451 |
| 4 | 60,000 | 0.5717 | 34,305 |
| 5 | 60,000 | 0.4972 | 29,831 |
| | | NPV | **41,130** |

Alternative solution

| Year | Cash Flow ₦ | Discount Factor@15 % | Present Value ₦ |
|---|---|---|---|
| 0 | -160,000 | 1 | -160,000 |
| 1-5 | 60,000 | 3.352 | 301,120 |
| | | NPV | 41,130 |

**Notes:**

a. The purchase of a machine is a cash outflow, which is why it is shown as negative.

b. The discount factor of 3.352 is cumulative for the five years, and is used to save the effort of multiplying ₦60,000 by the individual figures (in the present value table) for years 1 to 5;

c. The Net Present Value of the project is positive ₦51,120 (i.e. the difference between the initial outlay on the machine and the discounted cash inflows).

Since the project has a positive NPV, this means that Alhaji Aminu will be getting a return on his investment of more than 15%; as such, the project is worthwhile.

i. If his NPV had been nil, his return would be 15%.

ii. If the project had shown a negative NPV, the return would be less than 15%, and therefore, it would not be worth undertaking.

**Illustration 11.5**

Garko Industries is considering an investment in a fleet of ten delivery vans to take its products to customers. The vans will cost N30, 000 each to buy, payable immediately. The annual running costs are expected to total ₦40, 000 for each van (including the driver's salary). The vans are expected to operate successfully for 6 years, at the end of which they will all have to be scrapped, with disposal proceeds expected to be about ₦6, 000 per van. At present, the business uses a commercial carrier for all of its deliveries. It is expected that this carrier will charge a total of ₦460, 000 each year for the next 6 years to undertake the deliveries.

Calculate the ARR and NPV of the project. Use a 15% discount rate.

**Solution**

Net savings before depreciation:

Ten vans will be purchased for N60,000, and the savings to be made by Garko industries per annum are as shown below:

| | |
|---|---|
| ₦ Cost of the carrier (p.a.) | 460,000 less annual running costs |
| of the vans (40,000 x 10) | 400,000 |
| Net savings before depr. | 60,000 |

Thus, the inflows and outflows will be;

| Year | Details | ₦'000 |
|---|---|---|
| 0 | Cost of vans | (300) |
| 1 | Net savings before depreciation. | 60 |
| 2 | " " " " 60 | |
| 3 | " " " " 60 | |
| 4 | " " " " 60 | |
| 5 | " " " " 60 | |
| 6 | " " " " 60 | |
| 6 | Disposal proceeds from the vans | 60 |

Assuming a straight-line approach, the total annual depreciation expense will be ₦40,000 (i.e., ₦300 - 60)/6.

While the average annual savings after depreciation is ₦20,000 (i.e., ₦60 - 40). The vans will appear in the balance sheet as follows:

| | | ₦'000 |
|---|---|---|
| At the end of the year | 1 | 260(300-40) |
| | 2 | 220(260-40) |
| | 3 | 180 |
| | 4 | 140 |
| | 5 | 100 |
| | 6 | 60 |

The average investment at (balance sheet values) will be ₦160,000 (260 + 220 + 180 + 140 + 100 + 60)/6)

Thus, the ARR = ₦20,000 / 160,000 x 100% = 12.5%

Calculation of the NPV of the project

| **Year** | Details | **Cashflows** | **Discount Factor** | **Present Values** |
|---|---|---|---|---|
| | | ₦'000 | (15%) | ₦'000 |
| 0 | Cost of vans | (300) | 1.000 | (300) |
| 1 | Net savings before Depr. | 60 | 0.870 | 52.20 |
| 2 | " " " " | 60 | 0.756 | 45.36 |
| 3 | " " " " | 60 | 0.658 | 39.48 |
| 4 | " " " " | 60 | 0.572 | 34.32 |
| 5 | " " " " | 60 | 0.497 | 29.82 |
| 6 | " " " " | 60 | 0.432 | 25.92 |
| 6 | Disposal proceeds | 60 | 0.432 | 25.92 |
| | | | NPV = | (46.98) |

The project is unacceptable as it has a negative NPV.

## 11.6 CHAPTER SUMMARY

Simple investment appraisal techniques were examined in this chapter. The first two methods examined (Accounting Rate of Return and Pay Back Period methods), also known as traditional methods, were found to be seriously imperfect by their failure to take full account of the time value of money. The other three methods, i.e. Net Present Value, Profitability Index and Internal Rate of Return, are known as discounted cash flow methods and are rated to be superior to the traditional methods, because they take full account of the time value of money.

## MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS

1. Which of the following is NOT crucial to the decision to commit huge funds into capital asset acquisition?
   - A. Finance availability
   - B. Changes in the quality of the asset over time
   - C. Installation costs
   - D. Useful life and scrap value of the asset
   - E. Maintenance expenses

2. Which one of the following states the reason why opportunity cost is considered in capital budgeting decisions?
   - A. The benefit lost in the alternative use of the funds to be committed is important
   - B. So that if the decision turned out to be faulty, the assets could be quickly disposed of and the proceeds diverted to alternative uses
   - C. There may be problems defending the decisions at the company's Annual General Meeting
   - D. The banks that will finance the capital project will demand for it.
   - E. It is the government policy

3. Which one of the following is NOT a method of appraising investment proposal?
   A. Payback period
   B. Feasibility study
   C. Net Present Value
   D. Internal Rate of Return
   E. Accounting Rate Return
4. A system of expenditure planning which facilitates the decision making process as regards committing resources to a project where benefits would be spread over long time period is called ……..
5. The number of years required to recoup the amount originally invested in a project is known as .....
6. Using the Net Present Value (NPV) investment appraisal method, what would indicate that a project is worthwhile and therefore should be accepted?

**SOLUTION**

1. C
2. A
3. B
4. Capital budgeting
5. Payback period
6. A Positive NPV

**EXAMINATION TYPE QUESTIONS**

(1) Several methods are employed by management accountants for ascertaining the profitability of capital expenditure projects. Discuss briefly one of the following methods:
   (a) Pay-back method
   (b) Accounting Rate of Return method.

(2) A company wishes to expand its production. Two proposals of capital expenditure are being considered; each of the two proposals requires more or less the same outlay. You, as the management accountant of the Company, are required to furnish a profitability estimate to guide the Board in its decision. Describe briefly two methods that can be used for this purpose.

(3) Give a comparative description of the Pay Back Period and Net Present Value approaches of investment appraisal. Which one of the two do you consider superior?

(4)(a) Describe briefly the Pay Back Period method of investment appraisal techniques and state five advantages and five disadvantages of the method.

(b) Using the information given below, prepare tables showing the Net Present Values of each of the projects based on applicable rates of 12% and 15% per annum, respectively:

| | Year | Project X ₦ | Project X ₦ |
|---|---|---|---|
| Initial Investment | | 100,000 | 100,000 |
| Cash flows | 1 | 5,000 | 60,000 |
| | 2 | 20,000 | 40,000 |
| | 3 | 100,000 | 100,000 |
| | 4 | 10,000 | 5,000 |

Compare briefly the two projects and comment on which of the two projects would be profitable and should be undertaken.

(5) Electro PLC, a manufacturer of high-quality metal cables, has a 15% market share for one of its products, ECM-1. The standard cost and profit margin for one unit of ECM-1 are given below:

| | ₦ | ₦ |
|---|---|---|
| Sales | | 10.00 |
| Less Direct cost: | | |
| Material | 4.00 | |
| Labour | 2.00 | |
| Overheads | 1.00 | 7.00 |
| Contribution | | 3.00 |
| Less: Fixed costs | | 2.00 |
| Profit | | 1.00 |

The budgeted sales of product ECM-1 during 1994 are ₦3,000,000, and the marketing director proposed that the market share could be increased to 20% and maintained at that level for the next three years if the following promotion budgets were agreed to be spent at the beginning of each of the next three years:

| | |
|---|---|
| Year 1 | ₦200,000 |
| Year 2 | ₦150,000 |
| Year 3 | ₦150,000 |

The fixed cost of ₦2 per unit includes 50% for the depreciation charge. The remaining 50% would be expected to increase by ₦60,000 with the increase in the market share. The company has an overall profit return objective of 30%, which they would apply as a hurdle rate to this type of project with the type of risk involved.

**You are required to:**

(a) Evaluate whether or not the project should be adopted, assuming the above are the only variables to be considered.

(b) Show with suitable calculations what the outcomes will be if only a 4% increase in market share is achieved.

Present value factors:

| Yr. | 10% | 20% | 30% | 40% |
|---|---|---|---|---|
| 1 | 0.9091 | 0.8333 | 0.7692 | 0.7143 |
| 2 | 0.8264 | 0.6944 | 0.5917 | 0.5102 |
| 3 | 0.7513 | 0.5787 | 0.4552 | 0.3644 |
| 4 | 0.6830 | 0.4823 | 0.3501 | 0.2603 |
| 5 | 0.6209 | 0.4019 | 0.2693 | 0.1859 |

(ICAN Nov/Dec, 1993).

(6) As the financial consultant of the Bourbon Can Company, a medium-sized manufacturing business, you have been asked to advise on the current year's investment proposals. The proposed projects, shown below, are not mutually exclusive:

PROJECT A: To replace existing data processing equipment. The Initial Cost is ₦250,000. The expected life of new equipment is 6 years. Expected annual after-tax cash inflow ₦72,500.

PROJECT B: To develop a new type of container, costing ₦70,000. The total cost to be incurred in the current year. The expected life of the product is 5 years. Expected annual after-tax cash inflow ₦30,000, the inflows to commence in one year's time.

PROJECT C: To install safety equipment. The initial cost is ₦90,000. Expected life is 3 years.
Expected annual after-tax cash inflow ₦45,000.

PROJECT D: To construct a new factory building. The initial cost is ₦210,000. Expected life is 8 years. Expected annual after-tax cash inflow ₦57,000.

PROJECT E: To extend the existing loading equipment. The initial cost is ₦170,000. Expected life is 4 years. Expected annual after-tax cash inflow ₦70,000.

PROJECT F: To purchase patent rights to a new process. The initial cost is ₦135,000. Expected life is 7 years. Expected annual after-tax cash inflow ₦36,000.

You may assume that, except for project B, all cash inflows would commence in the current year. For calculation purposes, you may assume that the annual cash inflows are always received on the last day of the year. Tax can be assumed to have been paid in the year in which the profits are earned.

The company's weighted average cost of capital for the coming year is estimated at 16%. The company is, however, in a capital rationing situation, and its estimates show that it will only have ₦600,000 to invest in capital projects in the current year.

**Required:**

(a) Which projects would you recommend the company undertake? Give reasons for your choice.
(b) What factor, if any, other than those given in the question, would you advise the company to consider before making a decision? (ACCA, Financial Management)

(7)(a) Define any two "traditional" investment appraisal techniques, showing briefly their advantages and disadvantages.
(a) Demonstrate your knowledge of the more acceptable technique using cash flows expected from two projects being negotiated by the General Manager of Usman & Co. Ltd. Both projects require the initial cash investment of ₦25,000 each. Project one will earn ₦8,000 in the first year ₦10,000 and ₦6,000 in the next 2 years, and ₦1,000 each in years four and five.

(8). Modern Tech. Services Ltd is considering two alternative projects for a business expansion programme in the Northern part of the country. The projects have the following Naira cashflow profiles according to the data supplied by the company's Accountant:

| Year | Project 1 | Project 2 |
|---|---|---|
| | ₦ | ₦ |
| 0 | -1 million | -3 million |
| 1 | -2 million | .20 million |
| 2 | .95 million | .50 million |
| 3 | .85 million | .65 million |
| 4 | .78 million | 75 million |
| 5 | .62 million | .80 million |
| 6 | .40 million | 1.90 million |
| 7 | .10 million | .20 million |

**Required:**

(a) Calculate the payback period for each project.
(b) Based on payback periods, advise which of the two projects should be chosen.
(c) State the advantages and disadvantages of the payback period criterion of investment appraisal.

*(PE II May/June, 1994).*

## CHAPTER TWEIVE: PERFORMANCE MEASUREMENT

## BUDGETING AND BUDGETARY CONTROL

## CHAPTER CONTENTS

a) Explain budgeting and budgetary control.
b) State and explain types of budget including functional, cash, capital and master budgets.
c) Describe the stages in the budgeting process.
d) sPrepare different types of budgets with emphasis on principal budget factor.
e) Differentiate between fixed and flexible budgeting.
f) Explain zero-based budgeting.

### 12.0 OBJECTIVES

After studying this chapter, readers should be able to explain the:

a) concept of budgeting and budgetary control
b) types of budgets
c) concept of principal budget factor
d) difference between budgeting and forecasting
e) objectives of budgeting
f) factors to be considered in introducing budgets for the first time
g) concepts of budget committee and budget manual.
h) behavioral factors in budgeting
i) concepts of functional budgets, cash budgets and the master budget and the preparation thereof
j) concept of budgetary control
k) objectives of budgetary control
l) budgetary control process
m) concepts of flexible budgeting and fixed budgeting and the differences between the two concepts
n) concepts of Zero Based Budgeting and Incremental Budgeting

## 12.1 INTRODUCTION

At the beginning of the financial period of every organisation, whether publicly or privately owned, needs to develop its budget to guide its operations for the year ahead. Budgeting is therefore an important process of every organisation. How effectively the budgeting process is handled in an organization could define i t s success or failure. Budgetary control encompasses the systems of budgeting, standard costing and other control techniques that ultimately aim at positioning the organisation to achieve its objectives. There is no point in developing budgets if no control steps will be installed to achieve the set budgets, hence the need for budgetary control.

## 12.2 Definition of Budget

A budget is a quantitative and financial plan of the operations of an organisation or an individual. It identifies their sources and the commitments required to fulfil the organisation's or individual's goals for the budget period. Thus, a budget includes both financial and non financial aspects of the planned operations.

The process of preparing budgeting is termed budgeting.

### 12.2.1 What a Budget is:

A budget has the following features:

a) It is a plan in quantities and financials
b) It has objectives, and a clearly expressed means of achieving the stated objectives
c) The means to achieve the objectives are the resources required to fulfil those objectives. Thus there must be definite resources set aside to achieve the set objectives.
d) The set objectives must be achieved within a particular time frame. This time frame is termed the budget period.

**12.2.2 Misconceptions about Budgets** A budget is not only:

a) narrative plan
b) involving price increases in the budgeted period
c) developed for states, governments, public or private organization's but may be developed for individuals prepared by economists or accountants but everyone across the globe.

## 12.3 Types of Budgets

There are a variety of budgets; some of them are as stated below:

### 12.3.1 Long Term Budgeting

This is also referred to as long range planning, strategic or corporate planning. It is a systematic and formalised process for purposely directing and controlling future operations towards desired objectives for periods extending beyond one year.

The planning process involves the following sequential steps:

a. Identification of objectives
b. Search for alternative courses of action
c. Gathering of data about the various alternatives
d. Evaluation of the alternatives on the basis of cost-benefit relationships
e. Selection of the optimum alternative and the formulation of thereof into plans
f. Implementation of long range plans through annual budgets
g. Monitoring of actual results
h. Responses to variances

### 12.3.2 Short Term Budgeting

These are operational plans which generally cover one year and which form the basis of the annual budgets. They are detailed, containing well-defined targets, which translate into work plans and performance measurements for each unit in the business. Short term budgets are a means of implementing strategic plans.

These are developed after due consideration of:

a) The prevailing business environment viz: internal and external
b) The available resources viz: physical, human and financial.

### 12.3.3 Annual Budgets

The annual budget translates the company's long-range plan into short-term operating targets. These are usually developed by involving all people responsible for achieving the targets and objectives of the company.

### 12.3.4 Functional Budgets

The functional budgets are the budgets developed for the operations of each function in the organisation. It is described as the budget for the function that has the binding or effective budget factor or limiting factor consequently it is usually developed first bearing in mind that it is the budget factor.

All the other functional budgets are subsequently developed from the budget of the function with the binding constraint. Most often, sales is the limiting budget factor. However, this is not necessarily always the case. The following are examples of functional budgets:

a. Sales budget in quantity and in value
   Other sales related budgets:

   i. Selling and distribution cost budget
   ii. Advertising cost budget
   iii. Sales promotion cost budget etc.

b. Production budget in quantity, and cost
   Other production related budgets:

   i. Labour usage and cost budget
   ii. Materials usage and cost budget
   iii. Material purchases budget
   iv. Production overhead budget.
   v. Production cost budget

c. Service function budgets

   vi. Administration cost budget
   vii. Personnel cost budget

c. Policy Related Budgets

   viii. Capital expenditure budgets
   ix. Research and development budgets
   x. Advertising and promotional budget
   xi. Training and staff development budgets

### 12.3.5 The Cash Budget

The cash budget summarises the cash flow effects of all of the functional and policy related budgets by coordinating them into one summarised statement.

### 12.3.6 Master Budget

The master budget summarises all the budgets into a coordinated projected financial statement comprising of:

i. budgeted trading, profit and loss account (Income Statement) or income and expenditure account as the case may be;
ii. budgeted balance sheet or statement of affairs; and
iii. budgeted cash flow statement for the period.

Other forms of budgeting such as fixed and flexible budgets, periodic and continuous budgets, incremental and zero- based budgets, etc. will be explained in the later part of the chapter.

### 12.3.7 Principal Budget Factor

The principal budget factor or limiting factor or key budget factor is a factor, which at any time is an overriding planning limitation on the activities of the organisation. Examples of principal budget factors include: Staffing limitations, scarcity of materials, other logistics, limited financial resources, low sales demand, limited storage facilities, etc.

### 12.3.8 Forecasting in Budgeting

A forecast is a prediction or estimate of the events which are likely to occur in the future. The forecast becomes a budget only when management accepts it as the objective. Frequently, consideration of the forecast sales leads management to make adjustments to their plans so that the agreed sales budget differs from the original sales forecast.

Forecasting is very essential for facilitating budgeting. Budgeting for future profit or cash flows compels forecasting future costs and revenues at different level of activities.

**12.4 The Objectives of Budgeting**: Budgeting is a critical tool set out to:

a. Aid the planning of annual operations;
b. Coordinate the activities of the various parts of the organization and to ensure that

the parts are in harmony with each other;

c. Communicate plans to the various responsibility centre managers;
d. Motivate employees to strive to achieve target;
e. Control activities; and
f. Evaluate the performance of managers.

Each of these objectives is further explained below:

### 12.4.1 Planning

Budgets compel or force planning to take place. The result is that problems are anticipated before they arise so that hasty decisions made on the spur of the moment based on expediency rather than reasoned judgment are minimised. The annual budget also leads to the refinement of strategic plans.

### 12.4.2 Co-ordination

Budget is a vehicle that synchronises and harmonises separate plans of different units, sections and segments of an organisation. Budgeting compels an examination of the relationships between various operations and hence the resolution of any conflict.

### 12.4.3 Communication

Top management communicates its expectations to lower management through the budget, which is an executive order. Much vital information about the organisation is communicated through the process of preparing the budget. The budgetary reports are a feedback to management on the performance levels of each unit of the organisation.

### 12.4.4 Motivation

Realistic and attainable budgets serve as a standard of good performance and thus motivate staff to strive to achieve them. However, for a budget to serve as a motivating factor, all staff must participate in its development.

### 12.4.5 Control

Variance reports generated from budgetary control system are a good control mechanism facilitating management by exception, which is efficient and cost effective. Budgets clarify the authority and responsibility of managers by setting budget limits to their spending powers and setting targets to be achieved.

### 12.4.6 Performance Evaluation

Performance could be evaluated by measuring the success of achieving budgeted results. The budgets based on the accepted standards are the benchmarks for measuring performance.

## 12.5 The Budgeting Process

### 12.5.1 Introducing Budgeting for the First Time

When introducing a budgeting system for the first time, it is important to ensure that the following are in place:

a. A sound financial accounting system
b. Top management backing for the system
c. A budget committee
d. A budget manual.

### 12.5.2 Necessary Conditions for Successful Budgeting

a) Top management support
(i) Involvement of line managers
b) Appropriate accounting and information systems
c) Readiness to flexibly administer the budget
d) Realistic organisation structure with clearly defined responsibilities
e) Clearly defined long term corporate objectives
f) Regular budget reviews when necessary

### 12.5.3 The Budget Committee

The Budget Committee is made up of members of senior management that oversees all budgetary matters. A typical budget committee includes the chief executive officer, heads of strategic business units and the chief finance officer. The committee sets or approves the overall budget goals for the organisation while its major business units, directs/coordinates budget preparation, approves the final budget, monitors operations as the year unfolds, and reviews the operating results at the end of the period.

The budget committee also approves major revisions of the budget during the period. This committee usually consists of sectional or departmental managers and is usually serviced by the Budget Officer who normally is the finance officer. Other functions of the committee may include the following:

a. Determination of the Budget Policy Guidelines and selection of budget policies compatible with the organisational goals and objectives;

b. Establishment of budget timetable;
c. Review of budget estimates submitted by sectional heads;
d. Facilitating the co-ordination of budgets;
e. Suggesting amendments to budgets and revising budget estimates when necessary;
f. Approval of budgets after amendments;
g. Facilitating the generation of budgetary control reports;
h. Analysing budget reports and recommending changes where necessary;
i. Examination of variances, suggesting investigation of variances and recommending solutions to remedy off-standard performance; and
j. Offering advice to top management on all matters concerning the budget.

### 12.5.4 The Budget Manual

The budget manual sets out the budget guidelines and budgeting instructions. The Manual is intended to guide all departmental heads involved in the preparation of the budget and ensure they follow the goals cum objectives of the organisation. Furthermore, the manual indicates the technique, trend or method of estimating the income and expenditure variables.

Thus, the budget manual sets out instructions on the responsibilities and procedures of budget preparation.

The starting point in developing budgeting guidelines is the firm's strategy. In developing the initial budgeting guidelines, the budget committee needs to consider developments that have occurred since the adoption of the strategic business plan. Some of these include the general economic market trends; the goal of the organisation for the budgeting period; specific budgeting policies such as mandate for downsizing,

reengineering, and special promotions; the operating results of the year to date, etc. In summary, the manual should specify the following:

a. Objectives of the budget;
b. Functional budgets to be prepared;
c. Membership of the budget committee;
d. Organisational charts;
e. Procedure for budget preparation;
f. Budget time tables; and
g. Budget policy guidelines, etc.

## 12.6 The Process of Preparing Initial Budgets

### 12.6.1 The Initial Budget Proposal

Based on the initial budget guidelines, each responsibility centre prepares its initial budget proposal. A number of internal factors are considered by a budget unit in preparing an initial budget proposal. Some of these factors include the following:

a) Changes in availability of equipment or facilities;
b) Adoption of new manufacturing processes;
c) Changes in product design or product mix;
d) Introduction of new product;
e) Changes in expectations or operating processes of other budget units that the budget unit relies on for its inputs materials or other operating factors; and
f) Changes in other operating factors or in the expectations or operating processes in those other budget units that rely on the budget unit to supply their components.
g) In addition to the above, it is equally important to consider some external variables such as:

- Changes in the labour market;
- Availability of raw materials or components and their prices;
- The industry's outlook for the year ahead; and
- Competitors' action, etc.

### 12.6.2 Budget Negotiation

The officers in charge of budget units examine the initial budget proposals to see whether the proposal is within the budget guidelines. The officers also check to see if: budget goals can be reasonably attained; it is in line with the goals of the budget units at the next level; and whether the budget operations are consistent with the budgeted activities of other budget units.

Negotiations occur at all levels of the organisation. The role of the budget committee in the coordination process is very critical. This process is perhaps the core of the budgeting activity and takes the bulk of budgeting time.

### 12.6.3 Review and Approval of the Budget

As budget units approve their budgets, the budgets go through successive levels of the organisational hierarchy until they reach the final level, when the combined unit budgets become the budget of the organisation. This stage is closely followed by the review and final approval of the budget. The review involves examination of the budget for consistency with budget guidelines, attainment of the desired short-term goals, and fulfilment of the strategic plan.

### 12.6.4 Revision of the Budget

Procedures for budget revision vary from one organisation to the other. Once a budget has been approved, some organisations allow budget revisions only under special conditions; other organisations continuously update their budgets, by building into the budgeting system, quarterly or monthly revisions as may be deemed appropriate.

### 12.6.5 Behavioural Factors in Budgeting

We have already observed how important budgets are in the achievement of corporate goals. We must at the same time note that there are a lot of behavioural factors that influence the usefulness of budgets. Budgets must be effectively and flexibly administered by management so that these budgets may assist with planning, motivation and control. In order to foster positive behavioural conditions, there should be employee participation in the development of budgets. Especially, to be involved in the budgetary process, are employees who are responsible for the implementation of budgets.

Another important factor that promotes good behavioural consequences is when budgets are realistically attainable. Budgets must be strict but achievable. Unrealistic budgets

lower employee motivation to achieve the budgets.

If budgets are used as rewards or penal mechanisms, it may lead to conflicts and organisational goals may not be achieved. Budgets should be used as cost control mechanism rather than as cost-cutting mechanism. Budgets should facilitate achievement of goal congruence by harmonising corporate goals with those of individual employees in the organisation. Organisational objectives may not be achieved where there are significant conflicts between corporate and employee goals.

The following advantages may emerge where budgets are flexibly administered and the process is participatory:

- Employees tend to accept the budget as their own plan of action and goal congruence is achieved;
- Participation tends to increase morale among employees; and
- Employee cohesion increases morale and productivity.

The climate for operating the budgetary system should be well structured so that individuals will embrace it rather than being anxious and defensive. This requires that each individual be given freedom/authority required to influence/accept his or her own performance level and the responsibility for accomplishing it.

## 12.7 Preparation of Annual Budgets

### Preparing Functional Budgets

As already noted, functional budgets are the budgets developed for the operations of each function in the organisation. Usually, the budget for the function that has the binding or effective budget factor or limiting factor is first developed bearing in mind the budget factor.

All the other functional budgets are subsequently developed from the budget of the function with the binding constraint.

**Illustration 12.1**

Asuo Ltd, plans to produce and sell 8,000 sachets of sugar during the year 2023. The selling price is expected to be ¢30 per sachet.

A unit of sachet requires 4 units of Material A, 3 units of Material B and 2 units of Material C. Opening inventory of raw materials is as follows:

| | Units | Amount |
|---|---|---|
| | | € |
| Material A | 45,000 | 270,000 |
| Material B | 35,000 | 70,000 |
| Material C | 25,000 | 50,000 |

The closing inventory for each raw material is to be at a level which would meet the production requirement of 7,000 sachets of sugar. There is no opening or closing inventory of sugar.

Purchase prices for all raw materials during the year 2023 are expected to be 30% higher than the prices reflected in the opening inventory values.

Sales and purchases are on credit and the opening balances being as follows:

| | € |
|---|---|
| Debtors | 350,000 |
| Creditors | 270,000 |

The Company expects to receive ¢525,000 from debtors during the period and to pay ¢235,000 to creditors.

**You are required to:**

**a)** prepare a budget for raw material purchases; and
**b)** calculate the closing balances for debtors and creditors. (Show your workings) **(12 Marks)**

**SOLUTION 12.1**

(a) Raw Material Purchases Budget

| | Mat A | Mat B | Mat C |
|---|---|---|---|
| | ¢ | ¢ | ¢ |
| Quantity required to be used | 32,000 | 24,000 | 16,000 |
| Add planned closing inventory | 28,000 | 21,000 | 14,000 |
| Less planned opening inventory | 60,000 | 45,000 | 30,000 |

| | | | |
|---|---|---|---|
| Total units to be purchased | 45,000 | 35,000 | 25,000 |
| Planned unit purchase price | 15,000 | 10,000 | 5,000 |
| Total Purchases | x 7.8 | x 2.6 | x 2.6 |
| | 117,000 | 26,000 | 13,000 |

(b) Calculation of closing balances for Debtors and Creditors

| | Debtors | Creditors |
|---|---|---|
| | ¢ | ¢ |
| Balance b/f | 350,000 | 270,000 |
| Add Sales/Purchases | 240,000 | 156,000 |
| | 590,000 | 426,000 |
| Less Receipts/Payments | 525,000 | 235,000 |
| Closing balances | 65,000 | 191,000 |

Workings:

| | | | |
|---|---|---|---|
| (1) | 8,000x2 | 8,000x4 | 8,000x3 |
| (2) | 7,000x2 | 7,000x4 | 7,000x3 |

(3)

| | | |
|---|---|---|
| Material A | ¢270/45 x 130% | = ¢7.8 |
| Material B | ¢70/35 x 130% | = ¢2.6 |
| Material C | ¢50/25 x 130% | = ¢2.6 |

(4)

Total Sales

8.000 x ¢30 = ¢240,000

Total Purchases

| | |
|---|---|
| Material A | = 117,000 |
| Material B | = 26,000 |
| Material C | = 13,000 |
| | **156,000** |

**Illustration 12.2**

Sakyi Ltd. produces two products, Cake and Bluna, which use the same raw materials, oil and fat.
A unit of Cake uses 3 litres of oil and 4 kgs of fat. A unit of Bluna uses 5 litres of oil and 2kgs of fat.
A litre of oil is expected to cost ¢3,000 and a kg of fat ¢7,000.
Budgeted Sales for 2023 are 8,000 units of Cake and 6,000 of Bluna; finished goods in inventory at 1/1/2025 are 1,500 units of Cake and 300 units of Bluna, and the company plans to hold inventories of 600 units of each product at 31/12/2025.

Inventory of raw materials are 6,000 litres of oil and 2,800 kgs of fat at 1/1/2025, and the company plans to hold 5,000 litres and 3,500 kgs respectively at 31/12/2025.

The store's manager has suggested that a provision should be made for damages and deterioration of items held in store as follows:

| | | |
|---|---|---|
| Cake | - | loss of 50 units |
| Bluna | - | loss of 100 units |
| Oil | - | loss of 500 litres |
| Fat | - | loss of 200 kgs |

**You are required to:**

(a) Prepare a material purchases budget for the year 2025 (14 Marks)

(b) Distinguish between Functional Budgets and Master Budgets (6 Marks)

**(Total 20 Marks) ICAG NOV. 2004**

**SOLUTION 12.2**

(a) Production Budget

| | Cake | **Bluna** |
|---|---|---|
| | ¢ | ¢ |
| Units to be sold | 8,000 | 6,000 |
| Add planned closing inventory | 600 | 600 |
| | 8,600 | 6,600 |
| Less planned opening inventory | 1,500 | 300 |
| | 7,100 | 6,300 |
| Provision for deterioration or damages | 50 | 100 |
| Units to be produced | **7,150** | 6,400 |

**Materials Usage Budget**

| | Oil litres | | Fat Kgs |
|---|---|---|---|
| Cake (7,150 x 3) | 21,450 | (7,150 x 4) | 28,600 |
| Bluna (6,400 x 5) | 32,000 | (6,400 x 2) | 12,800 |
| | **53,450** | | **41,400** |

Materials Purchase Budget

| | Oil litres | Fat Kgs | |
|---|---|---|---|
| Quantity necessary to meet production Add | 53,450 | 41,400 | |
| planned closing inventory | 5,000 | 3,500 | |
| | 58,450 | 44,900 | |
| Less planned opening inventory | 6,000 | 2,800 | |
| | 52,450 | 42,100 | |
| Provision for damages/deterioration | 500 | 200 | |
| Total units to be purchased Expected | 52,950 | 42,300 | |
| unit purchase price | ¢3,000 | ¢7,000 | 42,300kg x |
| Total Purchases | 52,950kg x ¢7,000 | | |
| | 3,000 = ¢158,850,000 | =¢296,100,000 | |

**Illustration 12..3**

Cema Ltd. Manufactures two products $C_1$ and $C_2$. The products use the same raw materials $RM_1$ and $RM_2$ but in different proportions:

A bag of $C_1$ uses 20 kg of $RM_1$ and 30 kg of $RM_2$. A bag of $C_2$ uses 25 kg of $RM_1$ and 25 kg of $RM_2$.

The following estimates are relevant for next year:

(i) Raw Materials

| | **$RM_1$** | **$RM_2$** |
|---|---|---|
| - Cost per kilo | ¢5,000 | ¢4,000 |
| - Opening inventory (kg) | 10,000 | 8,000 |
| - Closing inventory (kg) | 2,000 | 1,000 |
| - Provision for loss (kg) | 500 | 300 |

(ii) Finished Products

| | | $C_1$ | | $C_2$ |
|---|---|---|---|---|
| - | Sales (bags) | 200,000 | | 150,000 |
| - | Selling price per bag | ¢30,000 | 40,000 | ¢22,000 |
| - | Opening inventory (bag) | 10,000 | | 25,000 |
| - | Closing inventory (bag) | | | 5,000 |

(iii) Direct labour hours required

per unit of each product 8hrs 5 hrs.

Standard wage rate of direct labour is ¢5,000 per hour.

You are required to prepare

a. Sales budget (2 Marks)
b. Production budget (3 Marks)
c. Material Purchase Budget for next year (4 Marks)
d. Direct labour Budget for next year (4 Marks)

**(Total 13 Marks)**

**ICAG NOV. 2003**

**SOLUTION**

(a) Sales Budget

| | C1 | C2 |
|---|---|---|
| Units to be sold Selling | 200,000 | 150,000 |
| price/unit | ¢30,000 | ¢22,000 |
| Total sales | ¢6,000 million | ¢3,300 million |

(b) Production Budget

| | C1 | C2 |
|---|---|---|
| Units to be sold | 200,000 | 150,000 |
| Add planned closing inventory | 10,000 | 5,000 |
| | 210,000 | 155,000 |
| Less planned opening inventory | 40,000 | 25,000 |
| Units to be produced | 170,000 | 130,000 |

Material Usage Budget

| | | RM1 (Kg) | | RM2(Kg) |
|---|---|---|---|---|
| C1 | (170,000 units X 20kg) | 3,400,000 | (170,000 units X 20kg) | 5,100,000 |
| C2 | (130,000 units X 25kg) | 3250,000 | (130,000 units X 25kg) | 3,250,000 |
| | | **6,650,000** | | **8,350,000** |

(c) Material Purchase Budget

| | RM1 (Kg) | RM2 (Kg) |
|---|---|---|
| Quantity **required** to meet production | 6,650,000 | 8,350,000 |
| Add planned closing inventory | 2,000 | 1,000 |
| | 6,652,000 | 8,351,000 |
| Less planned opening inventory | 10,000 | 8,000 |
| | 6,642,000 | 8,343,000 |
| Provision for loss | 500 | 300 |
| | 6,642,500 | 8,343,300 |
| Cost per kilo | ¢5,000 | ¢4,000 |
| | 6,642,500kg x | |

| | | |
|---|---|---|
| Total Purchases | ¢5,000 | 8,343,300 x ¢4,000 |
| | ¢33,212.5million | ¢33,373.2million |

(d) Direct Labour Budget

| | C1 | C2 |
|---|---|---|
| Budgeted Production | 170,000 units | 130,000 units |
| Hours per unit | 8 | 5 |
| Total Budgeted Hours | 1,360,000 | 650,000 |
| Standard wage rate/hour | ¢5,000 | ¢5,000 |
| Total wages | 1,360,000hrs x ¢5,000 | 650,000 x ¢5,000 |
| | = ¢6,800million | = ¢3,250 million |

## 12.8 CASH BUDGET

The cash budget summarises the cash flow effects of all of the functional and policy related budgets by coordinating them into one summarised statement.

### 12.8.1 Benefits of Cash Budget

Cash budget aids the timely identification of expected cash deficits/surplus in advance. Where deficit cash balances is likely to occur, the cash budgets will indicate when to raise additional funds to mitigate the shortfall.

Conversely where surplus cash balance is inevitable, cash budgets will guide managers as to the exact amount involved and for how long such could be invested for some return.

Cash budgets synchronise receipts and payments to ensure the carrying out of all operations as scheduled.

### 12.8.2 Preparation of the Cash Budget

To facilitate the preparation of cash budget, it will be necessary to first ascertain all expected cash receipts or inflows. These will include: Cash sales:

- Collection of customer debts
- Issue of shares and debentures
- Interest and dividend received
- Proceeds from sale of assets, etc.

The next step will involve the ascertainment of all expected payments or cash outflows.
These may include:

- Cash purchases
- Payments to suppliers
- Redemption of shares and debentures
- Payment of interest and dividend
- Purchase of fixed assets
- Payment for general expenses
- Payment of wages and salaries, etc

It is important to note that non-cash expenses such as depreciation, provision for doubtful debts, etc. are not treated in the cash budget. This is because they do not involve actual flow of cash. The timing of the various cash flows is of great importance and very significant to the entire process.

The third step is to ascertain the difference between the expected cash inflow and the expected cash **outflow.** Any resulting difference is either a surplus or a deficit.

The final step is to add the opening cash balance to the obtained surplus or deficit. The result of this addition will be the expected closing cash balance.

The Cash Budget is usually developed to cover monthly or quarterly control periods.

**Illustration 12..4**

Vaney Company Ltd. produces household items for the domestic market. In the previous year, company suffered theft in its offices, losing all cash on hand in the process. On the date of the theft, the company had an overdraft balance of ¢125 million with its bankers. Due to the good working relationship that exists between the company and its bankers, a short-term loan of ¢500 million will be granted to the company on 31st December, 2022, on the following terms:

Facility - Short Term Loan

Duration - One year

Interest - 27.5 % per annum simple interest

Repayment - Principal payment in 4 equal quarterly instalments

- Interest payable monthly in arrears

Additional information:

(i) Production in units:

| **2023** | | | **2024** | | | | |
|---|---|---|---|---|---|---|---|
| Nov | Dec | Jan | Feb | March | April | May | June |
| 5,000 | 6,000 | 5,000 | 8,000 | 6,000 | 5,000 | 5,000 | 6,000 |

(ii) Raw materials used for the production cost ¢10,000 per unit. 25% of this sum is paid in the same month as production and the balance of 75% in the month following production.

(iii) Direct labour which costs ¢1,000 per unit are payable in the same month as production whilst variable expenses is estimated at ¢6,000 per unit. 50% of the variable expenses are payable in the same month as production and the balance in the following month.

(iv) Sales at ¢250,000 per unit are as follows:

| **2023** | | | **2024** | | | | | |
|---|---|---|---|---|---|---|---|---|
| Oct | Nov | Dec | Jan | Feb | March | April | May | June |
| 200 | 600 | 600 | 768 | 720 | 640 | 500 | 500 | 560 |

(v) All sales are on credit and debtors take an average credit period of two months.

(vi) Fixed overheads are ¢18million per month and are payable each month.

(vii) A new packaging equipment costing ¢84 million is to be paid for equally in March and September 2023. It has a useful life of 5 years and a straight-line depreciation policy. The depreciation of this equipment has not been included in the fixed overheads above.

**(viii)** Rent income of ¢6 million per quarter is to be received at the end of each quarter

**Required:**

(a) Prepare a cash budget for the company for the six months ended 30th June, 2023.

(16 Marks)

(b) Briefly explain the purpose and importance of cash budgets in business planning and decision making. (4 Marks)

(Total 20 Marks)

ICAG NOV. 2004

**SOLUTION**

Vaney Company Ltd.

**Cash Budget for the six months ended 30 June 2023**

| | Jan. | Feb. | Mar | April | May | June |
|---|---|---|---|---|---|---|
| | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
| INFLOWS | | | | | | |
| Receipts (W1) | 150,000 | 150,000 | 192,000 | 180,000 | 160,000 | 125,000 |
| Rent | | | 6,000 | | | 6,000 |
| Loan | 500,000 | | | | | |
| **Total Cash Inflow** | **650,000** | **150,000** | **198,000** | **180,000** | **160,000** | **131,000** |
| OUTFLOWS | | | | | | |
| Raw Materials (W2) | 57,500 | 57,500 | 75,000 | 57,500 | 50,000 | 52,500 |
| Direct Labour (W4) | 5,000 | 8,000 | 6,000 | 5,000 | 5,000 | 6,000 |
| Variable Expenses (W3) | 33,000 | 39,500 | 42,000 | 33,000 | 30,000 | 33,000 |
| Fixed Overheads | 18,000 | 18,000 | 18,000 | 18,000 | 18,000 | 18,000 |
| Equipment | | | 42,000 | | | |
| Interest | | 11,460 | 11,460 | 11,460 | 11,460 | 11,460 |
| Loan Repayment | | | 125,000 | | | 125,000 |
| Total Cash Outflow | 113,500 | 134,460 | 319,460 | 124,960 | 114,460 | 245,960 |
| Net Cash Flow | 536,500 | 15,540 | -121,460 | 55,040 | 45,540 | -114,960 |
| Balance b/f | -125,000 | 411,500 | 427,040 | 305,580 | 360,620 | 406,160 |
| Balance c/f | 411,500 | 427,040 | 305,580 | 360,620 | 406,160 | 291,200 |

(W1) **Receipts from Debtors**

| | Oct | Nov | Dec. | Jan. | Feb. | Mar | April | May | June |
|---|---|---|---|---|---|---|---|---|---|
| Sales in Units | | 600 | 600 | 768 | 720 | 640 | 500 | 500 | 560 |
| | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m |
| Sales @ ¢250,00 / unit | 50 | 150 | 150 | 192 | 180 | 160 | 125 | 125 | 170 |
| Receipts | - | - | 50 | 150 | 150 | 192 | 180 | 160 | 125 |

(W2) **Payment made to procure Raw Materials**

| | Nov. | Dec. | Jan. | Feb. | Mar | April | May | June |
|---|---|---|---|---|---|---|---|---|
| Production in Units | 5,000 | 6,000 | 5,000 | 8,000 | 6,000 | 5,000 | 5,000 | 6,000 |
| | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
| Purchases @ ¢10,000 per Unit | 50,000 | 60,000 | 50,000 | 80,000 | 60,000 | 50,000 | 50,000 | 60,000 |
| Payments | | | | | | | | |
| 25% | 12,500 | 15,000 | 12,500 | 20,000 | 15,000 | 12,500 | 12,500 | 15,000 |
| 75% | - | 37,500 | 45,000 | 37,500 | 60,000 | 45,000 | 37,500 | 37,500 |
| Total | | **52,500** | **57,500** | **57,500** | **75,000** | **57,500** | **50,000** | **52,500** |

(W3) **Payment made in respect of Variable Expenses**

| | Nov. | Dec. | Jan. | Feb. | Mar | April | May | June |
|---|---|---|---|---|---|---|---|---|
| Production in Units | 5,000 | 6,000 | 5,000 | 8,000 | 6,000 | 5,000 | 5,000 | 6,000 |
| | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
| Variable Expenses @ ¢6,000 per Unit | | | | | | | | |
| Payments | | | | | | | | |
| 50% | 15,000 | 18,000 | 15,000 | 24,000 | 18,000 | 15,000 | 15,00 | 18,000 |
| 50% | - | 15,000 | 18,000 | 15,500 | 24,000 | 18,000 | 15,00 | 15,000 |
| Total | | 33,000 | 33,000 | 39,500 | 42,000 | 33,000 | 30,00 | 33,000 |

(W4) **Direct Labour**

| | Nov. | Dec. | Jan. | Feb. | Mar | April | May | June |
|---|---|---|---|---|---|---|---|---|
| Production in Units | 5,000 | 6,000 | 5,000 | 8,000 | 6,000 | 5,000 | 5,000 | 6,000 |
| | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m | ¢m |
| Labour costs @ ¢1,000/Unit | | | | | | | | |

### 12.9 Master Budget

As previously mentioned, the master budget summarises all the budgets into a coordinated projected financial statement comprising of:

- Budgeted Income Statement or Income and Expenditure Account, as the case may be; and
- Budgeted Position Statement or Statement of Affairs.

**Illustration 12.5**

Papyrus Brothers Ltd, a stationery/textbooks company, has been in business for the past ten years but has not practiced any system of budgetary planning and control. Upon the advice of a management consultant, the company decided to adopt this system, beginning from its 2021/2022 financial year. The company's balance sheet (Position Statement) as at 31st October 2021 is as presented below:

| | | | |
|---|---|---|---|
| Non-current Assets at book value: | ¢'000 | ¢'000 | ¢'000 |
| Freehold Land & Building* Furniture | | | 64,000 |
| Fittings & Equipment Motor Vehicles | | | 8,000 |
| | | | 24,000 |
| **Current Assets:** | | | **96,000** |
| Inventory | | | |
| Trade Debtors | | 6,400 | |
| Treasury Bills | | 40,000 | |
| | | 9,000 | |
| **Less Current Liabilities:** | | 55,400 | |
| Trade Creditors | | | |
| Bank Overdraft | 8,000 | | |
| Capital | 16,400 | 24,400 | 31.000 |
| | | | **127.000** |

The Land is valued at ¢24,000,000.

The company has a sales policy which blends both cash and credit terms in the following manner:

| Month of Sale | Cash Sales (%) | Credit Sales (%) |
|---|---|---|
| November | 30 | 70 |
| December | 45 | 55 |
| All other | 20 | 80 |

Credit customers in any particular month are expected to pay up in the following month.

This sales policy has contributed to the company's performance and the following sales projections have been provided:

| | Nov | Dec | Jan | Feb | March | April |
|---|---|---|---|---|---|---|
| ['000] | 32,000 | 32,000 | 40,000 | 80,000 | 24,000 | 24,000 |

The price structure of the company's goods covers its cost and profit patterns, namely, the stationery/books, wages and profit in the ratio 25%, 55% and 20% respectively.

A discount deal has been struck with all the Company's suppliers to the effect that the company enjoys 3% discount on half of each month's purchase paid on the spot. The balance is paid in full in the following month.

The company has decided that half of the inventory to be sold in December will be acquired in November because of excessive pressure that suppliers will face in December.

Wages have never been accrued beyond the month. Other expenditures will be ¢5,200,000 for the first 3 months and thereafter the vehicle running expenses which amount to 50% of these other expenditures will go up by 10% every month.

The company plans to purchase two computers on 31st January at a cost of ¢5,000,000. In anticipation of this, the company hopes to redeem ¢5,000,000 worth of its treasury bills at the end of December when such bills will be three months.

You are to prepare:

(a) A cash budget for the six months of the 2021/2022 financial year. (14Marks)

(b) An Income Statement for these six months. (7Marks)

(c) A Position Statementasat30 April2022. (7Marks)

(Total 28 Marks)

ICAG ADAPTED

**Solution**

**Cash Budget for the six months of year 2021/2022 Financial Year**

| | Nov | Dec | Jan. | Feb. | Mar | April |
|---|---|---|---|---|---|---|
| | ¢ | ¢ | ¢ | ¢ | ¢ | ¢ |
| INFLOWS | | | | | | |
| Receipts (W1) | 49,600 | 36,800 | 25,600 | 48,000 | 68,800 | 24,000 |
| T bills Received | | 5,000 | | | 6,000 | |
| **Total Cash Inflow** | **49,600** | **41,800** | **25,600** | **48,000** | **68,800** | **24,000** |
| OUTFLOWS | | | | | | |
| Payments to Suppliers (W2) | | | | | | |
| Wages | 17,600 | 17,600 | 22,000 | 44,000 | 13,200 | 13,200 |
| Other Expenses | 5,200 | 5,200 | 5,200 | 5,460 | 5,460 | 5,460 |
| Computers | | | 5,000 | | | |
| **Total Cash Outflow** | **36,620** | **30,740** | **39,050** | **64,160** | **31,570** | **24,570** |
| Net Cash Flow | 12,980 | 11,060 | -13,500 | -16,160 | 37,230 | -570 |
| Balance b/f | -16,400 | -3420 | 7,640 | -5,810 | -21,970 | 15,260 |
| Balance c/f | -3,420 | 7640 | -5,810 | -21,970 | 15,260 | 14,690 |

(W1) Receipts from Customers

| | Nov | Dec. | Jan. | Feb. | Mar | April |
|---|---|---|---|---|---|---|
| Sales in Units | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 | ¢000 |
| Sales | 32,000 | 32,000 | 40,000 | 80,000 | 24,000 | 24,000 |
| Cash Sales | 9,600 | 14,400 | 8,000 | 16,000 | 4,800 | 4,800 |
| Debtors | 40,000 | 22,400 | 17,600 | 32,000 | 64,000 | 19,200 |
| | 49,600 | 36,800 | 25,600 | 48,000 | 68,800 | 24,000 |

(W2) Payment to Suppliers

| | Nov | Dec. | Jan. | Feb. | Mar | April |
|---|---|---|---|---|---|---|
| | ¢ | ¢ | ¢ | ¢ | ¢ | ¢ |
| Purchases (8+4) | 12,000 | 4,000 | 10,000 | 20,000 | 6,000 | 6,000 |
| Payments | 5,640 | 1,880 | 4,700 | 9,400 | 2,820 | 2,820 |
| Payments (50%) | | 6,000 | 2,000 | 5,000 | 10,000 | 3,000 |

| | | 7,880 | 6,700 | 5,940 | 10,280 | 5,820 |
|---|---|---|---|---|---|---|
| Discount received | 130 | 60 | 150 | 300 | 90 | 9 |

**PAPYRUS BROTHERS LTD.**
**Income statement for the six months ended 30 April, 2022**

| | | **¢'000** |
|---|---|---|
| Sales | | 232,000 |
| Less Cost of Sales | **¢'000** | |
| Opening inventory | 6,400 | |
| Add Purchases | 58,000 | 64,400 |
| Gross Profit | | 167,600 |
| Discount Received | | 870 |
| | | 168,470 |
| Less: Wages | 127,600 | |
| Other expenses | 31,980 | |
| Depreciation Buildings (25% x 40,000) x 0.5 | 5,000 | |
| Furniture (25% x 8,000) x 0.5 | 1,000 | |
| Motor Vehicle (25% x 24,000) x 0.5 | 3,000 | |
| Computers (25% x 5,000 x 4/12) | 417 | 168,997 |
| Net Profit/(loss) | | (527) |

**PAPYRUS BROTHERS LTD.**
**Statement of Financial Position as at 30 April 2022**

| Fixed Assets | ¢'000 | ¢'000 |
|---|---|---|
| Freehold land and Building Furniture | | 59,000 |
| Fitting & Equipment Motor Vehicle | | 7,000 |
| Computer | | 21,000 |
| | | 4,583 |
| Current Assets | | |
| Trading Debtors | | |
| Treasury Bills | 19,200 | |
| Bank | 4,000 | |
| | 14,690 | |
| Less Current Liabilities | | |
| Trade Creditors | | |
| | 3,000 | 34,890 |
| Financed by: | | |
| Capital | | 126,473 |
| Net loss | | 127,000 |
| | | (527) |
| | | 126,473 |

### 12.10 The Meaning of Budgetary Control

Budgetary control can be described as the whole system of control established in organisations to plan the activities of the organisation and take steps to achieve the set plans. The control system involves the following steps:

- determination of standards;
- formulation of budgets;
- comparing actual results with standard performance to determine variances;
- analysing the variances; and
- Investigation of significant variances
- Taking corrective actions.

12.10.1 **Objectives of Budgetary Control**

Budgetary control serves the following important purposes:

12.10.1.1 aids planning of annual operations;

12.10.1.2 assists with the coordination of activities of various parts of the organisation to ensure synchrony or harmony between the various units within the organisation;

12.10.1.3 facilitates communication of organisational plans to the various responsibility centre managers;

12.10.1.4 aids the control activities within the organisation;

12.10.1.5 motivate s employees to achieve targets; and

12.10.1.6 facilities the evaluation of the performance of managers.

12.10.2 **The Budgetary Control Process**

This involves the following:

12.10.1.7 Establishment of budgets for various budget centres. A budget centre is a clearly defined part of an organisation for the purpose of operating a budget system;

12.10.1.8 Creation of control periods. A budget period is the period for which a budget is prepared and over which such control aspects take place. The following factors influence the determination of the control period;

12.10.1.8.1 The nature and size of the business;

12.10.1.8.2 The part of the business for which the budget is being prepared;

12.10.1.8.3 The basis of control; and

12.10.1.8.4 The velocity of operations, etc.

12.10.1.9 Keeping records of actual results through the financial accounting system;

12.10.1.10 Comparing the actual results with the budget;

12.10.1.11 analysing the reported variances; and

12.10.1.12 Investigating and reporting reasons for the observed variances.

## 12.11 FIXED AND FLEXIBLE BUDGETS

### 12.11.1 Fixed Budget

A Fixed Budget is one that is developed for a set level of activity at the time of preparing the budget. An example will be a direct material cost budget for Department X set at say ¢100,000.00 for the planned production level of 50,000 units of a product.

### 12.11.2 Flexible Budget

This is a budget developed for different and alternative production levels. It could also be an adjustment of an original budget which has not been achieved to the actual level of activity.

If actual production in the example stated in 12.11.1 above, is 55,000 units, it implies that the budget of 100,000 is unrealistic, therefore, the budget must be adjusted to a level of production of 55,000 units that has been achieved.

Alternatively, if at the time of developing the budget, production levels of say 45,000; 50,000 and 55,000 units could have been considered so that budgets would have been developed for each of those activity levels.

The flexible budgeting approach is based on cost behavior patterns with respect to activity level.

In this regard, cost maybe classified as:

12.11.2.1 Fixed cost;

12.11.2.2 Variable cost; and or

12.11.2.3 Semi-variable or semi-fixed cost;

Note that only the variable cost and the semi-variable cost is varied to the actual activity level in order to flex a budget.

**Illustration 12.6**

A company manufactures a single product and has prepared the following flexed budget for the year:

| Level of activity | 70% | 80% | 90% |
|---|---|---|---|
| | ¢ | ¢ | ¢ |
| Direct Material | 17,780 | 20,320 | 22,860 |
| Direct labour | 44,800 | 51,200 | 57,600 |
| Production Overheads | 30,500 | 32,000 | 33,500 |
| Administration Overheads | 17,000 | 17,000 | 17,000 |
| ***Total cost*** | **110,800** | **120,000** | **130,960** |

You are required to:

a) Prepare a budget flexed at 45% level of activity; and
b) State the draw backs of a budgetary control system.

**SOLUTION**

From the flexed budgets provided and using the High and Low method of forecasting, we can segregate the fixed cost from the variable cost as follows:

| | Percentage of Activity level | Direct Material Cost |
|---|---|---|
| | % | ¢ |
| High activity level | 90 | 22,860 |
| Low activity | 70 | 17,780 |
| Change in activity level and cost | 20 | 5,080 |

Variable material cost per percentage of activity level = $\frac{5,080}{20}$

$$= ¢254 \text{ per } 1\% \text{ of activity} \qquad \frac{57{,}600 - 44{,}800}{90 - 70}$$

Using the High activity level, we have the following:

Total direct material cost at 90% = 22,860

Variable material cost at 90% (254 x 90) =

22,860 Fixed cost =

nil

What we have just done is to test whether there is any fixed cost component in direct materials. We observe that there is none. It will be the same with direct labour since all of direct labour is variable. This can be tested as follows:

Variable labour cost per activity level =

= ¢640 per percentage of activity level.

The whole of the administrative overheads is fixed since it remains at the same amount for each of the activity levels. This implies that only the production overheads can be separated into the fixed and variable components.

Using the High and Low method, we have the following:

| | Percentage of Activity level | Production Cost | Overheads |
|---|---|---|---|
| | % | ¢ | |
| High activity level | 90 | 33,500 | |
| Low activity | 70 | 30,500 | |
| Change in activity level and cost | 20 | 3,000 | |

Variable production overhead cost per percentage of activity level $= \frac{3,000}{20}$

$= ¢150$ per 1% of activity

Using the High activity level, we have the following:

| | ¢ |
|---|---|
| Total production overheads at 90% | 33,500 |
| Variable production overheads 90% (150x90) Fixed | 13,500 |
| production overhead cost | 20,000 |

Please note that if we had used the low activity level, we would have derived the same answer. This can be verified by the reader. We can now draw a flexible budget at 45% activity level as follows:

| Activity level | 45% |
|---|---|

| | ¢ | ¢ |
|---|---|---|
| Direct Material cost (254 x 45%) | | 11,430 |
| Direct Labour cost (640 x 45%) | | 28,800 |
| Production overhead: | | |
| Variable cost (150 x 45%) | 6,750 | |
| Fixed cost | 20,000 | 26,750 |
| Administrative overheads | | 17,000 |
| **Total cost** | | **83,980** |

Draw backs from a budgetary control system include the fact that:

a) It may be perceived as punitive tool rather than one that is meant to enhance performance;
b) It usually meets with resistance from management;
c) It may stifle motivation and negatively affect staff morale especially when behavioural aspects of the budgetary system is ignored; and
d) implementation of a budgetary control system involves some cost and time.

## 12.12 BUDGETARY CONTROL IMPROVEMENT TECHNIQUES

### 12.12.1 Zero Based Budgeting ( ZBB)

Zero-based budgeting is a formalised system of budgeting for the activities of an enterprise as if each activity were being performed for the first time (i.e. from a zero base). It involves the process of identifying a number of alternative levels of provision for each activity, evaluation of same in terms cost and the benefits derivable from them.

ZBB is based on the philosophy that management should be required to justify existing activities in exactly the same way as new proposals. Thus established activities will have to be compared with alternative applications of resources that they would use during the budgetary planning period. Implicit in Zero Based Budgeting is therefore the concept of opportunity cost.

For practical purposes, managers do not have to budget from zero, but can start from the current level of expenditure and work downwards, asking what will happen if a particular aspect of the current expenditure and current operations were discontinued and removed from the budget. This way

a cost-benefit analysis is carried out on every aspect of the budget and this encourages the selection of better alternatives. Zero Based Budgeting applies Activity Based Costing (ABC) Principles.

### 12.12.2 **Approaches to Zero Based Budgeting:**

Managers within the organization are asked to specify the decision units within their area of authority.

A decision unit is a program of work or capital expenditure programme or a program for a given activity, which can be individually operated.

Each of the decision units or separate activities is described in a decision package.

A decision package is a document, which identifies and describes the specific activities in such a way that management can evaluate and rank them in an order of priority against other activities.

Decision packages are usually prepared by the managers of budget centres, however, in order to facilitate the construction of decision packages, managers should be given a formalised set of assumptions by top management on items such as activity levels for the coming year, wages increases projected for the coming year and other similar management policy directions.

There are two types of decision packages. They are:

a) Mutually Exclusive Decision Packages
b) Incremental Decision Packages

## 12.13 Mutually Exclusive Decision Packages

These are a set of decision packages each of which contains an alternative method of getting the same job done. For implementation purposes, only one of the packages is selected using cost benefit analysis. Once a package is selected, alternative packages are shelved or abandoned.

### 12.13.1 **Incremental Decision Packages**

These are decision packages that divide one aspect of work or activity into different levels of efforts. The "base" package describes the minimum amount of work that must be done to carry out the activity, and other packages describe what additional work could be done, stating the costs and benefits. Each activity in the decision package is evaluated and ranked using cost-benefit indices.

### 12.13.2 **The Ranking Process in ZBB**

The ranking process provides managers with the technique to allocate scarce resources between different activities. Minimum work requirements and those activities that are essential to get a job done including the work that meets legal obligations will be given priority. For example, in the finance department, such

activities may be the minimum requirements needed to operate the payroll, purchase ledger, sales ledger and to maintain and publish a set of accounts that meet the expectations of external auditors.

The ranking process involves the following steps:

- Cost centre managers will be asked to rank the packages of their own cost centres;

12.13.2.1 The manager of the next level up to the hierarchy of seniority will consolidate the rankings of all his subordinates into a single ranking list for the group of cost centres using the rankings of each cost centre as a guide;

- These consolidated rankings will pass in turn from one stage further up the management hierarchy for further consolidation. At higher levels of consolidation, the ranking process might be done by a committee of managers rather than an individual manager;

12.13.2.2 Activities which will cost more than they are worth both in quantitative and qualitative terms, will be dropped; and

12.13.2.3 Resources in the budget are then allocated according to availability of funds and the ranking of competing packages. Junior managers can handle packages involving small expenditure but senior managers must make decisions involving higher amounts. The ZBB process must however run through the entire management structure.

12.13.3 **Advantages of Implementing Zero Based Budgeting:**

a) Identification and removal of inefficient or obsolete operation(s);

b) increase in psychological impetus to enable employees avoid wasteful expenditure;

c) It obliges an organisation to look closely into its cost behaviour patterns in order to decide the effect of alternative courses of action. (It will, for example be necessary to identify fixed cost, variable cost, and mixed cost, step cost, directly attributed cost, etc.);

d) It serves as a budgeting and planning tool which responds to changes in the business environment; (for instance, obsolete items of expenditure are identified and discountenanced);

e) The documentation required makes a coordinated in-depth knowledge of an organisation's operations available to all management; and

f) It results in an efficient allocation of resources to activities and departments of the organisation. (by implication the organization gets value for money).

12.13.4 **Disadvantages of Implementing Zero Based Budgets:**

- The volume of extra-paper work created by decision packages is enormous and cumbersome;

12.13.4.1 It emphasizes short-term benefits to the detriment of long-term benefits;

12.13.4.2 It encourages the false idea that all decisions have to be made in the budgets.

Management must be able to meet unforeseen opportunities and threats at all times and must not feel restricted from carrying out new ideas simply because they were not approved by a decision package, cost benefit analysis and a ranking analysis;

12.13.4.3 It calls for management skills in decision analysis to construct decision packages, which the organization may not possess;

12.13.4.4 It might be difficult to sell the approach to managers as a useful technique because:

- Incremental costs and benefits of alternative courses of action are hard to quantify accurately,
- Employees or trade union representatives may resist management ideas for changing the way in which work is done,
- There is the difficulty of the ranking process because of the following:
- Large volume of packages
- Some items are difficult to rank
- Sometimes prioritising can be difficult

## 12.14 INCREMENTAL BUDGETING

This is a system of budgeting where the previous period's budget is used as basis for preparing the current period's budget by making incremental adjustments as may be influenced by factors such as inflation, expansion needs, growth needs, etc.

Merits of Incremental Budgeting:

a. It is simple to apply in practice;
b. It is most suited to such costs as wages and salaries etc; and
c. It is widely accepted and used.

Demerits of Incremental Budgeting:

a. The budget perpetuates past inefficiencies;
b. The budget preparation process is not sufficiently critical to diagnose the cost behaviour of the enterprise; and
c. It does not lead to the optimal and efficient allocation of budgetary resources.

## 12.15 Activity Based Costing Systems in Budgeting (ABC)

An Activity Based Costing System is a system of managing cost by controlling activities that drive cost or activities that cause cost to be incurred. Examples of cost drivers include:

a. Machine set up hours;
b. Number of purchase orders raised;
c. Number or part numbers maintained;
d. Number of suppliers' statements processed;
e. Number of debtors' statements processed;
f. Number of man hours worked; and
g. Quantity of materials input into a process.

### 12.15.1 The process of Activity Based Costing System

The process of Activity Based Costing System breaks the organisation down into activities. Activities are tasks that people or machines perform to provide a product or a service. Examples of activities in a bank for example include the following:

i. Processing deposits;
ii. Issuing credit card;
iii. Setting up a loan account;
iv. Opening an account;
v. Processing monthly statements;
vi. Evaluating projects for financing;
vii. Processing funds transfer; and
viii. Processing customers' withdrawals, etc.

Create a cost centre or cost pool for each activity for the purpose of pooling together all the costs of each activity;

i. Identify the factors that influence the cost of a particular activity. These are the cost drivers; they are the events or forces that are significant determinants of the cost of activities; and
ii. Trace the cost of activities to products using the most relevant cost driver as a basis of absorption.

### 12.15.2 Periodic or Continuous Budgets:

Periodic budget **or** Continuous budget **or** Rolling budget can be described as a budget prepared for a defined time horizon and maintained for the said time frame, before a new budget is prepared for the following period of time. In order words, it is a budget prepared for a set time horizon and at the same time subjected to a continuous review in order to ensure it covers the set length of time at any particular point in time. For example, a budget that covers a six-month period will be reviewed monthly by including the month following immediately and deleting the previous or past month, so that the budget will always cover a six- month period.

## 12.16 CHAPTER SUMMARY

The chapter covers the concept of budgeting in three sections. It defines budgeting and treats various types of budget. It emphasised the need to first formulate the budget for the function with the principal budget factor before formulating other functional budgets. Budgeting and forecasting appear to be same; however, differences between the two were stressed. A budget is a determined plan of what is desired whereas a forecast predicts what is likely to occur in the future, given the other variables. The first section concludes with an outline of the objectives of budgeting.

The chapter explains the process to be followed when budgeting is introduced for the first time. Concepts such as the budget committee, the budget manual, and the budget period were also explained. In this chapter, the process of preparing annual budgets and the behavioural influences on budgeting were touched while the last section used illustrations to demonstrate how budgets are prepared.

This chapter delved into budgetary control and clearly stated the meaning and its objectives. Budgetary control is intended to ensure that the whole budgeting system employed by an organisation is not only sound in principle but also effective in practice. An efficient budgetary control system will positively impact an organisation and ensure its success. Other areas looked into include the budgetary control process; fixed and flexible budgeting; Incremental budgetin**g** and the concept of Zero Based Budgeting

The content discussed in this chapter reveals that business organisations without budgets will not be able to employ the resources at their disposal in the most optimal way.

**MULTIPLE-CHOICE AND SHOR-ANSWER QUESTIONS**

1. Which of the following options best describes ’long-term budget’?

   A. A budget for the life time of the company

   B. A budget of six months’ duration in a fast moving goods industry

   C. A budget which duration could not be determined

   D. A budget for periods extending beyond one year

   E. A past budget retrieved for use in the current period

2. What is the position of depreciation in a cash budget?

   A. It is in the cash inflow section

   B. It is in the cash outflow section

   C. It is not reckoned with

   D. It is deducted from the forecast capital expenditure

   E. It is broken down into straight line and reducing balance

3. Why should the sales budget be synchronized with the production budget?

   A. The sales manager and the production manager are on the same level

   B. They are the most important budgets

   C. The sales manager and production manager are two senior members of the budget committee

   D. Budgeting starts from sales and ends with production

   E. The production budget flows into the sales budget

4. Which of the following ensures the achievement of a production budget?

   A. Material purchase should provide for sufficient raw materials

   B. A very efficient production manager should be employed to run the company

   C. Existing machines should be discarded and new ones installed

   D. The factory should run 24 hours

   E. All the company's debtors should pay up immediately

5. What is the usual cause for budget revision?

A. When the managing director changes his mind

B. When conditions existing at budget preparation time change midway in the implementation

C. Machine breakdown

D. Decisions taken at the Annual General Meeting

E. When the budget has been fully achieved in the first month

6. What does budgetary control seek NOT to do in an organization?

A. Planning of annual operations

B. Controlling company's activities

C. Communicating plans to responsible manager

D. Evaluating performance of managers

E. Punishing lazy operators

7. Which of the following best describes a “Budget Centre”?

A. A centre where budgets are prepared

B. A place where budgets are implemented

C. A clearly defined part of an organisation for the purpose of operating a budget system

D. An office where the budget committee holds its meetings

E. The centre where the performance of budgets is controlled

8. Flexible budgeting is a method of budgeting where…….

A. Managers are allowed to operate without stringent conditions

B. Budget is developed for different and alternative production levels

C. Revisions are permitted on a monthly basis

D. Actual performance is worked back to the budget

E. The budget period is not fixed

9. What does “Incremental budgeting” mean?

   A. Using a current period's budget as the basis for preparing the succeeding period's budget by making incremental adjustments as necessary

   B. Increasing the budget in line with actuals

   C. Preparing a budget by adding a flat rate percentage increase on actuals

   D. Not allowing any figure to be unchanged but increased from year to year

   E. Not allowing any figure to be reduced but increased from year to year

10. Responsibility accounting is a system where……

   A. Only responsible managers are involved in budgeting and implementation

   B. Every manager is jointly and severally responsible for the company's activities

   C. The company's Chief Executive Officer has the final responsibility

   D. Each manager is actively involved in drawing up the budget for his own sphere of activity and is also responsible for achieving the targets set

   E. Responsibilities are shared between each manager and his subordinates

11. A budget which summarises all other budgets into a coordinated financial statement is called ……..

12. A factor which places a restriction over the activities of an organisation is known as ...

13. Until a budget is approved by management, it is called ……

14. A method of forecasting which selects the periods of highest and lowest activity levels and comparing the changes in costs resulting from the changes in the two activity levels is known as ……..

15. The document which sets out the budget guidelines which guides the departmental heads in preparing their budgets is referred to as …….

16. A method of budgeting where the budget is developed for a given level of activity is known as ……

17. A system of budgeting for the activities of an enterprise as if each activity is being performed for the first time is called ……..

18. A budget that is prepared for a set time horizon and continuously reviewed to cover the set length of time at any particular point in time is known as ……...

19. The person responsible for the overall coordination of budgeting and budgetary control activities in an organization is called ……

20. A situation where the budget for a period is changed substantially midway into that period due to some unforeseen circumstances is known as ……

**SOLUTION TO MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS**

1. D
2. C
3. E
4. A
5. BE
6. C
7. B
8. A
9. D
10. Master Budget
11. Limiting Factor/Principal Budget Factor
12. Forecast
13. High and Low Method
14. The Budget Manual
15. Fixed budget
16. Zero-Based budgeting
17. Continuous/Rolling budget

18. The Budget Officer

19. Budget review

**EXAMINATION TYPE QUESTIONS**

1 (a) Write brief notes on the following:

(i) Budget Manual

(ii) Budget Committee **(4 Marks)**

(b) A company plans to produce and sell 4,000 units of product X during the next period. The selling price of product X is ¢10 per unit.
A unit of product X requires 4 units of material A, 3 units of material B and 2 units of Material C.

Opening inventory of raw materials are as follows:

| | Units | Total Value |
|---|---|---|
| Material A | 20,000 | 40,000 |
| Material B | 15,000 | 15,000 |
| Material C | 10,000 | 5,000 |

The closing inventory for each raw material is to be at a level which would meet the production requirements for 3,000 units of product X. There is no opening or closing inventory of product X.

Purchase prices for all raw materials during the period are expected to be 20% higher than the prices reflected in the opening inventory values.

Sales and purchases are on credit, and the opening balances being as follows:

| | |
|---|---|
| Debtors | ¢30,000 |
| Creditors | ¢20,000 |

The company expects to receive ¢45,000 from debtors during the period and plans to pay ¢23,000 to its creditors.

***You are required to*:**

(i) Prepare the budget for raw material purchases

(ii) Calculate the closing balances for debtors and creditors (12 Marks)

ICAG NOV., 2003

You have just assumed the post of cost accountant at Ayehu Ltd. Financial controller gives you the data below:

**Budgeted Data**

| | Product X | Product Y |
|---|---|---|
| | ¢000 | ¢000 |
| Sales for the year: Region | | |
| Greater Accra | | |
| Central | 5,000 | 7,500 |
| | 3,200 | 6,400 |
| Ashanti | 3,000 | 2,400 |
| Eastern | 3,800 | 3,000 |
| Standard data per unit of production | | |

| Direct Material | Standard price | Product X | Product Y |
|---|---|---|---|
| | Per kg | Kg | Kg |
| | ¢ 2 | | |
| Ceca | 4 | 10 | 5 |
| Yede | | 5 | 10 |

| | Per hour | Standard hour | Hours |
|---|---|---|---|
| | ¢ 25 | | |
| Plant 1 | 20 | 4 | 4 |
| Plant 2 | | 6 | 3.5 |

| Finished goods inventory, valued at standard production cost | Product X | Product Y |
|---|---|---|
| | ¢000 | ¢000 |
| January 1, 1998 | 640 | 600 |
| December 31, 1998 | 320 | 840 |

Fixed production overhead per annum, ¢3,055,000. Direct labour hours per annum ¢763,750. All production overhead is fixed and is absorbed on direct labour hour basis. Administration, selling and distribution expenses are absorbed on budgeted basis of 20% of production cost. Profit is budgeted at 25% of selling price. The company is preparing an annual budget for 2022.

**You are required to prepare:**

a) Production budget
b) Direct materials cost budget
c) Direct materials purchase budget
d) Direct labour utilisation budget
e) Direct wages budget

**(15 Marks)**

**ICAG NOV. 1998**

3. Below are the summarised final accounts of Korkor, a business woman for the year ended 31 December, 2021.

Income Statement

| | ¢000 |
|---|---|
| Sales | 480,000 |
| Cost of sales Gross | 360,000 |
| Profit | 120,000 |
| General expenses (including depreciation) | 106.000 |
| Net prodit | 14,000 |

Balance Sheet (Extract)

| | ¢000 | ¢000 |
|---|---|---|
| Fixed Assets Less | | 100,000 |
| depreciation | | 30,000 |
| | | 70,000 |
| Current Assets: | 45,000 | |
| Inventory | 40,000 | |
| Debtors | 2,000 | |

| | | |
|---|---|---|
| Bank | 87,000 | |
| Current Liabilities | 60,000 | |
| Creditors | | 27,000 |
| Net Current Assets | | 97,000 |

The following additional information is provided:

i. Korkor proposed to buy new premises at a cost of ¢50 million to be paid for in April.

ii. There is a heavy demand for her product and she is confident that from 1st April, monthly sales will increase by 20% if the new premises are acquired.

iii. The gross profit percentage would remain unchanged and general expenses, inclusive of allowance for bank interest at 15% would increase by ¢1.2 million per month from 1st March.

iv. Korkor expects creditors to increase in proportion to sales; and the inventory-in-trade balance will remain at ¢45 million.

v. Depreciation charge on existing fixed assets will remain unchanged at 20% straight line. The additional building will be depreciated on the straight-line basis with a 20-year life. For this purpose, the buildings are considered to comprise ¢40 million of the planned expenditure on the premises referred to above.

vi. General expenses are paid monthly for cash and Korkor draws ¢1 million from the business bank account each month to cover personal expenses.

Korkor intends to employ two sources of finance for the planned expansion:

i. She has asked for a bank overdraft facility of ¢20 million for the next 12 months.

ii. She will offer customers a discount of 2% for immediate payment on all sales from 1 April, 2022 onwards. She expects half of the customers to take advantage of this offer. The remaining customers will take the same period of credit as in the previous year.

**Required:**

(a) Calculate the estimated maximum overdraft Korkor will require during the year to 31 December, 2022 and the bank balance or overdraft at 31 December, 2022.

(10 marks)

(b) Prepare the estimated Profit and Loss Account for the year ended 31 December 2022.

(6 Marks)

(c) Prepare the estimated Balance Sheet as at 31 December, 2022. (4 Marks)

(Total 20 Marks)

ICAG MAY 2002

The Position statement of Nsapa Breweries as at 31/12/2021 is as follows:

| | | ¢'000 | ¢'000 |
|---|---|---|---|
| Stated capital | | | 200,000 |
| Income surplus | | | 89,200 |
| | | | 289,200 |
| Financing | | | |
| Fixed Assets (cost – 280,000,000) | | | 252,000 |
| Current Assets | | | |
| Inventory | 50,000 | | |
| Trade Debtors | 49,200 | | |
| Bank | | 6,000 | |
| | | 105,200 | |
| Current liabilities | | | |
| Trade creditors | | 50,000 | |
| Expense creditors | | 18,000 | |
| | | 68,000 | 37,200 |
| | | | 289,200 |

The following forecasts have been derived:

| | Nov 2000 | Dec 2000 | Jan 2001 | Feb 2001 | March 2001 | April 2001 | May 2001 | June 2001 |
|---|---|---|---|---|---|---|---|---|
| Sales (in units ) | 4,000 | 4,200 | 4,500 | 4,600 | 4,800 | 5,000 | 3,800 | 3,000 |
| Purchase (¢'000) | 24,000 | 26,000 | 28,000 | 36,000 | 32,000 | 28,000 | 24,000 | 24,000 |
| Wages and Salaries(¢'000) | 16,000 | 16,000 | 16,000 | 20,000 | 20,000 | 20,000 | 24,000 | 24,000 |
| Overheads (excluding Depreciation) (¢'000) | 14,000 | 14,000 | 14,000 | 14,000 | 14,000 | 16,000 | 16,000 | 16,000 |

| | |
|---|---|
| Purchase of fixed assets (¢'000) | 60,000 |
| Issue of shares | 40,000 |
| Dividend | 20,000 |

You are given the following additional information:

i. The selling price in November 2021 was ¢12,000 per unit and this is to be increased to ¢16,000 per unit in May 2022. 50% of sales are for cash and 50% on credit to be paid two months later.

ii. Purchases are to be paid for, two months after purchase.

iii. 75% of the wages and salaries bill relate to management salaries and 25% relate to wages of operatives. Managers are paid in the same month but wages of operatives are paid in the first week of the following month.

iv. Overheads are paid in the month after they are incurred.

v. The fixed assets are to be paid for in three equal installments in the three months following purchase.

vi. Dividends are to be paid three months after they are declared and the receipts from the share issues are budgeted to be received in the month of issue.

vii. Fixed assets are depreciated 10% per annum on a straight-line basis on those assets owned as at 30th June, 2021.

viii. Closing inventory at 1st January, 2021 was equal to the previous two months' purchases. At 30th June, 2022, it would be equal to three months purchase.

**Required**

(a) Prepare the following budgets for six months ended 30 June, 2022:

i. Cash budget

ii. Budgeted Profit and Loss Account

iii. Budgeted Balance Sheet

(b) Comment on the results, highlighting those areas that you wish to draw to the attention of the budget committee. [Total 20 Marks]

MAY 2001

Several assumptions are commonly made by accountants when preparing or interpreting budgetary information.

You are required to explain why each of the following three assumptions might be made by accountants when designing a system of budgeting and to set out in each case also, any arguments which, in your view, raise legitimate doubts about their validity:

(a) Budgeted performance should be reasonably attainable but not too loose. (5 Marks)

(b) Participation by managers in the budget-setting process leads to better performance. (5 Marks)

(c) Management by exception is the most effective system of routine reporting. (5 Marks)

(Total 15 Marks)

6 (a) Outline the duties of a Budget Officer in a large establishment (6Marks)

(b) Briefly list down the main contents of a Budget Manual. (8Marks)

[Total 14 Marks]

ICAG JULY 1993

7(a) Explain **FIVE** (5) objectives of budgeting (5 Marks)

(b) Briefly explain the following terms as used in budgeting

i. Budget manual

ii. Budget committee

iii. Master budget

iv. Principal budget factor

v. Functional budget (15 Marks)

(Total 20 Marks)

8. A product manager has responsibility for a single product and is in the process of submitting data to be compiled into budgets for 2022. The manager has performance targets set in relations to sales volume, profitability levels and a target cash surplus from the product. Shown below are the agreed budgeted sales for the product for December 2021 to May 2022.

| | Dec | Jan | Feb | March | April | May |
|---|---|---|---|---|---|---|
| Units | 14,000 | 16,000 | 22,000 | 17,000 | 20,000 | 24,000 |

The company policy is that, at each month end, the closing inventory of the finished goods should be 25% of the following month's forecast sales and the inventory of raw material should be sufficient for 10% of the following month's production. Inventory levels currently conform to this policy. One unit of raw material makes one unit of the finished inventory, there is no wastage.

Raw material purchases are paid for during the month following the month of purchase. All other expenses are paid for as incurred. All sales are made on credit and the company expects cash receipts for 50% of sales in the month of sale and 50% in the following month.

The company operates an absorption costing system which is computed on a monthly basis. That is, in addition to direct costs it recovers each month's fixed and variable manufacturing overhead expenses in product costs using the budgeted production and budgeted expenditure in the month to establish an absorption rate. This cost is used to place a value on the inventory holding. Opening inventory is valued at the unit cost which was established in the previous month. At 1st January 2022 finished inventory should be assumed at ¢40 per unit. A flow of cost based on FIFO is assumed.

Sales are made at a price of ¢58 per unit.

Estimated costs to be used in the budget preparation for the product are:

Manufacturing costs:

| | |
|---|---|
| Material | ¢10.00 per unit produced |
| Variable overhead and labour Fixed | ¢16.00 per unit produced |
| overhead costs | ¢210,000 per month |
| (including depreciation of ¢54,000) | |
| Selling costs: | |
| Variable Fixed | ¢7.00 per unit sold |
| | ¢164,000 per month |

Required:

a) Compute the monthly budgeted production and material purchases for January to March 2022

(6 marks)

b) Prepare a budgeted profit and loss account and a statement of cash receipts and payments for January 2022.

(10 marks)

ACCA Dec 1998

9.

(a) Outline six (6) objectives of a Budgetary Control System and explain how a budgetary control system can achieve those objectives (12 Marks)

(b) In the context of budgeting, explain the following terms:

i. Aspiration levels. (2 Marks)

ii. Budgetary slacks. (2 Marks)

iii. Responsibility accounting. (2 Marks)

iv. Zero-based budgeting. (2 Marks)

**(Total 20 marks)**

ICAG MAY 2005

10. Budgeting is concerned with the implementation of an approved programme within the long-range plan and capital budget. A budget therefore translates the long-range plan and capital

budget into an annual operating plan of an organisation.

The budgeting process cannot therefore be viewed as being purely concerned with the current year but must be considered as an integrated part of the long-term planning process.

**Required:**

(a) Identify **FOUR** major reasons for producing budgets and two (2) possible limitations of budgets. (6 Marks)

(b) Identify the stages involved in the budgeting process. (6 Marks)

(c) Distinguish between Budgetary Control and Standard Costing. (4 Marks)

**(d)** Identify **FOUR** areas which should be of concern to management in budget behavioral implications. (4 Marks)

(Total 20 Marks)
ICAG MAY 2004

11. As a management Accountant, you are required to:

(a) Distinguish between cost control and cost reduction,

(b) Explain the **TWO** basic approaches to cost reduction,

**(c)** Itemize and briefly explain techniques, methods and major difficulties of cost reduction programmes (20 Marks)

ICAG MAY 1999

12. The Managing Director of Ozone Ltd. recently attended a seminar on budgetary control systems where he gained the impression that the introduction of such a system offered opportunities not only to improve control of the company's operations but also to improve staff motivation. Unfortunately, he did not fully understand some of the points made at the seminar and has approached you, the company's management accountant, for a clearer explanation.

**Required:**

Provide concise explanations to the following queries made by the Managing Director:

Traditional budgetary systems are based on certain basic mechanisms for achieving control.

What is the traditional basis for achieving control? (3 Marks)

b. Many early management theorists believed that employees were motivated solely by financial rewards. However, in recent years, other writers such as Maslow have suggested that in certain circumstances the offer of additional financial rewards may not provide much (if any) improvement in motivation. Why does Maslow believe that in some circumstances the offer of more money may not be particularly effective in motivating staff and workers?

**(2 Marks)**

c. It has been suggested that 'zero based budgeting' can be particularly effective in improving control and staff motivation.

i. What is meant by the term 'zero based budgeting'?

ii. What are the advantages of 'zero based budgeting' over traditional budgeting systems and how can it be used to help improve managers' motivation?

(10 Marks)

[Total 15 Marks]

ICAG JULY 1993

13. The concept of control occupies an important place in systems theory and is a major part of functions of the management accountant.

**Required:**

a. List the basic elements of control and discuss how these are implemented in an effective control system.

b. Describe the safeguards which must be incorporated in systems design to ensure the continued effectiveness of a control system.

(Total 20 Marks)

ICAG JAN. 1992

14. (a) Explain the concept of Planning Programming and Budgeting Systems as used in budgetary control (8 marks)

Outline **three** advantages and **three** disadvantages of Planning Programming and Budgeting Systems as a system of budgeting. (12 Marks)

**(Total 20 marks)**

15. A company operates a system of quarterly rolling budgets. The budgets for the next three quarters have been prepared. The figures below reflect the likely cost behaviour of each element of cost. Quarter 4 is being developed based on these budgets and other information available.

Budget Quarters 1 to 3

| | Q1 | Q2 | Q3 |
|---|---|---|---|
| | ₦ | ₦ | ₦ |
| | (000) | (000) | (000) |
| Activity: | | | |
| Sales (units) | 18 | 34 | 30 |
| Production (units) | 20 | 40 | 30 |
| Costs: | ¢000 | ¢000 | ¢000 |
| Direct Materials cost | 50 | 100 | 75 |
| Production Labour | 180 | 280 | 230 |
| Factory overheads (excluding indirect Labour) | 170 | 200 | 185 |
| Administration | 30 | 30 | 30 |
| Selling and Distribution | 29 | 37 | 35 |

In the current planning stage for quarter four, flexible budgets are to be developed for low, most likely and high sales volumes (38,000 44,000 and 50,000 units respectively). The company wishes to have a closing inventory (end of quarter four) equal to the opening inventory of quarter one. Management will therefore adjust the production levels to fall in line with this policy.

Cost Structures as for quarters one to three will apply to quarter four except that:

i. Raw material prices are expected to rise by 10%.
ii. Production labour rates will increase by 2.5%. However, management have declared that all labour rate increases must be matched by increased efficiency so that labour costs (both total

fixed and variable per unit) are unaltered.

iii. A quarterly bonus payment of 50% of the variable labour cost per unit, will apply for all production above 40,000 units.

iv. Fixed factory overheads and fixed selling and distribution expenses will rise by 5%.

The expected selling price per unit is ₦ 18,000. Inventory is valued at full factory cost of 13 per unit. This has been established using absorption principles and based on long run cost and capacity predictions. Small fluctuations in cost prices or volumes will not cause this unit cost to be amended.

**Required:**

a) Explain what is meant by a 'rolling budget' and what additional benefits may be claimed for this compared to the annual type style of budget (4 Marks)

b) Summarize the variable cost per unit and the total fixed cost for each cost heading that will apply to quarter four, for production below 40,000 units. (6 Marks)

c) Prepare detailed flexed budget profit statements for quarter four under the separate assumptions of low, most likely and high levels of sales and corresponding production volumes. (8 Marks)

**ACCA June 1998**

16.

For many organisations in both the private and public sectors, the annual budget is the basis of much internal management information. When preparing and using budgets, however, management and the accountant must be aware of their behavioural implications.

**Required:**

(a) Briefly discuss four purposes of budgets. (8 Marks)

(b) Explain the behavioural factors which should be borne in mind and the difficulties of applying them in the process of budgeting and budgetary control. (12 Marks)

**(Total 20 Marks)**

**ACCA June 1999**

17.

**Budget Flexible**

a. Write in brief, fixed budget and provide two of its limitations
b. Lumco for the first time is introducing flexible budget to improve its control system.

Budget for two operational levels are given as follows:

Flexible budget for August 2022

PRODUCTION CAPACITY

| | 40% | 50% |
|---|---|---|
| | 80,000 | 100,000 |
| | Units ₦ | Units ₦ |
| Direct material and labour costs | 800,000 | 1,000,000 |
| Production Overhead | 520,000 | 600,000 |
| Administrative Overhead | 400,000 | 400,000 |
| Selling Overhead | 280,000 | 300,000 |

Profit is fixed at 40% of Mark-up.

**Required**

Prepare the flexible budgets for 60% and 80% production capacity. If the company intends to still realise profit mark up of 40%

**CHAPTER THIRTEEN**

**STANDARD COSTING TECHNIQUES**

**CONTENTS**

i. Types of standards and bases of setting standards;
ii. Methods of determining standard cost and the uses of standard cost;
iii. Standard cost card;
iv. Basic variances including sales, materials, labour and overheads;
v. Standard ratios including activity, capacity and efficiency; and
vi. Causes of variances.

## 13.0 Objectives

After studying this chapter, you should be able to:

i. Explain the concept of standards;
ii. Explain types of standards;
iii. Explain the uses of standard cost, the merits and limitations;
iv. Explain methods of determining standard cost;
v. Compute standard cost card;
vi. Compute basic variances including sales, materials, labour and overheads;
vii. Compute standard ratios including activity, capacity and efficiency; and
viii. Analyse causes of variances.

## 13.1 Introduction

Establishment of standards as a basis for planning and as a yardstick for measuring performance enhances organisational planning. Furthermore, when standards are set and effectively used, budgeting becomes relatively easy to establish. Besides, the control process which compares the standards set with the actual achievements ensures that variances are determined and corrective action are taken to remedy deviations and drive the business closely toward its objectives.

## 13.2 Standard Costing

### 13.2.1 Standards and Standard Cost

**Standard as a Concept**

Standards are control mechanisms which establish predetermined estimates of costs of products and services and then compare these predetermined costs with actual costs as they are incurred to ascertain variances which are analysed into their various components and investigated.

### 13.2.2 Types of standards

There are four types of standards; namely: basic, ideal, attainable and current.

**a) Basic Standards**

These are long term standards remaining unchanged over several years. They are standards established for use over a long period from which a current standard can be developed. Basic standards can be used:

i. to show trends over time for items like;
   - materials prices;
   - labour rates;
   - efficiency level; and
   - the long term effects of changing methods

ii. as a basis for setting current standards.

Basic standards cannot be used to highlight current efficiency or inefficiency and would not normally form part of the reporting system except as a background to statistical exercise

**b) Ideal Standards**

These are standards which can only be attained under the most favourable working conditions. Ideal standards are based on optimal operating conditions such as:

- no break downs of machinery
- no wastage of materials
- no stoppages
- no idle time
- no spoilage
- no shrinkage of materials etc.

Ideal Standards are used:

i. As long term targets;
ii. For long term development purposes; and
iii. For investigative purposes

**c) Attainable Standards**

These are standards which can be achieved if a standard unit of work is carried out efficiently, a machine properly operated or a material properly used. Allowances are made for normal losses and machine breakdowns. It represents future performances and objectives that are reasonably met.

Attainable standards can be used:

i. for routine control with the purpose of either
   - providing tough but realistic targets; or
   - motivating staff and management to achieve targets

ii. for stock valuation purposes especially in respect of:
   - product costing
   - cost control

Attainable standards should be revised periodically to reflect:
   - new conditions;
   - new prices, methods, technology, etc.

**d) Current Standards**

These are standards set for use over a short period to reflect current conditions. Where conditions are stable then current standards will be the same as attainable standards. But where a temporary problem exists then a current standard could be set to deal with the problem.

Current standards can be used:

- where there is a temporary problem with material quality
- where there is unexpected price rise due to inflationary circumstances
- in periods of energy crises
- for significant standards containing some subjective elements etc.

**Important points to note**

a) The type of standard directly affects the amount of variances and the meaning of the variance

b) Every standard contains some subjective elements.

### 13.2.3 Standard Cost Defined

Standard cost is a predetermined calculation of how much cost should be under specified working conditions. In other words, it is a standard that is expressed in monetary terms. It is built up from an assessment of the value of cost elements. Standard cost provides bases for performance

measurement. It is used for control by exception reporting, valuation of stock and for fixing product's selling prices

**Misconception about Standard Cost**

Standard cost is not an average of past costs. First, past cost may contain results of past mistakes and the inefficiencies thereof. Second, comparison of standard cost with past cost is considered in appropriate due to likely changes in methods, technology, and prices

### 13.2.4 Uses of Standard Costing

Standard cost:

a) provides a basis for assessing performance and efficiency
b) controls costs by establishing standards, analyses variances and ascertains the reasons for observed variances
c) facilitates the practice of management by exception at the operational level
d) eases the budget preparation process
e) can be used for :
   - valuation of stocks and work in progress
   - Profit planning and decision making
   - Pricing policy (especially where cost plus pricing systems are used)
f) motivates staff and management

### 13.2.5 Merits of Standard Costing

Standard cost;

a) assists in assessing performance and efficiency
b) enhances cost control by encouraging re-appraisal of methods, materials and techniques makes budgeting easier
   a. facilitates the prudent practice of management by exceptions
   b. provides guidance on possible ways of improving performance
   c. provides a basis for estimation and forecasting
   d. serves as a source of motivation (if full participation and involvement is sought)
   e. simplifies stock valuation and pricing policy
   f. assigns responsibility for non-standard performance

### 13.2.6 Limitations of Standard Costing

Standard cost suffers the following limitations.

a) The philosophy of standard costing is challenged as being inappropriate
b) Standard costing can only exist realistically within the frame work of a budgetary system

c) Variance analysis merely directs attention to the cause of off-standard performances.

d) It does not solve the problem nor does it establish the reasons behind the variance. Solutions to the identified problems are considered as management tasks

e) All standards involve forecasting which is subjective with the inherent possibility of error.

f) The process is a bit complex and difficult and is not understood by most line managers

g) Variance analysis are at best adjudged post mortem. It is based on past events and does not correct past error or wrong.

h) Standard costing system may be expensive and time consuming to install

i) In volatile conditions with rapidly changing methods, rates and prices standards quickly become out of date losing their control and motivational effects

j) The usefulness of a number of variances is questionable

## 13.3. **The Process of Standard Costing**

The steps involved are as stated below:

a) Set the standard cost (building the total cost up from the individual elements of cost)

b) Measure the actual cost (this should be done for each of the elements of cost and in total)

c) Compare the standard cost with the actual cost, any difference is called a variance which can either be:

i. Favourable or of better performance than planned; or

ii. unfavourable or of worse performance than the set standard.

d) Analyse the variance (variance analysis) into its smallest parts or element

e) Investigate the real causes for the variances (only for material or significant variances)**.**

Greater value is obtained from standard costing techniques when it is applied to a production process involving a substantial degree of repetition (mass production and repetitive assembly work)

## 13.4 Setting Standard Costs

To be able to set standards, we take every element that contributes to the operations of an organisation and set standards for each of them. We consider the following:

i. types, and prices of material and parts

ii. grades, rates of pay and time for the labour

iii. production methods and layouts

iv. tools, jigs and machines to be used

### 13.4.1 Setting Standards for Materials

a) The materials content of each product is derived through engineering studies. Examples may include:

   i. Raw materials

   ii. Sub-assemblies

   iii. Piece parts

   iv. Finishing materials

b) Determine the quantities required of each of the inputs, making provision for normal losses in production which may arise from occurrences like:

   i. Machining loss

   ii. Evaporation

   iii. Expected breakages

   iv. Expected rejections, etc.

c) Obtain the prices of materials from the purchases or procurement department

   i. These prices should be the forecast or expected prices for the relevant budget period and not past costs

   ii. the prices should reflect:

   - trend in material prices
   - anticipated changes in purchasing policies
   - quantity and cash discounts
   - carriage and packing charges

On the basis of the variables obtained above, you may then set standards for materials as standard quantities of each material at the standard material price per unit.

### 13.4.2 Setting Standards for Labour

The following steps are required:

a. specify the exact grades of labour to be used. This can be by the use of work study projections, techniques of work measurement, etc.

b. determine the time to be taken for a unit of output. The standard hour which is the quantity of work achievable at standard performance in an hour or minute

c. the personnel department should make a forecast of the relevant wage rates for the control period

d. set the standard labour cost as standard hour at standard wage rate

13.4.3 **Setting Standards for Overheads**

The predetermined overhead absorption rates are the standard rates for overheads for each cost centre using the budgeted standard labour hours as the activity base

$$\text{Variable Overhead Rate (VOR)} = \frac{\textit{Budgeted variable overheads for cost centre}}{\textit{Budgeted standard labour hours for cost centre}}$$

$$\text{Fixed Overhead Rate (FOR)} = \frac{\textit{Budgeted fixed overheads for cost centre}}{\textit{Budgeted standard labour hours for cost centre}}$$

13.4.4 **Setting standard for selling price**

This activity is a top management activity, which is done after considering the following factors:

i. Anticipated market demand
ii. Competing products
iii. Inflation estimates
iv. Elasticity of demand, etc.

13.4.5 **Standard sales margin**

Standard sales margin is the difference between standard selling price and standard cost. Where a standard marginal costing system is in use, it follows that the standard contribution margin will be the difference between the standard selling price and the standard marginal cost.

13.5 **Computation of standard cost card**

Once all standards have been set, they could be summarised in a standard cost card as follows:

| | ₦ |
|---|---|
| Standard direct materials cost per unit | xx |
| Standard direct labour cost per unit | xx |
| Standard direct expenses per unit | xx |
| Standard overhead cost per unit | xx |
| Standard cost per unit | xx |
| Standard profit margin | xx |
| Standard selling price | xx |

Using the standard cost card, budgets can easily be established, variances computed and analysed.

## 13.6 Variance Analysis

### 13.6.1 The Concept of Variance

A variance is the difference between a standard level of performance that ought to have been achieved and the actual activity achieved.

### 13.6.2 Analysis of Variances

Variance analysis is the breaking down of the total profit variance into its smallest parts.

**Illustrative example 13.1**

**Use the information provided below to compute various variances discussed in subsequent sections**

Ghartex Ltd. specialises in the manufacture of Kente for export and the local market. The following information relates to the Company's budget for a given month.

| | |
|---|---|
| Budgeted Production Quantity | 120,000 metres |
| Direct material: | |
| Material A | 0.5 kg (¢1,200 per kg) |
| Material B Direct | 0.5 kg (¢1,600 per kg) |
| labour: | 30 mins (¢6,000 per hour) |
| Variable Overheads | ¢30 million |
| Fixed Overheads: | ¢15 million |

Production overhead is absorbed on the basis of direct labour hours. Actual Results achieved during June 2000 are as follows:

Production - 130,000 metres Director Material Purchased:

| | | | |
|---|---|---|---|
| Material A | 80,000 kg | Costing | ¢88 million |
| Material B | 70,000 kg | Costing | ¢98 million |

Direct Labour Hours 80,000 hours worked costing ¢400 million.

| | |
|---|---|
| Variables overheads | ¢28 million |
| Fixed Overheads | ¢18 million |

### 13.6.3 **Basic Cost Variances**

The basic are the difference between standard cost allowed for actual output and the actual cost incurred for that actual output. The total cost variance can further be analysed into variances for each of the elements of cost, namely: material cost variance, labour cost variance, variable overheads variance and fixed overheads variance. Each of these variances will now be discussed in turn.

### 13.6.4 **Material Cost Variances, MCV**

This is the difference between standard materials cost and actual materials cost of the actual production volume achieved. The material cost variance can further be analysed into materials price variance and materials usage variance.

**Material Cost Variance** (MCV) formula is:

MCV = (SQ of production achieved x SP) – (Actual quantity used x Actual unit price)

Where:

SP denotes unit selling price

SQ denotes quantity use in producing one

SQ of production achieved = SQ X Actual output

**Note:** Actual quantity used = Actual quantity purchased where there is no opening and no closing inventories.

**a). Direct Material Price Variance**

This is the difference between the standard price and actual purchase price of the actual quantity of materials purchased or used.

**Direct Material Price Variance** (AQP) formula is:

AQP (SP– AP)

Where:

AQP=Actual quantity purchased AP=Actual unit price

**Possible Causes of direct material price variances**

i. Paying higher or lower prices than planned. This could be due to:
ii. Efficiency or inefficiency in purchasing
iii. Changes in market conditions
iv. Emergency purchases at short notice due to poor stock control procedures
v. Different discounts than planned
vi. Difference in material quality (superior or inferior)
vii. Buying substitute material for non-availability of planned materials.

b). **Direct Material Usage Variance**

This is the difference between the standard quantity specified for the actual production and the actual quantity used for the actual production expressed at standard price.

**Direct Material Usage Variance** formula is: SP (SQ of production achieved – AQ used)

Where:

AQ used denotes Actual quantity used

**Possible Causes of Direct Material Usage Variances**

**i.** Good or careless handling of materials by production personnel
**ii.** Superior or inferior quality of materials
**iii.** Pilferage of materials
**iv.** Actual scrap different from anticipated or normal scrap
**v.** Changes in quality control measures
**vi.** Gain or losses due to the use of substitute materials of higher or lower quality than planned.

**Using Illustration13.1**

**Direct Material Total Cost Variance**

(Standard Material Cost of output produced– Actual Cost of material purchased)

| Material | ₦ |
|---|---|
| A (0.5 x130,000)×1200)–88,000,000 | 10,000,000A |
| B (0.5 x130,000)×1600)–98,000,000 | 6,000,000 F |
| Total material cost variance | 4,000,000 A |

**Direct Material Price Variance**

(Standard cost of the actual quantity purchased) –(Actual Cost of the quantity purchased) Or

AQp (SP– AP)=(AQp x SP) – (AQp x AP)

| Material | ₦ |
|---|---|
| A (80,000× 1200)–88,000,000 | 8,000,000 F |
| B:(70,000× 1600)–98,000,000 | 14,000,000 F |
| Total material price variance | 22,000,000 F |

**Direct Material Usage Variance**

SP (SQ of actual production achieved – Actual quantity used)

| Material | ₦ |
|---|---|
| A:(65,000-80,000) | 18,000,000 A |
| B: (65,000-70,000) | 8,000,000 A |
| Total material usage variance | 26,000,000 A |

13.6.5 **Labour Cost Variances**

This is the difference between the standard direct labour cost and actual direct labour cost for the actual production achieved. This variance can further be analysed into labour rate variance and labour efficiency variance.

**a. Labour Rate Variance**

This is the difference between the standard direct labour rate and actual direct labour rate per hour for the actual total hours worked and paid for.

**Possible Causes of direct labour rate variances**

- Higher rates being paid than planned due to wage award
- Higher or lower grade of workers being used than planned.
- Payment of unplanned overtime or bonus

**b. Direct Labour Efficiency Variance**

This is the difference between the standard hours for the actual production achieved and the hours actually worked expressed at the standard rate.

**Possible causes of labour efficiency variances**

- Use of incorrect grade of labour
- Poor workshop organization and supervision
- Incorrect materials and or machines.

a) Idle Time Variance is determined as idle time x standard direct labour rate per hour.

**Possible Causes of idle time variance**

-Machine breakdown

-Power outages

-Work stoppages

-Ineffective coordination among staff

**Using Illustration13.1**

**Direct labour total cost variance**

(Standard Direct labour Cost for the production achieved–Actual Direct Labour cost)

(SHs per unit x130,000) x 6,000) – AHw x AR)

$[\frac{30}{60} \times 130{,}000) \times 6{,}000] - [400{,}000{,}000] = 10{,}000{,}000A$

**Direct labour Rate Variance**

Actual Hour's worked (Standard Rate –Actual Rate) 80,000

(6,000 – 5, 000)= 80,000,000 F

Note:

AR = ¢400,000,000 ÷ 80,000hrs = ¢5,000 per hr

**Direct labour Efficiency Variance**

Standard Rate (Standard hours allowed for production – Actual hours worked) Or

SR (SHs for130,000 units – 80,000)

6,000 (65,000–80,000) = 90,000,000 A

### 13.6.6 Total Overhead Cost Variance, TOCV

This is the difference between the production overhead received at budgeted or standard rate and the actual total overhead incurred.

Note:

The total overhead is the sum of variable and fixed overheads.

It is computed as:

TOCV = {(Standard hours for the production achieved x standard total overhead rate) – (Actual total overhead incurred)}

Overhead Cost Variance can be split into

a. Variable Overhead Cost Variance
b. Fixed Overhead Cost Variance

**Using Illustration13.1 for TOCV**

**Total Overhead Cost Variance, TOCV=**

{(SHs x Production Achieved) x Standard Total Overhead Rate) – (Actual Total Overhead incurred)} Or

$$\left\{(30/60 \times 130{,}000) \times \frac{45{,}000{,}000}{60{,}000} - \mathbf{46{,}000{,}000}\right\}$$

48,750,000 - 46,000,000 = 2,750,000F

Where STOR = (¢30,000,000 + ¢15,000,000) ÷ 60,000hrs

= ¢750per hr

13.6.6.1 **Variable Overhead Cost Variances**

This is the difference between the actual overheads incurred and the variable overhead absorbed. This variance can further be analysed into overhead expenditure variance and variable overhead efficiency variance.

a. Variable Overhead Expenditure Variance:

This is the difference between the actual variable overheads incurred and the allowed variable overheads based on the actual hours worked.

b. Variable Overheads Efficiency Variance

This is the difference between the standard variable overheads and the allowed variable overhead for the actual production achieved. It is determined as: (Standard hours produced less Actual hours worked) x VOR per hour.

**Using Illustration 13.1 for VOCV**

**Variable Overhead Cost Variance**

Standard Variable Overhead Cost – Actual Variable Overhead Cost

$$\text{VOR} = \frac{30{,}000{,}000}{60.000} = ¢500\text{per hr}$$

**Variable Overhead Cost Variance** = $(500 \times 130{,}000 \times 30/60) - 28{,}000{,}000$

32,500,000- 28,000,000

= 4,500,000 F

**Variable Production Overhead Expenditure Variance**

(Actual Hours Worked × VOR) – Actual Variable Overhead cost incurred or

(80,000× 500)– 28,000,000

40,000,000 – 28,000,000

= 12,000,000 F

**Variable Production Overhead Efficiency Variance**

(Standard Hours Allowed × VOR) –(Actual Hours Worked × VOR)

(65,000× 500) – (80,000 × 500) 32,500,000 – 40,000,000

= 7,500,000A

Total, VOCV=12,000,000F + 7,500,000A = 4,500,000F

NOTE

VOR = Variable Overhead Rate

13.6.6.2 **Fixed Overhead Variance**

This is the difference between fixed overhead absorbed at the actual production and the actual fixed overhead for the period. This variance can be analysed into fixed overheads expenditure variance and fixed overheads volume variance.

a. Fixed overhead expenditure variance

This is the difference between actual fixed overheads incurred and budgeted fixed overheads for the period

b. Fixed overhead volume variance

A measure of the over-or-under-absorption of overhead cost caused by actual production volume differing from that budgeted (CIMA).

It is computed as: (Actual production in standard hours x Process overhead absorption rate per hour)

- (Budgeted fixed overhead).

The fixed overhead volume variance is further analysed into fixed overhead volume efficiency variance and fixed overhead volume capacity variance.

i. Fixed overhead volume efficiency variance is measured as: Standard hours achieved for production less Actual hours used at FOAR

ii. Fixed overhead volume capacity variance is measured as:

Budgeted hours less actual hours at FOAR

**Using Illustration 13.1**

**Fixed Production Overhead Cost Variance**

FOAR = $\frac{15{,}000}{250}$

= ¢ 60,000 per hrs

Standard Fixed Overhead for Production achieved – Actual Fixed Overhead Cost

$0_{3_0}6$ (130,000×/× 250) – 18,000,000

16,250,000 –18,000,000 =1,750,000 A

**Fixed Production Overhead Expenditure Variance**

Budgeted Fixed Overhead – Actual fixed Overhead cost (15,000,000–18,000,000) = 3,000,000A

**Fixed Production Overhead Volume Variance**

(Budgeted Hours – Standard Hours Allowed) x FOAR

(60,000 – 65,000)× 250 =1,250,000 F

NOTE: TOCV = VOCV + FOCA = 4,500,000F+1,750,000A = 2,750,000F

**Illustration 13.2**

As part of a strategy to improve profitability by controlling costs, Mukaila, the Financial Director of Lagos Ltd. decided to introduce a system of standard costing, to take effect from 1st January 2022. For this purpose, the following standard cost for product J was compiled:

| | ¢/unit |
|---|---|
| Direct Materials [5kg @ ¢1,000 per kg.] | 5,000 |
| Direct Labour (1 direct labour hour) | 9,000 |
| Overheads - Variable {per direct labour hr | 1,000 |
| - Fixed | 5,000 |
| Total | 20,000 |

The fixed overhead charged was based on estimated fixed overhead expenditure of ¢19,000,000 and a total output of 3,800 units during the 50-week year.

During the first four weeks of the year, 3,000 units of product J were produced and the following actual costs incurred:

| | ¢000 |
|---|---|
| Direct Materials [16,000kg @ ¢1,100 per kg.] | 17,600 |
| Direct Labour | 27,000 |
| Overheads - Variable {2800 DL Hrs. @ 900 per hr.} | 2,520 |
| - Fixed | 16,000 |
| | 63,120 |

Mukaila recognised that some variances from plan had occurred but was unsure as to which ones he should investigate further.

**Required**

a. What is a Variance? Identify and discuss briefly the two basic causes of variances.

**(6 Marks)**

b. Compute the following variances: Direct Material - total, price and usage
Variable Overhead - total, rate and efficiency Fixed Overhead - total, expenditure and volume.

**(11 Marks)**

**c.** Outline the factors that Mukaila should take into account when deciding which variances he should investigate further.

**(8 Marks)**

**{Total 25 Marks}**

Adapted from ICAG JAN. 1993

**SOLUTION 13.2**

(a) A variance is any deviation from what is expected to be the level of performance under given working conditions. It is the difference between the standard cost margin allowed for the actual output and the actual cost incurred or actual margin earned for the actual output.
The two basic causes of variances are:

(i) Using more or less quantity of materials or labour hours per unit of output than expected. This may be due to using superior or inferior inputs or more/ less efficient use of inputs.

(ii) Paying a higher or lower rate per unit of input material or labour. This is influenced by the quality of the inputs.

(b) Direct Material total cost variance:
Standard Material Cost for actual output – Actual Material Cost

(3,000× 5,000) – 17,600,000 = 2,600,000A

Direct Material Price variance
(Actual Price–Standard Price) Actual Quantity of materials purchased

17,600-(1000× 16,000) = 1,600,000A

Direct Material Usage Variance
(Actual Quantity – Standard Quantity) Standard Price

(16,000 –15,000) 1000 = 1,000,000A

Variable Overhead Total Cost Variance.
Standard Variable Overhead Cost Actual Variable Overhead Cost

(3000× 1000) – 2,520,000 = 480,000F

Variable Overhead Expenditure Variance (Actual Hours worked @ VOR)– Actual Variable Overhead Cost (2,800× 1000) –2,520,000 = 280,000 F

Variable O/h Efficiency Variance

(Standard Hours Allowed -Actual Hours Worked) x VOR (3,000 – 2,800)× 1000 = 200,000F

Fixed Overhead Cost Variance

Standard Fixed Overhead Cost for the actual output Actual Fixed Overhead Cost

(5,000 x 3,000)–16,000,000 = 1,000,000 A

FOAR $= \boxed{\frac{Budgeted\ Fixed\ Overhead}{Budget\ Labour\ Hours}}$

$= \frac{19{,}000{,}000}{3{,}800\ hours}$

= 5,000 per direct labour hour

Fixed Overhead Expenditure Variance

Budgeted Fixed Overhead – Actual Fixed Overhead

(19,000,000 –16,000,000) = 3,000,000 F

Fixed Overhead Volume variance

(Budgeted Hours– Standard Hours Allowed) x FOAR

(3,800–3,000) x 5,000 = 4,000,000 A

Fixed Overhead Volume Efficiency variance

(Standard Hours Allowed – Actual hours Worked) x FOAR

(3,000–2,800) x 5,000 =1,000,000 F

Fixed Overhead Capacity variance

(Budgeted Hours - Actual Hours) x FOAR

(3,800 –2,800) x 5,000 = 5,000,000A

(b) The following factors should be taken into account in deciding whether or not variances should be investigated:

a. The materiality of the variance. Variances considered material should be investigated.

b. The reliability of the standards set. Where the standards are considered to be very reliable, then even small variances should be investigated

c. Variances that have a known history of frequent recurrence may not be investigated but those that are unusual should be investigated.

d. The cost benefit implications of investigation of the variances should be considered.

e. The implications of investigation of the variances on future operations of the organization

**Illustration**

The following information has been extracted from a flexible budget:

| Standard Labour Hours | Overhead Cost (¢000) |
|---|---|
| 90,000 | 7,000 |

The increase in overheads between these two levels of activity is deemed to be entirely due to variable expenses. The actual activity results for the period are:

| | |
|---|---|
| Output in standard Hours | 80,000 |
| Clock Hours Worked Overhead | 90,000 |
| Costs 7,500 | |

**Required**

a. Compute the total Variable overhead cost variance showing the expenditure variance and efficiency variance separately.

(5 marks)

b. Compute the total fixed overhead cost variance showing the expenditure, volume, capacity and efficiency variances separately (9 marks)

**ICAG JULY 1999**

**SOLUTION 13.3**

Workings

| Budgeted Data: | Activity level | Cost |
|---|---|---|
| | Labour Hours | ¢000 |
| High activity | 100,000 | 7,500 |
| Low activity | 90,000 | 7,000 |
| | 10,000 | 500 |

Change in Cost as a result of change in standard labour hours

$= \frac{500,000}{10,000 \text{ hrs.}}$

- ¢50 per standard labour hours

Using the high activity as that for the original budget we have the following:

| | |
|---|---|
| | ¢ 7,500,000 |
| Total Overhead Cost | 5,000,000 |
| Variable Overhead (100,000 hrs.@ ¢50) | 2,500,000 |
| Fixed Overhead | |

Thus:

Budgeted Fixed Overhead = 2,500,000

VOAR = ¢50 per labour hour

FOAR $= \frac{2,500,000}{100,000hr} = ¢25 \text{ per hour}$

(a) Variable Overhead Cost Variance:

Standard Variable Overhead Cost – Actual Variable Overhead Cost

(80,000 x 50) - 4,800,000 = 800,000 A

Variable Overhead Expenditure Variance:

(Actual hours worked @ VOAR) – Actual Variable overhead

(90,000hours x 50) -4,800,000 = 300,000 A

Variable Overhead Efficiency Variance

(Standard Hours allowed – Actual Hours worked) x VOAR

(80,000-90,000) x 50 = 500,000A

(b) Fixed Overhead Cost Variance:

Standard Fixed Overhead – Cost Actual Fixed Overhead Cost

(80,000 hours x 25) – 2,600,000 = 600,000A

Fixed Overhead Expenditure Variance

(Budgeted Fixed Overheads – Actual Fixed Overheads)

(2,500,000 -2,600,000) = 100,000A

Fixed Overhead Volume Variance

(Budgeted hours – Standard Hours Allowed) x FOAR

(100,00080,000) x 25 = 500,000A

Fixed Overhead Efficiency Variance:

(Standard Hours Allowed – Actual Hours) x FOAR

(80,000 – 90,000) x 25 = 250,000A

Fixed overhead Capacity Variance

(Acual Hours – Budgeted Hours) x FOAR

(90,000 – 100,000) x 25 = 250,000A

## 13.7 Other Variance Analysis Issues

### 13.7.1 Sales Margin Variances

Sales margin variance is calculated in terms of profit or contribution margins rather than price. This is because the objective of the sale function is to influence profits favourably. Besides this, the cost variances have already been computed and so the focus should now be on the profit or contribution margins.

a) Total sales margin variances

This is the difference between the actual margin and the budgeted sales margin based on standard profit or contribution margin. The total sales margin variance can be analysed into sales margin price variance and sales margin volume variance.

b) Sales margin price variance

This is the difference between the actual margin per unit and the standard margin per unit for the actual volume of sales made.

c) Sales margin volume variance

This is the difference between the actual sales volume and the budgeted sales volume at the standard margin per unit.

d) Methods of computing sales margin variances:

We can use either the marginal costing approach or the absorption costing approach.

Marginal costing approach:

| | | |
|---|---|---|
| Sales margin variances are evaluated on the contribution margin which is calculated as: Actual contribution margin: | | xxx |
| Actual selling price | xx | |
| less standard marginal cost | xx | |
| direct material direct labour | xx | Xxx |
| direct expense variable overhead | | xxx |
| Actual contribution margin | | xxx |
| standard contribution margin: standard selling price | xx | |
| less standard marginal cost: | xx | |
| direct material | | xxx |
| direct labour direct expenses | xx | xxx |
| variable overhead | | xxx |
| **Standard contribution margin** | | |

e) **Absorption Costing Approach**

Sales variances are evaluated on the basis of profit margins not contribution margin

The profit margins are calculated as

| | | |
|---|---|---|
| **Actual profit margin:** | | |
| Actual selling price | | xxx |
| Less **standard total cost**: | | |
| Direct materials | xxx | |
| Direct labour Direct | xxx | |
| expenses Variable | xxx | |
| overheads Marginal | xxx | |
| cost **Fixed** | xxx | |
| **overhead** | xxx | xxx |
| Actual profit margin | | xxx |

| | | |
|---|---|---|
| **Standard profit margin:** | | |
| standard selling price Less | | xxx |
| **standard total cost**: Direct | | |
| materials | xxx | |
| Direct labour Direct | xxx | |
| expenses Variable | xxx | |
| overheads Marginal | xxx | |
| cost **Fixed** | xxx | |
| **overhead** | xxx | xxx |
| Standard profit margin | | xxx |

Please note that in computing profit or contribution margins standard cost isused whether it is actual margin or standard margin that is being computed.

**Illustrative Example 13.4**

Dzifa Manufacturing Co. Ltd. Is located at the Ghana Free Zone and registered to produce for the export market. The company produces three types of plastic related products. The budget estimates submitted by the management accountant provided a budgeted sales

mix of:

| | | |
|---|---|---|
| R | - | 30% |
| S | - | 50% |

Based on the following budgeted sales for the year:

| Product | Units Selling | Price | Standard Cost | Profit Margin |
|---|---|---|---|---|
| | '000 | ¢'000 20 | ¢'000 10 | ¢'000 10 |
| Q | 200 | 15 | 9 | 6 |
| R | 300 | 10 | 6 | 4 |
| S | 500 | | | |
| | 1,000 | | | |

During the year, the following results were achieved:

| Product | Units Selling | Price | Standard Cost | Profit Margin |
|---|---|---|---|---|
| | '000 | ¢'000 | ¢'000 | ¢'000 |
| Q | 150 | 22 | 10 | 12 |
| R | 350 | 15 | 9 | 6 |
| S | 600 | 9 | 6 | 3 |
| | 1,100 | | | |

(A) From the above information, you are required to calculate:

(i) Sales Quantity Variance (4 Marks)

(ii) Sales Margin Price Variance (4 Marks)

(iii) Sales Margin Total Variance (2 Marks)

**SOLUTION**

**13.4** (i) Sales Quantity Variance

| Product | Budgeted Units | Actual Units | Quantity Variance | Standard Profit Margin | Sales Quality Variance |
|---|---|---|---|---|---|

| | | | | | ¢000 |
|---|---|---|---|---|---|
| Q | 200,000 | 150,000 | -50,000 | 10,000 | 500,000A |
| R | 300,000 | 350,000 | +50,000 | 6,000 | 300,000F |
| S | 500,000 | 600,000 | +100,000 | 4,000 | 400,000F |
| | | | | | 200,000F |

(ii) Sales margin price variance

| Product | Standard Profit Margin | Actual profit margin | Difference in margin | Actual quantity | Sale margin price variance |
|---|---|---|---|---|---|
| Q | 10,000 | 12,000 | +2,000 | 150,000 | 300,000F |
| R | 6,000 | 6,000 | - | 350,000 | Nil |
| S | 4,000 | 3,000 | -1,000 | 600,000 | 600,000A |
| Total Sales margin price variance | | | | | 300,000A |

(iii) Sales margin Total Variance:

| Product | Budgeted profit units | Standard Profit margin ¢000 | Total margin expected | Actual Units | Actual profit margin | Total actual profit margin ¢000 |
|---|---|---|---|---|---|---|
| Q | 200,000 | 10,000 | 2,000,000 | 150,000 | 12,000 | 1,800,000 |
| R | 300,000 | 6,000 | 1,800,000 | 350,000 | 6,000 | 2,100,000 |
| S | 500,000 | 4,000 | 2,000,000 | 600,000 | 3,000 | 1,800,000 |

Sales margin total variance is the difference between the total profit margin expected and the total actual profit margin. Thus, we have:

| Product | Total margin expected ¢000 | Total profit margin ¢000 | Sales margin total variance ¢000 |
|---|---|---|---|
| Q | 2,000,000 | 1,800,000 | 200,000A |
| R | 1,800,000 | 2,100,000 | 300,000F |

| | | | |
|---|---|---|---|
| S | 2,000,000 | 1,800,000 | 200,000A |
| | | | 100,000F |

13.7.2 **Reconciliation of budgeted profit and actual profit**

This process deals with reconciling the budgeted profit and the actual profit. We start with the budgeted profit and adjust the computed variances with the budgeted to arrive at the actual profit.

Notes on calculation of variances

1. all cost variances are based on actual level of production achieved rather-than the budgeted level of activity
2. when reconciling budgeted profit to actual profit there are two approaches
   a. marginal costing approach All variances are outlined except fixed overhead volume variances. Only fixed overhead expenditure variance is computed and used for the reconciliation.
   b. absorption costing approach all variances must be calculated. Fixed overhead and its sub-variances should also be computed and included in the reconciliation.

**Illustration13.5**

ABC manufacturing company produces a single product which is known as alpha. The product requires a single operation and the standard cost for this operation is presented in the following standard cost card:

**Standard cost card for product alpha**

| | ¢ | ¢ |
|---|---|---|
| Direct material: | | |
| 2 kilos of Aat ¢1 per kilo | 1 kilo | 2.00 |
| ofB at ¢3 per kilo | | 3.00 |
| Direct Labour: | | |

| | | |
|---|---|---|
| 3 hours at ¢3 per hour | | 9.00 |
| Overheads: | | |
| Variable-3 hours at ¢2 per direct labour hour Fixed -36.00 | | |
| hours at ¢4 per direct labour hour | 12.00 | 18.00 |
| Total Standard Cost Standard | | 32.00 |
| Profit Margin | | 8.00 |
| Standard Selling Price | | 40.00 |

ABC limited plans to produce 10,000 units of alpha in the month of May and the budgeted costs based on the information contained in the standard cost card are as follows:

| | ¢ | ¢ |
|---|---|---|
| Sales (10,000) units of alpha at ¢40 per unit) | | 400,000 |
| Direct materials: A 20,000 kilos at ¢1 per kilo | 20,000 | |
| B 10,000 kilos at ¢3 per | 30,000 | |
| kilo | 90,000 | |
| Direct Labour: 30,000 hours at ¢3 per hour | 60,000 | |
| Variable Overheads: 30,000 hours at ¢2 per D.L.H. | 120,000 | 320,000 |
| Fixed Overheads | | 80,000 |
| Budgeted Profit | | |

Budgeted fixed overheads are ¢120,000 per month and are charged on the basis of direct labour hours, giving a fixed overhead rate of ¢4per D.L.H. (¢120,000 divided by 30,000 direct labour hours).
The actual results for May are:

| ¢ | ¢ |
|---|---|
| Sale (9,000 units at ¢42 per unit) | 378,000 |
| Direct Materials: A 19,000 kilos at | 20,900 |
| ¢1.10 per kilo | 28,280 |
| B10,000 kilos | 91,200 |
| at ¢2.828 per kilo | 52,000 |

| | | |
|---|---|---|
| Direct Labour: 28,500 hours at ¢3,20 per hour | 116,000 | |
| Variable Overheads | | 308,380 |
| Fixed Overheads | | 69,620 |
| Profit | | |

Manufacturing overheads are charged to production on the basis of direct labour hours. Actual production and sales for the month were 9,000 units.

**Required:**

(a) Compute all the relevant variances (except mix and yield variances) (15 marks)

(b) Reconcile the budgeted and actual profits for the period. (5 Marks)

**[Total:20Marks]**

ICAG MAY 1997

**SOLUTION 13.5**

Calculation of Variances

Sales margin variances

Sales margin price variance

| Actual Sales qty @ actual contribution | ¢ |
|---|---|
| 9000 units @ ¢22 | 198,000 |
| i.e 9,000 (42 – 20) | |
| Less Actual sales qty @ std contribution per unit | |
| 9,000 units @ ¢20 | 180,000 |
| | 18,000 |

Note: Both actual contribution margin and budgeted contribution margin are based on budgeted variable cost of sales

| Sales Margin Volume Variance | ¢ |
|---|---|
| Actual Sales Volume @ Std contribution 9000 @ ¢20 | 180,000 |
| Less Budgeted Sales volume @ Std contribution 10,000 @ ¢20 | 200,000 |
| | 20,000A |

Direct Cost Variances

| Material Price | ¢ | ¢ |
|---|---|---|
| A actual qty @actual price actual qty @ Std price | 20,900 | |
| ¢1 @ 19,000 | 19,000 | 1900A |
| B. actual qty@ actual price actual qty @Std price | 28,280 | |
| 10,000@ ¢3 | | |
| | 30,000 | 1,720F |
| | | 180A |

Material Usage Variance

| Material Price | | ¢ | |
|---|---|---|---|
| A | Actual qty @ std price 19,000 @ ¢1 | 19,000 | |
| | std. Qty of production achieved @ std. price | | |
| | 18,000 @ ¢1 | 1,8000 | ¢1,000A |
| B | Actual qty @ std price | | |
| | 10,000 @ ¢3 | 30,000 | |
| | Std qty of production achieved @ std price | | |
| | 9000 @ ¢3 | 27,000 | 3,000A |
| | | | 4,000A |

Labour Variances

Rate

| Actual hrs. worked @ std rate | ¢ |
|---|---|
| 28500 hrs @ ¢3.2 | 91200 |

| | |
|---|---|
| Actual hrs. @ std rate | 85,500 |
| 28500 hrs. @ ¢3 | 5,700A |

Efficiency Variance

| | |
|---|---|
| Actual hrs. worked @ std rate | ¢ |
| 28500hrs @ ¢3.2 Actual hrs. @ std rate | 85,500 |
| 27,000 hrs. @ ¢3 | |
| | 81,000 |
| | 4,500A |

Variable overhead variances Expenditure

| | |
|---|---|
| Actual hrs. worked @ VOR | ¢ |
| 28500@ ¢2 | 57,000 |
| Expenditure incurred | 52,000 |
| | 5,000F |

Efficiency

| | |
|---|---|
| Actual hrs. worked @ VOR | ¢ |
| 28500@ ¢2 | 57,000 |
| Std hrs. produced @ VOR 27,000 hrs. @ ¢2 | 54,000 |
| | 3000A |

Fixed overhead Variances

| | |
|---|---|
| Expenditure | ¢ |
| Budgeted overhead Actual overhead | 120,000 |
| | 116,000 |
| | 4,000F |

Efficiency

| | |
|---|---|
| Actual hrs. @ FOAR | ¢ |
| 28500@ ¢4 | 114,000 |
| Std hrs. produced @ FOAR 27,000 hrs. @ ¢4 | 108,000 |
| | 6000A |

Capacity Variance

| | |
|---|---|
| Actual hrs. @FOAR | ¢ |
| 28500@ ¢4 | 114,000 |
| Less Budgeted hrs @ FOAT 30,000 hrs. @ ¢4 | 120,000 |
| | 6000A |

If absorption costing system is adopted, the Sales margin variances would have been calculated as follows:

Sales margin price variance

| | |
|---|---|
| Actual qty@ ASP: | ¢ |
| 9,000units @ ¢42 Less actual qty x BSP | 378,000 |
| 9,000 x 40 | |
| 9,000 units @¢2 | 360,000 |
| | 18,000F |

Sales Volume Margin Variance

| | |
|---|---|
| Actual qty@ std margin | ¢ |
| 9,000 @ ¢48 | 72,000 |
| Budgeted qty@ std margin | |
| 10,000 @ ¢8 | 80,000 |
| | 8,000F |

AYEDIN LTD. STATEMENT OF RECONCILIATION OF BUDGETED AND ACTUAL PROFIT (MARGINAL COSTING METHOD)

| | ¢ | ¢ | ¢ | ¢ |
|---|---|---|---|---|
| Budgeted Profit | | | | 80,000 |
| Sales margin variances | | | | |
| Price Volume | | 18,000F | | |
| Direct Cost Variance: | | 20,000A | 2,000A | |
| Material Cost Variances: | | | | |
| Price A B | | | | |
| Usage Variance: | 1,900A | | | |
| Mix A B | 1,720F | 180A | | |
| Labour Cost Variance: | | | | |
| Rate Efficiency | 1,000A | | | |
| Variable overhead cost variance | 3,000A | 4,000A | 4,180A | |
| Expenditure | | | | |
| Efficacy | | 5,700A | | |
| Fixed overhead expenditure | | 4,500A | 10,200A | |
| Variance | | | | |
| Actual profit | | 5,000F | | |
| | | 3000A | 2,000F | |
| | | | 4,000F | 10,380A |
| | | | | 69,620 |

AYEDIN LTD.

STATEMENT OF RECONCILIATION OF BUDGETED AND ACTUAL PROFIT

(ABSORPTION COSTING METHOD)

| ¢ | ¢ | ¢ | ¢ |
|---|---|---|---|
| Budgeted Profit | | | 80,000 |
| Sales margin variances Price | | | |
| | 18,000F | | |

| | | | | |
|---|---|---|---|---|
| Volume Direct Cost Variance: | | 8,000A | 10,000F | |
| Material Cost Variances: Price | | | | |
| A | 1,900A | | | |
| B | 1,720F | 180A | | |
| Usage Variance: | | | | |
| Mix A B | 333F | | | |
| | 999A | | | |
| | 666A | | | |
| Yield | | | | |
| | 3334A | 4,000A | 4,180A | |
| Labour Cost Variance: | | | | |
| Rate Efficiency | | | | |
| | | 5,700A | | |
| Variable overhead cost variance | | 4,500A | 10,200A | |
| Expenditure | | | | |
| Efficacy | | | | |
| | | 5,000F | | |
| Fixed overhead expenditure | | 3000A | 2,000F | |
| Variance Efficiency | | | | |
| Capacity | | | | |
| Actual profit | | 4,000F | | |
| | | 6,000A | | |
| | | 6,000A | 8,000A | 10,380A |
| | | | | 69,620 |

13.8 **Computation of activity, capacity and efficiency variances**

In place of the volume, capacity and efficiency variances, ratios could be computed which will bring similar implications.

13.8.1. Activity ratio = $\frac{\textit{standard hours produced}}{\textit{Budgeted labour hours}} \times 100$

13.8.2. Capacity ratio $= \frac{\text{actual labour hours worked}}{\text{Budgeted labour hours}} \times 100$

13.8.3. Efficiency $= \frac{\text{standard hours produced}}{\text{actual labour hours worked}} \times 100$

**Illustrative example 13.6**

The following information has been extracted from a flexible budget:

| Standard Labour Hours | Overhead Cost (¢000) |
|---|---|
| 90,000 | 7,000 |
| 100,000 | 7,500 |

The increase in overheads between these two levels of activity is deemed to be entirely due to variable expenses. The actual activity results for the period are:

| | |
|---|---|
| Output in standard Hours Clock | 80,000 |
| Hours Worked Overhead Costs | 90,000 |
| -Fixed | |

**Required**

Compute the following ratios for the period:

i. Efficiency Ratio
ii. Activity Ratio
iii. Capacity Ratio (6 marks)

**ICAG JULY1992**

**Q3**

**SOLUTION 13.6**

(i) Efficiency ratio $= \frac{standard\ hours\ allowed}{actual\ hours\ worked} \times 100$

$= \frac{80,000}{90,000} \times 100$

$= \underline{88.89\%}$

(ii) Activity ratio $= \frac{standard\ hours\ allowed}{budgeted\ hours} \times 100$

$= \frac{80,000}{90,000} \times 100$

$= \underline{88.89\%}$

Or

$= \frac{80,000}{100,000} \times 100$

$= \underline{80\%}$

(iii) Capacity ratio $= \frac{actual\ hours\ worked}{budgeted\ hours} \times 100$

$= \frac{90,000}{90,000} \times 100$

$= \underline{100\%}$

Or

$= \frac{90,000}{100,000} \times 100$

$= \underline{90\%}$

## 13.9 Causes of Variances

Variances may be caused by many factors. These factors can only be established using investigating skills. Investigation of variances relates to finding out the real cause of a variance. The skills of audit investigations are employed in investigating variances. The process involves: observation, enquiry, checking, etc.

a. **Factors that could bring about the decision to investigate variances would include:**

i. History of the variance

- the more frequent a deviation in a particular item occurs the less likely investigation will be undertaken.
- the less frequent the deviation, the more likely that it will be investigated.

ii. The reliability of the standard

- the more reliable a standard is thought to be, the higher the chances of initiating investigations for deviations and vice versa

iii. The significance of the deviation

- Immaterial variances are more likely not to be investigated but material variances are most likely to be investigated

iv. Cost benefit analysis

- when benefits from investigation are more than the cost, then investigation is more likely to be undertaken.

v. Effect of investigation on future planning and control

- In spite of cost, if investigation will help future operations and control then investigation may be carried out.

When variances are to be investigated, management should set upper and lower ranges of variances above the standard figure used. Management needs to only investigate where the actual results obtained falls outside the limit.

## 13.10 CHAPTER SUMMARY

Standard costing is a useful control tool that assists business organisations to ensure that their budgets are achieved by pin pointing standard performance areas for corrective action to be taken.

The chapter explained standard costing and variance analysis. The study was done in three sections.

Section one explained the concept of standard, the types of standards, standard costing, the uses, merits and limitations

Section two concentrated on analysis of various variances such as materials cost, labour cost and overheads variances. The last section continued the discussion on variance analysis by explaining sales margin variances. The section also dealt with reconciliation of budgeted profit with actual profit. The use of activity ratios was also explained. The chapter concluded with the explanation of causes of variance, and the factors that influence the investigation of variances

**MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS**

1. Which of the following conditions explain when attainable standards can be achieved?
   A. Under the best possible conditions
   B. Under the most possible conditions
   C. When there is no idle time
   D. When the same trends have been maintained over a long period of time
   E. When allowances are made for normal losses and possible machine breakdown
2. What is the purpose of computing variances?
   A. To detect fraud
   B. To punish inefficient managers
   C. To investigate why standards are not met, and take corrective actions
   D. To maintain good records
   E. To settle long-standing issues

3. What type of variance can result from machine breakdown, power outages and work stoppages?
   A. Direct labour efficiency variance
   B. Machine breakdown variance
   C. Material wastage variance
   D. Idle time variance
   E. Electricity wastage variance

4. When we record substantial favourable variances,
   A. It shows that we are working hard
   B. We investigate and break it down because it may be concealing some hidden weaknesses
   C. We give bonuses to employees
   D. We raise the standards.
   E. We proudly publish it for others to see

5. If the cost of investigating a particular variance is higher than the benefit to be derived from it, …….
   A. The investigation is dropped
   B. We still go ahead and investigate for the records
   C. We keep the investigation till the cases pile up
   D. We spread the cost over a number of years
   E. The investigation is done in piece meal

6. Standards which can only be attained under the most favourable working conditions are called ……

7. Which costing concept best describes situations when actual costs are compared with standard costs?……

8. The official document showing the components of the standard cost of a product is known as …….

9. The addition of direct material price variance and direct material usage variance w i l l give us a…….

10. The difference between the standard direct labour rate and actual direct labour rate per hour for the actual total hours worked and paid for is called ……

**SOLUTION**

1. E
2. C
3. D
4. B
5. A
6. Ideal standards
7. Variances
8. Standard Cost Card
9. Direct material cost variance
10. Direct labour rate variance

**EXAMINATION TYPE QUESTIONS**

(a) Outline the factors that should be considered when deciding whether or not to investigate variances revealed in a standard costing and budgetary control system (5 marks)

(b) Pass Company Limited operates a standard marginal costing system. The company produces a standardized product called "success" using a single raw material called material papa.

(c) During the month of August 2021, 5000 units of "success" were produced. The following data relates to actual performance of Pass Company Limited during the month of August 2021.

i. The company purchased 50,000 kg of material papa at a cost of ¢144,000,000.
ii. Stock levels of material papa were as follows:1/8/2021 20,000kg 31/8/2021 15,000kg
iii. The company paid workers for 26,000 hours at a total wages cost of ¢48,360,000.
iv. The company incurred ¢16,800,000 for variable production overheads.

The standard cost card below is given

| **Cost per unit** | ¢ |
|---|---|
| Direct material cost 12kg@ ¢3000 per kg | 36,000 |
| Direct labour cost 5 hours @ ¢1800 per hour | 9,000 |
| Prime cost | 45,000 |
| Variable production overhead cost 5 hours @ ¢600 | 3,000 |
| Variable production cost | 48,000 |

**Required:**

Calculate the following cost variances:

i. Direct material cost variance (analysed into price and wage variances)
ii. Direct wages cost variance (analysed into rate and efficiency variances)
iii. Variable production overhead cost variance. (analysed in to expenditure and efficiency variances) (15 marks)

**(Total: 20 marks)**

Adapted from **ICAG NOV.2005**

(a) State and explain briefly **FOUR (4)** different types of standards (8 marks)

(b) Nsawam Poultry Products Ltd. Employs different grades of labour on its farm. The direct Labour cost of its product is as follows:

Three (3) hours grade A labour at ¢250 per hour resulting in to ¢750 per unit of the product. During the month of March 2000, 300 units of the product were made, and the labour cost of grade A was ¢220,000 for 910 hours. During the month, there was a machine breakdown and 40 hours were recorded as idle time.

**You are required to calculate the:**

(i) Direct labour total cost variance
(ii) Direct labour rate variance
(iii) Idle time variance
(iv) Direct labour efficiency variance.

(12 marks)

**(Total 20 marks)**
**Adapted from ICAG NOV. 2000**

**13.3** A Company produces a product which has a standard variable production cost of ¢8,000 per unit made up as follows:

| | **unit** | |
|---|---|---|
| Direct materials | 4600 | (2 kg x ¢2300 per kg) |
| Direct labour | 2100 | (0.7 hours x ¢3,000 per hour) |
| Variable overhead | 1300 | |

Fixed manufacturing costs are treated as period costs. The following information is available for the period just ended:

| | **¢** |
|---|---|
| Variable manufacturing cost of sales (at standard cost) | 263,520,000 |
| Opening stock of finished goods (at standard cost) | 120,800,000 |
| Closing stock of finished goods (at standard cost) Direct | 146,080,000 |
| material price variance | 2,571,000 adverse |
| Raw material used in manufacture (at actual cost) | 170,310,000 |

**Required:**

(a) Determine for the period just ended:

i. the number of units produced,

ii. the raw materials usage variance,

iii. the total actual direct labour cost; and

iv. the total actual cost per kg. Of raw material used

(b) Outline the possible causes of the raw material variances

(20 marks)

ICAG MAY 2000

13.4(a) Explain the terms below:

i) Ideal standard

ii) Attainable standard

iii) Standard hour or minute

(3marks)

(b) List **FOUR** main steps in setting up a standard costing system

(6 marks)

a) Identify **THREE** causes of each of the following variances:

(i) Direct material price variances

(ii) Direct labour efficiency variances

(iii) Overheads expenditure variances

(6 marks)
**(Total: 15 marks)**

**ICAG NOV. 1997**

**13.5** End Times Limited manufactures and sells a single product, the standard cost of which is as follows:

| Standards Per Unit | ¢ | ¢ |
|---|---|---|
| Selling price Standard | | 107,200 |
| cost: | | |
| Materials: 64 units @ ¢400 | 25,600 | |
| Labour: 16 hours @ ¢300 Overheads | 4,800 | |
| 16 hours @ ¢2,400 | 38,400 | 68,800 |
| Standard Profit margin | | 38,400 |

The following information relates to the activities of End Times for the month of September 2021.

| | Budgeted | Actual |
|---|---|---|
| | ¢ | ¢ |
| Production and Sales Units | 60,000 | 50,000 |
| Direct labour hours | 960,000 | 20,000 |

| | | | |
|---|---|---|---|
| | - wage rate | ¢300/hour | 450/hour |
| Fixed overheads | | 768,000,000 | 800,000,000 |
| Variable overheads | | 1,536,000 | 1,614,000 |
| Material units | | 3,840,000 | 3,840,000 |
| | - price per unit | 400 | 300 |

The company's actual selling price during the period of September 2021 was ¢105,000 and there were no opening and closing balances of stocks and finished goods.

End Time Limited uses absorption costing system.

**Required:**

(a) Prepare profit statement to ascertain the Budgeted Profit and Actual profit of End Times Limited for the month of September 2021. (6 Marks)

(b) Calculate all the relevant variances and reconcile the budgeted and actual profits for the period. (14 Marks)

**(Total: 20 Marks)**

Adapted from ICAG NOV.2005

**13.6** Outline the procedure for setting standard costs of the following cost elements

(i) Materials cost

(ii) Labour cost (15 marks)

**13.7**

(a) Outline the benefits which a company may derive from a standard costing system. (5 Marks)

(c) Success Company Ltd., a manufacturer of aluminum roofing sheets, operates a standard marginal costing system. It produces a standardized roofing sheet using a single raw material, Alumina. Standard costs relating to one unit of roofing sheet have been calculated as follows:

| | | ¢ |
|---|---|---|
| Direct Material: | 10 kg. @ 3,000 per kg. | 30,000 |
| Direct Labour | 30,000 5 hrs. @ ¢1,800 per hr. | 9,000 |
| | | 39,000 |
| Variable production overhead | 5 hrs. @ ¢600 per hr. | 3,000 |
| | | 42,000 |

During the first week of November, 1000 units of roofing sheets were produced. The relevant information on actual performance is as follows:

(i) Operating stock of Raw Material, Alumina was 4,000 kg.

(ii) 10,000 kg of Alumina were bought during the week at a cost of ¢28,800,000

(iii) Stock at the close of the week was 3000 kg.

(iv) Stocks of Raw Materials are valued at the standard price of ¢3,000 per kg.

(v) 5,200 direct labour hours were worked during the week at a total wage bill of ¢9,672,000

(vi) Actual variable production overhead for the week was ¢3,360,000

**Required:**

Compute the following cost variances which arose in the week.

(i) Variable production cost variance.

(a) Direct wages cost variance analysed into rate and efficiency.

(b) Direct material cost variance analysed into price and usage.

(c) Variable production overhead variance.

(15 Marks)

ICAG Man Accounting NOV. 2000

(a) Discuss the relevance to management of the following variances and for variance; give a specific example of how it could arise:

(i) Direct material usage variance
(ii) Direct material price variance
(iii) Direct wages cost variance
(iv) Direct labour efficiency variance
(v) Fixed production overhead expenditure variance.

**(15 Marks)**

(b) Briefly explain the statement that in standard costing systems, variances are interrelated and should not be considered in isolation'.

**(5 Marks)**
**(Total 20 Marks)**

10.9

a. Define Standard Costing **1 Mark**

b. Mention the variance that is attributed to under and over absorption and interprets the meaning of under absorption

3½ Marks

c. TYT uses standard costing technique for determining its operational performance. The firm uses overhead absorption rate to estimate its overhead costs. Relevant data provided areas follows:

**Budgeted figures:**

| | |
|---|---|
| Output | 10,000 units |
| Production hours to utilise Fixed | 20,000 |
| Production Overhead | ₦40,000 |

**Actual figures achieved**

**Required:**

a. Compute variances for:

i. Fixed overhead cost variance

ii. components of fixed overhead cost variances

b.. Interpret variance that identifies under and over absorption

**(8 Marks)**

## CHAPTER FOURTEEN

## DEVELOPMENTS IN COST ACCOUNTING

## CHAPTER CONTENTS

a. Just in time Systems;

b. Kanban inventory control;

c. Back flush accounting;

d. Life cycle costing;

e. Target costing;

f. Advanced Manufacturing Technique (AMT);

g. Lean accounting;

h. Throughput accounting;

i. Data analytics, predictive analytics, and cost transparency;

j. Sustainability cost accounting and sustainable cost management;

k. Environmental cost accounting and carbon footprint accounting;

l. Cloud-based solutions and cost accounting; automation, integration; and

m. Blockchain technology and cost accounting: Transparency, security and smart contracts

14.0 **Objectives:**

After studying this chapter, readers should be able to update their knowledge of the newly developed approaches in cost accounting and explain the following costing related terms:

a. Just – in- time (JIT);

b. Kanban inventory control;

c. Backflush accounting;

d. Life cycle costing;

e. Target costing; and

f. Advanced Manufacturing Technique (AMT);

g. Lean accounting;

h. Throughput accounting;

i. Data analytics, predictive analytics, and cost transparency;

j. Sustainability cost accounting and sustainable cost management;

k. Environmental cost accounting and carbon footprint accounting;

l. Cloud-based solutions and cost accounting; automation, integration; and

m. Blockchain technology and cost accounting: Transparency, security and smart contracts

## 14.1 **Just-in-time (JIT)**

JIT system was developed in Japan, and has been widely acclaimed as a major contributor to the country's success in manufacturing processes. As the name suggests, it seeks to eliminate the need to keep inventories thereby saving the costs associated with holding inventories. The overall objective is to produce or procure inventory right at the time they are required without compromising quality.

The following are the main goals of JIT:

a. No opening or closing inventory. Items are purchased or produced "just in time" in the required quantities and quality;

b. Elimination of non-value adding activities;

c. No production wastages;

d. 100% on time deliveries;

e. Batch sizes of one;

f. Demand-pull manufacture i.e. the production chain is activated real-time by demand;

f. No breakdowns;
and

h. Short set- ups.

JIT costing approach can be subdivided into two, namely:

i. Just in time purchasing

ii. Just in time production

Under JIT purchasing, arrangements subsists with reliable suppliers for regular or frequent deliveries of inventories as and when they are required, thereby eliminating the need to tie down capital on inventories. The approach assists in saving costs associated with keeping inventories (raw material inventory and work in progress), save factory space and paper work relating to large and long-term orders. Under JIT purchasing, material procurement is executed in accordance with production required.

However, a successful JIT Production system is predicated upon two assurances viz:

(a) Prompt delivery by the supplier at the exact time required; and
(b) 100% quality with zero reject.

JIT production works on similar principles with JIT purchasing. Production is triggered by a confirmed order from the customer, hence the term "demand-pull".

Within the shortest possible time that the order is received, material supplier is notified and the order is executed, thus shortening the production set-up time.

JIT has two major limitations viz:

a. Lack of flexibility which makes it vulnerable to possible disruptions in the

supply chain; and

b. Difficulty in predicting demand patterns.

## 14.2 KANBAN INVENTORY CONTROL

Where supply time is lengthy and demand time is difficult to forecast, the best that can be done is to respond quickly to observed demand. This is achieved using Kanban system.

Kanban system is a variant of JIT but a step ahead of it. It is used as a demand signal which spreads through the supply chain. The aim of Kanban principles is to make supply processes as lean as possible.

Kanban can be used to ensure that intermediate inventory held in the supply chain which (usually small) are better managed.

Kanban is a concept which attempts to maintain minimum inventory. It involves more than fine-tuning production and supplier scheduling systems. It entails minimisation of inventories by supplying them when needed in production, and monitoring of work in progress.

The supplier or warehouse delivers components to the production line as and when they are needed, so that there is no storage in the production area.

Kanban follows strict rules of use and the close monitoring of these rules is a never ending one. This is done to ensure that Kanban does what is required.

Kanban can be implemented electronically. e-Kanban simplifiesmonitoring process. Basic features of e-Kanban operations include:

Elimination of common manual entry errors;

Provision of quick and precise information;

Quick response to changes;

Avoidance of over production;

minimization of waste; and

Maintenance of low costs associated with information transfer.

Traditionally, Kanban systems have been used by manufacturers to control the amount of inventories held on the production line. The manufacturing process is made quicker due to on-time delivery of inventory to production from the warehouse.

The use of the Kanban system in warehouses to aid just-in-time replenishment of pick bins makes perfect sense. It takes manufacturing/production process using JIT principle one step further to incorporate warehouse replenishment process. Same benefits are also reaped by the warehouse and the manufacturing process.

Benefits of Kanban Inventory Control

a. Warehouses do not overstock goods/

b. Suppliers have an instantaneous picture of inventory levels held;

c. the process moves from one of stocking goods to one of perpetual inventory;

d. Visibility of inventories movement is available to management, operators and suppliers alike; processes are as lean as possible;

e. Inventories are delivered when needed;

f. Integrating e-kanban systems into warehouse management systems allows for real- time demand signaling across the supply chain and improved visibility;

g. Data pulled from Kanban systems can be used to optimize inventory levels ; and

h. It aids better tracking of suppliers' lead and replenishment times.

## 14.3 **Backflush Accounting (Backflush Costing)**

According to the terminology, Backflush Accounting is defined as:

*A method of costing, associated with a JIT production system, which applies cost to the*

*Output of a process. Costs do not mirror the flow of products through the production process,*

*but are attached to output produced (finished goods inventory and cost of sales), on the*

*assumption that such backflushed costs are a realistic measure of the actual costs incurred*

.(CIMA).

In a nutshell, backflush accounting is a simpler cost accounting system designed to reduce

or eliminate detailed accounting entries. Instead of the traditional detailed, tracking of material movement through stores and production, backflush costing starts from the finished goods and works backwards to attribute costs between cost of goods sold and finished goods inventory and/or materials inventory with no separate accounting for WIP and stores.
The reduction in accounting entries leads to reduction of clerical cost.

The number of entries required in backflush costing depends on the nature of production and its complexity.

**Backflush costing entries**

**When raw materials are purchased**

Debit Raw-in-Progress A/c Credit Creditor

for RM or Bank

**For conversion cost**

Debit conversion cost control account

Credit wages payable account

**For cost of goods sold** (the entry closes finished goods account to cost of sales account Debit cost of

sales account

Credit finished goods account

## 14.4 Life Cycle Costing

The total "life cycle" cost of a product captures all costs incurred on that product from

"cradle to grave", or from "womb-to-tomb" ( i e. from costs of research and

development, plant and equipment, manufacturing costs, product development costs

and promotion costs). The purpose of lifecycle costing is to ensure that the firm recovers all costs over the estimated number of units expected to be sold over the lifecycle of the product.

Life cycle costing assists a firm to compete favourably as far as pricing is concerned most especially when similar products are introduced by competitors.

## 14.5 Target Costing

Target Costing is defined as the process whereby new products are to be designed in accordance with the unit prices that expected potential customers can afford or would offer.

A Target cost is defined as:

"A product cost estimate derived by subtracting a desired profit margin from a competitive market price". (CIMA).

Target costing is widely used in Japan and is gradually gaining ground in Europe and the USA. It is a market-driven approach where the target selling price of a proposed product is determined with a focus on gaining the desired market share. The required profit margin is deducted from the target selling price to arrive at the target cost for the product.

Emphasis is laid on the design stage to achieve target cost. This is because the design is considered as one that consumes the major chunk of the costs, prior to the release to manufacturing. Hence, product designers, purchasing and manufacturing specialists work together to determine the product and process features which will enable the long-run target costs to be achieved.

## 14.6 **Advanced Manufacturing Technology (AMT)**

AMT is a general term for manufacturing techniques which differ from the traditional production methods. Traditional production methods have been criticized by modern day accountants for:

its obsolescence;

Accommodating and rewarding inefficiencies; and Lack of

incentives for improvement

A sophisticated technique called Advanced Manufacturing Technology (AMT) has therefore been developed in replacement of the traditional methods.

AMT encompasses techniques like Computer Aided Design and Manufacturing (CAD/CAM), Flexible Manufacturing System (FMS), Total Quality Management (TQM), Material Resources Planning (MRP), Just-In-Time (JIT) and others.

AMT techniques promises effectiveness and healthy competition capability to produce high quality goods at low cost and thereby maximisation of customers' satisfaction. For short, AMT aids effective competition in a technologically driven world.

### Lean Accounting

### Definition and Principles

Lean accounting is a methodology that integrates accounting practices with the principles of lean manufacturing and lean thinking. It emphasizes the elimination of waste, improvement of efficiency, and

enhancement of customer value. Unlike traditional accounting methods, which may not fully support lean principles, lean accounting aims to simplify financial reporting and provide relevant performance metrics that aid decision-making.

**Key Components of Lean Accounting**

1. **Simplified Financial Reporting**: Lean accounting seeks to make financial statements easier to understand and relevant to specific business operations. This involves streamlining the information flow and accounting procedures.

2. **Value Stream Costing**: Instead of traditional product costing methods, lean accounting focuses on value streams, which encompass all actions required to bring a product from concept to customer.

3. **Lean Culture**: It promotes a culture of continuous improvement within the accounting department, encouraging teamwork and effective communication to enhance overall productivity.

4. **Lean Tools**: Various techniques are employed to reduce waste and improve efficiency within the accounting function, ensuring that financial management aligns with the company's lean strategies.

**Benefits of Lean Accounting**

- **Enhanced Decision-Making**: By providing clearer and more relevant financial information, lean accounting supports better decision-making processes within organizations.
- **Increased Efficiency**: Lean accounting practices help streamline accounting operations, reducing unnecessary steps and focusing on value-added activities.

- **Alignment with Business Goals**: Lean accounting ensures that financial practices are closely aligned with the overall goals of the business, promoting a unified approach to achieving operational excellence.

**Conclusion**

Lean accounting represents a significant shift from traditional accounting practices, focusing on value creation and waste elimination. By adopting lean accounting principles, organizations can improve their financial management processes, enhance customer satisfaction, and drive overall business success. This

approach not only simplifies accounting but also fosters a culture of continuous improvement, making it a valuable strategy for modern businesses.

**Throughput Accounting**

Businesses uses throughput accounting as an intelligent mechanism to plan production. The purpose of using this accounting approach is profit maximization. This concept is applicable when you have a certain constraint in the production process of multiple products. Hence, its more about production plan and manufacturing.

For instance, you may need to use raw material (bottleneck) in multiple products of your business. Bottleneck refers to the material or any other factor limiting your production capacity. The bottleneck may be some scarce raw material, scarce labor resources, scarce machine hours or any other production-related factor. So,

This type of accounting is only applicable when you have the following circumstance.

1. Your business has a certain bottleneck.
2. Your business produces more than one product, and these products use bottleneck resource in different proportion.

In such a situation, throughput accounting can help make a production plan that maximizes your company's profit. Let's understand how it's done.

**Detailed understanding**

Suppose there is a higher demand for the company's products. However, your production capacity is limited. It means the business does not have to worry about selling aspects, but making is a problem. Further, the business has multiple products, and all in are in equal demand.

Detailed analysis of the production facility reveals that production is limited because of machine hours. And it's not feasible for the business to purchase an additional machine. Further, all three products use different machine hours.

First of all, we need to understand that machine hour is a bottleneck limiting the business's production and profit. Secondly, we need to identify a product that produces the highest contribution using the lowest bottleneck resource (machine hours). Let's understand with steps how it can be done.

Step -1

**Identification of bottleneck resource –** it's about knowing your production facility. So, this stage involves a detailed analysis of production stages. There can be different bottlenecks for different businesses. For instance, a scarce resource can be labor hours, machine hours, raw material, or any other

factor of production. So, identifying scarce resources is more about knowing where the problem actually arises.

Step -2

**Calculate contribution per unit of all products –** It's about doing some math with the product. Simply deduct all variable costs from revenue.

Step -3

**Divide contribution per unit with scarce resource –** Again, it's simple math, divide contribution per unit with the scarce resource. It's called ***throughput value***.

Step -4

**Rank step-3 values in the highest priority** – The highest-ranked values in step-3 (***throughput value***) indicate the highest utilization of scare resources to generate profit. In simple words, we have identified a product that uses the lowest scarce resource to produce the highest contribution. So, priority should be given to the highest-ranked products. However, the maximum demand for each product should be considered while making product plan.

**So, what have we done?**

We have identified the product that produces the highest contribution by utilizing the lowest scarce resource. Further, we have ranked products in terms of contribution utilized by scarce resources. Hence, this information can be used to produce a production plan with the highest overall contribution.

**Throughput accounting ratio formula**

Following is the formula of throughput accounting

**Throughput accounting ratio = throughput value / fixed cost per unit**

Where, throughput value = contribution / scarce resource unit

Fixed cost per unit = Total fixed cost / total production

So, if the value calculated is more than one, it means that the throughput value is able to cover fixed cost. On the other hand, if the throughput ratio is less than one, it means that throughput value does not cover the business's fixed expenses. Hence, it can be helpful for the business to understand if it's viable to make product.

**Throughout accounting and theory of constraints**

Theory of constraint refers to the situation where some of the limiting factors halt production. These limiting factors are also called bottleneck or scarce resources, and production plan needs to be managed in line with the constraints.

Throughput accounting provides us with a calculation method effective in planning for production.

**Throughput accounting vs. cost accounting**

Throughput accounting is limited to the maximum utilization of scarce resources and achieving higher profitability. On the other hand, cost accounting is a broader concept that is focused on operational aspects of overall cost, price, production, contribution, cost behavior, cost analysis, process accounting, and manufacturing aspects.

**Advantages of throughput accounting**

Following are some of the advantages of throughput accounting.

1. Helps to formulate the best possible production plan. This plan leads to maximum utilization of scarce resources and maximum profit.
2. It helps closely understand contribution, production procedures, fixed cost behavior, and variable costs.
3. Throughput concepts can be applied quickly; it's flexible in terms of application and greatly help in decision making.
4. It's rational and considers product demand while making a production plan.

**Disadvantages of throughput accounting**

Following are some of the disadvantages of throughput accounting.

1. It can be difficult to understand the concept of throughput accounting.
2. It's not easy to compile small details like variable and fixed production elements.
3. The concept is only applicable when you have bottleneck resources and multiple products.
4. It's a short-term approach to profitability that does not consider long-term objectives that might be achieved by selling some other product mix rather than suggested by throughput.
5. The application can be more complicated if there are multiple bottlenecks

**Data Analytics**

Data analytics is the process of examining raw data to draw meaningful insights, which can inform decision-making and drive business strategies.

Data analytics involves collecting, transforming, and organizing data to uncover patterns, trends, and insights that can help organizations make informed decisions. It encompasses various techniques from statistics, mathematics, and computer science to analyze data sets effectively. The field includes data analysis, data science, and data engineering, and is crucial across many industries, including finance, healthcare, marketing, and technology.

**Types of Data Analytics**

There are four primary types of data analytics, each serving a different purpose:

1. **Descriptive Analytics**: This type answers the question "What happened?" by summarizing past data to identify trends and patterns. Techniques include data aggregation and mining.
2. **Diagnostic Analytics**: This type explores "Why did it happen?" by investigating anomalies and identifying the causes behind certain trends or events.
3. **Predictive Analytics**: This type forecasts "What is likely to happen in the future?" by using historical data to predict future outcomes.
4. **Prescriptive Analytics**: This type advises on "What should we do?" by recommending actions based on data analysis to achieve desired outcomes.

**Importance of Data Analytics**

Data analytics is essential for organizations to optimize performance, improve decision-making, and gain a competitive edge. By leveraging data, businesses can better understand their customers, enhance operational efficiency, and identify new opportunities for growth. The insights derived from data analytics enable organizations to make data-driven decisions rather than relying on intuition or guesswork.

**Cost Transparency**

Cost transparency is the practice of making the costs of products, services, or processes visible and understandable to stakeholders, enhancing trust and decision-making.

**Definition and Importance**

Cost transparency refers to the practice of disclosing the costs associated with products, services, or activities to relevant stakeholders, including customers, suppliers, employees, and investors. This transparency is crucial for several reasons:

1. **Informed Decision-Making**: Stakeholders can make better choices when they have access to clear and accurate cost information. For instance, customers can compare prices and value propositions, while suppliers can negotiate better terms based on transparent cost structures.

2. **Trust and Accountability**: By openly sharing cost information, companies can build trust with their customers and partners. This transparency fosters accountability, as stakeholders can hold each other responsible for their actions and results.
3. **Enhanced Efficiency**: Cost transparency can lead to improved operational efficiency. Organizations can identify areas for cost reduction and optimize resource allocation, ultimately leading to better financial performance.

**Benefits of Cost Transparency**

- **For Customers**: Access to detailed cost breakdowns can enhance customer satisfaction and loyalty. When customers understand what they are paying for, they are more likely to feel confident in their purchases.
- **For Suppliers**: Suppliers benefit from cost transparency by being able to align their operations with customer expectations and negotiate fair contracts. This can lead to collaborative efforts to reduce costs and improve quality.
- **For Companies**: Organizations that embrace cost transparency can improve their market competitiveness. For example, research shows that consumers are more likely to purchase products when they understand the costs involved in their production.

**Challenges in Achieving Cost Transparency**

Despite its benefits, achieving cost transparency can be challenging due to factors such as:

- **Complex Financial Systems**: Many organizations struggle with disparate financial systems that make it difficult to obtain clear cost data.
- **Data Quality Issues**: Poor data quality and lack of methodological rigor can hinder the effectiveness of cost transparency initiatives.
- **Resistance to Change**: Organizations may face internal resistance when attempting to implement new transparency practices, especially if it involves significant changes to existing processes.

**Conclusion**

Cost transparency is a vital practice that can significantly impact stakeholder relationships and organizational efficiency. By making costs visible and understandable, companies can foster trust, enhance decision-making, and improve overall performance. However, achieving this transparency requires overcoming various challenges, including data quality issues and organizational resistance

## Sustainability Cost Accounting

Sustainable Cost Accounting (SCA) has attracted increasing interest from both business practitioners and policy makers in recent times because of the ever growing awareness on the inclusion of environmental perspectives in financial reporting. Because traditional systems of accounting essentially relate to the area of financial success, in many cases it may not allow the consideration of environmental or social costs that have arisen from a corporate operation; On the contrary, companies do feel a developing urge to account for and disclose their environmental repercussions. This is because of the increased concern about global warming, resource depletion, and the sustainability of businesses as well.

SCA ensures that an enterprise accounts for the financial impact of its ecological footprint-things like carbon emissions, resource utilization, and ecosystem destruction. SCA takes a more holistic approach in considering externalities, which is a cost or advantage not well reflected in the market price of a commodity

1. The Importance of Sustainable Cost Management

1. Cost management is a vital aspect of any business, as it directly impacts profitability and long-term sustainability. However, in today's world, it is equally important to consider the environmental and social implications of our cost management strategies. sustainable cost management aims to strike a balance between financial efficiency and responsible resource usage, ultimately ensuring the long-term viability of both the business and the planet.

2. One of the key reasons why sustainable cost management is crucial is its impact on the environment. By adopting sustainable practices, businesses can reduce their carbon footprint, minimize waste generation, and conserve natural resources. For example, companies can implement energy-efficient technologies, recycle and reuse materials, and optimize transportation routes to reduce emissions. These measures not only benefit the environment but also often result in significant cost savings.

3. Another important aspect of sustainable cost management is its positive influence on a company's reputation and brand image. Consumers today are increasingly conscious about the environmental and social impact of the products and services they purchase. By demonstrating a commitment to sustainability, businesses can attract environmentally conscious customers and build long-lasting relationships based on shared values. This can lead to increased customer loyalty, positive word-of-mouth, and ultimately, higher sales.

4. To effectively implement sustainable cost management strategies, businesses can follow some key tips and best practices. Firstly, conducting a thorough assessment of current operations and identifying areas where sustainable practices can be integrated is essential. This could involve analyzing energy usage, waste management processes, and supply chain practices. Once potential areas for improvement are identified, businesses can set specific goals and targets to guide their sustainability efforts.

5. Collaboration and partnerships with suppliers, customers, and other stakeholders are also crucial in achieving sustainable cost management. By working together, businesses can leverage collective expertise and resources to develop innovative solutions and drive systemic change. For example, companies can collaborate with suppliers to source sustainable materials or work with customers to promote responsible consumption habits.

6. Case studies provide valuable insights into successful sustainable cost management strategies. For instance, Interface Inc., a global manufacturer of modular carpets, implemented a radical sustainability program called "Mission Zero." This initiative aimed to eliminate the company's negative environmental impact as fast as possible.

Through innovative approaches such as recycling old carpets into new ones and using renewable energy sources, Interface achieved significant cost savings while advancing their sustainability goals.

7. In conclusion, sustainable cost management is a critical consideration for businesses aiming for long-term success. By integrating sustainability into cost management strategies, companies can reduce their environmental impact, enhance their reputation, and create a more resilient and profitable business. With careful planning, collaboration, and the adoption of best practices, businesses can achieve a harmonious balance between financial efficiency and responsible resource usage.

2. The Benefits of Sustainable Cost Management

1. Increased profitability:

One of the primary benefits of implementing sustainable cost management strategies in business is the potential for increased profitability. By identifying and eliminating wasteful practices, businesses can reduce costs and improve their bottom line. For example, a manufacturing company may analyze its production processes and find ways to reduce energy consumption, leading to significant savings on utility bills. Similarly, a restaurant may focus on reducing food waste by implementing portion control measures or partnering with local food banks, resulting in cost savings and improved profitability.

2. Enhanced brand reputation:

Adopting sustainable cost management practices can also contribute to a business's brand reputation. Many consumers today are becoming more conscious of the environmental and social impact of the products and services they purchase. By showcasing their commitment to sustainability, businesses can attract environmentally and socially conscious consumers. For instance, an apparel company that utilizes eco-friendly materials and ethical manufacturing practices can differentiate itself from competitors and appeal to consumers who prioritize sustainability.

3. Reduced risk and increased resilience:

Sustainable cost management strategies can help businesses mitigate risks and enhance their resilience. By diversifying supply chains and reducing dependence on limited resources, companies can better withstand disruptions caused by factors such as natural disasters, political instability, or price fluctuations. For example, a food processing company that sources ingredients from multiple regions can minimize the impact of crop failures in a particular area, ensuring a consistent supply and avoiding potential financial losses.

4. Compliance with regulations:

Implementing sustainable cost management practices can also help businesses comply with increasingly stringent environmental and social regulations. By proactively integrating sustainability into their operations, companies can avoid penalties and legal issues associated with non-compliance. For instance, a construction company that adheres to green building standards and waste management guidelines can avoid fines and legal disputes related to pollution or improper waste disposal.

5. Employee engagement and retention:

Sustainable cost management strategies can contribute to higher levels of employee engagement and retention. Employees are increasingly seeking purposeful work and are more likely to be motivated and satisfied when working for a company that aligns with their values. By prioritizing sustainability, businesses can attract and retain

top talent who are passionate about making a positive impact on the environment and society. For example, a tech company that offers flexible work arrangements and promotes sustainable commuting options can attract environmentally conscious employees.

In conclusion, implementing sustainable cost management practices in business can yield numerous benefits, including increased profitability, enhanced brand reputation, reduced risk, regulatory compliance, and improved employee engagement and retention. By taking a proactive approach to managing costs sustainably, businesses can achieve a balance between financial success and environmental and social responsibility.

3. Key Principles of Sustainable Cost Management

1. Identify and prioritize cost-saving opportunities: In order to effectively manage costs in a sustainable manner, it is crucial to identify and prioritize areas where cost-saving opportunities exist. This can be achieved through conducting a thorough analysis of the organization's operations and processes. For instance, a manufacturing company may identify energy consumption as a major cost driver and implement energy-efficient technologies and practices to reduce costs. By prioritizing such opportunities, organizations can focus their efforts on areas that have the greatest potential for cost savings while also aligning with sustainability goals.

2. Implement sustainable procurement practices: Sustainable cost management involves not only reducing costs but also considering the environmental and social impact of procurement decisions. By implementing sustainable procurement practices, organizations can reduce costs associated with waste, energy consumption, and resource depletion. For example, a restaurant chain may choose to source ingredients from local suppliers to minimize transportation costs and support the local economy. Additionally, organizations can prioritize suppliers that adhere to ethical labor practices and environmental standards, ensuring that their procurement decisions align with sustainability objectives.

3. Foster a culture of cost-consciousness and sustainability: Sustainable cost management requires a shift in mindset and a culture that values both cost-consciousness and sustainability. This can be achieved through employee education and engagement programs that promote awareness and understanding of the importance of managing costs sustainably. For instance, a technology company may conduct training sessions to educate employees on the energy-saving features of office equipment and encourage them to adopt energy-efficient practices in their daily work. By fostering a culture that values cost-consciousness and sustainability, organizations can empower employees to actively contribute to cost-saving initiatives while also promoting sustainable practices.

4. Continuously monitor and measure cost-saving initiatives: To ensure the effectiveness of sustainable cost management strategies, it is essential to continuously monitor and measure the impact of cost-saving initiatives. This can be done through regular tracking and analysis of key performance indicators (KPIs) related to cost reduction and sustainability. For example, a retail store may track the reduction in energy consumption and waste generation as a result of implementing energy-efficient lighting and waste management practices. By monitoring and measuring the outcomes of cost-saving initiatives, organizations can identify areas for improvement and make data-driven decisions to further enhance their sustainability and cost management strategies.

5. Collaborate with stakeholders and share best practices: Collaboration with stakeholders, including suppliers, customers, and industry peers, can provide valuable insights and opportunities for sustainable cost management. By sharing best practices and lessons learned, organizations can collectively identify innovative solutions and drive industry-wide improvements in cost management and sustainability. For instance, a construction company may

collaborate with suppliers to explore the use of recycled materials in building projects, reducing costs associated with raw material procurement and waste disposal. By actively engaging with stakeholders and sharing best practices, organizations can leverage collective knowledge and expertise to achieve sustainable cost management goals.

In conclusion, sustainable cost management requires a holistic approach that balances cost reduction with environmental and social considerations. By following key principles such as identifying cost-saving opportunities, implementing sustainable procurement practices, fostering a culture of cost-consciousness and sustainability, continuously monitoring and measuring initiatives, and collaborating with stakeholders, organizations can effectively manage costs in a sustainable manner while driving positive environmental and social outcomes.

4. Integrating Sustainability into Cost Management Strategies

1. Identify areas for sustainable cost management: The first step in integrating sustainability into cost management strategies is to identify areas within your organization where sustainable practices can be implemented. This could include reducing energy consumption, optimizing waste management, or sourcing materials from sustainable suppliers. By pinpointing these areas, you can focus your efforts on implementing cost-effective sustainability measures.

2. Conduct a cost-benefit analysis: Before implementing any sustainability measures, it is important to conduct a thorough cost-benefit analysis. This analysis will help you determine the financial impact of implementing sustainable practices and whether the benefits outweigh the costs. For example, investing in energy-efficient equipment may have a higher upfront cost, but the long-term savings in energy consumption could offset this initial investment.

3. Set clear sustainability goals: To effectively integrate sustainability into cost management strategies, it is crucial to set clear sustainability goals. These goals should be specific, measurable, attainable, relevant, and time-bound (SMART). For instance, a goal could be to reduce water consumption by 20% within the next year. By setting these goals, you can track your progress and ensure that sustainability becomes an integral part of your cost management strategies.

4. Engage employees and stakeholders: Successful integration of sustainability into cost management strategies requires the engagement and support of employees and stakeholders. This can be achieved through training programs, awareness campaigns, and incentives. By involving everyone in the organization, you can foster a culture of sustainability and encourage innovative ideas for cost-saving measures.

5. Collaborate with suppliers and partners: Sustainability should not be limited to internal operations but should extend to your supply chain as well. Collaborating with suppliers and partners who share your commitment to sustainability can help drive cost savings. For example, sourcing materials from suppliers who prioritize sustainable practices can result in reduced costs and improved brand reputation.

6. Monitor and measure progress: Regular monitoring and measurement of sustainability initiatives are essential to ensure their effectiveness. This can be done through the use of key performance indicators (KPIs) such as energy consumption, waste reduction, or carbon emissions. By tracking progress, you can identify areas for improvement and make informed decisions regarding your cost management strategies.

Integrating sustainability into cost management strategies is not only beneficial for the environment but also for the financial health of an organization. By identifying areas for sustainable cost management, conducting cost-

benefit analyses, setting clear goals, engaging employees and stakeholders, collaborating with suppliers, and monitoring progress, organizations can strike a balance between sustainability and cost management, leading to long-term success.

6. Overcoming Challenges in Sustainable Cost Management

Managing costs while also implementing sustainable practices can present various challenges for organizations. However, with the right strategies and approaches, these challenges can be overcome, leading to long-term benefits for both the environment and the bottom line.

1. Resistance to Change: One of the significant challenges in implementing sustainable cost management practices is resistance to change within the organization. Employees and stakeholders may be hesitant to adopt new processes or technologies, fearing disruptions or increased costs. To overcome this challenge, it is crucial to communicate the benefits of sustainable practices effectively. Highlighting cost savings, improved efficiency, and positive environmental impacts can help build support and alleviate resistance. Additionally, involving employees in the decision-making process and providing training and education can help overcome resistance and foster a culture of sustainability.

2. Limited Budget and Resources: Limited financial resources can be a significant barrier to implementing sustainable cost management strategies. However, it is essential to recognize that sustainable practices can lead to long-term cost savings. Organizations can start by identifying low-cost or no-cost initiatives that can deliver immediate benefits. For example, implementing energy-efficient lighting, optimizing waste management systems, or encouraging employee engagement in sustainability initiatives can yield significant cost savings without substantial upfront investments. By gradually reallocating funds saved from these initiatives, organizations can further invest in more comprehensive sustainable cost management strategies.

3. Lack of Data and Metrics: Accurate data and metrics are essential for effective cost management and sustainability tracking. However, many organizations struggle to collect and analyze the necessary data to measure the environmental and financial impacts of their initiatives. To overcome this challenge, investing in data collection and management systems can provide valuable insights into energy consumption, waste generation, and other sustainability metrics. These insights can enable organizations to identify areas for improvement, track progress, and make data-driven decisions.

4. Supplier Collaboration: Sustainable cost management extends beyond an organization's internal operations and involves collaboration with suppliers and partners. However, ensuring that suppliers adhere to sustainable practices can be challenging. To overcome this challenge, organizations can establish clear sustainability criteria when selecting suppliers and partners. Regular communication and collaboration with suppliers can help align goals and drive sustainable improvements throughout the supply chain. Additionally, sharing success stories and best practices with suppliers can motivate them to adopt and implement sustainable practices, further strengthening the organization's overall sustainability efforts.

5. Regulatory Compliance: Meeting *regulatory requirements* can pose challenges for organizations striving to implement sustainable cost management strategies. However, compliance with environmental regulations is crucial for long-term sustainability and avoiding penalties. Staying updated with changing regulations and ensuring adherence is essential. Collaborating with industry associations, attending sustainability conferences, and engaging with regulatory bodies can provide valuable insights and guidance on compliance. Organizations can also leverage sustainability certifications and standards to demonstrate their commitment to sustainable practices, gain a competitive edge, and build trust with stakeholders.

overcoming challenges in sustainable cost management requires a proactive approach, collaboration, and the willingness to embrace change. By addressing resistance, leveraging available resources, collecting and analyzing data, collaborating with suppliers, and ensuring regulatory compliance, organizations can successfully integrate sustainability into their cost management strategies, leading to a more sustainable future.

7. Tools and Technologies for Sustainable Cost Management

1. Cost Tracking Software: Implementing cost tracking software can greatly enhance the efficiency and accuracy of cost management processes. These tools allow businesses to monitor and analyze expenses in real-time, helping to identify areas of overspending or waste.

2. Energy Management systems: Energy costs can be a significant expense for businesses, especially those operating in energy-intensive industries. Energy management systems, such as smart meters and building automation systems, help monitor and control energy consumption, enabling businesses to identify energy-saving opportunities and reduce costs.

3. Life Cycle Assessment (LCA) Tools: LCA tools enable businesses to assess the environmental impacts associated with a product or service throughout its entire life cycle. By considering factors like raw material extraction, manufacturing, transportation, use, and disposal, companies can identify areas where sustainability improvements can be made, while also reducing costs.

4. Supply Chain Management software: effective supply chain management is crucial for sustainable cost management. supply chain management software provides visibility into the entire supply chain, helping businesses optimize logistics, reduce waste, and streamline operations.

5. Data Analytics and Business Intelligence Tools: Data analytics and business intelligence tools can play a vital role in sustainable cost management by providing insights into operational inefficiencies and cost drivers. These tools enable businesses to analyze large datasets and identify patterns or trends that can inform decision-making. For example, using Microsoft Power BI or Tableau, companies can visualize and analyze energy consumption data, enabling them to identify areas of high usage and implement targeted energy-saving measures.

6. Green Building Design Tools: For businesses operating in the construction or real estate sector, green building design tools can help optimize energy efficiency and reduce costs. These tools, such as Autodesk's Building Information Modeling (BIM) software, enable architects and engineers to simulate and analyze the environmental performance of buildings during the design phase. By incorporating sustainable design principles and energy-efficient technologies, businesses can minimize energy consumption and operational costs over the building's lifespan.

7. cost Benchmarking tools: Cost benchmarking tools allow businesses to compare their costs and performance against industry peers, helping to identify areas of improvement and optimize cost management strategies. These tools provide valuable insights into cost structures, pricing models, and best practices, allowing businesses to stay competitive while reducing costs. For example, tools like Costimator or ProBenchmark enable companies to benchmark their manufacturing costs against industry standards, supporting informed decision-making and cost reduction efforts.

Incorporating these tools and technologies into cost management strategies can help businesses achieve a balance between sustainability and cost efficiency. By leveraging data, automation, and innovative solutions, companies can identify cost-saving opportunities, reduce environmental impact, and drive long-term profitability.

8. The Role of Leadership in Driving Sustainable Cost Management

Effective leadership plays a crucial role in driving sustainable cost management within an organization. Leaders set the tone for the entire company and are responsible for creating a culture that prioritizes both sustainability and cost management strategies. By aligning these two objectives, leaders can drive long-term financial success while also minimizing the organization's environmental impact. In this section, we will explore the key ways in which leadership can contribute to sustainable cost management and provide examples, tips, and case studies to illustrate their impact.

1. Championing a Culture of Sustainability:

Leaders must champion a culture of sustainability by integrating it into the company's values and mission. By consistently communicating the importance of sustainability to employees, leaders can foster a sense of responsibility and encourage individuals to find innovative ways to reduce costs while minimizing environmental impact. 2. Setting Clear Goals and Targets:

Leadership should establish clear goals and targets for sustainable cost management, providing a roadmap for the organization to follow. These goals can include reducing energy consumption, minimizing waste, or optimizing supply chain processes. By setting specific targets, leaders can hold themselves and their teams accountable for achieving sustainable cost management objectives. One notable example is Unilever, which has set ambitious sustainability goals, including halving their environmental impact while doubling their business by 2030. This clear vision has guided the company's efforts in reducing costs and improving sustainability across their operations.

3. Empowering Employees:

Leadership should empower employees to contribute to sustainable cost management efforts. By fostering a collaborative environment and encouraging employee engagement, leaders can tap into the collective knowledge and creativity of their workforce.

4. Investing in Training and Education:

Leadership should invest in training and education programs to enhance employees' understanding of sustainable cost management principles. By providing the necessary knowledge and skills, leaders can equip employees with the tools they need to identify cost-saving opportunities and implement sustainable practices. For example, Walmart developed its "Sustainability Leadership Academy" to educate associates about sustainability-related topics, resulting in significant cost reductions and environmental benefits across their supply chain.

5. Monitoring and Reporting Progress:

Leadership should establish robust monitoring and reporting mechanisms to track progress towards sustainable cost management goals. By regularly reviewing performance and sharing results with stakeholders, leaders can identify areas for improvement and celebrate successes. This transparency fosters accountability and motivates employees to continue their efforts. A case in point is Interface, a global carpet manufacturer that tracks its progress through their "Mission Zero" initiative, which aims to eliminate any negative environmental impact by 2020. Through their reporting system, Interface has been able to continuously improve their sustainability practices and reduce costs.

**Environmental Accounting**

Environmental accounting refers to the method that involves organizations utilizing conventional principles of accounting and finance to factor in environmental costs in their business results. The primary goal of this approach is to assist organizations in managing and understanding the potential trade-offs between conventional environmental and economic goals.

This technique is essential for businesses to be responsible toward society and remain sustainable. It allows them to comprehend how their actions impact the environment and know the associated costs by factoring in the environmental problems in their financial decisions and reports. This makes businesses more accountable and transparent. The approach has different types like sustainability and full cost accounting.

- Environmental accounting refers to an approach taken by organizations to track the effects of their actions on the environment. It incorporates environmental conservation and management principles into cost-benefit analyses and reporting practices.
- There are different kinds of environmental accounting, like environmental management and full cost accounting.
- Some environmental accounting benefits include the disclosure of resource utilization and environmental protection.
- There are various disadvantages of environmental accounting, such as the non-recognition of the expenses a company incurs to protect the environment and the inconsistent approach concerning the treatment of natural capital.

Environmental accounting refers to a kind of accounting that involves including environmental costs in organizational results. It allows businesses to provide their external stakeholders with the necessary information concerning financial performance. In other words, this accounting approach helps prepare financial reports for lenders, investors, and other parties interested in such information.

Typically, businesses utilize three methods to carry out this accounting process. They are managerial, financial, and national income accounting. Organizations, regardless of their size, can implement this technique. That said, a business's senior management team needs to support such practices. This team is best served if different cross-functional teams are developed to administer the procedure.

The cross-functional teams comprise employees across every business line, including sales, finance, procurement, and manufacturing. Such teams ensure all environmental accounting procedures and policies are followed and communicated.

## Objectives

Some key objectives of this approach are as follows:

- Fostering stakeholder participation and engagement with regard to environmental decision-making.
- Encouraging accountability and transparency concerning the management of environmental effects and utilization of natural resources
- Offering a detailed view of the economic activities' true benefits and costs by incorporating social and environmental considerations
- Supporting the switch to a green economy and promoting sustainable development
- Integrating the environmental benefits and costs into decision-making procedures and national accounts.

## Types

The different types of this accounting approach are as follows:

### #1 - Environmental Management Accounting

It focuses on resource management and internal decision-making through the identification, analysis, and measurement of environmental costs and the benefits of an organization's operations. Moreover, it concentrates on incorporating the data into the management systems to enhance environmental performance.

### #2 - Sustainability Accounting

This involves measuring and reporting on an organization's social, environmental, and economic performance, utilizing indicators like energy and resource, social impact assessments, and greenhouse gas emissions. This accounting method's objective is to offer a holistic view of an organization's sustainable development. Moreover, it aims to promote stakeholder engagement and spot areas for improvement.

### #3 - Full Cost Accounting

This technique involves computing a service or product's overall social, environmental, and economic costs, including hidden charges, which are generally not considered when conducting financial accounting. This process enables a more detailed assessment of the actual consumption and production cost. Moreover, it can help organizations make more informed business and policy decisions to minimize negative environmental effects.

Besides these, there are some forms of this accounting approach, like —

**# 4 - Environmental National Accounting**

This national-level accounting process specifically focuses on externality costs, environmental costs, and more.

**#5 - Environmental Financial Accounting**

It specifically focuses on reporting the environmental liability costs besides other important environmental costs.

**Advantages And Disadvantages**

**Advantages**

- This accounting method encourages innovation and enhances resource efficiency.
- It discloses details regarding the utilization of a nation's natural resources. Simply put, it reveals the cost of the total amount of resources a company uses and the income it generates by using them.
- This technique helps gauge overall contributions made by multiple companies or corporations to fulfill their social responsibilities.
- It offers environmental protection by helping organizations to become more environmentally friendly.
- This approach can improve customer loyalty and enhance corporate reputation.
- Another key environmental accounting benefit is that it helps follow environmental regulations the government sets.

**Disadvantages**

- A noteworthy limitation of environmental accounting is the non-recognition of the expenditure incurred to safeguard the environment.
- The national accounts system treats the slow depletion of any physical asset, like equipment or a plant, as depreciation. That said, the exhaustion of natural capital, specifically the forest, is considered income. Thus, the approach is inconsistent.
- There isn't any clear definition of environmental costs. This means companies are not reporting and monitoring such costs.

**Difference Between Conventional Accounting And Environmental Accounting**

Individuals new to the accounting field can find conventional and environmental accounting confusing. They must know their key differences to understand their meaning in purpose fully.

| Conventional Accounting | Environmental Accounting |
|---|---|
| It deals with the economic unit's physical aspects. | This concept deals with the relationship between the environmental and financial aspects. |
| Its objective is to save money and time against serious business mistakes. | This form of accounting aims to determine the favorable and unfavorable effects of the actions conducted by an organization. |
| The output includes management accounting reports and financial statements. | This accounting method's output includes environmental reports. |

**1. Elements of environmental accounting**

There are various elements of this accounting approach.
Environmental management accounting
Life cycle assessment
Environmental costs and benefits
Carbon Foot printing
Environmental performance indicators
Another key element of this accounting method one must know to comprehend and execute the process is environmental auditing.

**2. Basic characteristics of environmental accounting**

- Identification of costs
- Comprehensive data collection
- Performance tracking and benchmarking
- Stakeholder engagement
- Meeting the environmental regulations
- Risk evaluation
- Merger with financial accounting
- Identification of the benefits

**3. Internal and external functions of environmental accounting**

The internal ecological accounting systems can accumulate information regarding ecological systems for the management's internal use. Such information is expressed in physical units. On the other hand, external ecological accounting systems can accumulate and reveal information for external shareholders having an interest in environmental issues.

## Carbon footprint accounting

### Introduction

The phenomenon of global warming caused by extensive greenhouse gas emissions has become a widely recognized issue of climate change, drawing global attention. There is increasing evidence that carbon dioxide emissions result in higher social costs  Statistics indicate that nearly three-quarters of greenhouse gas emissions are derived from energy consumption. The manufacturing industry has become a core sector of analysis for achieving the “dual-carbon" (carbon peak and carbon neutrality) goals, and accounting for the carbon footprint of the manufacturing industry serves as the foundation and core of energy conservation and emission reduction efforts.

The carbon footprint refers to the collective greenhouse gas emissions caused by enterprises, organizations, activities, products, or individuals, and is used as a measure of the extent of global warming impact. Currently, the main carbon footprint accounting methods include input-output analysis (IOA), life cycle assessment (LCA), IPCC method, Kaya carbon emission identity equation, among others. All data related to carbon footprint accounting are collectively referred to as carbon data, which are characterized by uncertainty, dynamics and coupling, and have a significant impact on the results of carbon footprint accounting.

The LCA method tends to analyze simple scenarios under certain assumptions (e.g., comparing different assessment methods, results of multiple electricity combinations and evaluating key parameters and sensitivities, etc.), However the reliability of the LCA assessment results is not high for the complex and dynamic processes of change in actual production, mainly due to the fact that manufacturing product carbon footprint accounting has long been affected by the uncertainty of carbon data. The uncertainty of LCA results can mislead decision makers. Therefore, addressing and minimizing the uncertainty associated with carbon data has become crucial for accurately accounting for the carbon footprint of products.

In recent years, domestic and international scholars have conducted extensive research and analysis on carbon footprint models during the product manufacturing stage and throughout the entire product lifecycle stages. The research on the existing product carbon footprint quantification model can calculate the static carbon footprint at each stage of the product's entire life cycle. However, in addition to the changes in the organization's normal business activities, the product will also be affected by production

disturbances (equipment failure, delayed/incorrect material delivery, urgent orders, etc.) during the manufacturing process, and the product carbon footprint will then change dynamically. In recent years, reducing the impact of disturbances, optimizing manufacturing processes, and reducing carbon emissions have become hot research topics analyzed that the flexibility and variability of task flows are the main influencing factors for dynamic changes in energy consumption during the production process. Zhou et al. (2022) proposed a workshop dynamic scheduling strategy under implicit interference to reduce additional carbon emissions caused by energy waste. Cao and Li (2014) discussed the effects of disturbances such as machine failures, rework, and tool failures on the dynamic carbon emission characteristics of manufacturing systems. In traditional static carbon footprint accounting for products, the dynamic changes of carbon data caused by the overlay of business activities and disturbances are often overlooked, and the result of the product carbon footprint is only a fixed value under a specific situation. Therefore, analyzing the dynamic of carbon data from organizations to products to reduce the uncertain impact on the product carbon footprint is particularly important.

For carbon data in the product lifecycle, the accessibility and allocation of carbon data are crucial issues. If there is a complete measurement and acquisition method for carbon data, there is no need to worry about the accuracy of carbon data. However, due to the difficulty and cost of obtaining carbon data, the availability of carbon data will be limited. In the inputs and outputs of the product life cycle, carbon data of different sources and types are coupled to each other and cannot be directly used for product carbon footprint accounting, and carbon data allocation is required. The commonly used methods internationally are the three methods for inventory data allocation provided by the International Organization for Standardization (ISO): (1) Whenever possible, use subdivision or system expansion methods to avoid allocation issues; (2) If allocation issues cannot be avoided, use methods based on physical (chemical or biological) relationships between inventory data for allocation; (3) If the above methods are not feasible, use allocation approaches based on other methods such as economic value (ISO 14040, 2006; ISO 14044, 2006). At the same time, Guinée et al. (2009) studied the production chain of bioelectricity using different allocation methods such as mass, energy, and economic value, and pointed out that in the case of prioritizing consistency, the allocation method based on physical relationships is the first choice, but based on economics or system expansion methods are more effective, although they may be affected by insufficient data. DU et al. (2022) examined the advantages and disadvantages of the seven allocation methods for the different LIB products and applied them to the LIB recycling and remanufacturing process, resulting in different LCA results. Ma et al. (2021) proposed a multi-level data collection and allocation method for part level, workshop level, and enterprise level, matched each level with each life

cycle stage of the product, and established a machine tool product life cycle inventory acquisition model. The issue of carbon data allocation has long been a focus of scholars, but Schrijvers et al. (2016) indicate that allocation has however always been one of the most debated issues in LCA, and it is difficult to identify the correct allocation method.

Meanwhile, the existing ISO 14040-ISO 14044 standards also face challenges, particularly in the allocation and merging of multi-level carbon data from the organizational-level to the product-level, and reducing the uncertain impact caused by the allocation method is the most concerned issue. Therefore, it is essential to integrate the acquisition and allocation of carbon data to ensure its usability, reduce the coupling between carbon data and the uncertainty brought about by carbon data allocation.

To sum up, the uncertainty, dynamics, and coupling of carbon data are significant factors that cannot be ignored, as they contribute to the current issues of carbon data scarcity and poor quality. Reducing the uncertainty of carbon data from its source, considering the dynamic impact of activities and disturbances on carbon data, and dynamically allocating coupled carbon data based on activities can effectively improve the quality of carbon data and enhance the reliability of product carbon footprint accounting results. Therefore, it is important to carry out research on the carbon footprint accounting method based on the dynamic allocation of multi-level carbon data from organizations to products in manufacturing organizations, with the aim of reducing the carbon data problems in carbon footprint accounting from the perspective of carbon data acquisition, classification, allocation, and merging.

**Cloud-based Solutions**

Cloud-based solutions leverage internet-based services to provide scalable, flexible, and cost-effective resources for businesses, transforming how they manage data and applications.

**What Are Cloud-Based Solutions?**

Cloud-based solutions refer to services and resources accessed over the internet rather than being hosted on local servers or personal computers. This model allows users to store, manage, and process data remotely, providing significant advantages in terms of accessibility and efficiency. Cloud solutions typically fall into three main categories:

1. Infrastructure as a Service (IaaS): This provides virtualized computing resources over the internet. Users can rent servers, storage, and networking capabilities without investing in physical hardware. Examples include Google Compute Engine and Microsoft Azure.

2. Platform as a Service (PaaS): PaaS offers a platform allowing developers to build, deploy, and manage applications without dealing with the underlying infrastructure. This streamlines the development process and enhances productivity.
3. Software as a Service (SaaS): SaaS delivers software applications over the internet, eliminating the need for installation and maintenance on local devices. Popular examples include Google Workspace and Salesforce.

**Benefits of Cloud-Based Solutions**

- Cost Efficiency: Organizations can reduce IT costs by avoiding the need to purchase and maintain hardware. Instead, they pay for the services they use, often on a subscription basis.
- Scalability: Cloud solutions allow businesses to easily scale their resources up or down based on demand, ensuring they only pay for what they need.
- Flexibility and Accessibility: Users can access cloud services from anywhere with an internet connection, facilitating remote work and collaboration.
- Enhanced Security: Many cloud providers offer robust security measures, including data encryption and regular backups, which can be more effective than traditional on-premises solutions.

**Deployment Models**

Cloud solutions can be deployed in various ways, depending on the needs of the organization:

- Public Cloud: Services are offered over the public internet and shared among multiple users, making it cost-effective and scalable.
- Private Cloud: Dedicated to a single organization, providing greater control and security, often necessary for businesses with strict compliance requirements.
- Hybrid Cloud: Combines public and private clouds, allowing businesses to maintain sensitive data in a private environment while leveraging the scalability of public resources for less critical operations.

In summary, cloud-based solutions represent a significant shift in how businesses operate, offering flexibility, cost savings, and enhanced capabilities that traditional IT infrastructures cannot match. As organizations continue to adopt these technologies, understanding their benefits and applications becomes increasingly important.

## Blockchain Technology and Cost Accounting

The introduction of distributed ledger technology, or what is known as blockchain technology, has challenged current methods of doing business. One of these challenges is the significant shift in the roles of the cost accountant. Because cost accounting functions are closely connected to organisations' transactions which are translated into monetary values, they are affected by any new technology adopted. If the adoption of new technologies such as blockchain changes the process of conducting transactions, cost accountants should understand the changes to correctly reflect them in their work. If they do not, they will not be able to understand the controls included in the technology and trace financial transactions recorded using the blockchain, meaning they would not be able to correctly perform most of their work.

## The Effect of New Technology

The basics of cost accounting are not affected by the implementation of blockchain technology; however, blockchain does add risk to consider and controls to test. To prevent this, cost accountants should learn about blockchain technology and include it in their functions. There are two assertions that cost accountants should give assurance on: completeness of recording all transactions; and the existence of costs of recorded transactions. Completeness is achieved when all transactions that occurred during the period are presented in the accounts, and this assertion can be duly verified.

For example, an entity's use of cryptocurrency in its operations can be verified by reviewing the transaction history of that entity in the blockchain. Similarly, auditors can verify that assets exist in the entity by tracing the transaction hash in the blockchain to identify the receiving party of the asset inside the entity and then confirming that the asset resides in the location described. The process of verifying other assertions can be done using the data stored in the blockchain; the difference is that these data are verified and can thus be considered more trustworthy than the traditional data provided by one party.

## Smart Contracts

A smart contract is a code that reflects the terms and conditions of a contract between the entity and the supplier and executes itself without any human intervention. For operations records, the cost accountant may review the flow of procedures by tracing all internal transactions in the blockchain that occurred during a specific period and ensuring that all objectives were met. In addition, they may test for governance issues in the blockchain by verifying who is responsible for creating transactions, who can verify transactions, and which party is responsible for reviewing transactions.

In cost accounting, blockchain technology can offer organisations more verifiable transactions with internal parties. This kind of verification can also be useful to auditors when they trace transactions to verify their authenticity.

## How Smart Contracts Work

Smart contracts work by following simple "if/when...then..." statements that are written into code on a blockchain. When predetermined conditions are met, a network of computers executes the actions specified in the contract. For example, a smart contract could automatically release funds to a party once a service is confirmed as completed.

**Key Characteristics**

1. Self-Executing: Smart contracts automatically enforce and execute the terms of the agreement without the need for human intervention.
2. Immutable: Once deployed on the blockchain, smart contracts cannot be altered, ensuring that the terms remain unchanged.
3. Distributed: All parties on the network have access to the contract, which enhances transparency and trust.
4. Deterministic: The outcome of a smart contract is consistent and predictable, as it will execute the same way every time under the same conditions.

**Applications**

Smart contracts are commonly associated with cryptocurrencies, particularly Ethereum, which introduced a more robust framework for creating complex contracts using a Turing-complete programming language. They are fundamental to decentralized finance (DeFi) applications, non-fungible tokens (NFTs), and various other blockchain-based solutions.

Smart contracts represent a significant innovation in how agreements are executed and enforced, providing a secure, efficient, and transparent method for conducting transactions in the digital age. They are reshaping industries by automating processes and reducing reliance on traditional legal frameworks.

Limitations of current accounting practices allow for payments to sometimes be made to fake suppliers or transferred to fake bank accounts. For example, a fraudster can pose as an organisation's supplier and send an invoice with its own account number to the organisation. Or it could hack the email system of a supplier and use it to send invoices and bank details to the victim organisations. However, this risk is removed when payments are processed by blockchain technology because all payments are verified by the supplier and recorded in the block with a transaction hash for later reference. Risk is also minimised because both the supplier and the organisation can see the history of all of their transactions in the network, which gives them the ability to identify any fraudulent activity.

Another accounting aspect that blockchain has changed is related to the delivery of goods and services to customers. The receipt of goods and services can now be confirmed by customers directly by verifying the delivery transaction in the blockchain. In addition, the execution of contracts with suppliers can be completed using smart contracts in the blockchain. This execution is done according to the terms and conditions of the contract embedded in the code of the smart contract in the blockchain.

## Conclusion

Blockchain technology is like any new technology utilised by organisations: It necessitates the development of the cost accountant's skill sets to understand, manage and assess new types of transactions. Entities implementing blockchain technology should train accountants to use it to initiate transactions, verify these transactions and trace the transaction histories. Failure to do so will prevent the cost accountant from recording and reviewing transactions when new technology becomes the sole technology adopted by organisations to record and verify financial transactions. In addition, cost accountants will not be able to test controls embedded in the blockchain if they are not competent in using the technology; this also will affect competency.

**MULTIPLE-CHOICE AND SHORT ANSWER QUESTIONS**

1. In - relation to Just-in-Time (JIT) systems, what does "demand-pull" mean?

   A. Each process in production is linked to the other

   B. The system is based purely on demand and supply

   C. The production chain is activated real-time by demand

   D. Goods are produced before demand

   E. Production deadlines are strictly monitored

2. Kanban systems are a step ahead of JIT because it ensures that……..

   A. Intermediate inventory held in the supply chain are better managed

   B. Materials suppliers are kept on their toes at all times

   C. There are no opening or closing inventory

   D. Much factory space is saved

   E. There are no production wastages

3. In backflush costing,……..

   A. Every accounting entry is flushed out of the system

   B. There is no need for detailed tracking of material movement through stores and production

   C. There is a separate account for work in progress

   D. Financial records are not necessary

   E. The cost ledger is fully automated

4. **Which** of the following costs is **NOT** part of the life cycle costing?

   A. Product promotion costs

   B. Manufacturing costs

   C. Research and development costs

   D. Product developments costs

   E. Public relations costs

5. Which of the following is the main objective of Advanced Manufacturing Techniques (AMT)?

   A. Production of high quality goods at low cost, thereby maximizing customer satisfaction

   B. Large scale production of durable goods

   C. Uninterrupted production

   D. Reduction of cost of labour

   E. Industry cooperation

6. A system which seeks to eliminate the need to keep inventories by producing or procuring inventory at the time they are required is known as ……...

7. A feature of Kanban systems which eliminates common manual entry errors and provides quick and precise information is called ……..

8. A system which seeks to reduce or eliminate detailed accounting entries by starting from the finished goods and working backwards is known as ……

9. The total cost of production from research and development stage through to the end of the product can be recovered over the estimated number of units it expects to sell through a costing system known as ……..

10. A product cost estimate derived by subtracting a desired profit margin from a competitive market price is called ……..

**SOLUTION**

1. C
2. A
3. B
4. E
5. A
6. Just in-Time systems
7. Electronic Kanban (e-kanban)
8. Backflush Accounting
9. Life cycle costing
10. Target costing

## CHAPTER FIFTEEN: COST REDUCTION AND CONTROL

## CHAPTER CONTENTS

a. The concepts of cost control and cost reduction

**b.** Cost reduction

c. Differences between cost reduction and cost

d. Value analysis and value engineering

e. Organisation and Methods (O&M)

f. Work Study

g. Difference between method study and work measurement

h. Cost Audit

**15.0 Objectives:** After studying this chapter, readers should be able to:

a. Explain the terms "cost reduction" and "cost control";

b. Differentiate between cost reduction and cost control. c

Explain value analysis and value engineering.

d. Explain Organisation and Methods (O & M).

e. Explain work study and differentiate between method study and work measurement.

f. Explain cost audit

### 15.1 INTRODUCTION

The change in business environment has compelled organisations to adopt methods that will assist them in keeping costs to the barest minimum so as to remain competitive and profitable. The important consideration here is that new technology lowers cost and improves quality.

## 15.2 Cost Reduction

Periods of economic recession are usually characterized by reduction in products' demand and cost control will not be sufficient to improve demand. What will be required for a firm to remain in business and compete favourably at such times, is cost reduction.

According to the terminology, cost reduction programme is defined as "the real and permanent savings in project operation cost by achieving a reduced cost without impairing consumers' satisfaction". Cost reduction programme can take the form of:

a. Crash programme
The programme is not planned.
Examples of crash programme include expenditure cutting, staff retrenchment and capital expenditure deferment; and

b. Planned Programme.
In planned programme, operations are planned and monitored to ensure compliance.

### 15.2.1 Comparison between cost control and cost reduction

Cost control ensures that actual costs incurred are in conformity with the target set. Cost control technique comprises standard costing, budgetary control and capital expenditure. Cost reduction is concerned with permanent reduction in cost without impairing the quality of products and the efficiency of operations. The major reason behind the introduction of cost reduction is because cost control alone cannot guarantee the survival of a firm in a period of economic recession.

### 15.2.2 Where to Apply Cost Reduction

Generally, when a new product is to be designed, the firm should introduce cost reduction at the product design stage. Thereafter, design will also be introduced during the production process stage. At product design stage, it is important to also determine the appropriate facilities to acquire for production. What to reduce will include:

i. the use of cheaper materials that will not impair product specification.

ii. reduction in ordering and holding costs.

iii. reduction in re-order period or lead time.

iv. control of overtime work
v. adoption of merit rating for promotion

vi. adequate utilisation of production capacity

vii. elimination of non-value time

## 15.3 Value Analysis

Value analysis is an approach to improving the value of a product or process by understanding its constituent components and their associated costs. It then seeks to find improvements to the components by either reducing their cost or increasing the value of its functions. The terminology defines value analysis as "A systematic interdisciplinary examination of factors affecting the cost of a product or service, in order to devise means of achieving the specific purpose most economically at the required standard of quality or reliability.

The following are the key concepts of value analysis:

a. Value is the ratio between a function for customer satisfaction and the cost of that function;

b. Function is the effect produced by a product or one of its elements, in order to satisfy customer needs; and

c. Need is something that is necessary or desired by the customer.

Value analysis is therefore a method of increasing the value of an object. The object to be analysed could be an existing one or a new product or process. It is usually accomplished by a team following a work plan. The main purpose of value analysis is to improve every component in a product.

Value analysis follows the following process in steps:

i. Orientation/preparation;

ii. Information;

iii. Analysis;

iv. Innovation/creativity;

v. Evaluation; and

vi. Implementation and monitoring.

The application of value analysis only needs the use of basic techniques such as matrixes, pareto chart, pert and gantt diagrams, etc, in most of the steps.

The producer measures value analysis in terms of the relationship between product worth and product cost i.e (product worth ÷ price paid). The higher the outcome of this, then the higher is the satisfaction to the customer.

Advantages of using value analysis techniques include:

- A high customer orientation, focusing on those aspects of the product/service that better satisfy customer needs;
- Cost reduction by eliminating functions that do not supply specific advantages to satisfy customer requirements;
- New ideas that arise from the creativity/innovation phase and may add radical changes and therefore competitive advantages that will be regarded by the market; and
- A new systematic mentality to be taken into account for next designs of new products or to systematically improve the existing ones.

The problems that may arise during the application of value analysis can be of different nature. In order to arrive at a successful completion of the process, one has to bear in mind the following rules:

- Avoid making generalisations and superficial statements. It is important to be precise at every moment;
- Collect, determine and examine all costs involved only when one is cost conscious will it be possible to determine the value of the thing being assessed; and
- Make use of information from the best possible sources.

## 15.4 Value Engineering

Value engineering is an organized approach to providing the necessary functions at the lowest cost. From inception, the concept of value engineering was seen to be cost validation exercise, which did not affect the quality of the product.

The fact that it does not affect the quality of the product in question has now been seen as a shortcoming hence the need to adopt a better definition. The concept is now described as:

*an organized approach to the identification and elimination of unnecessary cost.*

According to the terminology, it is defined as "A redesign of an activity, product or service so that value to the customer is enhanced while costs are reduced (or at least, increased by less than the resulting price increase)".

**Unnecessary Cost:**

A cost which does not provide use, life, quality, appearance, of customer features or add to customers' satisfaction is an unnecessary cost.

Value engineering practice involves the following tasks:

a. Preparing and administering maintenance programmes;

b. Forecasting expenditure flows;

c. Advising on cost limits and preparing budgets

d. Advising on cash flow forecasting;

e. Advising on life cycle costing;

f. Cost analysis;

g. Cost benefits analysis;

h. Estimating;

i. Evaluating alternative designs;

j. Investment appraisal; and

k. Measuring and describing construction work but only in terms of cost planning

## 15.5 Organisation and Methods (O & M)

Organisation and methods originally came from the pioneers of scientific management (Taylor and Gilbreth). Their work influenced the early approaches and establishment of the function.

O & M is defined as *the systematic examination of activities in order to improve the effective use of human and other material resources.*

Essentially, it is a specialist function that has a primary objective of improving an organisation's efficiency and control. In this way, it can be seen as an essential function that should be part of the make-up of any organization. O & M and its associated techniques form the basis for:

i) Business Process Re-engineering; and

ii) Business Process Improvement.

Seeing that other line management do not have the time or skills required to provide the service, O & M therefore comes handy as a consultative service to management. Staff and management working daily on a process may not be able to think very far beyond the process thereby miss what may be

obvious improvement opportunities. Therefore the application of O & M principles will bring a fresh outlook into the business process(es).

Until recently, it was possible to find O & M departments established within as a clearly identifiable unit in organisations. However, in recent times, O & M function has been renamed Project Management or Business Improvement in many organisations. In many organizations today, O & M function has been subsumed into other functions, mostly IT.

O & M is indispensable seeing that it can provide a basis for the approach to almost any project. Steps to be followed:

a. Select the area/process that requires attention;

b. Record the current situation;

c. Examine and analyse the existing situation;

d. Develop, design and evaluate alternative solutions and recommend improvement opportunities;

e. Implement the chosen solution; and

f. Monitor and maintain the implemented solution.

## 15.6 Work Study

**Work study** can be described a system of increasing or maximizing the productivity of an operating unit by reorganising the work of that unit. Work study is sub-divided into two major methods namely:

**i)** methods study; and

**ii)** work measurement.

### a. Method study

This is described as the recording and critical examination of existing methods of doing work and comparing same with proposed methods with a view to coming up with easier methods which would be more effective and cheaper on the long run.

### b. Work measurement

As the name suggests, work measurement seeks to measure the time required for a qualified worker to complete a specific assignment at a specified level of performance.

## 15.7 Cost Audit

All organisations should strive to ensure that their cost structure should correctly reflect operational performance. However, this can only be achieved where cost audit is carried out as a continuous process. Cost audit assists immensely in the process of cost reduction.

The CIMA official terminology defines Cost Audit as "Verification of cost records and accounts, and a check on adherence to the prescribed cost accounting procedures and their continuous relevance".

Cost Audit is a continuous process of auditing the current operations and comparing the current audit results with the previous audit tasks. The two forms of cost audit are:

a. Efficiency or Performance Audit; and

b. Propriety Audit

**Efficiency Audit:**

The importance of efficiency audit lies in its ability to ensure that:

i) the organisation's plans have been carried out in an efficient manner; ii)

costs incurred are not overstated; and

iii) organisation's resources are properly utilised.

**Propriety Audit:**

This is concerned with investigating management plans with respect to financing. The core aspect of propriety audit is to ensure that the management:

i) takes right decisions; and
ii) establishes proper control and monitoring activities.

## 15.8 Differences between Cost Audit and Financial Audit

i. Financial Audit may be statutory in nature (compulsory for all PLCs) and in such instances must be carried out by external auditor whereas, cost audit is non- statutory, and can be carried out by persons appointed (internally or externally) by the management.

ii. Financial audit is based on specific past records, which is considered as a post-

mortem check, while cost audit is based on current and continuous operations.

iii. Financial audit tries to verify financial records while cost audit verify cost accounting records

## 15.9 Cost Audit Programme

A reliable cost audit programme must be designed for the entire organisation. The audit programme designed should be updated from time to time to cope with changes in organisation activities. The programme should cover all the three elements of costs and other relevant areas. For instance, the audit programme for material transactions may involve checking:

- procedures adopted in selecting suppliers.
- reliability of material classification and codification in order to ensure expenditure can be easily traced.
- whether materials are purchased economically.
- the control established in the store and the security established for store items • estimate for normal loss and ensuring that abnormal gain receive proper

  accounting treatment.
- methods adopted for issuing materials from stores to the factory.
- whether there is quality control unit for testing materials purchased and finished goods produced

## 15.10 Cost Control and Cloud Systems

As desirable as moving computing infrastructure to the cloud is, staying within budget can be a serious challenge. The variables include the amount of storage being used, computing demand, and the application running on each cloud instance, etc.

It is important to know all cloud costs and service requirements so that a good balance is struck between the money being spent on cloud infrastructure and the company's needs.

Cloud cost management is the process organisations use to keep cloud costs under control. Whether the organisation decides to leave all its cloud computing requirements to a single vendor or share them among multiple vendors, controlling of costs is very difficult to manage. The organisation may be lost in the spiral of what services it is enjoying and whether it is paying not too much for it.

Here's where cloud cost management comes in. Cloud cost management tools help companies control their spending by monitoring cloud use and demand. These tools can alert IT managers about cloud resources that are underutilised or that have become unexpectedly expensive.

**Factors that contribute to cloud costs**

- Resource usage: The more resources—computing power, storage, bandwidth—each cloud instance uses, the higher the bills will be. A company that doesn't keep an eye on resources may be surprised by unexpectedly high charges.
- Instance types: Virtual machines (VMs) and containers have different costs, and cloud providers charge depending on the contracted performance of each instance. A VM is a computing resource that uses software to emulate an instance of an operating system, rather than deploying an entirely new hardware server. A container is a prefabricated set of software that runs a particular application when needed. Several containers can run on a single VM, so containers are usually less expensive, although usage fees can eat up the difference.

- Storage costs: The amount of data stored online and the type of storage used will affect the size of the bill. Also, costs are associated with moving data into and out of storage. A data-intensive business—particularly one that frequently uploads and downloads data from the cloud—can incur much charges.

- Autoscaling: Some cloud services include automatic monitoring that scales available resources up or down on demand. This makes sense because surges can be expensive; think of a company that advertises on social or broadcast media and has a product or service go viral. Autoscaling, though expensive, can allow a company to handle traffic spikes that might otherwise cripple its servers when performance is needed the most.

- Geographic location: Many cloud service providers have data centers all over the world. Some offerings cost more than others, and a company's decision (or indecision) about where to host its services may affect what it pays. Hosting data far from where it's used—say, in Asia when the customer base is in the United States—may be less expensive than using geographically closer data centers because of lower-cost resources, utilities, and infrastructure in remote locations. Additionally, some countries offer tax incentives or government subsidies that can further reduce costs. However, it's important to balance performance trade-offs, such as increased network latency and slower response times, as well as security and regulatory issues with cost savings.

- Support: If a cloud contract includes support or if support is required on an ad hoc basis, it frequently comes at an additional cost. Support guaranteed by contractual service level agreements (SLAs) may be expensive—until you need it, at which point it can be priceless. Still, not every cloud instance requires 99.9% uptime with 24/7 telephone support.

**Advantages of Cloud Cost Management**

- Cost cutting: When a business tracks what it spends on cloud infrastructure, it can eliminate inefficiencies and optimise spending. Companies with poor IT governance often buy duplicative services and pay for resources that have long since outlived their usefulness. Cloud cost management can help find waste and, even better, prevent it in the first place.

- Cost allocation: Often, companies are unable to decipher which departments have incurred which IT expenses. Managing cloud spending lets a company charge costs back to the departments that are using the services. That gives individual departments an incentive to manage spending.

- Budgetary control: In companies where IT governance is lax and costs aren't allocated, casual and uncontrolled IT spending often happens because it doesn't count against any department's budget. Cloud cost management tools help companies understand what they spend on cloud services, making them better equipped to set and adhere to a budget or understand when and where budgets are being exceeded. For both cost allocation and budget control, chargebacks and showbacks—doing the accounting without the actual charges—are a solid way to help everyone understand what resources are being consumed.

- Forecasting: Knowing what you've spent in the past on cloud services provides some understanding of what's required to support corporate initiatives. The ability to manage cloud costs today enables companies to budget for the future.

- Visibility: Managing cloud costs isn't just about money; it's also about being able to control where and how data and other resources are used. Cloud cost management is an important first step in understanding corporate data usage, eliminating duplicative and conflicting databases, and securing corporate and customer information.

- Rightsizing: It's not uncommon for a cloud instance to be created with what turns out to be inappropriate resources—more memory or processing cores than are required to run an application, for instance. A cloud cost management tool can detect those inefficiencies and suggest that a company optimize the instance, saving it money.

**Cloud Cost Management Strategies**

Once a company decides to manage its cloud costs, several strategies can help optimize spending.

1. Remove unnecessary instances: There's no reason to pay for an idle cloud instance, and the meter is generally running whether it's being used or not. This is an easy win: Identify and turn off what you don't need.

2. Set organizational rules: Many companies have had significant success with internal processes to control shadow IT. A similar effort, such as rules that manage who is allowed to contract for cloud services, under what circumstances, and within what spending limits, can keep departments from spinning up unauthorized and wasteful cloud instances. If nothing else, mindfulness in departmental cloud spending, particularly in concert with cost allocation, can go a long way toward reducing costs.

3. Implement autoscaling: Rather than provisioning cloud instances for maximum load, companies can use autoscaling to allow those instances to efficiently accommodate demand surges and dips. Autoscaling may be expensive, but it's less costly than paying for more services than are necessary day in and day out.

4. Schedule demand: Not all instances need to be available 24/7. Shutting down or "sleeping" instances at night or on weekends when no one is likely to use them might save money without affecting service.

**MULTIPLE-CHOICE AND SHORTANSWER QUESTIONS**

1. Which of the following best differentiates cost reduction and cost control?

   A. Cost reduction and cost control are the same

   B. Cost reduction aims to slash costs whilst cost control aims to increase costs

   C. Cost reduction is a scientific method of controlling costs whilst cost control is arbitrary

   D. Cost control seeks to maximize the benefits of every amount spent whilst cost reduction seeks to generally reduce cost of products and services

   E. Cost control is used in respect of products whilst cost reduction is concerned only with services

2. Which of the following best describes Cost Audit?

   A. It is the audit of the cost of materials

   B. It is the rechecking and confirmation of suppliers' prices

   C. It is the process of ascertaining whether every aspect of the business is managed in the most cost-effective way

   D. It is the independent checking of material costs from different sources

   E. It is the complete building up of cost of manufacturing

3. Which of the following best describes Organisation and Methods?

   A. The total overhaul of an organisation and its methods of operation in order to improve profitability

   B. The systematic examination of activities in order to improve the effective use of human and other natural resources

   C. A complete review of production processes with a view to improvement

   D. An independent analysis of structures and policies of an organisation to ensure harmonisation

   E. A periodic review of the effectiveness of production machinery to ensure their optimum output at all times

4. An organised approach to the identification and elimination of unnecessary

costs of running a business is called ……..

5. An approach towards improving the value of a product or process by understanding its constituent components and the associated costs is known as …….

6. A system of increasing or maximizing the productivity of an operating unit by reorganizing the work of that unit is referred to as …….

7. The recording and critical examination of existing methods of doing work with a view to coming up with better and more effective methods is called ……..

8. The measurement of the time required for a qualified worker to complete a specific assignment is known as …….

**SOLUTION**

1. D
2. C
3. B
4. Value engineering
5. Value analysis
6. Work study
7. Method study
8. Work measurement